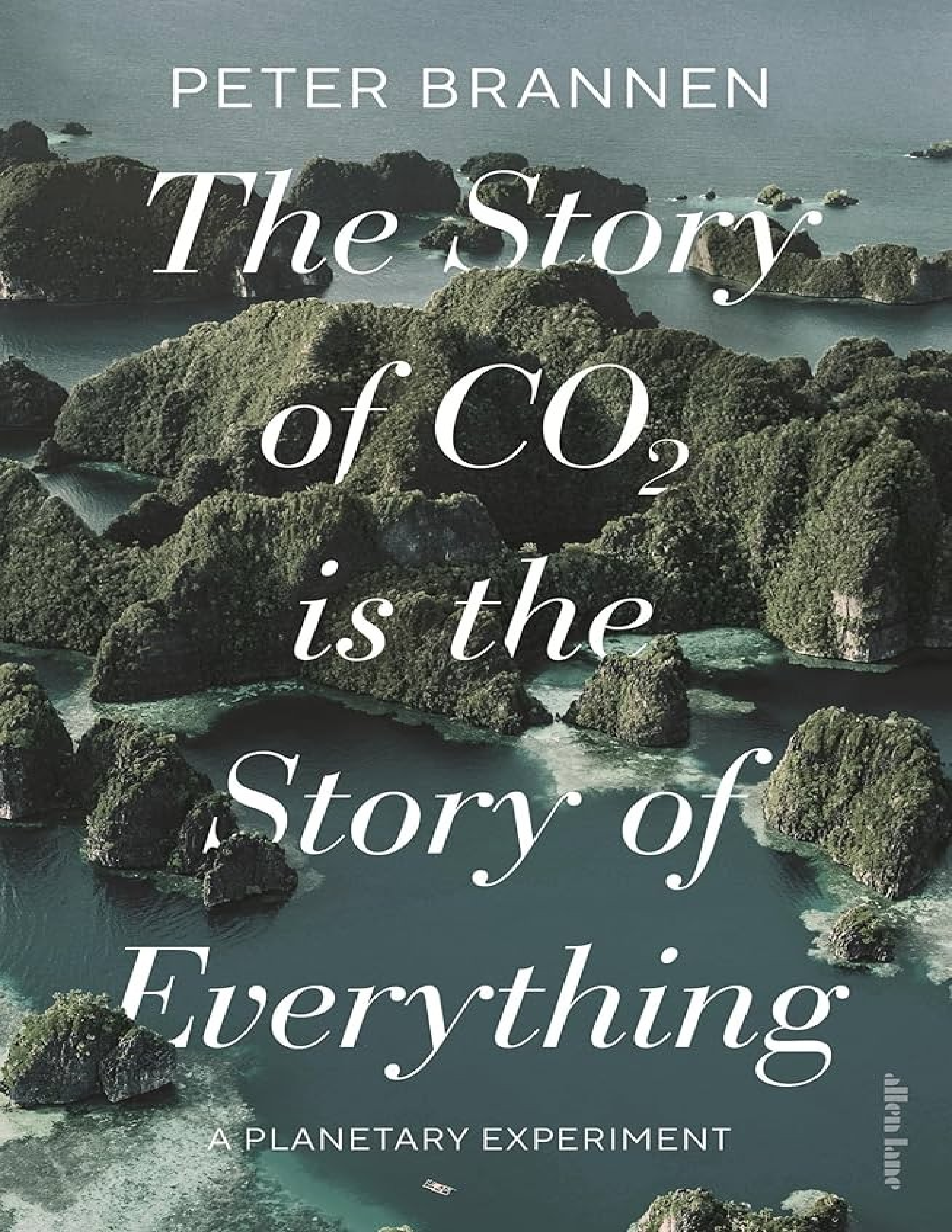

PETER BRANNEN

The Story of CO_2 is the Story of Everything

A PLANETARY EXPERIMENT

allen lane

About the Author

Peter Brannen is an award-winning science journalist and contributing writer at the *Atlantic*. His work has also appeared in *The New York Times,* the *Washington Post,* the *New Yorker, Wired,* and the *Guardian,* among other publications. His first book, *The Ends of the World,* was published in 2017. Peter was a 2024 visiting scholar at the Kluge Center at the Library of Congress, and is an affiliate at the Institute of Arctic and Alpine Research at the University of Colorado-Boulder.

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THE STORY OF CO_2 IS THE STORY OF EVERYTHING

A Planetary Experiment

Contents

To Dad

Carbon dioxide may be thought of as the most important substance in the biosphere …. It has supported the existence and development of life by serving as the source of carbon, the principal element of which all living beings, including geologists and geochemists, are made. In past times it was a source of the free oxygen in the air and the ocean that makes animal life possible. By absorbing and backscattering the heat radiated from the earth's surface … it maintains … a sufficiently high temperature in the air and the sea to allow liquid water, and therefore life, to exist. Earth's uniquely benign environment for living things depends fundamentally, of course, on its relationship to a small, steady star, the sun; but this relationship is modulated in essential ways by carbon dioxide.

—ROGER REVELLE, 1985

The flow of energy through a system acts to organize that system.

—HAROLD MOROWITZ

Introduction

THREE HUNDRED FIFTEEN MILLION YEARS AGO, A PHOTON SLIPPED FREE from the clutches of its parent star and raced 93 million miles across the void. Some 99.9999999 percent of its fellow photons kept on sailing past eternity. A rare few, after billions of miles, bounced off the craters and canyons of some savagely bleak world in the outer solar system. Still others—after a billion billion miles more—mutely scattered off disembodied nebulae, drifting through space. But our photon was headed to the most interesting place in the known universe: the Earth.

After eight minutes at light speed,[fn1] the photon slammed into a leaf and instantly found itself recruited into a frenetic, impossible labyrinth of electrochemical nanomachinery—a whirring, buzzing tangle of life. The sunlight had found itself, in other words, recruited into photosynthesis. The leaf was attached to a bizarre *Lepidodendron* tree—the length of its trunk seemingly sheathed in reptile skin—in an otherworldly Carboniferous Period jungle, haunted by centipedes the size of crocodiles. The leaf absorbed the photon, and many others, through tiny antennae. And it used this solar power, with the help of water, to transform carbon dioxide from the air around it into the stuff of life—effectively conjuring more tree out of thin air. In this way, energy, once forged in the thermonuclear center of the sun and radiated across the vacuum, was diverted into the world of the living, through the chemistry of CO_2. It was stored in wood, the leaves that wreathed this wood, the wax that coated these leaves, the sugars that fueled this tree. For a time at least. Trees like this, after all, get quickly consumed by the rest of the living world, where this solar power continues its flow through the biosphere—animating those same gigantic centipedes busily munching its leaf litter, or the fungi that would disassemble its trunk after death. These life-forms power themselves by burning plants like this in the controlled fires of metabolism, releasing that CO_2 back to the environment.[fn2] And it's a good thing too: without this vast metabolic exhaust of life on Earth, photosynthesis would fatally deplete CO_2 in the atmosphere in a matter of years, and freeze the planet to death.

But just then, with an earsplitting crack, all 160 feet of the lepidodendron fell into the black swamp below. And there it rested, buried in the mire, commended to the Earth for the ages. As the tree sank deeper and deeper into the muck, and then into the crust, the tournament of life carried on far overhead as the solar power captured at the surface by photosynthesis unendingly flowed and dissipated through these ecosystems. Whole regimes of animals came and passed over the face of the Earth—sail-backed dimetrodons, rhino-like reptiles, giant marine mosasaurs that dwelled in seas atop the continents, three hundred feet deep—all perched high above these former swamps, now buried deep in the crust.

The stories of these animal kingdoms on the surface of the Earth, their rises and falls, were shaped by an ever-shifting climate over geologic history. Sometimes these shifts came subtly over tens of millions of years, sometimes catastrophically, over the course of millennia. But this too was the story of CO_2. This is because CO_2 isn't merely the substrate of life; it's also, as one eminent group of climate scientists has called it, the "principal control knob governing Earth's temperature" or, more pithily, "Earth's climate driver." As a result, when CO_2 was high in the geologic deep past, the planet sweltered, and crocodiles and palm trees migrated to a balmy arctic. When CO_2 fell, ice sheets swelled on the poles instead, and the seas fell by hundreds of feet. But not only does CO_2 govern the planet's temperature; it also regulates the chemistry of the entire ocean. When CO_2 spikes faster than the ocean can accommodate, as it is doing today, the seas helplessly acidify—sometimes in Earth history, devastatingly so. Nevertheless, the delicate balance of carbon on this surface world, as it slips between the oceans, air, life, and rocks, has kept the planet habitable over these hundreds of millions of years.

One can imagine this unceasing movement of CO_2 as it wafts in the atmosphere, diffuses into the ocean, is chemically transformed,[fn3] and is ultimately incorporated into the shell of a nautilus bobbing along a reef wall before falling to the seafloor at death and settling into an ooze of other carbonate creatures that will someday become limestone. This seafloor rock, made by life, might then get subducted under the churning edge of a continent, where it dives down into the Earth, is cooked at 1,000 degrees Fahrenheit, and comes out the throat of a volcano as CO_2 once more. Then it does so all over again, this time through the way stations of air, water, plankton, and oil, before being subducted in Earth's tectonic churn and coming out as volcanic CO_2 again, hundreds of millions of years later. This unceasing movement of carbon—one that conscripts every part of our planet, and makes the Earth the *Earth*—is known as the carbon cycle.

Despite its vast and varied constituent parts, and the unfathomably complex interplay between those components—from rain forests and ocean-spanning eddies of phytoplankton, to the composition of the atmosphere and chemistry of the sea, to the burial of CO_2 in limestone and coal and oil—this carbon cycle has stayed remarkably stable, and thankfully within an astoundingly narrow window suitable for complex life, over hundreds of millions of years. During the few brief, terrifying interludes when the carbon cycle has spun out of control, the planet has witnessed cataclysmic mass extinctions. In the worst of these, a volcanic blast of CO_2 into the Earth system violently overwhelmed this stately flow of flows, and briefly sent the planet into plainly unearthly territory, nearly ending the age of animal life altogether.

But all along, our fallen *Lepidodendron* tree remained undisturbed, suspended in a lightless, stony slumber of the eons. The entire age of the dinosaurs passed silently above. Then the age of mammals. The ice ages fell upon the Earth, and in one of the brief springtimes from this freeze sprouted all of recorded human history.

Then, a month ago, at the bottom of an open-pit coal mine, this lepidodendron, from a quarter-billion years before *Tyrannosaurus rex*, was dug up by a hydraulic monster made of metal—a machine itself powered by the residue of ancient plankton—and sunlight fell on this ancient forest for the first time in more than 300 million years. The tree was retrieved from its rockbound slumber by a creature on a remote branch of the tree of life, a bipedal synapsid called *Homo sapiens*, who puttered around the floor of a mine pit in an excavator, like a giant hermit crab. Its steel claw dug into the ancient jungle that lined the pit walls and delivered this Carboniferous coal to train cars, which arrived at the mine like spent blood cells from industrial civilization's circulatory system of railroads, shipping lanes, and highways. The coal was then delivered to industrial mitochondria—power plants resembling the miniature ones that power our cells—where it was burned to release its ancient sunlight energy and finally turned back to CO_2 and water. As it was in the beginning.

This is where the story of our ancient photon ends. After 315 million years, the venerable tree—this unlikely emissary from deep time—was finally metabolized in a coal-fired power plant with the rest of its forgotten forest. The energy generated through this industrial-scale respiration—energy from the core of an ancient sun—was used to make steam, to power a gigantic turbine and generate electricity. The electricity was sent as electromagnetic waves over high-voltage transmission lines to a featureless warehouse on the side of the highway. There this prehistoric sunlight from some languid Carboniferous afternoon, first stored as chemical energy in a leaf, then frozen in rock in the Earth for the better part of an eon, transformed to mechanical, then electrical energy, was finally used to process a credit card transaction in an overheating server farm, and radiated back into outer space as waste heat—as much lower-energy photons than the one that, once upon a time, had landed on Earth. Meanwhile, the CO_2 captured by the tree to *store* that energy some 300 million years ago was let loose on the world again.

This is the strange metabolism of industrial civilization. Where all previous animal life in Earth history has been constrained in its ambition by the amount of photosynthetic energy available on Earth's surface,[fn4] this new human industrial superorganism now extracts hundreds of millions of years of photosynthetic energy, stored in all the life ever buried throughout Earth history, and respires it all at once in the furnaces of modernity. Consider the speed of a cheetah, or the burrowing power of a gopher, or the architectural ambition of a termite—all creatures burning organic carbon from their diet, with oxygen, to do work, breathing out CO_2 as exhaust. Now compare these projects with humanity's recent, sudden ability to propel many-ton planes up to several thousand miles an hour, or put mile-deep holes in the ground in search of minerals, or throw up the Burj Khalifa almost overnight. We've done this not by grazing the world's meadows or hunting more prey, but by pressing entire ancient seas and forests into service all at once and converting them en masse to CO_2.

The scale of this industrial respiration is difficult to fathom. This is because there's far more old life to burn, buried underground, than is available in the thin green photosynthetic film at the Earth's surface that we call the biosphere. The world's apex predators have always been limited in numbers by the energy captured in photosynthesis that manages to make its way up the food chain to their perch at the top, as it dissipates along every step of the way—almost all of it lost to heat, excretion, and decay. This is why ecosystems host far fewer predators than the herbivores upon which they feed. And why herbivores, in turn, are dwarfed by the biosphere's vast energetic foundation of plants. But humans have obliterated these ecological, energetic limits. To wit, every ten weeks humans burn through as much carbon as is contained in all the animals on Earth. In the process, we've disrupted the course of life on this planet, and the grand geochemical cycles that keep it habitable, on a planetary scale comparable only, in the history of biology, perhaps to the dawn of animal life or the first trees. This industrial respiration now injects over a hundred times more CO_2 into the atmosphere every year than all the volcanoes on Earth. Where photosynthesis might have started almost 3 billion years ago, and aerobic respiration perhaps some 2 billion years ago, transforming the planet over hundreds of millions, even billions of years, this new metabolism is barely two and a half centuries old and, given its consequences, is unlikely to last much longer—its end guaranteed in the next few decades by either a buzzer-beating technological miracle, a startling transformation of the political economy of the developed world, a ruinously tardy and incremental market-led energy transition, or, failing all that, corrective planetary catastrophe. It is a wholly novel, profoundly more energetic, and possibly catastrophic new metabolism on this planet—one that, even by the end of the eighteenth century, with the introduction and growing adoption of the coal-fired steam engine in Britain—had signaled its transformative, devastating power.

This unassuming machine, the steam engine—introduced into the context of a North Sea society organized by a new economic algorithm that conscripted everyone into production for increasingly global markets, and that was marked by unflagging growth—opened the geological floodgates, transformed western Eurasia overnight, and in the long run unwittingly exposed the climate and biosphere to potential ruin. But while the idiosyncratic political landscape of early modern Europe might have lit the fuse that detonated this planetary battery, the resulting radical transformation of the Earth's surface and biosphere since the Industrial Revolution was only made possible by this half-billion years of photosynthetic energy that had been building up in the Earth's crust since the dawn of animal life. Today, more than two centuries on, the resulting supereruption of carbon dioxide can be compared only with a few terrifying chapters in geologic history, when continent-scaled volcanism injected massive slugs of CO_2 into the Earth system. It should be more than a little concerning that, over the entire course of Earth history, these kinds of transient derangements of the carbon cycle—and on a similar scale as is currently underway on our planet—have reliably resulted in mass extinction.

Carbon dioxide, in other words, isn't some niche, industrial by-product that billows from smokestacks and just so happens to be a twenty-first-century regulatory bugbear—like ozone-thinning chlorofluorocarbons or methylmercury. CO_2 is the very stuff of life. As we'll see, the behavior of CO_2 on this planet is what makes it Earthlike. Its cycling through the crust, the oceans, and the atmosphere over hundreds of millions of years is what has kept this grain of sapphire in space so strangely habitable over the eons. And life, which makes itself out of CO_2, is a critical component of this system. It's a story that goes back to our very origins, when the animate carbon chemistry of the early Earth began distinguishing our planet from all the other lifeless rocks in the universe so far discovered. Today this life blushes in great heaving blue-green swirls in the oceans, and in planetary belts of forests that pulse green with the seasons, drawing down CO_2 and releasing it in the same measure, carpeting the seafloor in chalk, paving continents in leaf litter, diffusing through herds of bison and flocks of geese. These grand movements of carbon flow through, and are in turn transformed by, life. The biosphere often serves to stabilize these planetary flows of carbon around the Earth that keep it habitable. At other times, though, revolutions in the living world have been sufficient to reorganize or even derail this system all on its own. *Homo sapiens*, being part of this pageant of life some 4 billion years in progress, are now adding our own inscription to this legacy of biological revolutions that have remade the Earth.

Meanwhile, economists, technocrats, and futurists—overly impressed by the symbols of the synthetic world they've created—sometimes advance a view of our planet that manages to be provincial, immaterial, and megalomaniacal all at once, with human society operating in some platonic economic realm above, and beside, the rest of the biosphere. As one prominent advisor to the British government recently put it, "The environment is part of the economy and needs to be properly integrated into it so that growth opportunities will not be missed." This has it exactly, and dangerously, backward. Instead, like all life on Earth throughout its history, we are inescapably subsidiary to this movement of carbon around the planet. If we drive the carbon cycle far enough from equilibrium, it will respond in kind, no matter how hard we try to subordinate it to the market, or tame it with panicked legislation. The carbon cycle does not have to comply with our plans to temporarily "overshoot" our emissions targets, in the hopes they can be later drawn down, without setting off any tripwires in the Earth system in the meantime.

This cycle of carbon around the Earth and throughout time is a pulsing planetary web of self-amplifying *positive* feedbacks on the one hand (where small pushes can drive much larger, extraordinary changes to the planet) and dampening *negative* feedbacks on the other (that rein in this system when it threatens to careen out of control) all working at the microscopic, local, and global scales. Somehow, in aggregate, it keeps the planet within the narrow envelope of conditions suitable for complex life. We don't fully understand it. Would-be geoengineers from the tech and finance worlds, flush with cash but new to Earth science, often underestimate this deep complexity when glibly proposing quick fixes to our global planetary crisis, like artificially dimming the sun, or speculative carbon capture technology at a scale and effectiveness that defy the laws of thermodynamics. Or they ignore the uncertainties and error bars around projections of future climate change, and downplay the civilizational danger of the tremendous, almost unprecedented push we've already given the system.

Fortunately, though, there are reasons to believe in our planet's resilience. As geologic history teaches us, this Earth system can bend. But as Earth history also teaches us, with a large enough disruption it can also break. If you're going to get one thing right as a species that has designs on becoming a long-term tenant on this planet, then you need to adopt a civilizational lifestyle that's in strict accordance with, and humble submission to, the carbon cycle—not one that blithely warps it as violently and recklessly as possible.

An illusion, one crumbling with each coastal dead zone and lethal heat wave, is that we are somehow cordoned off from the planet's grand biogeochemical cycles; that human affairs and the mechanics of the biosphere somehow play out in separate realms. Instead, human economic and geopolitical systems have now become crucial parts of the Earth's carbon cycle. In fact, humanity has always been a component of, even an expression of, this system, not an addition to it. As we'll see, the CO_2-driven climate changes of the past 60 million years shaped humanity. And human-driven extinctions have long warped how carbon flows around the planet. The rise of agriculture and empires was mediated by the flows of plant energy through populations, organized in new ways and respired back to CO_2 in the labors of humans and animals alike. These novel flows of photosynthetic energy structured human societies for millennia, and those societies in turn structured the environments around them, perhaps even subtly warping the climate with agriculture long before the Industrial Revolution. As we're painfully being reminded, nothing has changed. Human society is still hopelessly embedded in, and structured by, these planetary flows of energy and material. They have been merely rerouted through us as we churn through the planet—the most energetic, voracious organism in Earth history, with the fog of CO_2 encircling the world as a testament to this unprecedented new metabolism.

Now, though, it's not the agricultural harvest and state granaries that are powering human society, but roaring rivers of plant energy from deep time, channeled through the nodes of a globally networked industrial civilization to transform the surface of the Earth almost as fast as has ever happened. This means that the future of the planet's life-supporting global carbon cycle is now governed less by tectonics or ocean circulation than by economic and geopolitical systems. The effectiveness and wisdom of the institutions that govern human affairs, and the irruptive forces of history that occasionally upend them, will ultimately determine not only our fate on this planet, but most other life on Earth. As we try to anticipate an uncertain future, the equally uncertain responses of this human system will be just as important to our climate projections as the feedback mechanisms intrinsic to the climate system itself. Capital flows and webs of debt that orchestrate the production of raw commodities, the supply chains that deliver them to centers of industrial production, the halting blushes of mass democracy and countervailing forces of reaction that vie for control of states armed with apocalyptic military might—these will all be as determinative to the amount of CO_2 that ends up in the atmosphere as the uncertain response of melting arctic permafrost. And just as understanding the sensitivity of the planet to carbon dioxide in its geologic past is key to understanding our climate future, so too is understanding the human history that produced these systems—a history that, after all, has sent the planet off on a trajectory almost unknown in the history of animal life.

IN THIS BOOK I WILL ATTEMPT TO EXPLAIN HOW WE GOT HERE—WHY, that is, the Earth system is derailing in the current moment—by starting at the very beginning of Earth history. This isn't an academic exercise: the truly cosmic nature of our current climate crisis can only be understood in the context of deep Earth history. By pulling on these threads, we find that carbon dioxide ties us deeply to what it means to be alive on this planet, and what this planet even is. All life is ultimately made out of CO_2. Life was continuously shaped, over its entire evolution, by paroxysms of climate and ocean chemistry change, driven by CO_2. As we will see, the capture of CO_2 by photosynthetic life, and its burial in the Earth's crust as fossil fuels, charged up the atmosphere with oxygen—quite surprisingly—and produced a kind of planetary battery that would eventually make the animal life we see on Earth today possible. And finally, at the end of history, came the appearance of one unassuming creature, whose evolution was intimately shaped by CO_2-driven climate change—a creature that in one specific place, at one specific time, began to exploit the massive cache of fossil fuel underground and release its photosynthetic energy and CO_2 at an exponential rate and on a planetary scale, sending the planet off on an almost unprecedented chemistry experiment.

To understand the significance of this chemistry experiment, to understand why the world looks the way it does, we need to build this story up, bit by bit, starting at the very beginning of Earth history. But before we do, we need to drop into a curio-filled eighteenth-century laboratory on the banks of the Seine, where a rather more literal chemistry experiment will anchor our journey and illuminate the intimate connection between CO_2 and life on Earth.

ON THE EVE OF THE FRENCH REVOLUTION, IN PARIS, ANTOINE LAVOISIER checked in on his shivering guinea pig. The aristocrat had placed the hapless rodent in the inner compartment of an insulated tub of ice, where he could carefully monitor the air going in and coming out of the contraption. The heat that the creature produced, meanwhile, was measured by collecting the meltwater that dripped from the device. What Lavoisier found was that oxygen was going in but that carbon dioxide[fn5] was coming out—and that the guinea pig was generating enough heat to melt a measurable amount of ice.

Curiously, when Lavoisier removed the animal from his "ice calorimeter" and repeated the experiment, this time not with a living creature at all, but rather a smoldering hunk of charcoal, he found that the exact same thing happened. It used up oxygen, and produced CO_2 and heat.

"Respiration is thus a very slow combustion process," he concluded, "very similar to that of coal." Two and a half centuries later, the biochemist Nick Lane would paraphrase the point, observing that "in the end, respiration and burning are equivalent; the slight delay in the middle is what we know as life."

It was an astounding discovery. The metabolism that powers animals and *fire* are the same process. Both use oxygen to burn plant stuff, turning it back to CO_2 and water, and release energy in the process. In biology, this energy is dissipated in the organized friction of life,[fn6] and animates and powers literally every single thing a living being does. In a fire, it simply escapes as light and heat.

A similar, if even more radical experiment, though, was still in its trial phase across the English Channel in Lavoisier's time. Since antiquity, coal had been used across the world, in almost every culture where it was readily available, to produce heat. But by pairing the combustion of this organic carbon to a crankshaft and flywheel, James Watt's new machine provided continuous rotational motion and effectively universalized the work that fossil fuel could perform. Where for hundreds of thousands of years plants had powered human and animal labor, the energy stored in all the ancient plant matter ever deposited in the Earth's crust was now, in the words of one historian, free to "blow smelting furnaces, turn cotton mills, grind grain, strike medals and coins, and free factories of the energetic and geographic constraints of animal or water power." Soon the poet William

Blake's "dark Satanic Mills" would seethe and clang from London and, before long, a half-billion years of old carbon life buried in the crust would explode into the atmosphere.

But Lavoisier would live to see little of this. While he all but invented the field of modern chemistry, dragging it out of centuries of mystical obfuscation (developing the prototype for the modern periodic table, and displacing alchemy's "phlogiston" with the reactions of oxygen), he would get his head chopped off for the trouble.

Like most Enlightenment-era scientists, Lavoisier was able to make his groundbreaking discoveries only thanks to abundant leisure time and education, afforded by the agricultural surplus produced by the labors of unseen masses. He had used an inheritance to buy his way into France's hated Ferme Générale (General Farm), a private group of financiers to whom the hopelessly indebted Crown had leased out the collection of increasingly onerous taxes, from which nobility was largely exempt. The Farm had a reputation for ruthlessness and profiteering, even lead_pfing Adam Smith to comment that the system could be recommended only to "those who consider the blood of the people as nothing in comparison with the revenue of the prince." Nevertheless, with his considerable profits from the Farm, Lavoisier bought a 1,400-acre estate in the Loire Valley and on Paris's Right Bank built, as one of his biographers puts it, "one of the most sophisticated—and expensive—laboratories in Europe." There, in his groundbreaking experiments, he found in living things a fundamental link between the production of CO_2, heat, and the work of life. But these discoveries were in turn subsidized by the productive output of his countrymen and women—their work. This manual labor in turn was powered by the solar energy stored in plants, captured by the country's crops—and dissipated in the muscles of humans and draft animals.

Along with these beasts of burden that humans enlist, and the free gifts of wind and water power, the work we can do on the surface of the planet has always been limited by the photosynthetic energy in our diet. In Lavoisier's day the global population produced about 100 gigawatts of this kind of power through their metabolism by turning plant energy back to CO_2. Today, though, industrial agriculture, subsidized by fertilizers synthesized from fossil fuels, has vastly swelled this civilizational project, and humans now generate 800 gigawatts of power in their metabolism alone, producing 2 gigatons of CO_2 every year just through our diet. Since this carbon originally comes from plants at the Earth's surface, and is taken up by the biosphere again, these emissions from our very breath do not add to the atmosphere's inventory of the gas. But the work humans can do on this planet has now been lavishly supplemented by 20,000 gigawatts of external power, mostly through burning ancient plant matter retrieved from the Earth's crust, in the form of coal, oil, and natural gas, in an industrial metabolism that generates almost 40 gigatons of CO_2 every year. We have, in only a moment, become unthinkably more energetic than anything that's come before in Earth history. In the past two centuries, as more of the natural world became subsumed into an emerging global "economy" and transformed by this energy into the stuff of industrial civilization, millennia-old ways of life were torn asunder, the human relationship to the land and sea—and even time itself—was reimagined, societies were continually toppled and remade, and wealth skyrocketed in industrialized countries, as did its unequal distribution, along with carbon in the atmosphere.

Ever the Enlightenment-era rationalist, Lavoisier had taken up with the Physiocrats, the first modern economic school, one that located the ultimate source of nations' wealth in this, the country's photosynthetic base—agriculture—not in gold or silver specie. Karl Marx would later credit the Physiocrats for laying "the foundation for the analysis of capitalist production" and his surplus value of labor. But Lavoisier was unable to detect the revolutionary undercurrents of his time, instead viewing his societal order as a fundamentally stable one, and money as "a fluid whose movements necessarily end up in a state of equilibrium."

But, like anything interesting in the universe, the society atop which Lavoisier sat was far from equilibrium, held up by flows of energy embodied in agricultural surplus—and represented by money—that were funneled toward Versailles to support the opulence of the royal court and service debt for ruinous wars. If a ruling class convinces enough people that this structure of society is legitimate—through violence, appeals to the supernatural, or a commitment to a civic order—then the particular flows of energy that define a political structure will remain. But, as we'll see, all self-organizing dissipative structures—whether hurricanes, life, or political and economic systems—require a continuous input of energy to maintain. When hurricanes lose their ocean heat source, they disintegrate. When bioelectricity is cut off, cells die. And when a polity no longer consents to maintaining the structures of energy flow through its society, it collapses.

Lavoisier was a brilliant scientist, but he was also an avatar of this unstable pre-Revolutionary hierarchy—one racked in the eighteenth century by decades of war, financial catastrophe, climate chaos, and crop failures. His profits from the General Farm (what would amount to tens of millions of dollars today) had secured his fate.[fn7] Soon the Reign of Terror began, the king would lose his head, the old order would dissolve, and in 1794 Lavoisier would be tried and convicted on bogus charges.

"It took but a moment to cut off that head, though a hundred years perhaps will be required to produce another like it," his contemporary Joseph-Louis Lagrange lamented.

In life, Lavoisier had commissioned an iconic oil painting of himself and his wife, Marie-Anne, surrounded by cutting-edge laboratory Sequipment and ancien regime splendor.[fn8] One day it would be sold off to the richest man in the world, John D. Rockefeller Sr., who gained his fortune by replicating Lavoisier's combustion experiments on a planetary scale. Rockefeller and his ilk, by conscripting armies of laborers, and implicitly backed in their adventures by the looming threat of the most powerful militaries in history, tapped into oceans of energy from deep time and powered the twentieth-century industrial machine. This explosion of planetary energy remade the Earth. When grafted onto centuries-old imperial rivalries, it killed tens of millions of people. It also lifted living standards in the industrialized world by unleashing a torrent of food, fueled the construction of massive waterworks, extended the human lifespan and literacy the world over, accelerated the exploitation of those at the imperial frontier, is now deranging the planet's atmosphere and oceans in a manner unseen in at least tens of millions of years and, by swelling the population by the billions and the ambition of the human project, has exponentially intensified a wave of extinctions that has followed our species around the world for tens of millennia. Today, as in the beginning, life is still made out of carbon dioxide. And the world's problems are made out of carbon dioxide as well.

Almost two centuries after Lavoisier's death, our planetary experiment with CO_2 may be nearing its end. By now the carbon cycle itself has been thoroughly, colossally provoked and threatens to terminate this strange civilizational project all on its own, if we don't relent. The two threads of CO_2's role in fashioning the Earth have catastrophically merged: the energy of all life has always flowed through the chemistry of carbon, at the same time that its delicate balance in the oceans and atmosphere has kept this planet habitable. By becoming an energetic leviathan, though, and metabolizing all the life we can find from deep time, industrial civilization has upended this balance. By burning through ancient life, we are effectively undoing millions of years of planetary evolution. In the meantime, the world that existed before this planetary experiment began is irretrievable. We can't go back. But if we are to endure on Earth for anything like geologic time, we need to find our civilizational rhythms in the pulsing flows and tributaries of this life-sustaining carbon cycle.

CO_2 is the ultimate source of carbon for all life, it's the primary knob of Earth's temperature, and it's the thing humans make more than any other substance on Earth. Some three decades after his imprisonment at Auschwitz, the Italian chemist Primo Levi—whose knowledge of carbon chemistry spared his life in the custody of the fossil fuel–and rubber-starved Nazis—would meditate on this contradictory miracle of CO_2, the stuff of life:

> Carbon dioxide … this gas, which constitutes the primary material of life, the permanent store that every growing thing draws on, and the ultimate destiny of all flesh, is not one of the principal components of air but, rather, a ridiculous scrap, an "impurity." … The air contains .03%.[fn9] … This, on the human scale, is an ironic feat of acrobatics, a juggler's trick, an incomprehensible display of omnipotence and arrogance, since it is from this constantly renewed impurity in the air that we come: we animals and plants, and we human species, with our four billion discordant opinions, our millennia of history, our wars and shames and nobility and pride.

Part I

1

CO_2, the Stuff of Life

> Have you descended to the springs of the sea or walked in the unfathomable deep? … Have you comprehended the vast expanse of the world?
>
> —JOB 38:16–18

“WITTGENSTEIN SAID DON’T ASK FOR MEANING, ASK FOR USE,” MICHAEL Russell of NASA’s Jet Propulsion Laboratory began, manically running fingers through sheaves of white hair.

“So what use is life?” he demanded of a half-filled auditorium at the University of Colorado in Boulder. “That is the question.”

The audience was eager for insights into the origins of life, the most intractable problem in the life sciences, and so had gathered in the lecture hall of the Benson Earth Sciences Building to receive the Scottish geochemist’s gnomic wisdom. By day, these fossil-spangled halls—like those of any geology department—host lectures on the staggering variety of climates, tectonic regimes, oceans, and life that this strange planet has hosted over four and a half billion years. In still other corners of the building, exchange students from Saudi Arabia and Malaysia glean from retired Exxon apparatchiks how to find and retrieve this same ancient life from its burial vault, and then oxidize it in the forges of pistons and power plants to move the world—the very metabolism and breath of modern industrial civilization.

“What does life do?”

Russell’s was the sort of question that drives popular interest in the sciences and inevitably lends itself to soaring Sagan-esque rhapsody. In this spirit, an audience member offered that life is “the universe looking back at itself.”

But Russell was impatient to dispatch this gloss of profundity.

“That’s some meaning there, but that’s not its use,” he said. “The universe is not egotistical.”

He shrugged off still a few more scattered guesses from the audience—each similarly lofty—before declaring life’s purpose.

“Life hydrogenates carbon dioxide,” he announced to deflated silence. “It takes carbon dioxide from the air and adds hydrogen from water … and makes organic molecules.”

Oh.

“That’s its job,” he said. “Everything you’re sitting on, everything you’re eating—it’s all carbon dioxide. Go down the evolutionary tree, and it’s carbon dioxide. Go down the food web, it’s all carbon dioxide.

“There’s only one prebiotic molecule, and it’s carbon dioxide.”

A LONG TIME AGO, HERE, THERE, AND EVERYWHERE ELSE, EVERYTHING was all together and unreasonably hot. One day, a very, *very* long time from now, everything will be very, *very* far apart, and incredibly cold, and nothing will ever happen again. But between those two intervals—on the descent from the Big Bang to the end of time—things can happen. We live in that liminal moment, after the cream has been added to the cosmic coffee, when galaxies convect, the larger swirls begetting smaller swirls, and fractals thread themselves all the way down through creation. But just as the tiny hurricanes of cream in your coffee don’t swirl endlessly over the course of breakfast, this filigree of physical reality—the galaxies, the stars, the planets, the cellular machinery of life—is temporary. It’s endlessly dissipating. In fact, it all exists in the service of getting us as quickly and efficiently as possible to that uniform, universal café au lait at the end of time. Toward tranquility, equilibrium.

While equilibrium might seem desirable in our own lives, in practice equilibrium is the end. Equilibrium is death. But though the universe as a whole might be straining toward that ultimate end—toward an exhausted state of uniformity and maximal disorder, when everything everywhere is the same temperature, when all debts have been settled, all contradictions have been resolved, and no more work can *ever* be done again—we still, thankfully, find ourselves far from that final state of equilibrium. The sun still shines, and so we make hay. In fact, we can only ever find ourselves in this brief moment, impossibly early in cosmological history when the universe is still so outrageously far from reaching equilibrium that interesting things can still happen. Life, love, everything we care about—these are all so-called far-from-equilibrium phenomena. And it was in this universal straining toward equilibrium, on a restless young planet, that life emerged. It was here that carbon dioxide was transformed to living matter, and the Earth became the *Earth*.

WE START OUR STORY ABOUT CO_2 IN THE MOMENT AFTER THE BIG Bang when the universe, as Russell puts it, "stood at an almost unimaginable distance from equilibrium." But within about twenty minutes of creation, the two lightest elements in the periodic table, hydrogen and helium, began to congeal from this primordial fire as it dissipated. And for 98 percent of the normal matter in the universe, that's about as far as the story goes.[fn1] Only about 2 percent of this original gas has ever been fused to anything else on the periodic table. Nevertheless, 100 million years after the beginning of time, stars began to form, and the nuclear furnaces at their hearts set about forging the heavier elements.

Stars, like the universe at large, are *very* far from equilibrium. They find themselves in the improbable state of endlessly trying to collapse in on themselves while exploding ever outward, stubbornly propped up by nuclear fusion. Obviously, this isn't a situation that can go on forever. Our sun, for instance, loses 4 million tons of mass every second through this perpetual explosion, just to maintain its existence. This is the price of being a star. And sure enough, someday it will exhaust itself. And someday after that—a future date that can only be written with the help of scientific notation—it will dissipate to nothing.[fn2] Eventually this will happen for all stars, and even for all black holes. Then forever happens. Equilibrium. But for now, and as long as the sun has fuel to burn, our star is still perched in a *far*-from-equilibrium state that seems both out of control and quasi-stable—persisting for a time, as long as it can keep exploding away.

In this way, in the collapsing cores of the first stars, hydrogen was fused to make more helium, helium fused with helium and still more helium to make carbon, carbon fused with helium to make oxygen, and so on. Some of this carbon and oxygen wafted off the stars like laundry escaping from a clothesline, catching a breeze into space. But some of the larger stars, taking great pains to ward off this collapse, kept on furiously burning through these elements, making it all the way up the periodic table to iron before exhaustion set in. Overwhelmed by the burden of being stars, they collapsed, wretched, doubled over by a quantum mechanical gag reflex, and exploded. This stuff, the detritus of fallen stars, then spread out into the universe, seeding it with metals. To make heavier stuff like gold, even more psychedelic alchemy was required, and space-time-warping collisions and detonations of neutron stars—that echo today in gravitational wave detectors here on Earth—have recently revealed themselves to be the missing cosmic philosopher's stone.

Now, stuff in our universe has the peculiar property that when you get a lot of it together it pulls other stuff toward it. Then it starts to spin and differentiate. By about 4.5 billion years ago, one of these eddies in the void—one of uncountable trillions of trillions—matured to a proper solar system: ours.

We pick up the thread in the aftermath of the unwitnessed celestial collision that formed the moon—when Earth collided with another planet 4.4 billion years ago, in a sublime, incomprehensible Ragnarök—and pan in on the more familiar scale of the terrestrial, in the age when geology became biology. When the dissipating energy of a restless young planet became the energy of life.

It's impossible to conceive how long ago this story picks up, but here's one way to think about it: If you took a walk back in time, with each step representing a century, then the last ice age—the one that saw North America trampled by mammoths and camels, and covered in an Antarctica's worth of ice—would arrive in only a few dozen footsteps. But reaching the beginning of Earth history would take almost four years of walking at a pace of twenty miles a day.[fn3]

Almost from its beginnings, this early Earth was a water world. Where scientists long imagined boiling incandescent seas of magma at the dawn of history, we now know that these oceans of liquid rock would have been surprisingly short-lived, stiffening perhaps within only a few million years after the planet recovered from the lunar apocalypse. On a planet whose age is counted in billions—that is, thousands of millions—of years, this was all but instantaneous. Thereafter, for the most part, and besides the odd cataclysmic impact here and there, the early Earth was a surprisingly wet, if utterly noxious, world.

By some 4 billion years ago, howling winds swept over an endless, green, metal-soaked, anoxic, acidic ocean. This was a truly dreadful sea, perhaps twice the depth of our modern ocean and corrugated in soaring waves. The high seas were pushed by endless winds that howled at all latitudes, at all times. There wasn't so much as a single cell of bacteria in all this roaring vastness. Had you dropped a fish into this putrid seawater, anywhere on Earth, it would instantly die. By day, between endless hurricane clouds, the skies were pink, lit wanly by a faint star, kept warm by extraordinarily high CO_2 levels and bereft of oxygen. A titanic moon loomed menacingly in the sky, tens of thousands of miles closer, hurling gigantic tides across the ocean planet. The days came and went in twelve hours, the night sky afire with strange constellations.

The surface of this violent, drowning planet was not a propitious place to start something as delicate as life, especially not in a "warm little pond," as Charles Darwin once proposed. For one thing, there might not have been any dry land on which to put these ponds. And even if there had been lonely volcanic island chains, meekly poking up above the abyss, they would have been subject to the ruinous environment of the surface world: searing UV radiation, never-ending hurricanes, and serial asteroid impacts of a magnitude sufficient to embarrass the dinosaur doomsday some 4 billion years later.

This was a planet under extreme duress, violently far from equilibrium. And these stresses needed to be dissipated. Hurricanes did the job at the surface, endlessly racing around the ocean planet, to wick heat off into space. Deep in the heart of this global ocean, though, there was another system growing far from equilibrium, in the turbulent chemistry where the churning young Earth met the deep sea. And whenever systems are pushed far enough from equilibrium, startling new phenomena can emerge.

"IF PEOPLE USE THE PHRASE 'ORIGIN OF LIFE' THEN THEY DON'T know what they're talking about."

Michael Russell occasionally pauses and clutches his head as if the next idea—if not channeled carefully enough through his mouth—will erupt with so much violence as to crack his skull.

"Nothing in the universe that's dynamic originates, it's all just one entropy generator after another," he told me, growing agitated. "Because you're writing a book you probably have to use the word *origin,* but it's all about emergence. And the emergence of life is a geological issue, it's not a biological issue. It's about nano-engines—we're all living off nano-engines!"

Backing up a bit. As a child in 1940s London, Russell bounded through the rubble of buildings bombed out by the Blitz, picked trilobites from local quarries, and obsessed over the anatomy of old steam locomotives—the iron viscera of their boilers, pistons, and coupling rods. Night school at a technical college secured him a job at a chemical plant in East London, where his profit-sharing Quaker bosses encouraged him to pursue higher education. A geology PhD sent Russell around the world, taking the temperature of dyspeptic volcanoes in the Solomon Islands and prospecting for silver in the Yukon. "If you really want to have good fun, be a geologist," he says. "Geologists often know two or three languages, and they always know the best places to drink."

In the 1960s, his research led him to an ore mine in Ireland to study rocks that had formed at the bottom of the ocean, hundreds of millions of years before. Upon arrival, Russell was turned away by a surly mine geologist, so he had to have his rock samples smuggled out by drillers who sympathized over pints of Guinness at the local pub. The rocks were made of strange, tiny, hollow structures, precipitated at hydrothermal vents at the bottom of the ocean, where an open wound in the earth was penetrated by the ancient deep sea. It was only decades later, as Russell's son played with chemical gardens, that he thought back to these strange rocks and had his career-defining epiphany. Here was the cradle of life. In unusual seafloor vents near the beginning of time, the stresses of a young planet came to be concentrated, and life came along to relieve them.

"Frustration is absolutely required for creativity," he told me. "Van Gogh was very far from equilibrium, and so was his painting. So is life, because that's how the universe works. So, the Early earth was far from equilibrium, and CO_2 was the problem. That's why life emerged."

As with all of Russell's koans, this will take some unpacking.

TODAY, AS IN THE BEGINNING, THE EARTH REMAINS FAR FROM EQUIlibrium. And this is a good thing. If it isn't immediately obvious, one could imagine what it would mean if dreaded equilibrium ever arrived. "Matter mixes, water flows downhill and wood burns into ashes," writes the earth system scientist Axel Kleidon. "If nothing else were to take place, sooner or later all matter would end up in a uniform mix of everything, water would collect in the world's oceans, and all biomass would be burnt to ashes. All processes would lead to a 'dead' Earth state with no gradients present to drive fluxes and no free energy available to run life."

In fact, all of the emergent structures on our planet—the shuffling tectonic plates, the hurricanes, the life—are working toward exactly this end. They exist to relieve planetary disequilibria, to mix and relax them, to dissipate energy, and in so doing, shepherd the planet toward rest. It's all energetically downhill.

Take tropical hurricanes, for example. Only one one-billionth of the energy emitted by the sun in its churning collapse actually hits the surface of our planet, and a further 30 percent of this one one-billionth is instantly reflected off the surface back into space. Nevertheless, that infinitesimal fraction of the sun's energy that actually warms the Earth is still sufficient to drive giant convecting air masses on a planetary scale, weather systems, and even hurricanes—all of which form to dissipate this solar energy as efficiently as they can. They exist because while the Earth is heated by the sun, it is an uneven warmth, leading to huge temperature differences between the equator and the poles—a disequilibrium that hurricanes help smooth out. But in a larger planetary context this energy from the sun needs to be reradiated back to space. When this buildup of heat in the upper ocean becomes too great, and the slow and steady diffusion of it from the sea into the atmosphere isn't fast enough, something has to give. The system is frustrated, out of equilibrium. And so the tropical hurricane emerges out of a jumble of chaotic weather and—with no one there to orchestrate it—reliably self-organizes into a towering rotating structure that sucks that heat out of the ocean, dissipates the energy in tremendous winds, and radiates much of it out of its cloud tops and back into space as infrared radiation, waste heat, relieving the planet of its frustration.

Crucially, then, tropical hurricanes, though tremendously powerful in their own right, don't represent an *addition* of energy to the system, but rather the opposite. They are a conduit for its dissipation, from a frustrated energy source in the upper ocean to the energy sink of outer space. In doing so, they help the planet along on its endless search for equilibrium. And they are quite good at their job: at its awesome height one hurricane dissipates more energy in this effort than it takes to power all of human industrial civilization.

Today, though, it's not just the Earth's restless atmosphere that gets driven into a strange new organized state by the influx of energy from the sun. The sun also drives carbon chemistry at the Earth's surface into an almost impossibly far-from-equilibrium state called "life." But life today, surprisingly, does much the same thing as tropical hurricanes—it self-organizes and puts the energy from the sun to work in the impossibly complicated cellular machinery of photosynthesis to power and make still more life out of CO_2 while dissipating this energy through a global metabolism that takes shape in the near-infinite forms of the biosphere—in the thundering, plant-powered journeys of vast wildebeest herds, in the stirrings of vast schools of fish (which dissipate up to a terawatt of energy whisking up the ocean), and in the entire heaving churn of life, as it goes on living and dying. "At the ecosystem level, biochemistry is simultaneously building itself up from CO_2 and breaking down to CO_2 again," the biochemist Kamila Muchowska writes. And in all this commotion, life reradiates the intense energy from the sun once captured by plants back into space as low-energy waste heat—infrared radiation. Dissipation.

As with hurricanes, this endless dissipation only persists because of the perpetual influx of free energy from the sun that pushes these strange systems far away from equilibrium. If the sun were to suddenly wink off, not only would hurricanes soon fizzle out, never to spin up again, but the surface biosphere, which dissipates 100 terawatts of this solar energy through the machinery of cells and bodies, simply could not exist to dissipate it either. It would cease to stir as surely as a pot of water stops boiling when the stove is turned off. But that day is far in the future. In the meantime, the Earth is allowed to persist in a state that, looking around, strikes us as perfectly natural but in fact is just about the strangest thing in the universe. The wind keeps blowing, volcanoes keep exploding, and life goes on.

Both hurricanes and life itself, it turns out, are trying to do the same thing: open up channels—one through the ocean and atmosphere, and the other through carbon chemistry—to dissipate this influx of energy, through motion, work, friction, and waste heat, relaxing the planet. From the sun to hurricanes to life, it's all cascades of dissipating energy.

The Earth, after all, like everything else in the universe, is trying to settle. In fact, its entire history can be seen as a frustrated search for equilibrium, on a planet continuously pushed far from it. And if it wasn't incessantly driven into this strange state, we might expect these energy-dissipating processes at its surface to soon produce that dreaded "dead Earth." But as long as the flows of free energy from the sun and the Earth's interior exist, then the Earth processes that emerge to dissipate them—atmospheric circulation, tectonics, geochemical cycling, life—will take on the character of a standing wave, ever perched on the edge of failure, without ever being allowed to do so.[fn4] Until of course, someday they do. In the meantime, though, they will go on perpetually straining in the same direction as everything else in the universe: toward equilibrium.

In their industry, these so-called dissipative structures like life or hurricanes produce far more disorder, or entropy, in the form of useless waste heat than is ever rescued in the production of their baroque structures. This waste heat, this low-energy radiation, leaves the planet a far degraded version of the high-energy sunlight that once landed on Earth's surface. And unlike the original sunlight, it can never be recovered or put to use to do the same thing ever again. Its production by Earth, by life, by weather, by earthquakes, by *whatever*—shepherds our little corner of the cosmos along in the great project of relaxing the thermodynamic tension produced by the Big Bang, when everything was dense and extremely hot, and toward the ultimate heat death of the universe, when everything will be far apart and extremely cold.

WHAT I'VE SKETCHED ABOVE IS THE SECOND LAW OF THERMODYNAMICS. The law is often popularly described as the tendency of things to get ever more disordered over time. But as demonstrated by the emergence of highly *ordered* way stations along this journey, like hurricanes, or the incredibly intricate machinery of life, the Second Law is a little more subtle in practice. And these subtleties are worth lingering on for just a moment longer if we're to better appreciate how its unforgiving constraints must have shaped the emergence of life on Earth some 4 billion years ago.

Crucially, the Second Law dictates that the amount of so-called free or useful energy in the universe only ever dissipates away. To keep it simple, this “free energy” is energy that you can use. The free energy in a can of gasoline, for instance, can be burned to push pistons up and down, and move your car down the street. But good luck trying to recover that energy in the disordered waste heat that radiates off your car hood, or from the patch of road briefly heated by friction with the car’s tires. Or in the turbulent air in the car’s wake. The original, dense, chemical energy from the can of gasoline hasn’t been destroyed, of course—energy can never be destroyed (the *First* Law of Thermodynamics). It has only been transformed into these more diffuse, more disordered, more useless forms, which can never be made to do the same work again.

When these insights are applied at large to our world, they explain nearly everything we see. They account for the overwhelming likelihood that hot soup will cool off until it reaches equilibrium with the room, as the room grows imperceptibly warmer. And why the reverse never happens (rooms don’t spontaneously cool off and heat up the soup in them). They account for the fact that cream mixes into coffee and never spontaneously unmixes, or that rusting struts will buckle, causing bridges to collapse, while a collapsed bridge never seems to springily reassemble itself from mangled steel and start de-rusting. None of these kinds of miracles are forbidden by the laws of physics; they’re just astronomically unlikely. The reason is that, like Tolstoy’s unhappy families, there are many, many more ways to be a broken bridge than a working one. Systems like bridges, left unattended, will tend to progress toward lower potential energy, more disordered states, like unceremoniously collapsing into a ravine and rusting away, that are far more probable than, say, spontaneously reinforcing themselves.

Energy, again, hasn’t been destroyed in any of these so-called irreversible processes—it has only been dissipated to more disordered, more useless forms. And in a universe that’s always falling apart, it takes a perpetual investment of free energy to put things back together, even as this kind of useful energy only ever dissipates away. In the long run, it’s a losing battle. *Entropy* is the measure of this tendency toward disorder, and just as free energy only ever dissipates away in our universe, entropy only ever increases.[fn5]

This means that generating and protecting order in this universe—in other words, keeping things far from equilibrium—whether it’s a healthy organism, a clean room, a new car, or a globally networked industrial civilization, means perpetually dissipating new sources of free energy to maintain these systems and guard against their steady erosion by those same, relentless forces of entropy. This is why rooms get messier without a continual investment of time and energy to keep them orderly. It's why you fall apart if you don't burn enough food to fuel your body's maintenance and operation. It's why empires often crumble in poor harvest years, when diminished flows of the energy in crops can no longer support the same level of social complexity. It's also why people who worry about the exhaustion of the world's oil supplies[fn6] —the largest source of civilizational free energy ever exploited—also fear that such an energy crunch would lead to the collapse of the (highly ordered) phenomenon of industrial civilization, the structure of which is maintained only through the dissipation of 600 exajoules of energy a year, and along with it, an unimaginable exhalation of waste heat and CO_2.

Perhaps it's not a surprise, then, that the rudiments of this Second Law—a principle that explains much about our cosmos—were first uncovered by a French military engineer during the investigation of coal-fired steam engines in the initial takeoff of the Industrial Revolution. A quarter century after his father had sentenced Antoine Lavoisier to death while serving on Robespierre's blood-soaked Committee of Public Safety, a young Sadi Carnot found his family and country humiliated in the wake of Waterloo. Carnot attributed this humiliation to his country's technological gap with its cross-channel imperial rivals, so he turned his attention to James Watt's peculiar steam engine—a technology that by now was insinuating itself into, and transforming, every aspect of modern life, from grain milling to mining, in the vertiginously industrializing Western Europe. To take away England's steam engines, Carnot perceived, "[w]ould be to dry up all her sources of wealth, to ruin all on which her prosperity depends, in short, to annihilate that colossal power." In this compulsive search for the physical principles that explained the power of the engines, he discovered that they were able to extract work by bringing a system back toward equilibrium that is far from it. And the further from equilibrium a system is—the hotter-burning the fuel, the higher the waterfall at a mill—the more work an engine can do as it dissipates this disequilibrium. But the energy that is inevitably lost along the way to entropy[fn7] can never be recovered, and in the end, you're left with less energy to do work than you had at the beginning.

The upshot is that the more we try to generate and maintain order in our world—whether in the highly improbable structures of biology, or the architecture of a global economy—the more free energy needs to be dissipated, and the more waste heat and disorder are ultimately generated in the environment. There's no winning against entropy. This has been the story since the beginning of time.

While it might seem like a paradox, then, that such an orderly phenomenon as life could ever emerge out of the rules of a universe that requires ever more *disorder*, it isn't. But life is able to make and maintain these little islands of order only at the cost of producing far more disorder in the environment at large. Life drives a bewildering network of energetically uphill reactions, but only by pairing them with much larger downhill ones, dissipating sources of free energy and generating waste heat. And on balance, life is, in fact, an exceptionally powerful entropy-generator. Just as the highly ordered structure of a hurricane dissipates free energy far more efficiently than the random churn of the atmosphere, so too with life when compared with the mere chemistry from which it emerged.

So if life today is inextricably bound by these constraints of the universe—if everything is, on balance, energetically downhill—then its origin story must follow these requirements as well. All of chemistry, including the chemistry that keeps us alive, relies on spontaneous reactions that only take place because, on net, they release energy and increase entropy—they are downhill. Even if locally they produce astounding little islands of order. This means that the first step in the origin of life wasn't some miraculous *uphill* roll of a rock, or some Frankenstein's monster moment of vivification—the addition of some new life force to inanimate matter—but rather a *dissipation* of energy. Somewhere on the restless young Earth, there must have been some system that was frustrated in its descent to equilibrium. And this frustration must have already existed before life ever came along to relieve it. In some sense this must have been life's "purpose," the reason for its emergence. Perhaps the first life wasn't a miracle, but a necessity. A hurricane of sorts, made of carbon dioxide, one that, once spun up in all its dissipative power, would prove stubbornly persistent over the eons, to the current day.

But where Carnot sought to understand the engines that launched the Industrial Revolution, and hurricanes—atmospheric physicists would later note—bear a striking resemblance to the perfectly efficient engines Carnot once theorized,[fn8] today's origin-of-life researchers are interested in the engines that launched life itself, and powered themselves by relieving the early Earth of a similar planetary disequilibrium. At its root, "life," as the molecular biologist Elbert Branscomb argues, "is concerned with developing nanoengines." As it was in the beginning.

"So we have to create an engine," says Russell.

GIVEN LIFE'S OUTRAGEOUS SUCCESS THRIVING AT THE EARTH'S SURface today, dissipating the energy of the sun, it is tempting to suppose that that's where it began as well. But where today the living surface world is powered by photosynthesis, the first life was not photosynthetic. In fact, the form of photosynthesis with which we're familiar, the one that produces oxygen in both plants and pond scum alike, was a far later invention, emerging many hundreds of millions of years in the future. Recall that, at the time life emerged, the surface of the Hadean planet was a drowned, hopeless place, streaked by endless hurricanes, pummeled by titanic asteroid impacts, and irradiated by lethal UV. Given how awful this surface world was, though, perhaps there was somewhere else where life—where an engine of dissipation—could have emerged. To some researchers, then, the depths beckoned.

In 1977, a team of geologists from the Woods Hole Oceanographic Institution on Cape Cod and the US Geological Survey set out for the Pacific Ocean by way of the Panama Canal, in search of hydrothermal vents at the bottom of the ocean. There were no biologists aboard the R/V *Knorr* because there would obviously be no biology to study on this trip. The vent world had been tentatively sniffed before, in soundings of hot water plumes near the seams in the seafloor crust, but never investigated up close. It was assumed to be an unpleasant place, where an angry Earth opened its guts into the ocean. There was obviously no reason to think it would be particularly hospitable to life. Then, 250 miles off the Galapagos Islands, where the great tectonic plates were rifting apart, and where there had only been barren seafloor for hundreds of miles, the geologists noticed the temperature spike as they approached the vents. But strangely, a camera towed along also revealed giant clams coming into view at the bottom of the sea. When the temperature dropped again and the vents had passed, the clams disappeared as well. They anxiously dropped into the plucky submersible *Alvin*, plunged to the deep, and, in the days to follow, came upon a wilderness of life: forests of tube worms, white crabs, and fish, all huddled around billowing plumes of warm brine, shimmering in the near-freezing water at the bottom of the sea. The bounty of life was so unexpected they had to resort to storing the animal specimens in Russian vodka stowed aboard. "Isn't the deep ocean supposed to be like a desert?" the geologist Jack Corliss radioed on one dive, perplexed.

In this midnight world they had happened upon one of the biggest discoveries in the history of science. Here was life powered not by the energy of the sun, which had been thought to underwrite the entire biosphere, but by the chemical energy of the Earth itself. When scientists returned to this, the Galapagos rift, two years later, *Alvin* came upon even more spectacular, seething, toxic, metal-rich black "smoke" furiously billowing from mineral chimneys. Still, these blistering, noxious jets were somehow encrusted in thickets of giant mussels and crimson featherduster worms. The discoveries instantly inspired speculation that life itself may have started in such a primordial setting.

The optimism, though, was misplaced. While these vents support an astounding bounty of life today, such cauldrons are far too hot and violent to nurse something as gossamer as early life. Besides, however strange the deep-sea bacteria that underwrote all this life on the "black smokers," its metabolism was still very much dependent on the chemistry of our modern, oxygen-rich world.[fn9] Even down here, the oxygen that suffuses the deep sea is still thanks to photosynthesis taking place at the Earth's surface, a process that hadn't yet evolved on the early Earth. Still, this was a decidedly alien ecosystem: where today plant life that supports the rest of the biosphere uses sunlight energy to build itself out of CO_2 this oddball bacteria was using the energy released by burning toxic sulfur compounds spewing out of the Earth to do the same. The discoveries revolutionized biology, pointing to power sources available to life beyond our star. It made life a more general phenomenon, one that thrived not off sunlight per se, but potentially off any kind of free energy.

In 1988, Russell, thinking back to his smuggled Irish rocks and his son's chemical gardens, predicted the existence of a peculiar, much gentler kind of hot spring at the bottom of the ocean than the raging "black smokers" discovered by *Alvin*. In fact, Russell's so-called alkaline hydrothermal vents didn't have that much to do with volcanism at all. Instead these vents would be driven by the heat-releasing chemical reactions of seawater that insinuated into cracks in the seafloor, where they met solid rocks from deep in the Earth's mantle, in places where they had been fortuitously exposed near the seafloor's surface. Because these iron-rich rocks are from the high pressure and temperature underworld of the Earth's interior, they are out of equilibrium with the surface world, and furiously react when introduced to the deep sea, producing somewhat tamer vents of warm water all on their own—vents that fizz with hydrogen.

This process, called *serpentinization*, is "as ancient as water on Earth," the geochemist William Martin writes. You've likely seen the marbled green rock produced by these deep-sea reactions—serpentinite—in countless corporate lobbies, and in the lapidary backdrop of the UN General Assembly dais. On a mushier primordial Earth, before the modern crust had time to distill out of an inchoate planet, the seafloor would have been seething with this kind of far-from-equilibrium geochemistry. Here in these geochemical gardens at the bottom of the ocean at the beginning of history, Russell predicted, the fundamental reactions of life could occur. Here, and perhaps here alone on the planet, CO_2 could be reliably transmuted to the stuff of life.

This fundamental stuff of life is, somewhat uninspiringly, known as "organic carbon." But don't let your eyes glaze over. Ignore all modern connotations of the word *organic*. For our purposes, all you need to remember is that organic carbon is what life makes. It's food, it's structure, it's everything. *Very* roughly speaking, life makes this stuff of life by taking in carbon dioxide and then jamming hydrogens on to the carbon that don't want to be there. This is what life does. Plants, for example, use sunlight to split water, and then (again, very roughly speaking) they take the H's that they tear off of this H_2O and use them to build plant stuff out of CO_2—stuff with lots of C's, H's, and O's in it. *This* is organic carbon. Wood is organic carbon (lignin: $C_{81}H_{92}O_{28}$), as is sugar (sucrose: $C_{12}H_{22}O_{11}$). The psychedelics in mushrooms; the spice in peppers; the caffeine in coffee; the booze in your cocktail; your eyeballs; the petals of a bougainvillea; the beak of a cassowary; the baleen, blubber, and brain of a blue whale; the feathers of a velociraptor; the silk of a spider; the spider itself; the scum on your bathtub; the mane of a lion—all are organic carbon, ultimately made from CO_2. Every step in the core metabolic cycles common to all life, which both build up and break down all biomass on Earth, is organic carbon. DNA is organic carbon, RNA is organic carbon. Your blood cells and the hemoglobin in those cells (every molecule of which has 2,952 carbons, 4,664 hydrogens, 832 oxygens) are organic carbon. Your neurons, your neurotransmitters, your fat, the carbohydrates and protein you eat, the clothes you wear (whether natural fibers or synthetic), this paper and the ink upon it (or the e-reader for that matter) are all organic carbon. Virtually anything you can find in a grocery store is organic carbon, including the packaging the food comes in, the customers and employees, and the linoleum floors.[fn10] Coal, gasoline, plastics, all made of ancient life, are thus all organic carbon too.

Simply put, life is organic carbon. It's almost all C's, H's, and O's.[fn11] To wit, 98 percent of the atoms in your body are C, H, and O. This incredible diversity of material that life can make out of carbon is owed to the element's singular, modular versatility, one that allows it to exist as a solid (wood, leaves, plastic, flesh, diamonds), gas (CO_2 methane, perfume), and liquid (jet fuel, whiskey, olive oil), all at room temperature. Silicon, meanwhile, at one row down in the periodic table, makes only rocks.[fn12] Though astronomers sometimes muse about the possibilities of silicon-based life elsewhere, most biologists remain profoundly skeptical. Life, wherever it emerges, is overwhelmingly likely to be an organic carbon phenomenon. And when you die, your remains—organic carbon in life—will eventually turn back to carbon dioxide and water.[fn13]

"Organic [carbon] chemistry is not an accidental stage on which abstract principles of life perform a play that could be performed elsewhere," write the biochemists Eric Smith and Harold Morowitz. Rather, life is fundamentally carbon, and the source of that carbon is CO_2—Primo Levi's "primary material of life, the permanent store that every growing thing draws on."

To make the building blocks of life, then—and ultimately life itself—we need a way to get hydrogen's electrons onto CO_2 in the first place, and start making this miraculous stuff of life, in a process called "reduction."[fn14] But taking this one fundamental step requires making CO_2 accept electrons that it really does not want. It's an extremely difficult reaction to pull off on our world—one that rarely happens on its own, absent life. If we could synthetically transform CO_2 to organic carbon today as effortlessly as life does, we wouldn't have to dig *old* life out of the Earth by the gigaton as fossil fuels, and power our societies off the accrued interest of deep time.

"CO_2 is the problem," says Russell about the origin—er, emergence of life. "If it was easy to reduce carbon dioxide, you wouldn't need life, it would just be a geochemical reaction. It's because it's so goddamn difficult, that's why there's life."

Everything that happens in the universe, though, must do so spontaneously. It has to happen because it *has* to happen. That is, it has to be, on balance, energetically *downhill*. Things that can't happen, generally don't. Rocks don't roll uphill. The cream and coffee don't unmix. So there must have been somewhere on the early Earth where, despite its difficulty, this stuff of life could be reliably formed. Here, finally, is where the story comes together—free energy, dissipation, entropy, life—at the bottom of a sea, on a planet old-beyond-comprehension.

What's so felicitous about Russell's vents is that they provide bizarrely perfect conditions for making this stuff of life. In fact, they turn what would have been a nearly impossible reaction into an all-but-necessary one. When the ancient sea insinuated into cracks in the seafloor, and met rocks from deep in the Earth's mantle, the reaction first would have endlessly generated hydrogen, before seeping back out as a much cooler, alkaline brew, into what would have been an acidic, CO_2-rich ancient sea on the early Earth. Chemistry in a world with no oxygen—which would otherwise voraciously snap up this hydrogen to make water—works far differently than it does in ours. And freed from this competition carbon dioxide can, in theory, actually react with this hydrogen and release energy. Though it takes a lot to get going, it is ultimately downhill. So in these anoxic seas these two water masses would have desperately wanted to react. But there was a hitch.

Just because a reaction *can* happen doesn't mean it *will*. Even reactions that ultimately release lots of energy can be held back for a time, by the smaller, up-front inputs of energy needed to get them going. And when these energetic barriers are particularly high, catalysts can lower them, vastly speeding these reactions up. For instance, an unusual enzyme in yeast known as OMP decarboxylase catalyzes a chemical reaction in 18 milliseconds that would otherwise take 78 million years to occur on its own. And on the early Earth, even if CO_2 and hydrogen could react and release energy, it didn't mean that they would. In fact, they were still happily stable on their own, even in this cauldron—and even as it grew ever further from equilibrium. And the cracks in the crust where these two chemistries met—where the odd chemistry of these gentle vents met the acidic, CO_2-rich seas of the time—is where this disequilibrium would be focused. This was frustration, key to all creativity, as Russell says.

Just as a raging fire first needs help to ignite—even if, once started, it will release far more energy—so too with reducing CO_2. The reaction needed a catalyst. Otherwise it would not have happened, however frustrated. Perhaps this is what life was for. And the tiny mineral vesicles that precipitated out of these vents—not unlike the chemical gardens Russell's son once played with—might have provided the ideal reactors in which to concentrate and experiment with this new organic chemistry. Now the origins begin to come into view.

Or so Russell thought. His prediction was mostly ignored by the origin-of-life research community for over a decade.

THEN, IN 2000, AN EXPEDITION LED BY THE OCEANOGRAPHER Deborah Kelley of the University of Washington came upon an otherworldly travertine[fn15] drip castle at the top of a mountain at the bottom of the Atlantic Ocean, thousands of miles from any shore. Kelley plunged to the deep in the trustworthy *Alvin*,[fn16] piloted to the bottom of the ocean by the steady hand of an offshore oil industry veteran, where she found unearthly spires that reached almost two hundred feet from the seafloor. Kelley called the unexpected ghostly forest of stone the "Lost City" and described it in the kind of lyrical prose uncommon to academic journals. There were pinnacles "that resemble upturned hands," "very delicate, snow white, branching 'fingers,'" "anastomosing … veins"; there was a gigantic rampart called "Poseidon"; structures that looked "like bowls turned upside down" and that cupped a "mirror-like surface (similar to the surface of a lake on a calm day, but upside down)." "Fluids spill out over the lips of the flanges," she wrote, and slowly added to the strange sculpture as they did so. There were "stalagmite-like growths," "delicate deposits reminiscent of wasp nests," "fallen chimneys," and older vents, "knobby with the appearance of badly poured cement." The Gaudi-like edifice, wreathed in wreckfish and corals, had fluids "weeping actively from many of the steep cliffs." This limestone *sagrada familia* had been precipitating here for over 100,000 years, produced by vents that were far tamer than the violent black smokers discovered decades earlier, but that almost exactly matched Russell's decade-old predictions. Here in the middle of the Atlantic Ocean, where the great tectonic plates pulled apart, iron-rich rocks from deep in the Earth's mantle had risen up to meet the deep sea, and the water gently simmered and fizzed with this strange chemistry.

Woods Hole Oceanographic Institution geochemist Susan Lang was on the 2003 expedition that returned to these sunken cathedrals with Kelley, in search of Russell's proto-metabolism, even probing the edifice up close after a half-hour plummet to the bottom of the sea in—what else—the venerable *Alvin*.

"You're at the top of this underwater mountain, and the currents were so strong there that we kept running into the chimneys," says Lang about one particularly perilous dive. "That wasn't fun."

She pulls up a video from a recent cruise and the spectral white towers come into view. Other than a gormless fish gawking at the camera, though, it could have been a scene from the Hadean.[fn17] This megafauna was a diversion.

"We have no idea what that thing is," she says about the fish. "There were no macrobiologists on the cruise."

Lang is herself an organic geochemist. The field studies the rumors of old life in the rocks, written in carbon. It's a discipline that lends itself not only to the study of the origins of our living world, but also to the search for that other kind of old life in the rocks: fossil fuels. In fact, the field's roots date back to Nazi Germany, when that country was fatally starved for fuel and rubber, and the search for synthetic substitutes took on existential importance as it geared up for war. It was the Munich-based chemist Alfred Treibs who first discovered that crude oil and coal were, in fact, rich with organic carbon compounds today found only in chlorophyll.[fn18] It was plant stuff, in other words. Fossil fuels, he realized, were just old life.

"Our annual conference is joint between environmental people and the petroleum industry," Lang says. "It's really the only time I ever overlap with those folks. Oh, my God, they have money. It's insane."

Here at the vents of Lost City, as well as in laboratories reproducing the chemistry of these deep-sea chimneys, Lang and others have detected the kinds of organic carbon, synthesized out of CO_2—chemicals with boring names like acetate and pyruvate—that nevertheless lie at the heart of all core metabolic reactions that make life on Earth. And this whole affair—the entire seething verdant riot we call the living world—fundamentally goes through just five such intermediates made from CO_2.[fn19] These compounds give rise to every such process that both "builds up" and "breaks down" life. And this is in *all* life, mind you—from anaerobic bacteria in the deep sea, to elephants on the savanna. All of these compounds have been either found at the vents or shown to plausibly form there. Where virtually no rocks remain from the primordial Hadean Eon,[fn20] having all been eroded away and destroyed in the intervening sweep of deep time, this *core* metabolism common to all life has endured for billions of years, passing through the bodies of quadrillions of organisms, maintaining its ancient form as a kind of ever-kindling, living fossil—the oldest one.

THIS IS TO MAKE IT ALL SOUND FAR TOO SIMPLE, THOUGH. LIFE IS much more than mere chemical reactions, and simply generating some organic carbon from CO_2 though a major discovery at the vents, is still a long way from the ridiculously complex network of reactions that build up life. And metabolism as it exists in each of our cells today—the reactions that power and build us—is so ridiculously overengineered that it has to be reckoned with first if one is to come up with a theory of how this engine of dissipation could have ever emerged on a lifeless planet.

Today we get our energy by burning food in our metabolism. This "burn" expresses itself as the flow of electrons through our cells—literally electricity. And this electricity flows through mitochondria, the cell's so-called powerhouse, as we methodically shuttle the electrons from our food downhill (energetically speaking) in an absurd electronic bucket brigade, before they're finally united with an electron-starved oxygen at the end of this chain.[fn21] But as we do so, this electric current coursing through us powers and maintains an unlikely disequilibrium that keeps us *alive*. Namely, as the electrons head energetically downhill they are used at the blistering rate of 100–200 electrons a second, to push *positively* charged protons energetically *uphill* to the other side of a membrane through an astoundingly intricate mechanism that has been described as "akin to the coupling rod of a steam engine."[fn22] All this positive charge, now held back behind the dam of a membrane, is suddenly far from equilibrium, and available to do work. The work of life. To do that work, though—and here's where things truly start getting ridiculous—the protons are painstakingly let back *down*, but only through the gears of a *rotating turbine*. This turbine is pushed as surely as water through a mill, spinning at the rate of more than 100 revolutions a second. This scarcely believable piece of cellular Swiss clockwork delays the fire of life even *longer* by using the power from this spinning turbine to manufacture the universal energy currency for all life on Earth (something called adenosine triphosphate, or ATP), which is finally sent to the far reaches of the cell to be broken apart again for energy, to do the things of life.

All this strange machinery, though, is useless unless the far-from-equilibrium voltage of the cell is maintained, with protons improbably all pushed to one side of a membrane. And ridiculous gradients like these are present not only in our cells, but in all life today.[fn23] So important is this disequilibrium to keeping you alive that even a minuscule dose of cyanide —which cuts off the power to this machinery, grinds these pumps and turbines to a halt, and brings dreaded equilibrium to the cell—kills you almost instantly. And just as a dam with turbines can't operate if the water on each side of it is at the same level—in equilibrium, that is—when you die, the electric fields in every cell in your body disappear, these cellular pumps and dams fail as well, and you soon equilibrate with the environment. But as long as you're alive, equilibrium is kept blessedly far at bay. That we can live for more than five seconds without this absurd, infinitely baroque gearwork falling apart, much less several decades, is a kind of secular miracle.

Why would life go through all the trouble of inventing this impossibly convoluted, seemingly miraculous machinery? Creating the kind of far-from-equilibrium electrical chemistry of the cell from scratch would have been an incredibly steep uphill step for the first life to make, and it seems almost impossible to have been invented ad hoc. The problem is solved, though, if it was in fact the disequilibrium—that strange electrochemistry that keeps us tenuously alive at every moment—that came first. Perhaps the first life merely fed off a gradient in the environment that was similar to the one so assiduously maintained by the cell today. By this account, life as it exists today, with its bewildering nanoscopic labyrinth of cellular kluges, might merely echo the old chemistry of the vents.

"The idea is that geo-energetics transitions to bio-energetics on the early Earth," says Lang. "Rocks are carrying out these reactions, and then biology just mapped on top of that."

Today life makes the stuff of life out of CO_2 but only with the voltage generated by pumping protons uphill, in the absurd machinery of the cell described above. Amazingly, this far-from-equilibrium chemistry is almost exactly reproduced by the vents.

On the early Earth, proton-rich, acidic oceans outside the vents, separated from the alkaline water produced inside, would have almost exactly replicated the kind of voltage found across cell membranes, produced by life. In fact, the early Earth as a whole, with the electron-rich rocks of the Earth's interior and the relatively electron-hungry surface world of the ocean and atmosphere, could be thought of as a kind of cell itself, or a one-volt battery that the first life helped discharge through the chemistry of carbon dioxide. And at the vents, where this disequilibrium was focused, thin semiconducting walls of metallic minerals—of the sort still found at the center of our most important enzymes today, including those that guide electrons through our cells—would have shuttled hydrogen and its electrons onto carbon dioxide, reducing it at last, releasing energy.

While this reaction might be "goddamn difficult" on our world today, it is made almost miraculously effortless in this strange realm of the vents. Across these mineral membranes, in these seafloor chimneys, the disequilibrium of the early Earth was thus relieved in this engine of dissipation, in unassuming but monumental reactions. Here organic carbon was endlessly generated, and these reactions happened not as some astronomical fluke, not the result of some guiding world spirit, or élan vital, but because they *had* to. In the process they got the bizarre improbable chemistry of life off the ground, making the kinds of high-energy compounds and building blocks from CO_2 that make up and energize all life to this day, and for free. While economists might like to say that there is no such thing as a free lunch, the first life got, as the biogeochemist Everett Shock called it, "a free lunch you are paid to eat."

"These reactions *need* to happen," writes University College London biochemist Nick Lane. "They are the only way of dissipating the unstable disequilibrium of reduced, hydrogen-rich, alkaline fluids entering an acidic [CO_2-rich], metal-rich ocean. The only way of reaching blessed thermodynamic equilibrium."

In relieving the early Earth of its frustration, this emergence of life has even been described as a "collapse" into a more stable order, and a "path of least resistance" for the planet. By this account, without life coming along to relieve it, the stressed early Earth would be a bit like a buzzing storm cloud with no lightning bolts to discharge its energy. This is an inherently unstable regime. Far from equilibrium. The lopsided charges of a thundercloud need to be dissipated. This might have been the case on an early Earth as well, where energy flows that now course through life were frustrated.

"Seen from this perspective," writes complex-systems scientist Anne-Marie Grisogono, "life appears as a thermodynamically necessary mechanism to relieve the continuous production of free geochemical energy on Earth … more efficiently than abiotic processes could. If this astonishing hypothesis is correct, then it could be that wherever sufficiently steep free energy gradients exist and conditions permit, life will arise not as a rare, fragile, and contingent accident, but inexorably."

We might not be so lonely in the cosmos after all.

In her search for this kind of protometabolism at Lost City, Lang has found some of these precursors to life. But these are still very simple organic compounds. And life has been so successful in the intervening eons, taking over the entire planet, that it now drapes the deep sea in snowdrifts of organic carbon—some of it thousands of years old—making it difficult to tease out what the vents are producing all on their own out of CO_2 without the help of life.

This is why Lang is headed back to Lost City, via the Azores, on a converted oil-drill ship called the *JOIDES Resolution*, the jewel of the seventy-five-year-long International Ocean Discovery Program. The program has done as much as any other to illuminate our understanding of Earth history, from pulling up cores of sediment from the ocean floor that have revealed the last 70 million years of climate history in astounding detail, to drilling into the asteroid crater that killed the dinosaurs. But Lang's cruise, which will drill deeper into the environs of Lost City than ever before to get beyond the pollution of the living world in search of the earliest reactions of life, will be one of the last. After almost a half century, the US government pulled its funding for the drill ship *JOIDES* in 2024. Meanwhile, in a world increasingly desperate for metals and minerals of the exact sort that precipitate out of these strange vents, the location has become all but irresistible for deep-sea mining. As such, the alien world of Lost City and the vent world around it have been auctioned off to a Polish minerals company. While claiming a pedigree as old as the Earth itself, this vent system may soon be put in contact—as with many geological formations before it, from the coal of Appalachia to the copper of Australia—with the strange new civilizational system that emerged on the Earth's surface in only the past few centuries. And it is more than a little ironic that, while these kinds of primordial chimneys might have been fundamental to the development of all metabolism on Earth, they may be strip-mined in the decades to come, in service of a new kind of industrial metabolism, based not on fossil fuels, but on metal-hungry solar panels and batteries. A return, of sorts, to life's origins.

A PLAUSIBLE MAP CAN BE SKETCHED FROM THESE HUMBLE ORIGINS at the vents, some 4 billion years ago, to the continuously unfurling complexity of a living world that now distinguishes our planet fromall the others. And unlike other origin-of-life hypotheses, where me tabolism is invented out of whole cloth after life emerges, the story we have traced puts metabolism right at the start: there was a disequilibrium that needed to be relieved at the vents, an unending stream of free energy to dissipate, CO_2 available to build the stuff of life (even desperate to do so), and entropy to generate. There were tiny cavities within which to experiment with this curious chemistry of life, where the products of one reaction could be fed back into the next, to fuel still more complex biochemistry. And in the outflow to the ocean, there was even a way to get rid of the waste.

If life is, as Russell likes to say, a kind of engine, then this last bit is key. Life, in the very act of living, helplessly generates entropy. It unavoidably makes waste, exhaust. To stave off equilibrium—or death, in this case—life needs to continually expel this exhaust. For example, today even the simplest cells on Earth produce forty times more waste than they produce living matter. If they don't get rid of it, they die. Put a banana in a tailpipe for a demonstration of the same phenomenon. And early protolife, still an unguided chemistry experiment, would have been even more wasteful still. We tend to define life by the physical forms it takes, but we could perhaps more comprehensively think of it as this entire flow, with the organic carbon residue left over to make bodies merely serving as levees that keep the roaring rivers of energy and material channeled within its banks.

The details of this origin story still comprise vast lacunae in understanding. And even within this research community there are disagreements on which pathways the original metabolism might have taken. It is perhaps telling, though, that the simplest, most ancient way to make life on Earth out of CO_2[fn24] one still used by anaerobic bacteria and archaea today, and present in the last common ancestor of all life on Earth—a microbe that appears to have loved hot, anoxic environments and heavily relied on metals for its metabolism—is also the easiest to reproduce with the chemistry of the vents, without the help of biology whatsoever.[fn25] Whichever path life took, though, this vent-world theory for its origins reframes life not as a chance miracle, but a *necessity*—one that the planet needed to relieve the stresses of a young, restless, unstable chemistry. It's a creation story that obeys the thermodynamic requirements of life.

THIS IS LIKELY A DIFFERENT STORY THAN THE ONE YOU'VE HEARD about life's origins. A century after Darwin first suggested it, the alternative "warm little pond" still reigns supreme in many corners of the research community, as well as in the popular press. This persistence can in large part be traced back to 1953, when Darwin's intuition that life started in just such a putrid puddle of primordial soup at the planet's surface seemed to be validated with the success of the Miller-Urey experiment. The experiment was essentially a tabletop version of Darwin's pond, in which laboratory flasks dosed with ammonia, hydrogen gas, methane, and a spark produced amino acids—the building blocks of proteins. It was also an experiment that, it turns out, doesn't resemble the chemistry of the early Earth at all and, Lane laments, "has blinkered the field for two generations."[fn26] But along with the discovery of DNA's double helix that same year, and in the excitement of the postwar Information Revolution, many couldn't resist the idea that all you needed was to somehow fashion a self-replicating string of genetic code, and then you were off. Natural selection would take the wheel thereafter and, given the oceanic expanse of time provided by geology, the biosphere would eventually emerge, somehow.

Information, it was assumed, in the form of genetics, could bootstrap the rest of biology from scratch—including the preposterous, thermodynamically uphill Rube Goldberg machines of metabolism. Voilà. Here were life's origins. In contrast to the hydrothermal vent world—sometimes called the "metabolism first" theory for the origin of life—this is the so-called "information first" theory. In its most mature form, it is represented by the “RNA world” scenario, in which a soup of these strange complex molecules, formed somewhere on the surface of the early Earth, began replicating, catalyzing reactions today performed by enzymes, and eventually gave way to life.[fn27] Perhaps Darwin was right, then, and the protometabolism of the vent world is a red herring.

The RNA world, though, is extraordinarily strange. To make it in anything like a biologically relevant amount takes a continual flow of energy to stitch together, from a similarly continual supply of complex organic molecules. Rather than being driven toward this synthesis as might happen at the vents, nothing other than an extraordinary hopscotching series of chance events would drive these reactions. This is because conditions on the early water world were decidedly unfriendly to RNA’s formation de novo: after all, discrete puddles of primordial soup might not have even existed on this ocean planet. To explain how RNA could have been synthesized in such an environment, some versions of the theory appeal to an exquisitely timed barrage of lightning strikes, others to the exotic chemistry of cyanide, lethal to life today, and the presence of what today would be ruinous levels of UV radiation to drive a wide variety of crucial and perfectly coordinated reactions. Unfortunately, this cyanide would have to be concentrated to make the theory work, perhaps by intermittent, ocean-evaporating asteroid impacts (a leading proposal), or perhaps on lone volcanic islands, which might have poked up from the endless heaving oceans. Perhaps these steaming archipelagos provided ephemeral volcanic ponds to nurse life from this bizarre chemistry—along pathways that do not at all resemble how life constructs itself today—and would synthesize something like RNA at last, against all odds. Perhaps some of it would then get subsumed into a little frothy bubble at the edge of this poisonous puddle when it dried up and then … life?

To Russell, and a growing community of origin-of-life researchers, this picture hardly resembles life at all. “Evolution is not a synonym for magic,” one writes wryly. Life isn’t just the one-off synthesis of strange chemicals, in finely tuned conditions, on the surface of a deadly world, through unlikely reactions that don’t resemble biochemistry. Making an improbable molecule is not the same thing as life. Life is not a static recipe. It is fundamentally, and inescapably, a far-from-equilibrium, dissipative *process*. In life today, genetic molecules only make sense in the blazingly energetic, bewilderingly complex, networked, and dynamic context of a living cell.

But perhaps most problematic for the theory, a replicating molecule implies replication, and replication implies exponential growth. And growth is fundamentally an energetic, material affair.

A strange little molecule replicating in a noxious pond (provided such ponds existed on the primordial water world) can't do so without an unending supply of organic carbon to continually build more of itself, and a correspondingly endless reservoir of free energy to pull off this continual act of creation. The chemistry of this primordial soup, however interesting, is quickly exhausted in a puddle, and reaches equilibrium. It's inert. A flash in the pan. It relies on strange events and a narrow, unlikely series of steps, to hopscotch to an end product, RNA, predetermined by modern chemists, rather than creating the kind of naturally occurring rocket of protobiology found at the vents—an ongoing metabolic process that feeds off free energy. These warm little ponds, without a flow of reactants in one end and exhaust out the other, would have quickly grown stagnant and, before long, reached the dreaded state of equilibrium.

Alternatively, life instead opened up a channel of energy dissipation, perhaps first unzipped at the vents, one that it has kept propped open ever since. "Despite the remarkable complexity of living order," Smith and Morowitz write again, the "function of the biosphere is a simple one: it opens a channel for energy flow through a domain … that would otherwise be inaccessible to planetary processes." In a strong sense, the Earth *needed* life.

IT WOULD BE TO CREDIT LIFE FAR TOO LITTLE, THOUGH, TO DESCRIBE it as merely a dissipative process. Hurricanes aren't alive. Life is. Hurricanes do not adapt, reproduce, evolve, respond to, or remake the environment to their own ends. Life really is different. And even if metabolism is the foundation of life—and its very wellspring in the Hadean—at some point information really did take the wheel. It's an astounding feature of our planet: put genetic material in a suitable environment, and it will reorganize the world around it into a living vessel, an organism, for its own perpetuation. "Biological systems," the theoretical physicist Sara Imari Walker writes, "are distinctive because information manipulates the matter it is instantiated in." This fact, presented drily as such in an academic paper—that information somehow becomes causally effective in the physical world through life—is nevertheless surely one of the strangest in the universe.

The extraordinary thing about life is that it took the next step. If it did start in the deep, at some point protolife must have eventually “bubbled off” the vents and struck out on its own. Today life reproduces itself by translating strings of information encoded in DNA into proteins, an algorithmic process that eventually unfurls into entire bodies. Without this ability to reliably reproduce itself through genetic instructions (in molecules plausibly built up from the sort of precursor chemicals made by vent chemistry), such forays into the wider world would have been short-lived. This is because almost all of these emissaries from the deep would have been instantly destroyed by an indifferent Earth shortly thereafter. Stuff that is not good at sticking around, doesn’t. This is the principle of natural selection. In the shadow of the astounding successes of life are endless iterations of failed chemistry and biology, rejected by the world at large. What survives is what the world allows. And at the end of this never-ending winnowing is the strange stuff that, for whatever reason, did stick around—ourselves included. Over generations and uncountable blind experiments, this vast shadow of failed experiments left behind only those surviving inextinguishable hurricanes, spinning ever fiercer and more robust to decay. Compounded over countless generations, this selective process left behind a sophisticated regulatory regime that kept these emergent energy-dissipating flows propped open.

“Life forms that ‘go digital’ may be the only systems that survive in the long-run and are thus the only remaining product of the processes that led to life,” writes Walker.

And as we will see, novel information systems would become crucial to coordinating larger, more energetic organizations of life—from the more convolved genomes of animals to the modern-day torrent of information that ties together the energy flows of a globally networked society. But at base, energy still powers it all. Without it, information doesn’t do anything on its own. It’s a book sitting on a shelf.

“The exhaust drives your car, not the onboard computer,” Russell says. “The Jet Propulsion Lab did not start by making computers in case somebody might discover the idea of rocketry. Rockets came first.”

Similarly, your computer can perform dazzling operations and feats of calculation, but it needs to be plugged in. This isn’t a facile observation. There is a fundamental relationship between information processing and energy, and nothing about our own information age is possible in the first place without flotillas of overheating server farms chugging out of sight, generating colossal amounts of waste heat and entropy—themselves feeding off wholesale immolations of fossil fuel, in power plants that remain similarly out of view. We see the tiny payoff at the end: some logic gates switching on and off, operations that produce the dancing symbols on our screens. And the effect of living our modern lives on computers imparts to information a kind of unphysical, ethereal status, spawning transhumanist dreams of being "uploaded to the cloud." But in the end, we still live on a physical planet, and the nightmarish metabolism of our world—burning fossil fuels by the gigaton, dissipating energy by the exajoule, and sending electricity over high-voltage transmission lines by the gigawatt—still underwrites the entire synthetic fever dream of modernity. If the plug were ever pulled on industrial civilization, the whole thing would collapse almost immediately.

That we don't see this ongoing explosion of mostly fossil energy that subsidizes our information economy makes it easier to imagine that life at its origin also sprang from a kind of ethereal world of information, rather than from a fundamentally energetic and material base. But the digital view of life's origin, one that places genes at the very beginning, belies the fact that life is also plugged in. Even in its "simplest" forms, it is fundamentally, shockingly *electric*. Life itself is even sometimes described, quite literally, as a kind of electric current. The strength of the electric field across the membrane of a mitochondria is that of a bolt of lightning. A typical cell uses a million electrons a second. Your brain runs on 20 watts. Your body churns through its own weight in ATP every day. And so on. Interrupt the reactive flow of energy and material through your system only briefly—whether by depriving yourself of food, or the oxygen to burn it back to CO_2—and you will die.

"Life is electricity," says Russell. "Life is an engine."

It might seem degrading to think of life in such mechanical terms, but it becomes more pardonable when considering the absurdly convoluted, electric machinery common to all cells—whose components are often described, not even metaphorically, as "rotary motors," "crankshafts," "pumpjacks," and "hydroelectric turbines." Meanwhile, the global landscape of biological energy transactions—mediated by carbon dioxide and today powered by the sun—has even been described as a kind of global electron "market." When life struck out from the vents, to pull off the kinds of otherwise impossible uphill reactions that geochemistry once performed for free, it began to make energetic "loans" to itself against collateral electrons to "pay" for these reactions. As these sorts of financial descriptions of nature begin to pile up—along with the profusion of "motors" and "engines" in biology—it can all start to seem like a vulgar reduction of the natural world to the crass language of modern industrial capitalism. Or perhaps this homology could instead be seen as recognition that energy flows within society are subject to the same thermodynamic constraints of every other phenomenon on Earth.

TODAY LIFE BURNS MUCH HOTTER THAN THE FEEBLE METABOLISMS OF the vent world. Two billion years after the emergence of life, eukaryotic cells emerged that used 5,000 times more energy than the simple primeval microbes at the dawn of time. Some 2 billion years later, with the rise of oxygen, came a world of animal life that was vastly more energetic still. And when a cold-blooded reptile world was later supplemented by the kind of feverish hot-blooded mammal- and bird-dominated ecosystems we see today—an escalation in energy consumption mirrored by ocean life over the same time span—it represented a further increase in energy dissipation through life. As a result, and as life happened upon more powerful metabolisms, and ever-greater disequilibria to exploit, we finally find ourselves in an extraordinarily energetic world of biology, with organisms that are Saturn V rockets compared to the dimly kindling metabolism of protobacteria near the base of life's family tree.

Perhaps this shouldn't be surprising, though. If the final destination of everything is equilibrium, and the ratchet toward that end moves in only one direction, then dissipative processes like life, however seemingly improbable, that move the ratchet faster than it otherwise would might not strike us as so unlikely. In fact, in its singularly prodigious skill for dissipating sources of free energy and producing entropy, life might be seen as a natural consequence of the rules of the universe. Just as a rock careening down a hillside will find the steepest slope, or a straining reservoir will inevitably widen the cracks of a buckling dam and ultimately burst through it, if there is a route to dissipating more free energy—if there is a path to generating more *entropy*—life will take it.

Just as the water draining from a bathtub will self-organize into a whirlpool, creating *structure* to hasten the water's downhill flow to a lower point (and lower energy), so too with life. Life, in this case, isn't the water going down the drain, but rather the orderly whirlpool that emerges to facilitate this flow. If this is all beginning to sound rather metaphysical, there is an obvious evolutionary motivation for life to dissipate as much free energy as it can: in the winnowing tournament of the biosphere, life that is more efficient at capturing these flows of energy for itself—by catching a fleeing zebra in its teeth, say—and dissipating them through metabolism before *something else* does, in a very real sense, "wins." The genes of the lion that catches the zebra will get passed on to the next generation, while those of the starving lion will not.

"Far from struggling against entropy production, life maximizes it!" write the geoscientists Charles Langmuir and Wallace Broecker. "And this explains its great success."[fn28]

The strange life-giving fact about our universe is that one route to ever-more disorder is through these temporary way stations of extreme order, through which free energy is ever more efficiently channeled and dissipated. And if the universe happens upon these bizarre arrangements of matter, they will prove difficult to extinguish. Evolutionary jumps in Earth history have only served to widen this flow of dissipation. And with modern humans now using at least an order of magnitude more energy than animal metabolism alone, through the industrial metabolism of fossil fuels, it represents a nightmarish escalation of this trend that extends all the way back to life's origins.

Doing anything in this universe—doing *work*—requires dissipating energy and generating entropy. And it just so happens that, for life on this planet, these energy flows are mediated by organic carbon made from CO_2. We are no different. Today, some 4 billion years on, the competitive, distributed, algorithmic system to transform as much of the world's natural resources into low-entropy goods and services for *Homo sapiens* at lower and lower cost—called capitalism—is among the most effective generators of entropy at large ever devised. We shouldn't be surprised that in yoking this system—and the political, informational, cultural, and military apparatus required to maintain it—to *hundreds of millions* of years' worth of energy stored in ancient life underground as fossil fuels, and unleashing it all at once at Earth's surface, has transformed the surface of the planet in a geological nanosecond.

“We have this weird low-entropy situation,” I fumbled in my interview with Russell, launching into what I thought would sound like an irredeemably stoned musing, “where you have lots of reduced organic carbon buried underground and a highly oxidizing atmosphere above, and our industrial civilization between them is obviously very complex and unlikely, and is dissipating this planetary gradient in what seems to me to be sort of analogous to some of the stuff you’re talking about. Is that way off base?”

“I think that’s exactly right,” he said before leaving me with one last cryptic pensée. One that sat uneasily with me. I leave it to the reader to judge whether it is insightful and profound, meaningless, or even repellent and ahistorical.

“Capitalism is not an ideology, it’s a discovery,” he said, while describing himself as a man of the left and railing against the sorts of people rewarded by capitalism’s impersonal entropy-generating algorithm. Capitalism, he claimed, is continually relieving disequilibria: shortages and high prices reroute tankers of oil and barges of copper adrift in the middle of the ocean to the highest bidder. Clothes can be produced by the shipping containerful to meet the fashions of the week and delivered to a customer across the planet. Amazon’s warehouse is restocked before you knew you even wanted to click. “Capitalism is brilliant at making sure this equilibrium is satisfied all the time very fast. Just like life, it needs regulation. The fact that administrations are destroying the regulators is a sin on the present world. That’s dastardly sinful. You increase the entropy, you fuck the world over, it’s great while it lasts, baby. But capitalism is not an ideology. Capitalism is a discovery.”

AS THE FRENCHMAN CARNOT SHOWED, THE FURTHER A SYSTEM IS from equilibrium, the more work you can do by relieving it. Beginning in his day, and accelerating ever since, humanity began exploiting the biggest disequilibrium in Earth history: an ocean of organic carbon underground, built up by life out of CO_2 over hundreds of millions of years, on the one hand, and on the other the white-hot-burning atmosphere of oxygen above, separated from each other by only the thin mineral membrane of the Earth’s crust. Studding this membrane—like steampunk versions of the cellular turbines that dissipate the disequilibrium of the cell to do the work of life—are oil pumpjacks and gas turbines, forcing these reservoirs to react to do the work of industrial society. As a result, today humans are dissipating this extraordinary planetary disequilibrium as fast as possible. Our engines have gotten bigger, burn hotter, dissipate more energy, do more mechanical work, and generate more entropy than ever before in Earth history. The rest of the book will explain how we eventually got to this point, still 4 billion years in the future.

Stepping back, though, the biosphere as a whole can be thought of as "endless forms most beautiful and most wonderful," in Darwin's words, or perhaps less romantically as "a vast industry of cascading dissipation-accelerating engines," in Russell's. Humans are a part of this legacy, from our early days honing fire to the political and economic structures that now prescribe the flows of energy and material through a society that now dissipates almost 600 exajoules of energy per year. But at root, we're still made of the planet, the division between us and the Earth is still as porous as the chambers of an alkaline vent, and it is still carbon dioxide all the way down.

2

The Great CO_2 Freakout and the End of Eternity

> I obtain dim, shuddering glimpses into those Polar eternities; when wedged bastions of ice pressed hard upon what are now the Tropics; and in all the 25,000 miles of this world's circumference, not an inhabitable hand's breadth of land was visible.
>
> —HERMAN MELVILLE

THE DIMLY KINDLING FLAME OF LIFE MIGHT HAVE BEEN STRUCK ALMOST 4 billion years ago, but in the billions of years to follow there would little on this planet to remind us of Earth. Suffocating oceans, tinted an uncanny green—not with seaweed and blooms of plankton, but with iron—sprayed pink skies with methane-drenched surf. Islands of rock now breached this surf and hosted empty rivers coursing over dead land. The riverbeds shimmered with uranium and fool's gold. This was the chemistry of an exoplanet. And the faint life on this Archaean world would have been just as unrecognizable as the landscape. Where life at its origins might have relied on the alchemy of seafloor vents to turn CO_2 into the stuff of life, in these early days this repertoire grew to include stranger lifestyles still, with microbes learning to steal electrons from hydrogen sulfide and even iron to power similar reactions. Today, though, the entire recognizable living world springs from a completely different process. Rather than hugging to rare seeps at the bottom of the ocean, or scavenging the noxious and meager gifts of gasping seas, like these archaic microbes, plants and plankton instead ingeniously figured out how to take in CO_2 from the atmosphere, then use the energy of the sun to tear water apart, and finally use the electrons loosed from this water to power the reactions that make wood, sugars, and all manner of green stuff out of thin air. Photosynthesis as we know it, in other words.

Fortuitously left over from this reaction is oxygen. It's fortuitous because then animals like ourselves—and, somewhat surprisingly, perhaps, plants as well—are able to use this liberated oxygen to burn this plant matter back to CO_2 and water again, releasing its stored energy to do the work of life. As a result, when life first took this miraculous step—training the free energy of our local star onto water—the possibility of *Earth* arrived. This revolutionary potential of photosynthesis,[fn1] and its Earth-making gift of oxygen, would someday transform the planet.

You can find artifacts from the early days of this revolution in old battered hunks of oddly striped rock, hidden in the metal cabinets of any university geology department worth its salt. These relics bear the laminations of microbial life, banded in bows and swirls that pinch off in loops here and there. Some of them date back more than 3 billion years and represent the first physical fossils in Earth history. These are the stromatolites: petrified mounds of bacterial muck that thrived in the ancient shallow seas before animals. Some of these rocks even still bear the stuff of life itself, in faint residues synthesized from ancient CO_2 eons ago.

Many of these stromatolites were likely built by pioneering photosynthesizers called cyanobacteria. But their story didn't end with these ancient rocks. In fact, cyanobacteria are still so abundant that, to this day, massive 10,000-mile blooms of them shoot off the backside of South America like a fire hose, streaming out across the equatorial Pacific. They absorb so much sunlight that they change the color of the Earth, altering the structure of the ocean and climate on a planetary scale. Elsewhere, more modest cyanobacteria fester in stagnant ponds at the edges of office parks and foul the shores of Lake Erie, in vast slicks of green that bloom in summer. There they photosynthesize and produce oxygen, just as they've been doing since they invented the trick, billions of years ago. While horseshoe crabs and coelacanths might get all the hype as living fossils, the average industrial waste site, or stagnant golf course water hazard, hosts emissaries from far, far deeper in time.

Eventually the oxygen produced by this photosynthesis would power the frenetic, vastly more energetic fever dream of animal life. But that planet of the far future was still separated from this early, unrecognizable one by a yawning chasm of deep time. Strangely, perhaps unnervingly even, this kind of photosynthesis might have already arrived on the planet by 3 billion years ago. Unnervingly because even after it did evolve, oxygen would remain a virtual flatline in the atmosphere for hundreds of millions of years. This astounding invention, which would ultimately prove planet-transforming, was initially an inconsequential oddity, restricted to a few ocean refuges, with cyanobacteria pulsing here and there in transient seaborne oases atop dreadfully anoxic seas, and in mounds of bacterial muck draping the shoreline. But for some reason, there remained essentially no oxygen on the planet at all. As a result, there was also no ozone layer, itself a by-product of the later rise of oxygen, making the surface world not only suffocating but awash in lethal radiation. Even the rocks were unrecognizable on this planet: thousands of minerals now common to our planet, from the phosphorites that we grind up to fertilize our crops, to the turquoises and azurites that line the aisles of gem and mineral shows—and which can only form on an oxygenated planet—similarly didn't exist until well into this, the Earth's middle age. Oxidized red rocks, of the sort that today lend their hues to the desert arches and canyons of the American Southwest, couldn't, and didn't, exist yet. The planet at this time was an Earth-in-name-only.

It wasn't until around the time of an unthinkable global ice age, over 2 billion years ago—halfway into Earth history—that oxygen did at last haltingly blush for a few hundred million years. But this so-called Great Oxidation Event, though stunningly demarcated in the rock record by a suite of novel minerals, is not the final blossoming of a newly breathable planet that many imagine—or at least as it often gets discussed in the popular press. Despite the widespread pop-science trope, there was no mass extinction of hapless microbes that couldn't deal with the reactive gas. What actually happened was that oxygen soared to perhaps 2 percent of the atmosphere in these golden years (a level low enough to kill you in minutes) compared to 21 percent today, before crashing back down and remaining perhaps only a percent of the atmosphere or less for over a billion years afterward. In the dreadful expanse of time that followed—while the surface oceans, especially near the shore, might have been newly breathable for microbes—the oceans at large still suffocated, blushing with poisonous hydrogen sulfide, while the breathless deep sea remained suffused with iron. For much of this somewhat anticlimactic eternity after the Great Oxidation, oxygen on Earth might have remained too low even for alien observers to register it in our atmosphere. As a result, the living world continued to be adapted to a planet that by our standards hosted hardly any oxygen at all, even after the tantalizing whiff of it some 2 billion years ago.

Instead, the ageless doldrums that followed the Great Oxidation are known as the “Boring Billion,” a span of geologic time from 1.8 billion to 850 million years ago so seemingly uneventful that it sits in sublime, Zen-like equipoise, a cosmic void in Earth history. This was a planet stuck in second gear, in the tedious valley where most Earth-like planets in the universe might get stuck, stalling out as hopeless microbial worlds for eons until their cleansing fire by stellar catastrophe. From a very animal-biased perspective, and very roughly speaking, nothing happened for almost all of Earth history—and threatened to keep happening, forever. Landmasses that had emerged from the oceans drifted and collided over hundreds of millions of years, amassing continents that came and went, just as easily as they built the epic unseen mountain ranges studding their surfaces, only to be eroded to nothing. But with no consciousness to mark the tempo, these were ageless days—the untenanted eons of eternity.

Let’s wallow in the boredom a moment longer, if only to highlight the shocking cataclysm that would terminate it and the riot of animal life that would follow. It’s 1,165 million years ago, or it’s 973 million years ago—it doesn’t really matter. Limning the exposed wastes of the empty continents, crusted slicks of bacterial slime ripple the tidal flats, just as they did a billion years before. A dimmer sun rises and sets on these shores hundreds of billions of times, to no audience. Offshore, the deep languishes, vacant and unknown. On land is bare rock, wind, the dull roar of an odd rockslide now and—every few centuries—the muffled report of a distant volcano. But that’s it. In all this time, Earth’s crowning achievement after almost 4 billion years is pond scum. But put some of this muck under a microscope and all this planetary boredom begins to appear a little deceiving. In a handful of water one would find the frenzied makings of the world to come: restless eukaryotic cells squirm in this microcosm, representing strange chimeras of several older microbes, now all housed and made to work together in much larger, more rowdy cells that, for the first time ever, had sex with each other. Though some algae was already riffing on some very simple forms of multicellularity by this early date, someday these eukaryotes would join together to form bodies and give rise to all *complex* multicellular plant, animal, and fungal life, even if their evolution was painfully slow in the hundreds of millions of years after their origin. Still, though, even if the disparate pieces that would eventually make up our modern world were all here—if only dimly perceptible in this endless age before animal life—this was not a world that could host us yet. As is true for almost any time in Earth history, if you had parachuted onto this planet without the accoutrements of an astronaut, you would have died within minutes.

But what has this all got to do with CO_2? As we'll see in the next chapter, everything. The ultimate rise of oxygen, the one that finally allowed the energetic, planetary frenzy of animal life that we take for granted, and that occurred "only" a few hundred million years ago, was surprisingly less a story about photosynthesis per se—which, after all, had already been around for billions of years—than it was about life turning CO_2 into plankton and plants and, most crucially, burying them in the Earth as fossil fuels, charging up the atmosphere in the process. This planetary process would make the surface world breathable to animals like us, while putting in place the geological Chekhov's gun of carbon in the Earth's crust for industrial society to discharge. But in order to get to that story, we first need to find a way out of the geological doldrums of the Earth's endless preamble. Luckily, the fossil record provides an apocalyptic escape route.

All of a sudden, starting over 700 million years ago, things got *decidedly* nonboring. And it's at this point we turn to that *other* planet-making aspect of CO_2: it is not only the source of carbon for all life, but also the primary knob controlling the climate. Its behavior on Earth as it moves through the rocks, the air, and the oceans sets the very stage upon which the carbon chemistry of life can play out. And strangely, just before the rise of animal life, this entire earthly system went completely insane.

AS EARLY AS THE 1950S AND ’60S, CLIMATE SCIENTISTS WERE WORRIED. CO_2 had been known to be a greenhouse gas since the 1850s,[fn2] and by 1896, the Swedish physicist Svante Arrhenius even estimated that doubling the CO_2 in the air would result in about 5 degrees of warming—a value still on the outer range of our best understanding of Earth’s climate sensitivity. Decades on, as twentieth-century scientists witnessed the industrial supereruption of CO_2 out of the ground in the wake of World War II, this Victorian-era science was becoming concerningly relevant to the modern world. Working from the basic physical principles that were known about the gas, the scientists put two and two together, reasoning that burning fossil fuels could soon warm the planet enough to liquidate the world’s arctic sea ice. So they tried divining the future by describing Earth in a series of equations. That is, they made a climate model. One of these pioneering scientists was the Soviet Mikhail Budyko. In 1972, Budyko accurately predicted that industrial CO_2 emissions would result in 1 degree Celsius of global warming by 2019, and the loss of about half of the Arctic’s multiyear ice. Bingo. But when he and his colleagues turned the knob slightly in the other direction, instead *reducing* the amount by which the planet was warmed, they were shocked.

By cooling the Earth even slightly, they found an apocalyptic threshold hiding in their equations,[fn3] beyond which the planet would irretrievably enter a so-called white-Earth disaster. As the Earth cooled, the ice sheets predictably advanced. But once they had advanced far enough on their theoretical planet, they would pass a point of no return and inexorably swallow the whole Earth in an endless winter. This is because, as the ice sheets in their model grew, they reflected ever more sunlight back to space, made the world even colder, and propelled the march of the ice sheets even farther. Finally, the Earth tipped. A world-ending positive feedback kicked in and the advance of the glaciers became self-sustaining, unstoppable, irreversible. The runaway ice raced to the equator, encasing the whole planet in a seemingly permanent and fatal “ice-albedo catastrophe.” Unnervingly, this point of no return in their model kicked in fairly far from the equator; the ice merely needed to reach within several thousand miles of it on either side before the planet’s fate was sealed. After that, *alea iacta est*. And surely, Budyko and colleagues thought, once the Earth was interred in such an icy tomb, there would be no escape. Thereafter the planet was far too bright and reflective to ever thaw again. It was all over. Obviously such a global endgame never happened, the scientists reasoned, because we're here today, to make models of this fictional planet and talk about how it never happened. This was just a curiosity of the models, not an insight about Earth history—as this fatal excursion was, by definition, one that we couldn't find in our own geological past.

But meanwhile, an academic world away, and on every continent, geologists were finding the unthinkable: evidence for massive ice sheets, once upon a time, in the ancient tropics.

GEOLOGISTS ARE EVERY RENTAL CAR AGENCY'S WORST NIGHTMARE. In Las Vegas, much of the rental fleet circulates among a raft of retirees, who come to the desert to hop between islands of arctic air-conditioning, giving away their life savings. But occasionally these cars end up instead in the hands of the dreaded geologist, and in the red part of the insurer's spreadsheet. These are people like Carl Simpson and Lizzy Trower from the University of Colorado Boulder. We met at the Best Western by the Vegas airport and peeled out of the city. Before long our tires were rumbling over the Mojave scrubland and I was eavesdropping from the back seat on a debate about whether we should drive over this jagged mountain pass or that cactus-paved desert wash.

"Aren't there scorpions out here?" I asked, upon learning that we hadn't brought tents to sleep in for the next few days.

"Yeah," Simpson said. "I try not to think about them."

For hungover Angelenos driving back from Las Vegas, the Mojave holds a kind of uncanny terror. The harrowing sandscape out the window offers no consolation for the headache, and the car thermometer reads 110 degrees—a reminder that, for all the cheap comforts of modernity, one still lives on an indifferent natural world that would just as soon see you cooked alive in your car. As Vegas retreats in the rearview like a lurid mirage, the spare signposts of humanity that exist in this wilderness surrender to the wastes: industrial solar plants that sprawl across the desert like giant, mirrored lotuses—vaporizing birds that pass through their death rays—and strange casinos that get more tawdry with every exit. But soon these civilizational odds and ends give way to scorched playas, sand dunes, abandoned talc mines, and hillsides of desert-varnished scree. Geology, in other words.

Near Baker, California, and the World's Tallest Thermometer,[fn4] we pulled off Interstate 15 and drove into the desert.

The landscape out here is a jumble, a work in progress. About 20 million years ago, California finished swallowing its section of the Farallon ocean plate,[fn5] and an unyielding tectonic vise was released from the West Coast. As a result, the region has been slowly sorting itself out ever since—slumping, stretching, and wrinkling itself in basins and ranges as it settles. Unpaved roads that appear to go on forever, and to nowhere in particular, thread these desolate ranges, offering access to listing blocks of strata, exhumed from unthinkable ages. Finding the right mountain, we started hiking.

"This stuff is so weird," Simpson muttered to Trower, turning over a wrinkled chunk of dolomite in his hand. We found ourselves in the moments before the greatest climate disaster in Earth history, and things were getting odd.

Here, 720 million years ago, waves would have lapped gently at our feet, in the shallows off the coast of a doomed supercontinent. Supercontinents come and go in cycles, and just as the disparate landmasses of our own modern world are expected to reassemble 250 million years from now, our world was last united 200 million years ago in the form of a globe-bestriding colossus called Pangaea. But Pangaea was only the most recent supercontinent. Here in Death Valley we were on the shores of a supercontinent from *almost a half-billion years before* Pangaea, called Rodinia. To put this in perspective, it would still be a half-billion years until the first dinosaur evolved. And 650 million years until they would succumb to their fatal asteroid strike. Pangaea, then, still unthinkably ancient in its own right, was inconceivably far in the future to these rocks at our feet—twice as far as the next supercontinent is from us today, or as Pangaea is from us in the past. These rocks were 120 *times* older than the first incisions of the Grand Canyon by the Colorado River. Time is big. And at this remove you can forget trying to orient yourself on the planet. This was a world in which "North America" was capsized on its side, mostly south of the equator, and bordered by Antarctica *to the northwest*. Australia was farther north still. To the south, bordering the proto-Appalachians, was a barren Amazonia. Eastern New England and the Canadian Maritimes, meanwhile, were on the other side of Brazil, near the South Pole … east of

West Africa. Again, forget trying to orient yourself. A huge chunk of Argentina was somehow lodged in Arkansas.

It goes without saying that things like fish or land plants did not yet exist on this primeval world—and wouldn't for hundreds of millions of years. Rodinia was a kingdom of utterly desolate rock. Above it, a dilated moon, thousands of miles closer than today, loomed over the wastes, its craters shimmering in relief. Twenty-two-hour days whizzed by on Earth, and the year stretched thirteen months. A pale sun flickered in the sky, and the sea reached down from the supercontinent's rifting northern limb. High tide surged here over an equatorial Death Valley.

All along this shoreline, under the surf and atop tidal flats, slicks of bacteria patiently laid down mounds of rock, layer by layer—as they always had—in seaside gardens of slick mounds, like limestone toadstools. For billions of years this bacterial slime, untroubled by the churn of animal life (then not yet even a glint in the Earth's eye), had decorated the world's coasts in these stromatolites. Today, in Death Valley, whole hillsides are built of this timeless masonry. The old seabeds are incised by ravines, patrolled by tarantulas, and festooned in barrel cactuses. Stromatolites like these would have been familiar to any time-traveling beachcomber for almost all of Earth history. But immediately overlaying these placid coastal rocks of Death Valley are the vestiges of an apocalypse. This was a planet at a cosmic crossroads. At this startling intersection, when the Earth's uneventful eternities midwifed the riotous pageant of animal life, the planet nearly died. And it was this apocalypse that we had come to see.

In 2021, just down the road at Furnace Creek, the Death Valley National Park visitor center recorded the hottest reliably recorded temperature anywhere on Earth in human history: an astounding 130°F. But here at the edge of the park, Simpson and Trower directed my attention to a strange rock that indicated a climate catastrophe so extreme it has almost no parallel in Earth history. It represents, as one paper puts it, "the greatest disruption of the Earth system and the biggest divide in the Earth system history."[fn6]

On one side of that history was the Boring Billion—the great age of bacteria—and on the other, the provisional age of animal life we still enjoy. But in the liminal moments between these two worlds was this weird rock—gritty, stuffed with odd pebbles and boulders, like lithified head cheese. It was "glacial tillite" created by the grinding advance and retreat of massive ice sheets that plowed over the supercontinent, scouring the bedrock and dumping the debris here in the shallows of Death Valley, some 700 million years ago. Elsewhere in the park, massive boulders press into seafloor muck, dropped from above by glaciers that would have blocked out the ancient sun. This, after well over a billion years with no evidence for ice anywhere on the planet. But the existence of glaciers in the park is even more surprising in light of its earlier location: tiny magnets in these Death Valley rocks that once aligned with an ancient magnetic field like iron filings told us that we were at the equator. That is, the same rocks that bear the scars of massive, sunlight-blocking, Earth-clearing, Antarctic-style ice sheets, at *sea level*, were squarely in the tropics 700 million years ago. And sea-level ice sheets should never appear in the tropics.

Budyko's model, it turned out, had been right, and the story finally came together in the mind of geoscientist Joe Kirschvink at the California Institute of Technology. In 1992, in a short article hidden in the 1,348 pages of an academic tome intended only for the eyes of geologists, Kirschvink made the outrageous suggestion: the Earth, he claimed, near the very end of the Precambrian age—and just before the rise of animal life—suddenly plunged into an unthinkable total winter. This was very nearly the end of life itself. After over a billion years of equable weather, the planet flipped in a geologic instant. Shimmering white ice sheets that once meekly coalesced at the poles raced to the equator in a catastrophic positive feedback loop just as Budyko predicted, shrugging off sunlight and perhaps even clasping crystalline hands over the entire Earth—advancing thousands of miles in mere centuries and locking the whole world in ice thousands of feet deep. Where there were once warm tropical limestones laid down by bacteria, there was now only the windswept silence of endless glaciers. Alien telescopes keeping tabs on us—which, for eons, would have detected only a water planet, flecked with barren continents—now would have seen a glittering, dead, white orb. On land, the ice was up to almost four miles thick in places, and the global ice sheets sucked so much water out of the oceans that the seas dropped by perhaps a kilometer and became twice as salty, and iron precipitated onto the seafloor, draping it in bands of metal as it had only a billion years earlier—a bizarre signal from the rock record, the meaning of which is still debated but which indicates at the very least that global ocean chemistry briefly became completely deranged during this apocalypse. At the end of a listless eon, Earth had suddenly plunged, at terminal velocity, into an unthinkable ice age—a runaway glaciation so deep and severe that scientists long doubted whether it was even possible.

In the strange moments before the explosion of animal life into a fossil record bereft of anything but pond scum for almost 4 billion years, the planet broke. And it broke because the CO_2 thermostat that keeps our world habitable had failed catastrophically. This is the bizarre creation story of our modern planet—when the world was apocalyptically baptized in ice, then fire … then ice, then fire again, and born anew. This was Snowball Earth.

Recipe for a Disaster

No two supercontinents are alike. Pangaea, that more familiar landmass that would assemble some 500 million years later—and on a world, by then, utterly transformed by plant and animal life—resembled a massive crescent, its limbs reaching poleward from either side of the tropics. The far more ancient Rodinia, on the other hand, sprawled bloblike across the equator like a supercontinental amoeba. This accidental arrangement might have had cataclysmic consequences, since most of the sunlight that hits the Earth falls on the planet's midsection, and land—especially bare land—is much more reflective than the wine-dark seas. This means that in the age of Rodinia, not only was the sun more dim to begin with, but much of its light would have been reflected back to space by a barren tropical supercontinent, without ever warming the planet. The Earth was perhaps poised on the edge of catastrophe—the only thing keeping it from freezing over a tenuous, insulating wisp of CO_2 in the air.

We might similarly be under the heel of Snowball Earth at this very moment if not for CO_2. Our planet needs carbon dioxide in the oceans and atmosphere or else it won't be habitable. Without it, not only would plants and plankton have no feedstock of carbon from which to build up the entire surface biosphere, but the planet would be frozen to death as well. Again this is because of carbon dioxide's well-established role as the most important greenhouse gas in the atmosphere, an observation that traces back to what by the nineteenth century was already maturing as a science. While it's true that water vapor in the air traps more heat than CO_2—a fact endlessly recited by climate deniers—water nevertheless plays a subsidiary role to carbon dioxide in the atmosphere when it comes to warming. This is because water vapor, once evaporated, quickly rains out in a matter of days (nine on average) while additional CO_2 can linger in the atmosphere almost indefinitely. Moreover, water vapor responds to the long-term background amount of CO_2 acting as a positive feedback to *amplify* warming when CO_2 rises. Vaporize lots of water on a cold planet and it will condense and rain out, with the planet returning to the same state as when you started.

Put lots of CO_2 in the atmosphere, on the other hand, and it won't condense and rain out. Instead much of it will warm the planet for millennia, evaporating ever more water vapor—water that will continue to cycle through the atmosphere as long as CO_2 remains high. In fact, humans have already accelerated the planet's water cycle by almost a quarter since we started burning fossil fuels. This is why CO_2 not water vapor, is considered the "principal control knob governing Earth's temperature." Indeed, in simulations where our modern Earth is suddenly robbed of its CO_2 blanket, the temperature drops by more than 60°F (34.8°C) within only fifty years.[fn7]

On the other hand, too much CO_2 in the air gets you to the uninhabitable Venus, whose crushingly thick atmosphere tops 96 percent CO_2 and whose surface temperatures reach 900°F. So, for the sake of life on Earth, the planet needs to somehow find a balance: Just enough CO_2 to prevent a Snowball Earth catastrophe, but not so much as to produce a nightmare greenhouse inferno. This is a surprisingly narrow aperture the planet has to navigate. In recent geologic history, for instance, when CO_2 has varied between 0.1 percent and 0.018 percent of the atmosphere, it has been the difference between alligators in the Arctic Circle on the one hand, and an Antarctica's worth of ice burying North America on the other. If we lived our lives on geologic timescales, this exquisite sensitivity of the climate to CO_2 would be apparent, and we might not be so cavalier about injecting gigatons of the stuff into the atmosphere of the only known habitable planet in the universe.

So how does the planet navigate this tiny window of habitability? Well, getting the CO_2 into the atmosphere in the first place is no problem: volcanoes eternally issue it from their calderas, and from midocean ridges,[fn8] supplying the lifesaving gas to the atmosphere without interruption or recognition. Other planets spew different things from volcanoes, like ice, methane, and sulfur. But this is the Earth, where not only are the two most important greenhouse gases water and CO_2 and not only is life made mostly from water and CO_2 but so too is the gas emanating from Earth's volcanoes,[fn9] with CO_2 sometimes comprising 40 percent of volcanic gas, and water making up much of the rest.[fn10]

But before trouble starts, the planet also needs to scrub that same CO_2 from the air, and at about the same rate, before it builds up so much that it cooks the planet. In the long term, it does so by weathering rocks. This is what it sounds like: CO_2 in the air combines with rainwater and makes carbonic acid, and when this CO_2-rich rainwater—this *weather*—falls on rocks, it chemically breaks them down. Rock weathering today is aided by biology: plant roots secrete organic acids and tear apart whole mountainsides, assisted in their destruction by gravity, ice, and even rock-warping sunlight, which continually cracks open rocks and exposes them to the withering chemical assault of acidic rainwater. The runoff from these storms carries what was once carbon dioxide from the air and calcium etched out of the rocks—and washes it off the land into rivers, where it is eventually delivered out into the shallow seas. Today in the oceans, things like tiny shell-bearing plankton, larger shell-bearing things like oysters, and entire underwater worlds of coral reefs, cemented together with coralline algae, use this carbon from the air, and this calcium from the rocks, to build skeletons, quilting the seafloor in blankets of carbon-rich limestone.[fn11] In the run-up to Snowball Earth on a more impoverished biosphere, this limestone would have been instead deposited by stromatolites laid down by bacteria, or even precipitated out of the sea all on its own. And so this carbon dioxide from the air—once issued by a volcano, then rained out of the sky, then swept into the open ocean—is finally laid down at the bottom of the sea as limestone.[fn12] Over the long term, the planet stays habitable.[fn13]

Luckily for the planet, the higher CO_2 gets, the more effective this planetary CO_2-sucking mechanism becomes. That's because when there is too much CO_2 in the atmosphere, it inevitably makes the planet warmer and stormier, which breaks rocks down faster. This prevents the Earth from getting too hot in its hour of need by weathering ever more rock and drawing down ever more CO_2 … cooling the planet off. In fact, where the early Earth near its creation billions of years ago might have been kept from fatally freezing over thanks to an atmosphere extraordinarily thick with CO_2 under the regime of an otherwise faint young sun, it subsequently escaped the fate of Venus in the ensuing eons as the sun brightened, perhaps thanks to the emergence of the continents more than 3 billion years ago. As this rock emerged from the sea and offered itself to the stormy surface world above, it aggressively weathered, drawing down CO_2. Venus, on the other hand, slightly closer to the sun, might have seen its oceans boil off near the get-go and its plate tectonics stall out as it wandered off on its path of runaway-CO_2 certain planetary death.

Conversely, when the planet gets very cold indeed, the same weathering reactions instead slow down and allow the carbon dioxide that ceaselessly issues from volcanoes to slowly build up in the atmosphere. This warms the planet up again and prevents a plunge into an icy planetary tomb like Snowball Earth. These are the guardrails to the climate system—they have kept the planet habitable.

Today one actual idea for mitigating climate change, as CO_2 uncontrollably soars from industrial combustion, is to accelerate this rock-weathering process, which normally operates on a hundred-millennia timescale useless to humans, by simply crushing lots and lots of rock and spreading it all over the planet. Then, the idea is, you would just let it sit there and get weathered. Like crushing an ice cube to melt it faster, the more rock that is broken apart and weathered in this way, the more CO_2 will be accommodated in the ocean and ultimately buried in the rock record. However, this geologic process, even with human, technological aid, isn't fast enough to keep up with the overwhelming derangement of the carbon cycle currently underway. For example, to remove about one-fortieth of our yearly CO_2 emissions might require finely grinding up 8 billion tons of rock and covering an area larger than the United States in the stuff. At certain speeds, guardrails can be plowed through, and the human injection of CO_2 into the atmosphere is among the fastest in Earth history.

Thankfully, though, most of the time, in those long, languid, happier stretches of geologic time without incident, the Earth is responding not to transient, gigantic explosions of CO_2 out of the crust, like today, but to the ceaseless, trickling supply from volcanoes, and natural rock weathering suffices to stabilize the climate. And while this process might sound like a uselessly gradual process—etching the planet's surface only on the unhurried timescale of the gods—its maw is ravenous. As soon as mountains have the temerity to start rising on our restless planet, they are demolished, sloughing off kilometers of stone like snakeskin. When you happen upon a peacefully meandering muddy river, you're watching a mountain or highland somewhere else getting destroyed.

Take the Himalayas. Since they first hove into the sky 50 million years ago—and they have been continuously thrusting toward the stars ever since—they have shed *entire additional Himalayas' worth* of rock to weathering and erosion. This missing massif has been carried to the sea by the silty Ganges and Brahmaputra Rivers, where the mountains now lie flat, smeared far into the Bay of Bengal at the bottom of the ocean, piling up in seafloor sediment some ten miles thick. Similarly, the shallow continental shelf that frills the East Coast of the United States is what's left of the once-Himalayan-scale Appalachian Mountains, which have been smeared far into the Atlantic Ocean over the past 200 million years by weathering and erosion. Near me, in Colorado, there are even geologic rumors of the existence of what were once the "Ancestral Rocky Mountains"—a range just as majestic as the one that stands there now, but nevertheless completely worn flat long ago, long before the modern ones ever began to rise. They offer nothing to the local topography.

When you begin to see the world like a geologist does, mountains, rather than representing imperturbable monuments to solidity, are in fact among the most temporary features of the landscape. A mountain flecked with ponderosa pines, sinking their roots into fresh rock, generation after generation, becomes like the fast-motion footage of a whalefall carcass being relentlessly torn apart by eels and crabs until there's nothing left. And just as those animals recycle the whale's carbon back to the sea, this weathering and erosion move carbon from the air to the ocean, along with the alkalinity that makes it possible to draw CO_2 down from the atmosphere and keep the climate and oceans habitable.

When the buzz saw of weathering is trained on volcanic rocks, especially of the denser sort that ooze up from the mantle and pave the world's ocean floors, called basalt, and that occasionally get thrust up onto land, CO_2

drains from the atmosphere especially fast. This is why geoengineers are currently trying to inject CO_2 straight into Iceland, which rises above the surf from the volcanoes of the ocean's mid-Atlantic ridge. They want to artificially weather these rocks, transmuting the island's basalts directly into limestone,[fn14] storing that CO_2 permanently as geology. It's also why the islands that stretch from Indonesia to the Philippines, and that started rising from the seafloor 20 million years ago, provide about 10 percent of the world's carbon sink today: soaring volcanic islands that find themselves in the hot, storm-tossed tropics, perpetually lashed by typhoons and chewed apart by jungles, are veritable CO_2 sponges.

Without such continuous mountain-building and plate subduction the world over, the planet would be quickly worn flat. This would be devastating. In fact, this kind of global flatness in some of the more tectonically quiescent moments of Earth history has even been invoked to explain the planet's steamier climate regimes deep in its past. But when mountain ranges get pushed up, on the other hand, they invite weather, nature, and gravity to tear them apart, exposing ever more rock to weathering, and dispose ever more CO_2 elsewhere. For this reason, planetary scientists are increasingly recognizing plate tectonics as a requirement for climatic stability, and perhaps even complex life, on *any* planet in the universe. In short, if rocks didn't weather, CO_2 would build and build in the air as a result, and the living world would be eventually destroyed in the ensuing fever. This delicate feedback, then, between volcanic CO_2 plate tectonics, and rock weathering, provides the planet a thermostat, keeping it habitable over hundreds of millions of years. This thermostat has been almost unreasonably effective at keeping our planet from veering into climate doom. Almost.

So that's how the planet has worked, ever since the continents emerged from the sea perhaps 3 billion years ago. Snowball Earth, on the other hand, is how the planet breaks. For some reason, in the nightmare of Snowball Earth, this CO_2 thermostat failed and the planet nearly died.

AS LUCK WOULD HAVE IT, RODINIA WAS POSITIVELY PAVED IN MASSIVE expanses of black basalt rock of exactly the sort that geoengineers today salivate over for burying CO_2. These vast provinces of cooled lava mottled the supercontinent, produced by a series of eruptions that racked the planet in the 100 million years leading up to Snowball Earth. While these eruptions would have briefly resulted in pulses of CO_2-induced warming some several hundred millennia long, they laid the pieces for a longer-term, much more catastrophic CO_2 drawdown on this planet. Rodinia was in its death throes and beginning to fracture, slowly being blown to pieces by massive plumes of searing mantle that were rising from the middle of the Earth—and that were now arriving at the surface. The supercontinent heaved and flooded its barren land in millions of square miles of lava: 825 million years ago (in China and the Australian Outback), 800 million years ago (in South China), 775 million years ago (in Canada's Northwest Territories), and 755 million years ago (in Western Australia). And now that Rodinia was rending apart, these vast kingdoms of dark, weatherable rock found themselves liberated from the continent's brutally arid interior and freshly exposed to torrents of tropical rainfall invading the new coastlines cracking out of the supercontinent. These horizon-filling vistas of basalt highlands were hammered by hurricanes sweeping in off the trade winds of a global super-ocean and began weathering like mad. Here in Death Valley, Trower has even found the chemical signatures of such extreme weathering in the rocks, leading up to Snowball Earth. The rending world of Rodinia had become a CO_2 black hole.

With ever more coastline carved out of the fracturing supercontinent's interior, there was ever more space to dump all this carbon on the shallow shelves offshore. Moreover, there might have been a flood of Miracle-Gro washing into these seas as well, supercharging this explosion of carbon life. The reason is that, while life's main building blocks (CO_2 and hydrogen) aren't rare on the planet, other, more scarce nutrients can check life's growth, the most important of which is phosphorus. But where today industrial society mines phosphorus from rocks to spread on our crops and residential lawns to keep them neon green, the phosphorus rapidly weathering out of the crumbling rocks of ancient Rodinia, and washing into the ocean, would have just as effectively served as plant food. This nutrient-rich runoff fueled blizzards of photosynthesis in the seas that swirled off the coasts in sickly blooms of cyanobacteria, which promptly died and fell to the seafloor, pulling ever more carbon out of the surface world, effectively drawing down CO_2 and burying it in the deep. Encouraged by this mineral flood pouring off a splintering Rodinia, tiny microbes also began girding themselves in mineral armor, precipitated from the sea around them, for the first time. They soon found themselves using this armor to protect themselves from the first attacks ever from the first predators ever in all of Earth history.[fn15] Life was going mad. And all these inventive new microbes—far bigger than the long-regnant bacteria, and now ballasted in heavier mineral armor—sank ever more carbon to the depths. And so all this life—vivified by the breakup of Rodinia—fell to the seafloor, pulling ever more CO_2 out of the atmosphere with it.

But while drawing down CO_2 might be salutary in our own time to stave off global warming, for a Rodinian planet poised at the edge of a climate catastrophe, pulling still more CO_2 out of the air threatened the apocalypse. The engine of the carbon cycle had been revving since the supercontinent started breaking up over 800 million years ago, but now, 80 million years later, it was taken out of neutral, at the edge of the abyss.

And then, the coup de grâce. At the very last moment of the Boring Billion, over a million square miles of lava began erupting in towering fountains of fire, once more oozing out over the barren wastes of tropical Rodinia, building up massive barren highlands of basalt more than a mile high, like Ethiopia's but on a supercontinental scale. Today little but the volcanic plumbing remains of this cataclysm, the rest of the basalt eroded away. It comprises massive, coffee-colored columns of ancient magma from these eruptions that still loom over reindeer-mowed tundra in the uninhabited highlands of Canada's Victoria Island.[fn16] These titanic arctic cliffs are what's left of volcanic sills that once fed vast eruptions in the heart of a lifeless equatorial Canada 720 million years ago. Similar sills stripe the bedrock from Greenland to Alaska, where ancient pillow basalts of the sort that you might find burbling out of the Hawaiian Pacific today ornament the North Slope of the Brooks Range. Much later, in the age of animal life, these kinds of eruptions would cause most of the major mass extinctions. But there were no animals to kill yet on this microbial world. And just like the basalts that had been gurgling out of Rodinia in pulses over the previous 100 million years, these basalts, after a transient pulse of CO_2 into the atmosphere when they erupted, furiously weathered and weathered under the assault of tropical trade winds and hurricanes, drawing down CO_2 rapidly in the million years leading up to the end. Then, 716 million years ago, after an interminably mild climate lasting a billion and a half years, the music stopped. A curtain of ice engulfed the planet, and the Earth went on near-lifeless intermission for 56 million years.

There are always factors that conspire to make geologically historic disasters worse than just your run-of-the-mill very bad days, and at this point we've outlined an increasingly baroque derangement of the Earth system—one that brought our planet to the edge of the abyss over 700 million years ago. But even after the ice started forming on the poles, there was still a possible salvation. This is where the climate guardrails come in. When ice begins to cover up the continents, it shuts down rock weathering altogether and allows CO_2 to build back up in the atmosphere, which then nurses the planet back from hypothermia. Perhaps this lifeline could have saved this ancient world. But once again, the unique geography of Rodinia, clustered as it was around the tropics, was the planet's final undoing.

When the CO_2 plummeted and ice began to form on the poles, there was no land there to cover up. The ice instead formed over the polar seas of the Neoproterozoic world. This matters, because covering up land and shielding it from weathering could have at least slowed the drawdown of CO_2 and perhaps could have even kept the planet from freezing over. But if you're not covering up any land with your ice sheets, instead nucleating them uselessly over ancient polar oceans, then this potential rock-weathering safety switch is rendered useless, and CO_2 will continue to fall as the ice relentlessly advances. And that's what happened. Any weathering feedback that might have saved the planet became irrelevant. Instead, CO_2 continued being drawn down apace, while the ice sheets implacably advanced over the ocean. And by the time they ran aground on the supercontinent and began shielding it from weathering, it was too late. Budyko's white-Earth disaster threshold had been crossed. After reaching the all-is-lost boundary of around 35 degrees latitude, the ice became unstoppable, racing the final two thousand miles from either side of the equator in perhaps just over a century. And then, silence.

FOR 56 MILLION YEARS, SILENCE. IMPRISONED IN ICE SHEETS, THE planet fell far below freezing at all latitudes, for all seasons. The ice imperceptibly flowed toward the equator over hundreds of millennia, where it silently sublimated in wisps of ice clouds. And that was about as exciting as it got over an era of frozen stillness that lasted nearly as long as the age of mammals. The

Earth was pitilessly cold and dry, and sand seas swept over the continents where the glaciers missed. But *cold* doesn't quite capture this desolated planet. It was unearthly. Today the average temperature of the Earth—already cold compared with most of the planet's history—is about 60°F. But 700 million years ago, it might have dropped to almost *negative 60°F*. And that's the global *average*. While there were glaciers in the tropics that blocked out the sun, at the poles it was plainly a land of death. Some of the colder models that simulate the ancient polar winters of this Snowball planet drop to almost 100°F colder than the darkest months of central Antarctica today. This isn't the Earth anymore. It's so cold, in fact, that CO_2 would have formed seasonal clouds of dry ice at the poles, like it does on Mars. Though it might not sound like it, if this is true, then it was one of the most dangerous moments in out planet's entire history. Had the Earth's blanket of CO_2 snowed out from these clouds as dry ice,[fn17] sinking into the shell of *water* ice then covering the planet, it would have been too cold for the Earth's steady volcanic trickle of CO_2 to ever warm the planet again. Instead it would have instantly frozen out of the air as well, and for all but the rest of time. The Earth never would have recovered. The planet would still be frozen to this day, the age of animal life never would have happened, and you wouldn't be reading this sentence. In this preemptive apocalypse, our planet would have eventually thawed out, but perhaps only a billion years from now, when the sun turns into a red giant and boils away the oceans. Luckily, though, whatever CO_2 was frozen at the poles did in fact sublimate again during the wretched summer seasons of this Snowball Earth 700 million years ago, when the temperature would soar to a balmy −50°F.

Far beneath the global rime were anoxic, acidic, blood-red, supersalty, and utterly inhospitable oceans, buried in the dark, under thousands of feet of ice. There they hovered near freezing, warmed only by the leak of geothermal heat dissipating from the seafloor. Snowball Earth is a reminder that our planet's earthiness is contingent, unguided by any providential force other than physics. When things go haywire in the climate system, they can go *extremely* haywire. There is no Gaia to protect us, and if conditions conspire, there's nothing to stop this planet from being launched into a state just as alien as any other stupid rock in the galaxy.

Given how unearthly this planet *was*, an early and persistent challenge to the very idea of Snowball Earth was just how, exactly, it maintained its most salient earthly feature throughout it: life. How did anything survive?

When Phoebe Cohen, a paleontologist at Williams College, airdropped into the wilds of the Yukon-Alaska border to investigate the strange age before the Snowball, she enrolled in her own crash course in survival when a grizzly bear paid a visit to her camp. Cohen says the vulnerable experience brought her closer to the microscopic fossils she found there in the rocks—the first organisms in Earth history that ever had to protect themselves in mineral armor from the attack of a fellow life-form. But getting through Snowball Earth was the ultimate act of survival, and ever since it was first proposed, researchers have struggled to imagine the so-called refugia that might have existed on this alien world, where the flame of life could stay kindling until Earth proper returned.

When I caught up with Cohen at a paleontology conference, she said she had become fond of the idea that "watermelon snow" served as one such sanctuary for biology. Today this stained snow is familiar to springtime mountaineers in the Sierras, where carotenoid-loaded algae daubs the slopes crimson. In the Alps it's known as "glacier blood." There's something parsimonious about the idea that the best place to survive on Snowball Earth was in the snow itself, but other planetary foxholes might have included little pockets of meltwater on the surface of the ice sheets, where dust and volcanic ash landed. Some of these frigid pools might have even developed into full-blown moulins, where glacial meltwater flushed down through cracks in the ice and even into the sea far below—possibly serving as a refuge in the deep as well, in an otherwise dreadfully hostile ocean.

"Some of the models suggest that there would have been cold deserts in parts of Snowball Earth," Cohen said. "So you could imagine there might be communities surviving there."

"That doesn't sound like a very friendly place," I said.

"No, but none of this is very friendly," she replied.

THE SOVIET SCIENTIST BUDYKO HAD DOUBTED THAT THE EARTH HAD experienced Snowball state for this very lethality. He questioned whether any life could have made it through. But more troubling was its seeming irreversibility. There was simply no path out of the icy ruin for any planet unfortunate enough to have slipped over the edge. Three decades later, Kirschvink too worried about the prospect of a “permanent ice catastrophe.” But the models of atmospheric scientists had provided an exit for planet Earth, unknown to Budyko. The only problem was that this escape from the global winter required a violent thaw as apocalyptic as the Snowball itself.

Escape from Snowball Earth: Volcanoes Never Sleep

Today CO_2 sits at 421 parts per million (ppm), or 0.042 percent, far higher than it’s ever been in the history of *Homo sapiens*. In fact, it’s the highest it’s been since the Pliocene epoch over 3 million years ago, when evergreen forests stretched all the way to the shores of the Arctic Ocean, the Greenland and the West Antarctic ice sheets likely didn’t exist, and the shoreline of the United States pushed almost halfway into the Carolinas. A CO_2 level over 400 ppm, like it is today, then, is *very* high in the context of our modern world. Even keeping our planet at a lower level of 350 ppm indefinitely would likely raise our seas, perhaps by as much as seventy feet in the next few centuries. And in a worst-case climate scenario, humans could inject enough CO_2 into the air to push that number over *1,000 ppm*. This is a level not seen on this planet since there were crocodiles and palm trees in the Arctic Circle 50 million years ago—when there was no ice on Antarctica, and the tropics would have been totally unlivable for mammals like ourselves. With those numbers in mind, we now turn our attention to the outrageous end of the Snowball, 600 million years earlier.

When you cover the continents with mile-thick ice sheets like the Snowball did, and when you make it so cold and dry that you all but end the phenomenon of weather altogether, the CO_2-burying processes of rock weathering shut down entirely. And if you cover the oceans in a callus of ice, so too does almost all photosynthesis. As a result, you all but stop pulling CO_2 out of the air altogether. What never stops, though, is the eternal leak of CO_2 from the necks of volcanoes poking above the frozen wastes.

During the endless silence of planetary glaciation 700 million years ago, volcanic CO_2 was all the while building and building in the air, billowing from ice-choked calderas that poked out of the mile-thick rime. But with the planet encased in ice, this CO_2 wasn't ever getting scrubbed from the atmosphere, so it just kept piling up and piling up in the sky for tens of millions of years. Budyko was right to think that getting out of Snowball Earth was almost impossible, but he didn't account for just how bizarre and out of equilibrium this planetary state really was. When CO_2 at last passed a certain threshold on Snowball Earth, perhaps as high as an unfathomable 100,000 ppm or more, the ice world disintegrated almost instantaneously—launching the planet instead into a phantasmagoric supergreenhouse inferno.

The Snowball was suddenly obliterated by unimaginably high heat and CO_2 and the global ice sheets flooded the oceans with steaming meltwater. Sea level might have risen by *a kilometer* in just two thousand years. The ferocious melt was so rapid that it might have capped the oceans in a mile-deep lid of 130°F fresh water, one that still sat atop near-freezing, supersalty seas of the dearly departed Snowball. As the oceans began to mix, they rose an additional 150 feet just from the warming of the water alone. Absurdly, though, if you were standing on a Rodinian beach you might not have even noticed the seas rising at all. This is because, as the land bounced back from underneath the weight of the now-vanished ice sheets, and the gravitational pull of these titanic ice sheets on the oceans disappeared, the seas might have appeared to some Rodinian beachgoers to instead retreat from the coast, and even drop by over three hundred feet—despite the unthinkable rise in sea level globally. But anyone unlucky enough to find themselves on the supercontinent at the time would have had more pressing matters to attend to than their beachfront property's resale value: the outlandish CO_2 levels may have caused temperatures over land to soar more than 170°F. And when it was all over, one research group claims, the ice sheets had scraped a global average of three to five kilometers (about two to three miles) of rock off the surface of the entire planet. Everything about this epoch is superlative, bizarre, unprecedented.

In the Mojave, under the shade of Joshua trees, a strange pale carbonate rock caps the glacial deposits. It's the same strange rock that caps Snowball Earth deposits all over the world, and that is nearly unique in Earth history. Unlike the carbonates today that make up coral reefs, and that carpet the bottom of the ocean in seashells, life didn't lay down this rock. It simply crashed out of the oversaturated seas, violently precipitating out of the ocean all on its own, without the help of life, directly onto the seafloor, all at once, in massive sheets. In 1998, a team led by Harvard University geologists Paul Hoffman and Daniel Schrag held up these bizarre rocks as proof that Snowball Earth was terminated by an unimaginably high-CO_2 supergreenhouse. On a world that was just ground to flour by ice sheets, and then torched by absurd levels of carbon dioxide, this was a rock-weathering fever dream. These carbon-rich rocks simply crystallized out of the ocean, conjured by some of the most extreme ocean chemistry in Earth history.

"There is a need for something catastrophic," Oxford's Raymond Pierrehumbert writes about these strange febrile strata, visible here in Death Valley and sitting right atop the glacial rocks of the Snowball itself. "Something with a switch that can cause an avalanche of carbonate into the ocean …. There is no shortage of weirdness that needs to be accounted for."

AT THIS POINT EARTH'S CARBON CYCLE AND CLIMATE WERE COMpletely unhinged, and within about 15 million years so much CO_2 had been buried in this postglacial avalanche of carbon into the ocean—and by storms of resplendent green algae kindled by the flood of nutrients violently weathering out of Rodinia—that the Earth doubled back, and plummeted again into the complete and utter desolation of planetary glaciation. Snowball Earth, round two.

By this point you can guess what happened. During this encore of Snowball Earth, with the world imprisoned in ice once more, volcanic CO_2 built back up in the atmosphere again, unchecked over millions of hopelessly cold years and eventually reaching outrageously high, nightmarish levels. Then the reverse apocalypse kicked in and the planet melted "overnight" in a high-CO_2 inferno *again*—this time after being frozen for only 15 million years of utter planetary desolation. The bizarre carbonates from *this* post-Snowball supergreenhouse once more drape glacial rocks all over the world, sometimes in avalanches of limestone cement hundreds of feet thick. As the Snowball catastrophically collapsed once more, superhot oceans of meltwater, just offshore from rapidly disintegrating mile-thick ice sheets, drove some of the wildest weather in Earth history. Many of the strange postglacial rocks, some of which formed at the bottom of the ocean more than 1,000 feet deep, ominously bear the petrified ripple marks of gigantic waves that endlessly passed in the storm-tossed seas far overhead—the only signature known from the entire rock record on Earth of such extreme weather. "The late Neoproterozoic world was, quite literally, a different Earth," writes Smithsonian Institution paleontologist Doug Erwin.

Now for the weirdest part. Not only did the apocalyptic seesaw between the godforsaken wastelands of Snowball Earth and the chaotic, ultrahigh-CO_2 supergreenhouse aftermaths not exterminate all life on Earth, but animal life seems to have been somehow forged by it. Near the end of the age of the Snowball, sunbaked rocks in Oman bear the whiff of a fossil steroid[fn18] produced today only by sponges—the simplest and perhaps oldest lineage of animals. Jellyfish too likely sailed through these brave new seas. And in the tens of millions of years after Snowball Earth, the beginnings of a new planet became unmistakable.

After one final, halfhearted attempt at a Snowball 579 million years ago[fn19] —an aborted glacial reconquest that barely escaped the poles—the fossil record of large complex life astonishingly arrives with a provisional world of mysterious blob creatures, clinging to peaceful seafloor gardens. There were fractal blobs, six-foot-long ribbed fronds, and various forms pressed into the rock that could be most faithfully described as "doodads." Before long, though, these strange creatures would give way at last to the high-energy acid trip of animal life. Soon bizarre, three-foot-long "anomalocarids" darted through the seas like krill from hell, led by barbed arms that sprang straight from their faces. The creatures' slatted flanks beat rhythmically as they chased trilobites across the seafloor. Today these first monsters are imprisoned in the rocks, crumbling from mountainsides of scree in British Columbia, and paved over by suburban subdivisions in Pennsylvania. Bulbous worms, meanwhile, retreated from the Cambrian fever dream, and into the seafloor they burrowed, churning up muck that had been undisturbed for billions of years. As uncanny as these early creatures were, though, every single animal phylum alive on Earth today appeared in some form in this "explosion" of animal life in the Cambrian Period. Even our unimpressive ancestors, which appear as tiny flukelike fossil smudges in the rocks, and which bear the first tentative hints of a backbone, make a cameo here. Everything in Earth history to follow, then, is a variation on a theme—though there would be some rather elaborate variations, from *T. rexes* to blue whales.

At the end of almost 4 billion years, on a planet bereft of complex life, the Earth had been suddenly throttled by the most extreme convulsions of the carbon cycle in its entire history. It endured its most extreme climate catastrophes—the most extreme cold, the most extreme heat, the most extreme sea-level fall, the most extreme sea-level rise, the most extreme volcanoes, the most extreme erosion, the most extreme winds, the most extreme waves, the most extreme ocean chemistry, and the most extreme transformation in the planet's entire history. In Death Valley, this entire story is written in unassuming rocks—stromatolites, tillites, cap carbonates—whose drab textures are easily overlooked for the fireworks of desert flowers and collared lizards hosted on their surface, but that nevertheless signal a total planetary catastrophe almost 100 million years long. It was a complete destabilization of the Earth system, and it was one driven by carbon dioxide. All hell broke loose, and when it ended—for some reason—the riot of animal life exploded.

SO WHAT IS THE CONNECTION BETWEEN THE RISE OF ANIMALS AT long last—in the planet's old age—and the greatest climate catastrophe of all time? It might have been that in that long, languid Boring Billion that preceded it—before the breakup of Rodinia and the inception of the Snowball—plate tectonics had been mostly gummed up, and weathering slowed as a result, a situation that could explain both an almost eternal interval of balmy temperatures (high CO_2) over this span, as well as an ocean starved of nutrients weathering off the land. Such starvation conditions would have rewarded tiny bacteria, who could get by in far more meager conditions than the kinds of algae, like seaweed, that like us require a far more nutritious world to survive. Then, according to a group led by Jochen Brocks at the Australian National University, with the nightmare of the Snowball, when Rodinia was scarified by ice sheets and then pummeled in the ensuing fever by weathering and erosion, the flood of nutrients into the sea finally fueled the rise of this kind of algae. Algae, the ancestor of all plant life, and which had already evolved at least 500 million years earlier, is dramatically bigger than bacteria and would have encouraged the rise of even larger predators to graze on it. (Kelp, for instance, and all other seaweed, is algae.) Solar energy, now packaged in larger and larger globs of photosynthesis, could power larger, more complex food webs and larger organisms. And all of this larger life, in turn, would have sunk to the deep more easily, drawing down more CO_2 from the surface and ultimately, as we'll see in the next chapter, leaving still more oxygen behind. Big life begets big life.

Another idea, with an august pedigree stretching back to the voyages of Charles Darwin and Alfred Russell Wallace, is based on the principle that separation and isolation promote biodiversity. Like finches marooned on the islands of the Galapagos, life in the Snowball would have been driven back to increasingly rare refugia scattered across the globe. With life so cloistered, and subject to the extreme selective pressures of a planet at the edge of death, these rare habitable pockets would have been crucibles for evolution.

But we had come to Death Valley to investigate Carl's hunch that it was instead the cold itself that forced life to get big. His idea, he says, is "based on the observation that the most common substance in the ocean is, in fact, water." As the seas got drastically colder in the plunge into the Snowball, they also became far more viscous. If you're a tiny microbe, this would have spelled doom: you would have been suddenly stuck in an ocean of Jell-O. Getting multicellular, then, would have been a reasonable strategy just to get around. But the tiny world of water molecules, jiggling around in frenzied Brownian motion, is *much* different than the world at our macro scale—the big world of fluid dynamics, when water begins to act more like … water. And the evolutionary solutions for how to get around at our scale are *much* different than those for getting around in the microscopic world. But when you give evolution a challenge, strange things can happen. And when the Snowball was over, life that might have originally evolved multicellularity to get around in the cold viscous water had now opened up entire new landscapes of being. Being alive on this planet now meant not only providing for yourself, but also cooperating and coordinating with other cells, specializing, dividing labor, and distributing energy equitably among yourselves. It meant building societies of cells rather than rewarding selfish unicellular jacks-of-all-trades. Being an animal is hard, complicated work.

However the first animal evolved, to eventually get to a world like ours, filled with large active predators and prey, there was still something fundamentally missing from this planet: a preposterous amount of oxygen.

Even after Snowball Earth, it would take tens of millions of years for oxygen to accumulate to anything like modern levels. And one very strange idea holds that this too was the ultimate legacy of the Snowball—the long arm of the thaw. Even stranger, the more that is revealed in geologic investigations of the Great Oxidation Event, that first halting surge of oxygen almost 2 billion years earlier, the more that that episode—a similarly transformative event whose details are shrouded by an inconceivably thicker veil of deep time—also seems to have followed a similar sequence of events as this prefatory Snowball apocalypse to the age of animals. Namely, this unthinkably more archaic planetary spasm also seems to have been kicked off by a tropical supercontinent disgorging millions of square kilometers of lava that covered its surface in basalt, then it rifted asunder and weathered like mad, before temporarily being swallowed in ice, in what amounted to a far more ancient iteration of Snowball Earth—the only other known from Earth history. It was also one that, for some reason, temporarily blasted the Earth system with oxygen. And as we'll see, the coming age of animal life and the ultimate rise of oxygen might owe less to what the Snowball created than to what it destroyed.

NIGHT FELL IN DEATH VALLEY, THE PROPANE STOVE FAILED FOR inscrutable reasons, and the bracing desert cold set in. I retired to my patch of gravel and, remembering the scorpions, cinched the sleeping bag hood around my head to a half-dollar-sized blowhole around my mouth, hoping nothing would crawl in. Peering out of this nylon sphincter, I spied a small flash ignite among the stars and slowly flare across the otherwise limpid desert sky. It was the glint off the solar panels of one of the last Iridium satellites. Launched at outrageous expense in the 1990s, they serviced defunct luxury satellite phones that cost $3,000, weighed a pound, found no buyers, and quickly bankrupted the company. The most lasting legacy of the useless phones was this fleet of tacky satellites, which perennially annoyed astronomers with flares brighter than Venus, and which lit up the darkest skies of the most remote reaches of the globe. The resurrected company now pitches phones to oil and gas companies outside of network coverage, in those same far reaches of the globe where ancient seas of Mesozoic plankton slumber underfoot. Oil pumpjacks disturb this slumber, retrieving this fossil sunlight from far underground, where it's plugged into a distribution network that spiders across the Earth and delivered to the nodes of industrial society to be ignited and released again as CO_2 all over the Earth's surface. The world of animal life might have been bred by the Snowball some 600 million years ago. But the past few million years have surely bred its strangest.

3

Fossil Fuel Goes Down, Oxygen Goes Up

> Causes that tend to raise the proportion of oxygen contained in the air:
> A. Without diminishing the proportion of carbon dioxide
> 1. None B. In lowering the proportion of carbon dioxide
> 2. Formation of mineralized combustibles (coal, etc.)
>
> —JACQUES-JOSEPH ÉBELMEN, 1845

EARTH'S ABUNDANCE OF OXYGEN DOESN'T COME FROM WHERE YOU think it does.

"I would be willing to bet if you lined up a bunch of NASA scientists in a row and asked them the question, 'Why can you breathe the air?' most of them would say, 'Oh, because there are trees out there,' and that's just complete nonsense," says University of Wisconsin–Madison geologist Shanan Peters. The first time I met Peters he was fixing my flat tire in the middle of the Sevier Desert, somewhere near the Utah-Nevada border.

If you've thumbed through the brochures of almost any conservation nonprofit, which inevitably pitch the oceans or rain forests as the source of your next breath, Peters's quip might come as a surprise—or even an affront. Surely our abundance of oxygen is the benevolent largesse of plant life carpeting the continents, and swirling eddies of photosynthetic plankton alive on the Earth today. But it isn't so. Yes, photosynthesis[fn1] is a precondition for a breathable atmosphere, but it's not sufficient on its own. Not even close. This is because, to a first approximation, all of the oxygen produced by life at the Earth's surface is also used up by life at the Earth's surface. Essentially all the oxygen produced by a rain forest is used up in those trees' own undoing when they die and decay—by fungi, bacteria, bugs, animals—in the oxygen-consuming work of life and death. Eventually this process returns all this plant biomass, even that of an entire Amazon, back to the environment again as the CO_2 and water from which it was made—and uses up oxygen in equal measure. Furthermore, life isn't the only thing that draws down oxygen. Oxygen is an extraordinarily volatile gas that will react with anything it can on the Earth's surface, such as the volcanic gases and rocks that ceaselessly strip the atmosphere of it. All this gets you an atmosphere with zero oxygen—or extremely close to it—not the 21 percent surplus we enjoy today.

"Even on a world that's totally photosynthetic and everything's happy, oxygen cannot be sustained in that environment," says Peters. "I think when you fully appreciate that, it kind of changes your perspective on what's important."

Surprisingly, photosynthesizers living on the Earth's surface today can account for only a vanishingly tiny fraction of the vast reservoir of oxygen that is in our air. In fact, there is about 800 times more oxygen in the air than can be accounted for by the living world. So where did it all come from? Why can we breathe?

It turns out that to get an oxygen-rich atmosphere capable of supporting white-hot metabolic furnaces such as ourselves, yes, you need photosynthesizers pumping out the stuff, but then—*crucially*—you need to bury lots and lots of those plants in the Earth over geologic history, shielding them from consumption and decay. You might know this buried biomass in the crust more colloquially as fossil fuels.[fn2] In fact, if you know you're on a planet with photosynthesis, *and* that you can comfortably breathe the air, you could infer that there's a vast sea of fossil fuels underfoot. Oxygen, in sufficient quantities, is extremely difficult to maintain in the atmosphere, and its surprising presence on our planet only makes sense in light of geology—through processes that unfold, not on the timescale of the daily, annual, or even decadal flurry of forests, algae blooms, and biology, but over the hundreds of millions of years afforded by deep time.

THIS IS A SUBTLE STORY, SO LET'S START WITH THE OBVIOUS. YOU weren't taught wrong. Yes, oxygen is indeed produced by photosynthesis. All of it, more or less.[fn3] Today plants, algae, and cyanobacteria release it as a by-product of making the stuff of life. As we saw, cyanobacteria did it first, billions of years ago—in those microbial muck mounds, for instance, the stromatolites that we met in chapter 2. By at least a billion years ago—long before the atmosphere became breathable, or animal life arose, or its prelude when Earth became a snowball—algae recruited these cyanobacteria into their cells to do the trick for them. Seaweed, it turns out, is very old. And eventually, long after Snowball Earth, one lineage of this drifting algae crept ashore, evolving into all the land plants and tree life we know and love today. But even though these land plants evolved in "only" the past few hundred million years, they were still built upon the same crucial, irreducible bit of hardware, billions of years old.

As we've seen, making the stuff of life out of inert CO_2 is intensely difficult to pull off on our planet. To do it, cyanobacteria, algae, and plants take in CO_2 from the air and oceans, then split water apart with solar power to draw on the endless reservoir of hydrogen in H_2O (from the ocean, lakes, rain, water vapor), and use its electrons to power the transformation of CO_2 into energy-dense organic carbon[fn4] —life's job, as Michael Russell would have it. Voilà, plant stuff.

In this way, photosynthesis detains the energy of the sun here on Earth for a moment by temporarily binding it up in all manner of greenery. This strange sunlight-charged trick makes this solar energy available to the rest of the living world, which then feeds off and unravels this organic matter back into CO_2 through respiration, releasing this sunlight energy once again to power its own life. Thus photosynthesis breathes life into the bottom of the food chain and powers virtually the entire biosphere. And unlike other, older ways of making a living on planet Earth—like the microbes that got their hydrogen not from water but from far more rare fonts on the planet, like the ghostly alkaline vents at the bottom of the ocean we met in the first chapter[fn5] —photosynthesis potentially gave life access to all the hydrogen in all the H_2O in the ocean.[fn6] And the ocean, being made of water, contains a lot of H_2O.

The arrival of photosynthesis, then, meant the potential for a planet draped in far more life—one whose continents would someday be smothered in leafy greens, its oceans stirring with phytoplankton—than the meager world that came before. Even better, though, and unlike any other metabolism on Earth, the reactions that drive photosynthesis just so happen to leave behind oxygen, which gets released back into the environment, to the benefit of us all. Without this "by-product," as we'll see, hotter-burning, more energetic animals couldn't exist. With the invention of photosynthesis, then, life's ambitions suddenly swelled. Wherever you are, look around. Nothing you see is possible without it.

Upon nanoscopic investigation, though, this all appears to be something of a miracle. This is because the first step in photosynthesis—ripping water apart—is almost hopelessly difficult. H_2O simply does not want to be torn apart. The oxygen atom in water fiercely clings to the two hydrogens by their electrons—and doesn't let go without a fight. (If humans could find an efficient way of breaking these sturdy bonds in water, our energy problems would be over tomorrow: we could all drive hydrogen cars and power our steel plants with free hydrogen.) Even worse, even after tearing these hydrogens out of water, using them to make the stuff of life out of CO_2 is extraordinarily difficult on the surface of our planet.

This is why, it turns out, the first photosynthesizers on the planet weren't foolish enough to start with water. A single photon simply does not provide enough power to drive the kind of photosynthesis we're familiar with. The energetic price is just too steep. So instead, these ancient photosynthesizing microbes started with much more pliant, if diffuse, sources, like noxious hydrogen sulfide (H_2S), which needs less of a jolt from sunlight to split up and use. The only problem for the rest of the world, though, is that this kind of low-powered, primeval photosynthesis—though it still makes the stuff of life out of CO_2—doesn't release oxygen as a waste product. It makes sulfur. This was the unrecognizable world of Earth's earliest chapters, when the biosphere busily "photosynthesized," but oxygen figured nowhere on this strange planet for hundreds of millions of years.

Everything, from the Earth's deep crust to the top of its atmosphere, would be someday fundamentally remade by oxygen. But this planetary metamorphosis would begin in secret, at the invisible scale of the affairs of microbes. And at some point, somewhere on Earth, two of these ancient photosynthesizers—purple and green sulfur bacteria, perhaps, neither of which produced oxygen—somehow wired themselves together in a single effort, to team up and capture several photons at once, training all of this combined solar energy in one impossibly coordinated, continuous assembly-line effort to transform water and CO_2 into life. Perhaps already by well over 3 billion years ago, life had added this skill to its repertoire—finally harnessing enough power to drive the life-giving process that would ultimately make our world.

So how does life do it? Behold the insanity that is oxygen-producing photosynthesis, the process that endlessly rolls out the green, energy-rich living substrate of planet Earth. To reckon with it is to understand that human technology has nothing on biology. And that ours is a world of marvels.

Here's what we all take for granted.

When sunlight hits a leaf—a candidate for the seemingly most unremarkable event in the world—tiny antennae channel these photons of solar power, drawing this energy into the bewildering miniature landscape of a plant cell, passing it on through cities of chlorophyll, where it makes a ghostly transit, in a manner strangely reminiscent of the way energy flows through exotic matter near absolute zero. This energy, forged by nuclear fusion in the center of a star 93 million miles away, now finds itself ineluctably drawn, in this passage, toward the nanoscopic maw of a strange, water-eviscerating machine—one that it will soon power. Its journey thus ends at a special pair of chlorophylls (fittingly called "the special pair"), a dense tangle where this wave of sunlight energy finally crests, kicking the special pair's electrons into a furious state, before they are jolted loose one by one, then briskly dispatched and sent off down the circuits of the cell.

The special pair, though, now has a howling void where those electrons once drew close, and so it menacingly turns to two water molecules it meanwhile holds captive. This water was drawn up sometime earlier through the plants' roots, but it's in this transfiguring sanctum that it now finds itself face-to-face with the business end of a protein complex studded with impossibly well-calibrated manganese jaws, a mineral structure that appears nowhere else in nature, and one that finally consummates the elaborate, electrochemical sleight of hand that tears water to shreds. The special pair, meanwhile, is at peace again, newly sated with water's electrons. But as for those electrons just liberated from it, in the frenzy of its sunlit excitement, they now find themselves hopscotching downhill to power the pumping of that same hydrogen's *protons* to the other side of a membrane … which are then let back down through the same rotating turbines that almost all life uses to produce ATP. Finally, still more light, absorbed by still more tiny antennae, rezap these now somewhat spent electrons torn from water to reenergize them to such a high level that, at long last, it becomes energetically "downhill" to turn CO_2 into organic carbon, which plants do (with the energetic help of that new ATP it manufactured along the way) to power a never-ending metabolic vortex[fn7] that churns out energy-dense glucose ($C_6H_{12}O_6$) … and then from there on to everything else that makes a plant a plant—starches, cellulose, fats, bark, flowers, fruit. Whew.

Incidentally, this plant energy then goes on to power virtually everything else in the rest of the biosphere, ourselves included.[fn8] But staying zoomed into the atomic world for a moment longer, one finds at the final, Earth-making steps of photosynthesis outlined above, the monumental 13-nanometer-wide gates of a single enzyme, an ungainly bundle of thousands of vibrating atoms arranged *just so*, through which almost all life on Earth has passed at one point, since it's across this threshold that CO_2 is finally transformed to the stuff of life. This is the portal between nonliving and living realms. This alchemical, world-transforming molecule at the crossroads of biology and inanimate matter is called … ribulose-1,5-biphosphate carboxylase/oxygenase. The name is so unappealing, and the thing it names so important, that even scientists have had the good sense to give it the more felicitous and vaguely Mediterranean-sounding nickname "Rubisco." Despite its size—an order of magnitude smaller than *color* (400–700 nanometers)—Rubisco is the most abundant enzyme on Earth, with eleven pounds of the stuff for every person on the planet. Out of this unthinkably microscopic crucible springs virtually the entire biosphere. Meanwhile, the naked oxygen from water, having been stripped of its hydrogen and forgotten, is left behind. It escapes out of the plant and into the air and oceans—an angry, unstable molecule, looking for anything to react with—almost as an afterthought.

Of course, all of the above is hopelessly anthropomorphic, and some of the processes outlined here occur on a timescale in which, if one imagines them taking place over the course of a few seconds, an actual second would last some 30 million years.

Remarkably, all this nanochoreography happens spontaneously. These reactions can't help but happen, given the arrangement of atoms in a leaf. Other than the initial input of free energy from the sun—which zaps electrons into such a frenzied height that, in their inevitable descent, it actually becomes *favorable* to manufacture the stuff of life out of CO_2—every subsequent reaction is, on balance, downhill, including all the goings-on of virtually everything else alive on the planet. And like a river inevitably coursing down toward the sea, there's nowhere for agency to intervene in this ridiculous,[fn9] sunlight-powered, organic carbon-making trap—one that temporarily pauses the flow of energy from the sun like water behind tiny biological carbon dams and then releases it, thereby powering the rest of the biosphere.

That even the "simplest" life on Earth is stitched together from all this impossibly fine gadgetry is enough to discredit the idea of "simple life" altogether. Moreover, this isn't an isolated trick. Life repeats this sunlight-transfiguring miracle quintillions of times a second, in the chloroplasts of every parking lot weed and blade of grass in the median strip. And it's been doing it for billions of years. Amazingly, any explanation for the composition of the Earth at a planetary scale ultimately implicates these electron-scaled phenomena.

But despite the dizzying, whirring, whizzing complexity of all this madness, the end result of photosynthesis is rather simple: energy goes in, transforming CO_2 and water to plant stuff and oxygen. Ultimately, it's why Earth is the *Earth*.

BUT IF, AS THE BIOCHEMIST KAMILA MUCHOWSKA PUTS IT, LIFE IS continually "building itself up from CO_2 and breaking down to CO_2 again," the opposite reaction to photosynthesis must also take place, and in virtually equal measure. If it didn't, photosynthesis unchecked could drain the skies of CO_2 in less than a decade, by locking it all up in organic matter, and plunge the planet into Snowball Earth. Indeed, as soon as plant stuff is made, the rest of the living world sets about undoing this handiwork, using that same oxygen to burn that same plant stuff back to CO_2 in the controlled fires of metabolism. This process, as discussed at the outset of our journey, is called “aerobic respiration,” and the use of the word *burn* here to describe our energy metabolism, as Antoine Lavoisier found in his Paris laboratory over 230 years ago, isn’t a metaphor. When we eat plant stuff, or the things that eat plant stuff, or the things that eat the things that eat plant stuff, we ignite this organic matter with oxygen, and it is unraveled, just as in a fire. The original stored solar power that is released drives the machinery of life—and when we exhale, carbon dioxide is loosed upon the world once more. Ours is a somewhat delayed, distracted fire. It diffuses its power not in tongues of flame, but in the narrow tributaries of our physiology—down electron chains, turning tiny turbines, clenching muscles, torqueing bodies, expressing this dissipating energy in a raised eyebrow, the minting of a blood cell, a fleeting thought, a hope, a twinge of pain or pleasure. Consciousness is but one expression of the glow of this living fire—and our interior lives are etched in this dissipating former sunlight.

Less romantically, if you’re storing more organic carbon around your midsection than you’d like, you can go for a run. When you do, you’ll lose weight by breaking down those energy-storing fats—originally derived from photosynthesis—with oxygen and breathe them out as their original CO_2 again. You might notice your breath quicken to burn more of this lipid fuel (chemical formula: $C_{55}H_{104}O6$). You might also notice that your breath is wet, and that your body steadily radiates heat at almost 100°F. CO_2 water, heat. You are a fire. Hold your breath long enough, depriving yourself of oxygen, and you’ll be extinguished as surely as a flame under a glass.

This process *completely* reverses photosynthesis. The oxygen is used up, CO_2 is returned to the world, and the loop is finally closed when the disorganized waste heat from our metabolism, this former sunlight that we use to power our lives, eventually—and inevitably—gets reradiated back into outer space as an exhausted, diffuse, and imperceptibly dim glow of infrared radiation issuing from life. In the end, even after all this biological commotion, this is the only thing that has changed from the beginning: high-energy photons that once streamed onto the planet from the sun have been dissipated to lower-energy photons of waste heat headed back out into space, from whence they came—after a brief, if remarkable, detour through the chemistry of CO_2 as it's temporarily transformed into the stuff of life, and back again.

Oxygenic Photosynthesis and Aerobic Respiration

For those who would be aided by a visual demonstration of this stunning fact, first demonstrated by Lavoisier in his Left Bank laboratory, here are the two reactions.

The first—photosynthesis—takes carbon dioxide, water, and energy to make plant stuff and oxygen. Voilà, oxygen-producing photosynthesis:

$$\underset{\text{(carbon dioxide)}}{CO_2} + \underset{\text{(water)}}{H_2O} + \underset{\text{(sunlight)}}{\text{energy}} \longrightarrow \underset{\text{(organic carbon)}}{CH_2O} + \underset{\text{(oxygen)}}{O_2}$$

And in creatures like us its exact mirror-image opposite takes place. We break down plant stuff *with* oxygen to release that energy, producing CO_2 and water, again. Voilà, aerobic respiration—or, when it happens without life, combustion or oxidation:

$$\underset{\text{(plant stuff)}}{CH_2O} + \underset{\text{(oxygen)}}{O_2} \longrightarrow \underset{\text{(carbon dioxide)}}{CO_2} + \underset{\text{(water)}}{H_2O} + \text{energy}$$

It's essentially a perfect cycle. Virtually all of the oxygen produced by photosynthetic life on Earth's surface is used up by other life, in its own undoing. For every measure of tree that's made, a measure of oxygen is made, and no more. But when that tree dies and inevitably decays back to CO_2 *all* of the oxygen it produced in life is ultimately used up. A rotten log, one that produced oxygen in life as part of a living tree, is soon transmuted back to CO_2 and water by fungi and beetles. The former sunlight is meanwhile dissipated in the daily lives of the creatures disassembling it. Once the dead tree completely decays, and is reduced to nothing by the ecosystem, the legacy of surplus oxygen it once gifted to the world falls back down to virtually *zero*. Similarly, a whale carcass—which was ultimately made of photosynthetic plankton built from CO_2 at the ocean's sunlit surface, and which produced oxygen once upon a time—is quickly stripped clean and metabolized by oxygen-burning eels and crabs that breathe out that original CO_2.

This unraveling of photosynthesis, one that draws down oxygen, also happens without the aid of life or fire at all—only a bit slower. Gooey organic carbon stuff, like the sugars and starches made by plants, or even tougher stuff like wood or coal, simply doesn't want to stick around for long on a surface world filled with highly reactive oxygen. If carbon's astounding versatility owes in part to its easygoing relationship with other atoms—just as happy to bond with this one here as it is to break it off with that one there—then oxygen is extraordinarily greedy, vigorously ripping apart almost anything in its unending quest to fill its outer electron shell. This is why wood burns and gasoline ignites in the presence of oxygen. Energy-dense organic carbon, then, sitting around on a surface world filled with oxygen—and preferring to go back to being its original water and CO_2—is like the water perched above a waterfall or behind a dam. It "wants" to fall back down to that lower-energy state, releasing heat in the process.

In other words, plant stuff, and everything derived from plant stuff, including yourself, is out of equilibrium with our atmosphere. It was all once CO_2 and water, and it will be again. Failing any other intervention, all organic stuff will eventually oxidize all on its own. One way or another it will decay someday, or burn, because it has to. In the realm of Rubisco, solar power is merely temporarily diverted into the world of carbon chemistry—life—starting the downhill flow of energy through the rest of the biosphere (ourselves, and our machines included) on an inevitable journey back to CO_2—"the destiny of all flesh." Plant life has pushed this carbon chemistry away from equilibrium for a time. But it is this disequilibrium that we exploit whether we're breathing or driving our cars. We catalyze this inevitable, fundamental reaction back to CO_2 on the road to equilibrium.

“It’s the fate of wine to be drunk,” Primo Levi wrote. “And the fate of glucose to be oxidized.”

THIS WAS TRUE EVEN ON THE ENDLESS AGE OF THE EARTH WITHOUT much oxygen in the air at all. This is because it turns out there are lots of other, less powerful substances that can break the stuff of life back down, releasing its stored energy. These are called oxidants. Oxygen is obviously one of them, but today microbes can use lots of other “oxidants” when oxygen is scarce, like nitrate, sulfate, or forms of iron and manganese. Since the rise of oxygen, though, these microbes have been mostly relegated to the suffocating, shadowy recesses of the biosphere, in deep-sea muck, buried in fetid wetlands or in the Earth’s crust, hiding out in low-oxygen regions of the ocean or in stagnant ponds, or, in the case of truly archaic methane-spewing microbes, eking out lives in cows’ stomachs, or buried in landfills, eating trash.[fn10] But all these metabolisms provide a much dimmer “burn” than the oxygen-supercharged one that fuels us—even if they do suffice to power sufficiently humble microbes.

Animal life, however, doesn’t use any of these other strange substances to “burn” our food for energy. This isn’t because we surveyed the field of oxidants and expressed an aesthetic preference for oxygen. We couldn’t exist without oxygen. Compared with all the life that’s ever existed throughout Earth history, we have extreme and extraordinary energy needs, and burning food with oxygen is far more energetic than any other kind of metabolism on Earth—releasing up to 10 times more energy than anaerobic (or oxygenless) metabolisms, like the fermentation in yeast. As a result, with today’s atmosphere abundant in oxygen, animals are free to be far more energetic than any other kind of life-form in the history of Earth. And they—we—*are*. Sugar-powered hummingbirds zip across entire continents, sailfish bolt through the ocean at 70 mph, ibexes scale entire mountains on hooves, and humans crack calculus problems and navigate complex social lives with the 20-watt meat computers in their skulls—none of which would be possible on a world without lots and lots of oxygen to burn the organic carbon fuel in their diet. Otherwise there just wouldn’t be enough *power*[fn11] to support life this energetic. So too for your gas-powered car, which simply wouldn’t work on the stifling early Earth, and for the same reason.

It’s all just a matter of burning organic carbon with oxygen, whether in biological engines or mechanical ones. And with more energy that’s available to life, the more it can grow, in size and complexity. To that end, perhaps we could use even more energetic oxidants than oxygen, like fluorine, to burn our food, and get even *more* power out of metabolism and burn hotter still. But life doesn’t breathe fluorine because, in the words of geologist Peter Ward, “it explodes when it comes into contact with organic molecules.” Not only does this sound dangerous, but there would also be nothing for life to do with such an instantaneous reaction, one that happens all on its own. In any case, there isn’t much fluorine around (thank goodness). So oxygen it is. And oxygen is truly wonderful. At a full 21 percent of the atmosphere, it’s plentiful too.

BUT SOMETHING ABOUT THIS ENTIRE STORY SO FAR DOESN’T ADD UP. How could there possibly be so much oxygen in the atmosphere if it’s all used so quickly? And as amazing as photosynthesis is, if it’s been around for well over 2.5 billion years—and perhaps over 3 billion years—why is an oxygen-rich world such a vastly more recent phenomenon on Earth, appearing only in its latest, briefest, and most spectacular act: the age of animals? Creatures like us wouldn’t have been able to breathe comfortably for almost all of Earth history, finding relief only sometime in the wake of Snowball Earth—in the past few hundred million years, or only about the *last tenth* of Earth history. The rest of the time, our planet was not particularly “Earthlike,” despite the presence of photosynthesis over most of this span. This is strange—an eerie fact about the evolution of the Earth that scientists have long struggled to explain, invoking countless mechanisms that might account for the near eternal delay. But as we’ll see, an oxygen-rich ocean and atmosphere might be a very unlikely requirement for a planet. The incredibly reactive gas simply does not want to exist at the Earth’s surface at levels we would find at all breathable, and so for the most part over Earth history—the billions of suffocating years outlined so far in the book—it didn’t.

So how did we get from the long-reigning world of the Boring Billion to *our* world? If, in the end, life isn’t the ultimate control on oxygen, perhaps the nonliving world is. And perhaps the answer to our life-giving atmosphere isn’t found on Earth’s surface at all, but in the geology beneath us.

“LET’S DO A LITTLE EXPERIMENT,” SHANAN PETERS ANNOUNCED TO a room of geologists at the North American Paleontological Convention. “Let’s combust every living cell on the Earth right now—burn it to completion—all of it. Then ask the question: What happens?”

Peters pulled up the next slide, one that provided the answer. Burning every living thing on the planet—all the rain forests, the pond scum, the sequoias, the vast swirling blooms of phytoplankton, the bromeliads and boreal forests, the continents of corn rows, alfalfa, soy, sorghum, to say nothing of the shoals of herring, solitary blue whales, Portuguese man-of-wars, deathcap mushrooms, herds of wildebeest, and poison dart frogs—would only cause oxygen in the atmosphere to drop from 20.9 percent to, well, still about 20.9 percent, only ever slightly less. Meanwhile, the CO_2 released from burning every living thing on the planet’s surface was less than might be emitted by the end of this century anyway, just from burning fossil fuels alone.

“This is what we get,” he said. “Virtually no change.” And were one transported to this obliterated world, it would remain shockingly breathable. “Generations of humans could live out their lives breathing the air around them, probably struggling to find food, but not worrying about their next breath.”

This, Peters explained, is because for the most part, our abundance of oxygen doesn’t come from the living world. There isn’t nearly enough plant life at the surface to account for it all, given that, as noted, something like all the oxygen produced by photosynthesis is quickly used up by other life as it busily sets about unraveling that photosynthesis. And even if this organic matter weren’t consumed by life, it’s unstable in our atmosphere anyway, reacting with oxygen all on its own. As we saw above, over the full life-and-death cycle of a tree—or even an entire forest, or even an entire *planet* of forests—it produces virtually *zero* net oxygen. On a planetary scale it’s a wash. It doesn’t matter if you let plants do their thing for a billion years, you’ll never get an oxygen-rich atmosphere like our own: virtually all the oxygen that is created by the living world will be used up by the rest of the living world.

But what if only 99.99 percent of the oxygen produced by life is used by life? What if 0.01 percent of that plant matter manages to get buried underground and shielded from decay, and the corresponding oxygen it released goes unused? What if a plant, or a forest, or a bloom of plankton was spared after death from rotting and decomposition, and the oxygen that it made in life wasn't put to work in its own unraveling? What if, say, you buried a tree before it could be respired back to CO_2? It stands to reason that *that* oxygen would remain in the air, unused. And this is, in fact, what happens. Plants, trees, entire forests, and swirling blooms of algae in the ocean die and fall to the earth. Yes, 99.99 percent of them are respired back to CO_2 on the Earth's surface, or in the deep ocean, using up oxygen in the process. But the tiny remainder that aren't respired represent a leak of organic carbon into the crust. Before this photosynthetic stuff can be decomposed with oxygen, it is quickly buried and shielded from decay. And there it stays buried for hundreds of millions of years, while the oxygen it produced in life remains in the air. While this fate might account for less than 0.01 percent of the photosynthesis on Earth, such an infinitesimal amount, compounded over geologic time, can add up to quite a lot.

Therein lies the reason you can breathe. Over hundreds of millions of years, this tiny trickle out of the living world and into the rocks—one eventually responsible for our gigantic, globe-spanning reservoir of coal, oil, and gas—has allowed Earth's oxygen to build up to its absurd levels today. This is a fact widely known by geologists, but virtually no one else: the oxygen in the atmosphere ultimately comes not from the Amazon or blue-green hurricanes of plankton alive in the oceans today, but from the burial of fossil fuels in the rocks millions of years ago.

This explains Peters's surprising claim that if you burned down all the forests on the planet today, oxygen in the atmosphere would hardly drop whatsoever. After all, there are far, far more forests buried in the strata underground, from the entire sweep of Earth history, than exist on the Earth's surface. They're called coal. And indeed, if you *did* dig up and burn all the plant matter (and algae and cyanobacteria matter) ever buried in the planet's crust as fossil fuels over all time, oxygen *would* drop to zero.[fn12] But as a window onto just how much stuff is actually down there, today industrial civilization burns through the equivalent of all the animals on Earth every ten weeks. The living biosphere, then, is a rounding error compared to the planet's entire stock of organic carbon made by life over Earth history and buried in the crust. And even at our nightmarish pace there's no foreseeable future where we burn through all, or even a fraction of, the fossil fuel buried in the crust.

Thankfully this abundance means that while industrial civilization might push CO_2 up into dangerous territory from burning fossil fuel, it won't soon drain the skies of oxygen because of it. This is because, though oxygen has indeed dropped in the atmosphere in the past few decades as a result of the ongoing industrial bonfire of fossil fuel, in a mirror image to CO_2's rise, we measure oxygen in the air in parts per *hundred*, not parts per *million*, like CO_2—so we're safe for now. It might be extremely dangerous, possibly even civilizationally fatal, to push CO_2 up from 0.04 percent of the atmosphere to 0.09 percent (as we might do anyway by the end of the century), but it's not nearly as dangerous to drop oxygen by the same amount. Luckily, most of this organic carbon made by ancient life remains scattered diffusely enough throughout the world's mudstones[fn13] that even burning all the easily recoverable fossil fuels on Earth would not fatally wound, or even dent, our oxygen supply.[fn14]

So each life-giving breath we take is the cumulative legacy, not of the fizzle of life on Earth's surface, but of the vast reservoir of old life beneath it. We might take it for granted, but this is extremely strange and flammable air we're breathing. It shouldn't exist. It is made possible only by separating the reactive products of photosynthesis on a planetary scale and keeping them separate for hundreds of millions of years. This unlikely quarantining of plant stuff from the very oxygen it made in life has been, just like the regulation of the climate over Earth history, the work of plate tectonics. As the plates have smashed together, torn apart, and eroded, dumped their sediment in basins and subsided into the crust, accumulating the layer cake of geologic record over the eons—and as CO_2 has meanwhile cycled through the Earth's atmosphere, oceans, and biosphere—some of the plant life in this endless churn at the Earth's surface has been accidentally caught up in this geologic drama and entrained in the rocks as fossil fuels, eventually producing a crust now suffused with organic carbon. But if what's in the rocks beneath us is what matters most, then perhaps the control on oxygen is a fundamentally geologic process, not a biological one. This insight can lead you to some very strange places.

"THAT'S JUST ABOUT THE BEST VIEW OF THE GREAT BANK AS YOU'RE going to get," Peters announced to the group of geologists I'd tagged along with as we surveyed a two-thousand-foot-tall sheer wall of Cambrian-aged limestone called Notch Peak. It towered over the bleak, greasewood-patched scrubland of the Sevier Desert in western Utah. The astonishing precipice, made of stacked layers of seafloor from over 500 million years ago, is so steep and massive it's second in scale only to Yosemite's El Capitan in North America. Rocks like this, meanwhile, began stacking up in astonishing quantities around the world. Entombed in the ancient rockface before us, which started accumulating at the very dawn of the age of animals, trillions of trilobites lay lightlessly, in the exact position they died on the seafloor—each on some specific day, a half-billion years ago.

Everything about this half-mile-high colossus of frozen time in the desert, now slightly canted and tastefully striped like a Florentine cathedral, was on an inhuman scale. The ocean cliffs seemed menacingly indifferent to the world of the valley floor, far below. In only the past few *thousand* years, these desert basins have seen vast Ice Age lakes wither, mammoths disappear, and Japanese American internment camps spring up—all within an interval of time so short as to be indistinguishable to future geologists. But when this sunbaked seafloor that hung over us was still seafloor, the first insects were more than 100 million years in the future. The first trees too.

The "Great Bank" to which Peters was referring, and of which this range was only a sliver, was the Great American Carbonate Bank, a staggering, continent-spanning pile of seafloor, laid down in layers over the course of tens of millions of years atop North America and beyond. The bank now heaps up underfoot across most of the US in places more than *three miles* deep, spanning a breadth of almost 2 million square miles. It is representative of the huge packages of rock that would make up the age of animal life.[fn15] In comparison, our wafer-thin layer of soils, buildings, roads, forests, prairies, lakes, and creatures far above represent a strange film atop this yawning pile of strata.

The Great Bank's greatness was first recognized by petroleum geologists, who gave it its lofty nickname, and whose work describing this unbelievable stack of rocks from the dawn of animals, and the fossils within it, has helped them to extract billions of barrels of oil, trillions of feet of gas, and billions of dollars from it, from Quebec to Carbon County, Wyoming. It began to assemble tens of millions of years after the last gasps of Snowball Earth, when North America was covered no longer by mile-thick ice sheets, but instead by warm and shallow tropical waters and, just under the waves, by this vast and growing “carbonate platform,” one that perpetually sank under its own weight, allowing it to continuously grow to its monstrous proportions atop the continental crust. The best analogue for such a platform today is the much smaller Bahama Banks, but those modern islands—giant underwater mountains of limestone, capped by shoals of limestone sand and trimmed by limestone coral reefs—are almost exceptional on our modern planet.[fn16]

But early in the age of animal life, formations like the Great Bank were everywhere. In fact, much of North America might have remained almost completely drowned for tens of millions of years as these titanic stacks of limy bedrock piled up—not only here in Utah, where the edge of the Great Bank finally dove into the abyss of the Nevada deep sea, but across the entire continent and beyond, stretching from Nevada to northern Alaska to Svalbard, and from Greenland to Mexico and even Argentina[fn17] in one continuous structure.[fn18] The heart of the Great Bank lends its pale carbonate tones to waterfalls, road cuts, karst caves, lakeside cliffs, and municipal building façades across the Midwest. You could have sloshed across much of this shallow tropical kingdom without getting your hair wet, before finally running aground near Peters’s home in Madison, Wisconsin, where Cambrian aged beach sand—today mined for fracking—marks what was once the 500-million-year-old shoreline. But across the American Midwest and beyond, the Great Bank lies just underfoot—and in places fills out the lightless rocky depths miles below. It has been preserved over the eons from a planet that, though finally home to animal life, is still alien enough from our own, and distant enough in time, that it can be practically regarded as extraterrestrial.

These shallow American seas, like those that covered other continents, were patrolled by nightmarish one-off experiments at the dawn of animal life: first by vaguely crustacean-looking predators, patched together from a grab bag of unlikely and uncanny body parts—utterly unlike anything alive today[fn19] —then, tens of millions of years later, by twenty-foot-long, cone-shelled, squidlike creatures, and human-size sea scorpions. And everywhere by tentacles, carapaces, and antennae, sunlit through the waves. This feverish burst of evolution that startlingly punctuates the fossil record after an eternity bereft of animals seems to have been spurred by fleeting blushes of oxygen—which embarked on a halting, unsteady rise, not finally reaching modern levels until some 100 million years later—and a new mineral-rich ocean chemistry that encouraged skeletons.

It was further propelled by an arms race: appendages, armor, eyes, and chemoreceptors multiplied and elaborated, as more energetic animals evolved to be better at sensing, killing, avoiding, and protecting themselves from one another. The first complex nervous systems in the fossil record appear here in the Cambrian as well, to negotiate this newly dangerous world. The first dim flickerings of consciousness perhaps too. Light spots that helped creatures tell up from down in the water were modified to proper eyes that saw predators, while nerve nets developed into brains to process all this new information. With survival at stake, paths through the water to avoid these predators, or toward food, had to become optimized, made more energy-efficient. Energy is life, after all, especially in this new cutthroat world. What had been, for billions of years a static, two-dimensional microbial mat world suddenly became an information-rich 3-D arena filled with increasingly active, armored, armed, and intelligent animals. And it was through this ruthless process that life, though mere organic matter, developed interests, strategies, fear. This transformative interval, famously known as the Cambrian Explosion of animal life, has also been referred to as the “information revolution” or “cognitive revolution.” And these ever-more-energetic lifestyles, bodily possibilities, and even interior lives only became possible as oxygen tentatively rose and life could explore new possibilities, permitted by ever-more-metabolic power.[fn20] Our origin story, and that of every animal phylum on Earth today, is cataloged and archived in these titanic stacks of rock all over the world. And in the hundreds of millions of years to follow, these strata would ceaselessly pile up as well, as animal and plant life came to cover the Earth and swell in ambition and diversity.

But Peters was less interested in the strange creatures entombed in these rocks, and the remarkable story they told, than in the equally astounding rocks that hosted them.

The geologic record is not a trusty bookkeeping operation, a catalog of everything that’s ever happened, but rather a stunningly episodic product of the endless contest between creation and destruction. The Earth’s continents are restlessly heaving, smashing into each other, being uplifted, subjected to the planing forces of erosion, carried away by rivers, and piled up somewhere else on Earth to make *new* strata. Then destroyed all over again.

This is the rock cycle, which has been going on for as long as there have been continents. A long-held expectation of the geologic record is that these forces of creation and destruction should be roughly in balance. That is, the longer a rock sits around, the more likely it will find itself caught up in the Sturm und Drang of plate tectonics, and be destroyed and recycled. As a result, there should be exponentially less untouched rock left over, the further and further back we look in deep time. At least that's what one would expect. Instead, as the Great Bank attests, this isn't true at all. Peters and colleagues are now busy compiling the ages and volumes of all the rock, on all the continents, over all the Earth, from all of history—picking up the thread of twentieth-century Soviet scientist Alexander Ronov, who noticed that the actual rock record did not at all resemble the one described by theory. What they've confirmed so far is an astounding, unprecedented jump in the volume of strata laid down right at the beginning of the age of animals—a massive accumulation on the continental crust unlike anything that came before, in over a billion years. And these strata, which have survived to the current day in astounding quantities, still remain almost unmatched in their sheer scale in the Earth's crust. This is not at all what one would expect.

For some reason, after the near-eternal preamble of the Boring Billion, Snowball Earth, and the long breakup of Rodinia, rocks suddenly began stacking up on the continents like never before in Earth history. Even stranger, this switch was flipped more or less synchronously with the dawn of the age of animal life. And this extraordinary growth in Earth's strata has kept up, for the most part, in fits and starts, ever since. In Utah, where this legacy is spectacularly exposed—tilted up and exhumed from the deep—we are let in on the vast secret of Earth history, just beneath us. But elsewhere it's revealed where rivers cut deep canyons into the high country and time stripes the Triassic canyon walls, or when stacks of Jurassic sandstone strata prop up towering waterfalls, or when road cuts expose 10 million years of siltstone from the Devonian Period on the side of the highway. Indeed, if you similarly uplifted Indiana like the West, new rivers would incise a Grand Canyon out of its fathomless strata more staggering than the original.

On average, today's continents are covered in a blanket of sediments like this nearly two miles thick, one that has been piling up for the past half-billion years. This matters because it's in these kinds of sedimentary rocks that fossil fuels get sequestered. Not in the frozen magmas of igneous rocks, or in glittering veins of crystals and caches of jewels, or in deeply cooked and metamorphosed gneisses and schists.[fn21] These sedimentary rocks might begin their lives as chalky laminations of seafloor, or giant carbonate reefs like the Great Bank, but also as sandy point bars in river bends, great sandstones left by desert dunes—dunes once climbed by the dinosaurs who left their footprints behind—or the candy-striped clays of floodplains and river deltas that hold their bones, or forests of coal once visited by gigantic millipedes. But together, stacking up over the eons, these are what give us our fossil record. It's in these sedimentary rocks that we've reconstructed the history of life.

But these rocks do not only preserve the forms of ancient life as fossils; they also hold the very stuff of life itself, in the great reservoirs of organic carbon made, once upon a time, by plants at the Earth's surface. Fossil fuels, in other words. While things like sandstones and limestones can serve as capacious oil and gas reservoirs when those fuels migrate up into them from gently cooked layers below, they don't hold much of their own. Instead the most organic-rich rocks are things like shales and coals that form at the bottoms of sickly oceans and coal swamps. Just like politicians and ugly buildings, putrid swamp muck and seafloor ooze become more respectable with time, as they "lithify" to coal or shale, trapping organic matter from life in rock essentially forever.[fn22] And as we've seen, when fossil fuel makes its way into rocks like this, a surplus of oxygen is left behind in the air. But whiffs of organic carbon can be trapped in any kind of sedimentary rock. So if you build up strata, you will build up oxygen in the air. Strangely, Peters thinks it was the very accumulation of these rocks itself—this extraordinary swelling archive of Earth history over the past 500 million years—that was driving the equally extraordinary new world above, by pumping oxygen into the surface world. By this strange inversion of reasoning, the fossil record doesn't merely passively record the evolution of life on Earth, it drives it. The past is never dead, as Faulkner wrote—it isn't even past.

"You can think of it as a giant capacitor," Peters says about the coupled system of the modern atmosphere and the vast underground world that, counterintuitively, has oxygenated it over millions of years. "We're discharging it right now, using oxygen to metabolize our food, and digging up fossil fuels in the crust and burning it. But the original charging of that capacitor is fundamentally a geological process."

Peters thinks that this frighteningly discontinuous growth spurt of rock on top of the continents—a switch that seemed to be flipped to the on position only in the wake of Snowball Earth, for some reason—is why we can breathe. So what flipped the switch?

To answer this question, we first need to attend to another geological mystery: the astounding fact that just beneath these towering strata from the age of animals, and just before the explosion of complex life into the fossil record, something like a billion years have gone missing. Compared to our half-billion year run that would follow, hardly any rocks at all have survived from the long prologue to the age of animals. Darwin thought that this troubling gap in the rocks—this extraordinary defacing of the primeval geologic record by erosion—would forever obscure our origin. Peters thinks it explains it.

IN 1869, THE ONE-ARMED CIVIL WAR UNION ARMY VETERAN AND self-taught geologist John Wesley Powell embarked with nine other explorers on a mad, government-funded expedition to survey the roaring Colorado River in wooden rowboats,[fn23] on behalf of a swelling American empire eager to lay its eyes on its Manifest Destiny. In the quieter moments of the expedition, between death-defying plunges down rapids, Powell noticed in the looming walls of the Grand Canyon a startling break halfway up the rocks. Here rocks from vastly different ages sat directly atop one another, with some unknown eon separating them. "We have looked back unnumbered centuries into the past," Powell would write about the canyon's geology in almost ecstatic reverie, imagining a future day when it would again be "washed away into the Colorado Sea, and coral reefs, and shales, and bones, and disintegrated mountains, shall be made into beds of rock, for a new land, where new rivers shall flow." The line in the rocks that inspired this deep-time rhapsody would come to be known as the Great Unconformity. Strangely, Powell's observations about the dizzying break in time in the canyon walls would hold up not only for the Grand Canyon, but for the rest of North America and even the world. This Great Unconformity, it would turn out, represents a sizable erasure of Earth history, and even cosmologic history.

A general description of North America, and much of the rest of the world, is that the sometimes kilometers-thick stacks of strata that make up the age of animal life sit directly atop, and bury, crystalline “basement rock” often more than a billion years older—the deep, magmatic, and cooked roots of the continents. All across the West, and across much of the planet, from Utah to South Africa, this incomprehensible gap in geologic time is evident as an unremarkable line in the rocks, across which hundreds of millions of years have vanished. Behind the Red Rocks Amphitheatre in Denver, all across Colorado’s Front Range, and throughout North America, sandstones and limestones from the age of trilobites are laid down directly on top of mangled basement rock, from unthinkably deeper in time. A similar pattern exists in the rocks from Arabia to Siberia to Sweden to South Africa, and at the same juncture in Earth history.

What this all means is that, for some reason, the fossil record of animals—and the age of fossil fuels—seems to be preceded by an unrivaled period of global erosion, during which much of the world’s continents were worn threadbare, before being reburied by the titanic stacks of our more boisterous age. This vast erosion of Earth history gets expressed across the world as the Great Unconformity—the odd feature of the planet whose greatness lies in what’s missing. In other words, just before the age of animals was written in the rocks, much of the previous story was erased. Could it be that this extraordinary episode of destruction was required to kick off our extraordinary period of growth? Even stranger, could these missing rocks explain why I can breathe the air right now?

If the rock cycle behaved as it was long imagined—namely, that strata growing in one place were balanced by their destruction somewhere else on earth, then the story of fossil fuels and oxygen outlined so far might not work. This is because while, yes, the leak of plant matter into the crust might leave oxygen behind when it’s being sequestered, elsewhere these fossil fuel–rich rocks would be getting uplifted and exposed to the oxidizing ruin of the surface world. And when these kinds of rocks get eroded and destroyed they draw down oxygen just as readily as if they had never been buried in the first place. Indeed, where the current human industrial experiment with the atmosphere is often unfavorably compared to the natural CO_2 emissions of volcanoes, it turns out that even more CO_2 might be emitted every year by this kind of weathering of fossil fuel–rich rocks exposed at the planet’s surface today. Even worse, huge amounts of oxygen are used up, not only by weathering carbon that gets exposed at the surface again, but by reacting with things like iron minerals and volcanic gases as well. All of these so-called sinks for oxygen begin to tilt the balance against a breathable world.

So if growth and destruction of the rock record were in fact balanced, as they might have been during the Boring Billion, then oxygen would stay suffocatingly low. But if, for some reason, growth comes to dominate for a time, then this reservoir of ancient plant matter can grow in the crust. And when you build up this stack of strata on the continents that float above the Earth's mantle like Styrofoam, it can stay there for virtually all of Earth history. But the rock record obviously can't grow forever—strata can't just keep piling up toward the heavens indefinitely. So how do you kick-start the kind of far-of-equilibrium growth regime, as we have had, for the past several hundred millions of years? Well, one way is to clear a preposterous amount of crust off the continents to make room.[fn24] Peters thinks that the planetary-scaled erosion represented by the Great Unconformity is what did the trick. It effectively set the stage for the Earth to switch into growth mode. But what could cause such untold erosion on a planetary scale?

ENTER, ONCE MORE, SNOWBALL EARTH. ONE WAY TO GET RID OF A billion years of history, just before the age of animal life, is to scrape it off the face of the Earth. And the apocalyptic glacial plowshares of Snowball Earth might have scraped the top *three kilometers* off the continents across the entire planet over 600 million years ago. In doing so, they might have erased vast libraries of Earth history in their ruthless advance. Indeed, there is evidence that titanic quantities of sediment from the continental crust had ended up offshore and sent into the deep ocean trenches around the time of the Snowball, perhaps shoved there by the glaciers. In other words, the Great Unconformity, visible in rocks all over the world, could be the cataclysmic, destructive signature of Snowball Earth. And once Snowball Earth had cleared off these kilometers of rock, and the planet had recovered from its nightmarish thaw, sediment was now free to stack up again on the crust in gigantic formations like the Great Bank, and all the formations above it, burying untold quantities of life over hundreds of millions of years and, in doing so, pumping oxygen into the air and making our age of animal life possible.

If this is true, then something as singular and extreme as Snowball Earth is all but a prerequisite for intelligent life, at least on this planet. At the very least, it highlights the bizarrely contingent and improbable factors that go into our existence on Earth at this very moment. In that way it's not unlike the improbably large moon that stabilizes our orbit, the strange position of the gas giants in our solar system (an arrangement almost unlike any other observed in the galaxy), the size of the Earth, or even the ratio of magnesium to silicon in its crust. All of these have been invoked not only as oddities particular to our little corner of the cosmos, but as *prerequisites* for life on our planet. Had these conditions and events deep in our past diverged even slightly, you wouldn't be reading this book. No one would be reading anything.[fn25]

But Snowball Earth is a rather spectacular explanation for the Great Unconformity, and in recent geological meetings it has become an increasingly controversial, even acrimonious, topic. There might have been other, more subtle mechanisms at work as well.

The geologists Rebecca Flowers and Francis MacDonald have tested the hypothesis that the Snowball drove cataclysmic erosion all over the planet, using an ingenious technique called thermochronology to determine when, in an ancient rock's history, it was exhumed from deep in the crust. In places where they've tested the Great Unconformity, it appears that the vastly older basement rock beneath had kilometers of rock eroded off it in the tens of millions of years *before* Snowball Earth. In other places, the erosion appears to have occurred tens of millions of years after the global glaciation. In fact, they object to the very idea of one global Great Unconformity, instead viewing the strange boundary as a composite of many unconformities, all over the world.

Unconformities per se aren't unusual in the rock record. The planing forces of erosion are ceaseless, and as the Earth heaves, smashes together, and uplifts strata, they lathe these rocks to nothing. Indeed, most cities today, just sitting as they are at the Earth's surface and subject only to these withering forces, will leave virtually nothing behind in the fossil record whatsoever.[fn26] The pyramids, for instance, are estimated to weather away to nothing in about 100,000 years—impressive, no doubt, but falling far short of the million-year scales that geologists use to mark the tempo of deep time. We might flatter ourselves with our creations, but humanity has never created anything remotely as massive or sturdy as a mountain range, and erosion plus deep time casually obliterates entire Himalayas. Some rocks at the surface today, like the mangled gneisses and schists of New England, have been exhumed from miles of vanished rock above, testifying to this destruction. It is only in those rare moments in history that a plot of Earth finds itself in a subsiding basin, being covered in sediment, that might make it into the long-term rock record. But the intervals between will be marked by unconformities, and the vast chasms of missing time will be legible only as unremarkable lines across the walls of highway road cuts, where, say, a rock from the seafloor and rock from a desert dune tens of millions of years later abruptly meet. Unconformities like this are a dime a dozen in geology.

So unconformities are part of the rock record. They are geology's spaces between the notes. But the Great Unconformity, even if it was many unconformities, is still a seemingly global signal that covers such a vast, even cosmic sweep of Earth history and effaces almost a billion years. It seems to be altogether different and to require a common cause.

Instead of Snowball Earth, then, the continental crust might have been worn threadbare long before the global glaciation, and then continued to be denuded even after it—in an intermittent but continual destruction of the crust over hundreds of millions of years. This rather more protracted destruction might have been a product of the truly strange tectonics of the Earth's endless middle age, when virtually no mountains had been built up on Earth for the entire Boring Billion. It appears that Rodinia overstayed its welcome as an exceptionally long-lived supercontinent, with the barren landscape perhaps eroded down to its roots in the meantime. As subduction zones stayed strangely far offshore—along with their tectonic drama—the supercontinent was, well, boring. As it continually wore away it would have floated higher on the crust, subjecting itself to more erosion still. When it finally broke apart through the age of the Snowball, and the planetary system started going haywire, even more erosion followed. Perhaps, then, this litany of uniquely erosive events bracketing the global glaciation, as Rodinia assembled and broke apart, might have done the trick. But causality can be difficult to tease out of the rock record at this remove: remarkably, it is still unclear whether Snowball Earth was a consequence of this unprecedented period of erosion, or a cause. When the rocks are missing it can be tough to tell.

Or perhaps the Great Unconformity was a combination of all these things. Was it formed by pulses of erosion driven by the long assembly and breakup of Rodinia? Almost certainly. But surely an unthinkable global glaciation, one that dropped sea level by almost a kilometer, must have contributed *something* to the erosion. And besides, the later supercontinent, Pangaea, didn't leave anything remotely like a Great Unconformity behind.

Peters, though, is agnostic on the cause. It doesn't matter to him whether the Great Unconformity was ultimately the product of a cinematic obliteration of the crust by the globe-spanning glaciers of Snowball Earth, or of the somewhat more subtle, long erosive history of Rodinia and its breakup, or some combination of the two.

"How the surface eroded is not as important as the fact that it was," he says.

No matter what the case, in the tens of millions of years after Rodinia broke up, the sedimentation switch was flipped on again, the planetary capacitor began charging, the strata started piling up again, and oxygen began its unprecedented rise. This breath of life eventually made our far more energetic age of animal life possible.

NOW FOR THE SCARY PART. THE GROWTH PHASE CAN'T GO ON FOREVER. And on an old enough planet, everything has happened before. In his effort to compile all the strata ever, Peters thinks he has detected another episode far deeper in Earth history, eerily like our own—the only other comparable pulse of growth in the continental crust over all of Earth history. We touched on this episode in chapter 2. This was the Great Oxidation Event. Over 2 billion years ago—more than a billion years *before* our Snowball Earth shepherded the dawn of animal life—it appears as though another supercontinent covered in basalt broke up and weathered away in the tropics, around the same time as the only other Snowball Earth event in Earth history kicked off. It even left behind hints of a far more ancient Great Unconformity, along with what looks to be a massive pulse of growth of the rock record. And, as the name implies, this much older peak, at the beginning of the truly primeval Paleoproterozoic Era, is also linked to the only other sizable blush of oxygen in the planet's history—one that lasted hundreds of millions of years, on an otherwise noxious and stifling early Earth. Most of the Earth's commercial iron ore deposits—like the vast reserves in Minnesota's Iron Range—come from this interlude, when the metal was overwhelmed by oxygen and rusted out of the ocean.

This invigorating chapter of Earth history has long been attributed to the evolution of photosynthesis, but that might not make sense: photosynthesis had already evolved hundreds of millions of years, and possibly a billion years, earlier. Perhaps, then, this first bloom of oxygen in Earth history was also fundamentally driven by geology rather than biology. But it wouldn't last. After a few-hundred-million-year run, the Great Oxidation came to an ignominious end. The biosphere was seized by one of the greatest catastrophes in the history of life when this microbial world collapsed, the planet slid into the Boring Billion, and oxygen dropped to punishing lows for most of the Earth's vast middle age.

Could it be that these two Snowball-related blushes of oxygen—the Great Oxidation Event beginning over 2 billion years ago, which lasted a few hundred million years, and the one that we're living in now, which began over 600 million years ago—are temporary, and transient, oxygenated salad days? The Boring Billion then, the period *between* these planetary blossoms, when oxygen might have been extremely low again and life languished, starts to seem a little more interesting and ominous. The Boring Billion may be our future.

If oxygen is, in fact, controlled by freak and exceptional episodes of erosion of the continental crust—whether in cataclysmic snowballs or by a more stately attrition over the ages—and its subsequent renewal by fossil fuel–rich sedimentary rocks, then you can ultimately reach a steady state when the forces of creation and destruction are balanced, and the growth phase of the rock record is over. And steady state is death. This is because of those other sinks for oxygen mentioned before, like volcanic sulfur gases and iron minerals, which start to put the balance in the red.[fn27] And once you reach steady state, it's these oxygen sinks that start to dominate. As a result, in order to maintain a breathable surplus of oxygen in the air, the rock record on the continents needs to not only remain intact but continually *grow* to keep the planet alive. But this can't go on forever. It could be that when this growth phase ended around 2 billion years ago, the Great Oxidation Event ended as a result as well, oxygen crashed, and the Earth collapsed into the Boring Billion.

Today, once more, the growth phase in the fossil record may be over.

“DOES THE EARTH HAVE A KILL SWITCH?” WAS THE TITLE OF PETERS’S talk at the North American Paleontological Convention, a talk that appeared to leave some of the attendees dazed.

“Steady state is not an option,” he announced to his fellow geoscientists. “The moment the world enters the steady state, we’re in big trouble. In order to breathe you have to continually grow the sedimentary record. Maintaining an oxygen-rich atmosphere is dependent on this non-steady-state condition, which is not likely to stick around. This means that it’s quite possible that the [age of animal life] is transient. There might be some feedbacks that might buy us some time but maybe not.”

Peters’s simple rock model predicts the general shape of oxygen over Earth history, one that maps onto Earth’s major biological transitions. And if it can predict the past, perhaps it can predict the future as well. If it’s true that Snowball Earth cleared the top three kilometers or more off planet Earth—or even if the slow and steady, or episodic and irregular, erosion of Rodinia did the trick instead—then it should perhaps be a little concerning that the average thickness of sediments on the continents (at the time of writing) laid down over the entire age of animal life since then is more than three kilometers.

After a long growth period, the continental rock record might begin to look like the textbook depiction of it, where rocks get destroyed just as readily as they get made. This would be bad. There might be no room left for growth. That we’re even still able to breathe right now may be thanks in part to an unusual period of exceptionally high seas that surged over the continents during the dinosaur-haunted Cretaceous Period—one that laid down fossil fuel–rich rocks atop the crust, just before the age of mammals.

You’re likely familiar with this episode, whether you realize it or not—and not only for its spectacular fauna, but because it provided much of the oil on the planet, giving rise to the strange tangle of alliances and rivalries that delineate global geopolitics today. These high seas loaded with life account for the vast oil deposits tapped by bobbing pumpjacks all throughout the middle of North America—where great seas roamed by mosasaurs spilled over the continent from Texas through Colorado, all the way to Alaska’s North Slope—as well as the Persian Gulf, Venezuela, China, and Siberia. The glittering island cities of the Gulf of Mexico[fn28] tap into the offshore bounty of this age as well. These hot, stagnant, soaring seas draped their seafloors atop the continental crust with photosynthetic plankton—captured sunlight that still blazes night and day from the gas flares of oil derricks—and left behind an equivalent gift of oxygen as well. The fossil record of charcoal from the wildfires of the Cretaceous—the remnants of a more flammable age—testify to this dinosaur-aged injection of oxygen into the atmosphere.

This late flush of sedimentary growth, and subsequent bursts of fossil fuel burial in the age of mammals, might have bought our planet some time, Peters told me "somewhat tongue-in-cheek but maybe not too tongue-in-cheek." In 2016, Princeton scientists investigating Antarctic ice cores found that oxygen in our time is indeed dropping and had in fact dropped by almost a percent over only the last 800,000 years. The study's authors provided other mechanisms internal to the Earth system that might account for this somewhat alarming drop, but Peters (again only half jokingly) claims that his model provides a more sinister, eschatological cause. It is not uncommon to come across papers in which geochemists puzzle over the seemingly similar sizes of oxygen sources and sinks on Earth today, which should seemingly cancel each other out. Peters merely bites this bullet. They do. The age of energetic complex life—a relatively recent feature on a 4.5-billion-year-old planet—might be a passing fancy, a strange transient phenomenon not long for this world.

"That's the bleak implication," he says.

OF COURSE, PETERS COULD BE COMPLETELY WRONG. IN MY CONVERSATIONS with other geologists, many registered an instinctive skepticism at such a sweeping and singular mechanism for oxygenating the planet. His theory is both simple and elegant, the sort of idea that litters the graveyard of science. Suffice to say that, while the Earth is fundamentally oxygenated by the burial of organic carbon over its entire history, there are many other knobs to turn besides the changing fortunes of the continental crust. It is definitely the case that the sinks for oxygen, which robbed the surface of the stuff for much of Earth history, have waned over time. This was certainly true billions of years ago, when the planet first had to overwhelm iron-filled oceans with oxygen before it could begin to accumulate in the air. Perhaps, once these sinks are gone, oxygen is then free to build up in the air happily ever after. And maybe the amount of noxious volcanic gases that can strip oxygen from the air and oceans, or the kinds of minerals that can do the same, have declined thanks to far more obscured processes taking place deep in the Earth's mantle on our planet's one-way journey to oxygenation. For instance, even the subtly changing composition of deep seafloor crust, as it is continually recycled over all of Earth history, might have irreversibly tipped the balance toward this kind of breathable surface-world oxygenation. Indeed, one can find any number of planetary models that put the Earth on such an ever-climbing escalator to high oxygen. In this telling, abundant oxygen wasn't a contingent and freakish accident, but instead the inevitable destination of a photosynthetic planet. "A simple matter of time," as one paper puts it.

Besides, most geologists think that once oxygen is high—once it surpasses a threshold—feedbacks in the system might kick in to keep it high indefinitely. When oxygen starts to get low, for instance, plant matter doesn't decay as easily, so it gets buried, which … raises oxygen. This is another planet-saving negative feedback process. Just as the climate has guardrails, this give-and-take of the planetary respiratory system may as well. Indeed, there are models with all these feedbacks built in—and others too[fn29] —and one simply has to wind them up and let them run, to produce a pattern that resembles the broad sweep of Earth history just as faithfully as Peters's model does. And what if much of the organic matter that lands on the seafloor and then gets subducted into the mantle doesn't all come out of volcanoes as CO_2 again, but instead somehow survives its journey deep into the Earth for vast stretches of geologic time before being recycled? If so, then this would be an all-but-permanent resting place for plant stuff, and Earth's capacitor will remain charged—oxygen will remain high, that is—as long as carbon is delivered into the deep. Peters's kill switch would be moot.

In fact, maybe his entire approach is off the mark. Perhaps it isn't the volume of rock laid down that controls oxygen levels over time at all, but rather it's the huge fluctuations in the amount of photosynthetic stuff on the surface of the planet at any given time in geologic history, and that ends up making it into those rocks, that makes the difference. When nutrients like phosphorus flooded out of the shattering Rodinian supercontinent and from something quite amazingly called the "Transgondwanan Supermountain,"[fn30] these minerals from land breathed life into global blizzards of photosynthetic plankton offshore. Maybe it was the explosion of these global blooms in the ocean, and their persistence ever since, after an eon of nutrient starvation in the Boring Billion, that kicked the planet into a new steady state, increased the amount of organic stuff that ultimately got buried, and spurred the rise in oxygen at the dawn of animal life.

Or maybe, instead of phosphorus, the world was instead remade by vast, sparkling gypsum deposits on Rodinia that eroded out of this fracturing world. These were dried-up seas in the middle of the supercontinent that, when Rodinia broke up, would have flooded the oceans with sulfate. University College of London geochemist Graham Shields-Zhou thinks that this blitz of chemistry might have supercharged global blooms of strange sulfate-breathing bacteria in the ocean. In their planetary takeover, these microbes would have burned through, and exhausted, a gigantic reservoir of organic gunk in the stifling oceans, one that had posed a barrier to oxygenation over all of Earth history until then.

Something as unusual, and singular, as this global chemical cascade must have happened in the moments before the explosion of animal life because, just then, around 579 million years ago, the rock record is seemingly racked with the most profound disruption to the global carbon cycle in the planet's history. Spanning perhaps 7 million years, rocks around the world chemically record an almost physically impossible contortion of the Earth system unlike anything before or since. The bizarre geochemical signal, called the Shuram Anomaly, remains largely mysterious, and because of its place in the rocks—at the crossroads of the endless Precambrian Age and the dawn of animal life—NASA is keenly interested in its significance, underwriting geoscientists to dash off to outcrops around the world to study it, from Oman to Australia. One of the most prominent explanations for the strange signal (expressed as a ratio of carbon isotopes gone completely haywire)[fn31] involves a fundamental and permanent shift in the Earth system from the stagnant old seas steeped with organic carbon gunk and permanently low oxygen to new, clearer, more oxygenated ones, newly filtered by life. Whatever happened (and it's still entirely unclear), it is the most extreme deviation of the carbon cycle since the origin of life, and one that oddly accompanies the first large complex fossils in the fossil record. If it strikes you as scandalous that such fundamental questions remain as to how the only known habitable planet in the universe got to be that way—at the same time that the richest people on Earth make lavish bets on a hopelessly dead rock like Mars—it is.

Nevertheless, in each of these accounts, the rising oxygen early in the age of animals was a phase change for planet Earth, and once this planetary threshold was surpassed, internal feedbacks in the carbon cycle (and others) helped the Earth maintain this new, oxygen-rich steady state—a novel, dynamic, and fundamentally stable system that has served us well for over 500 million years, and that may have nothing to do with the unnerving nature of the sedimentary rock record at all. This new planet just operated *differently* than the old planet—perhaps irrevocably so. Maybe there was just no going back to the old breathless world of the Boring Billion once the age of animal life started. If this is the case, then we can rest assured that this transition to an oxygen-rich Earth was essentially a permanent system change, and that the days before us are many.

AND YET. RIGHT NOW, RATES OF PHOTOSYNTHESIS ON EARTH ARE extremely high, sufficient to double atmospheric oxygen in only a few thousand years. But for some reason oxygen is still falling. And even if one were to radically diminish photosynthesis on the planet, it would still be sufficient to oxygenate the world many times over, over a geologic time span. The Earth, meanwhile, has all the time in the world. All of this suggests to Peters that it is the geologic record itself, not the changing fashions of biology, that provides the main knob on oxygen. Fundamentally, to drive up oxygen, you need to put all that biology *somewhere*. And it might turn out that it's the storage space that's hard to come by, not the gooey green stuff.

"The Drake Equation is missing this ability to maintain this charged capacitor," said Peters, referring to the back-of-the-envelope equation used by astronomers to calculate the number of habitable worlds and intelligent civilizations in the universe. "You get a window on this planet and then it's over. You better figure out how to engineer yourself out of it."

LET'S RECAP THE ENTIRE BOOK TO THIS POINT. SO FAR WE'VE (1) TURNED CO_2 into life at the dawn of Earth history; (2) invented an impossibly intricate way of doing so by using sunlight, and which happens to produce oxygen (hundreds of millions of years later); and now (3) made space to build up the rock record and the planetary capacitor by burying fossil fuels in the Earth's crust. So now it's time to charge that planetary capacitor.

What this looks like in practice is the entire history of animal life to follow.

The fossil record, from the coal forests that stripe the mine pit walls of Wyoming to the ankylosaur-strewn oil-tar sands of Canada, is synonymous with a charging planetary capacitor. We owe each breath to this carbon in the ground, a sea of what was once CO_2 then transformed to organic carbon by life tens of millions, even hundreds of millions, of years ago. These slumbering fossil fuels are the very reason we're alive. And with every breath, and every molecule of fossil fuel we ignite, we discharge it, burning the stuff of life with this gift of oxygen to do the work of living, and return it to the air as spent CO_2.

In the long run, this is a closed system, even if it has been temporarily pushed into the far-from-equilibrium oxygen-rich state we enjoy today by the electron-scaled whirr of biology coupled with tectonic processes that play out over millions of years. But nothing stays far from equilibrium forever. And while this geologic legacy of fossil fuels might explain our very existence, it may ultimately be our undoing as well. As we'll see, the Carboniferous Period, some 300 million years ago, saw an unparalleled burial of life on Earth, when globe-spanning forests fell into the fossil record, oxygen reached outrageous highs, giant bugs ruled the planet, and wildfires swept across endless swampland. Hundreds of millions of years later, the legacy of this strange age—its vast forests of buried life—would be resurrected by one population of one species of primate, and immolated on a scale nearly unmatched in the geologic record. This was when humans started burning coal. But discharging the capacitor in this way, on a planetary scale, has been tried before in Earth history, and, as we'll see in the coming chapters, it very nearly brought down the age of animal life entirely—hundreds of millions of years before James Watt's steam engine ever drew its first violent breath.

Today we find ourselves at the end of an eons-long process, one that began very roughly sometime around Snowball Earth and, if Peters is right, may not last indefinitely into the future. But this is perhaps the only place we could ever find ourselves in time. If you wake up somewhere in the universe and you can breathe, look beneath you and you'll find interesting paleontology.

"Another way to put it is an earth that is habitable for [animals] will have a fossil record and it will be a rich one," Peters said. "Because without that you're not going to have a conversation about it afterwards."

4

CO_2 and the Great Age of Coal

"WE'RE JUST GEOLOGISTS AND YOU'RE NOT A WRITER," THE APPALACHIAN State University geologist Sarah Carmichael told me as we wound up switchback roads to the highest point in Kentucky. Bullet holes pocked the Virginia state line marker; hickory, sumac, and walnut trees hid roadside coal outcrops; and across the valley, a mountain was missing. "We're just looking at the rocks here, so it'll be fine, but I've had enough run-ins with pickup trucks trying to frighten me."

We hopped out at a turnoff, and I thoughtfully turned a rock hammer over in my palm as a truck went by. I was trying my best to look geologist-y.

"I would rather be in a university van with the Appalachian State sticker on it," she said. "I have no credibility in my car. There's bad memories. It's always a tense place here, not just now."

Before us was a lush vista of rolling Appalachia green, except for a massive terraced crater where a mountain once stood. Tiny dump trucks dotted the barren crater rim like fruit flies. The former mountain, which at one point would have been about eye level with us, had been subjected to mountaintop removal mining—the rare euphemism that's actually up to the task of evoking the thing it describes. The mountaintop had indeed been removed, the coal extracted, and the aftermath was like that of a monstrous beetle infestation that had stripped down not only the bark from the trees, but the contours from the Earth.

"They're not actually mining anymore," said Carmichael. "The trucks are just left on top to say that it's an active mine so they don't have to remediate the land."

This was the final, sputtering stage of a rapacious century-and-a-half-long process in central Appalachia—one kicked off by southern land speculators and consummated by wealthy northern industrialists. Miners flowed into these hills—from former slave plantations, war-torn Europe, and the surrounding mountains—and money flowed out, along with train cars of fossil forests. A bonfire of geologic history, one that began in England decades earlier, had jumped the Atlantic and proven inextinguishable as the modern industrialized world was born. The miners would soon be replaced by mechanical workers, and by the end of the twentieth century, the mine shafts had given way to the wholesale destruction of the mountains before us. As for the mountain we were standing on—the tallest in Kentucky—it had been similarly slated for "removal" but had been granted a stay of execution after local protests. After all, strip-mining the highest peak in the state would have been a bit on the nose. But all across Appalachia these mountaintops have gone missing, and in their place deep rocky wounds mark the landscape. How on Earth, by what force, under what spell or imperative, does an organism spread out across the land, tearing the tops off mountains?

IN THIS CHAPTER WE WILL SEE, IN ACTION, THE PROCESS OF CHARGING Earth's capacitor outlined in chapter 3. We'll see how the gun that would someday be discharged in the Industrial Revolution found itself latent on the stage of history—how all those fossil fuels got into the rocks in the first place. Then we will begin to trace how human society began releasing this extraordinarily vast cache of geologic energy from the subsurface back into the atmosphere, hundreds of millions of years after its capture in the Earth.

But to get to the great age of coal—the aptly named Carboniferous Period—from the explosion of animal life into the fossil record where we left off, over 500 million years ago, we have to traverse some 200 million years of intervening Earth history. The fossils from this span fill out the unloved antechambers of natural history museums, unfairly hurried past en route to the flashier dinosaur exhibits. But as with most of Earth history, this gap in the public imagination conceals an unfathomably long, and unfathomably rich, span of time. When one finally gets to the age of coal, an unrecognizably verdant and vital Earth had already bloomed in the meantime. Where for billions of years the continents had hosted only windswept wastelands of barren rock, all but bereft of oxygen, by the

Carboniferous Period (359–299 million years ago) strange flooded forests spanned the vast tropical lowlands of Appalachia. Where breathless eons came before, oxygen was now so high that these endless swamps throbbed with the deafening drone of gigantic insects and often exploded into wildfire—even waterlogged. This great age of coal[fn1] is a bizarre sliver of geologic time that accounts for a mere 2 percent of Earth's history but 90 percent of its coal. It also, not coincidentally, marks the high-water mark of oxygen in Earth history. This too was the result of all of this carbon going into the ground. While dinosaurs serve as a useful geologic mile marker in the age of animals, those celebrities of the fossil record were still almost 100 million years in the future. Nevertheless, by the Carboniferous, this was already a seasoned, battle-hardened world, one that had seen more than a dozen global extinction events and had been home to a roster of life that unsentimentally changed with the geologic seasons.

A quick tour of those epochal seasons between the dawn of animal life in the Cambrian and the Carboniferous coal period some 200 million years later illuminates just how CO_2 and its movement through life, the climate, oceans, and rocks shapes every element of our living planet. When animal life exploded onto the stage on a planet bereft of it for billions of years, in the Cambrian Period (541–485 million years ago),[fn2] it did so in the context of a largely forbidding, high-CO_2 world, with extraordinarily hot and stagnant seas that pulsed with deadly tides and waves of extinction, as complex life tried to dodge an early, anticlimactic end. But when the Cambrian begat the Ordovician Period (485–444 million years ago), thousand-mile arcs of volcanic mountain ranges—which would someday become the backbone of the Appalachians—furiously weathered away in the tropics, drawing down CO_2 and cooling the feverish planet. While the explosion of animals in the inaugural Cambrian marked a shocking rupture with all of Earth history before it, by this second, more hospitable age they had become established and confident stewards of this planet: the oceans grew more suffused with oxygen as life buried ever more carbon, and the diversity of life on Earth *tripled* in only 10 million years.[fn3] A menagerie of alien life—one that to our eyes, and at a remove of over 400 million years, can only be described feebly as squidlike, or vaguely horseshoe crab–like, or clamlike—now ruled the seas.

But the mountains kept eroding, CO_2 kept falling, and the good times abruptly came to an end when the world crossed a climate threshold and plunged into another ice age. Just as we're worried about CO_2 soaring today, falling CO_2 can similarly carry the planet over a catastrophic threshold in the other direction. While nowhere near as extreme as Snowball Earth, this smaller but still punishing ancient ice age, some 445 million years ago, saw glaciers plow over a polar Sahara and Arabia, the seas suddenly fall hundreds of feet, draining the shallows that covered the continents, and the circulation of the ocean upend in what would be the second worst die-off in Earth history: the End-Ordovician mass extinction.

As we've seen, the carbon cycle, when pushed toward extremes, is liable to whiplash, and a million years later the planet catastrophically rebounded, leaping into a sweltering *greenhouse* to kick off the ensuing Silurian Period (444–419 million years ago). Hot, algae-choked seas now soared hundreds of feet, drowning what were once the desolate glacial outwash plains of Saudi Arabia. This sickly high tide of extinction left behind so much dead life that today Silurian sunlight and CO_2 flare from the derricks of the largest gas fields in the world, from Libya to Iran. Over millions of years, the ancient world would eventually recover from this climate paroxysm too, and as our fish ancestors radiated in its wake, giant sea scorpions plied the warm seas. On land, rising from the once-desolate rockscape, knee-high sprigs of greenery at the edges of lakes and rivers suddenly gave way to thirty-foot-tall trees in the Devonian Period (419–359 million years ago)—the first forests ever.[fn4] It was another epochal change in the history of Earth and a similarly profound change in the carbon cycle, doubling the amount of photosynthesis on the surface of the planet. In effect, this also doubled the amount of energy now available to flow through the biosphere, raising again the ambitions of this strange planet. A motley crew of waddling fish and insect pioneers were lured ashore by this new terrestrial world, or perhaps *thrown* ashore by the heaving tides of a closer moon.

But the first trees had another profound effect on the carbon cycle, drawing down CO_2. Trees, after all, build themselves out of CO_2 in the air around them. By the end of the Devonian, these first forests would prove so successful that they nearly destroyed the world—several times, in fact. As the first tree roots plowed through the continental rock, and then the first seeds allowed these trees to push farther inland, the rocky surface of the

Earth was churned up in an unguided, open-ended geoengineering project that resulted in catastrophe. Their roots reworked the land, unearthing mineral phosphorus, which washed into the rivers and out into the seas, igniting global blooms of algae that also drew down CO_2 blooms that then rotted and robbed the oceans of oxygen as they decayed. These deadly tides, which pulsed throughout the global extinctions of the Devonian, fell to the seafloor, draping the deep in organic muck. Deprived of oxygen in these dreadful seas, much of this avalanche of carbon slime that fell onto the seafloor was spared from decay, and today this ancient dead life from an age of extinctions comprises most of the natural gas that we frack from New York to North Dakota—releasing that 380-million-year-old CO_2 back to the atmosphere.

By the end of the Devonian Period, in the wake of these extinctions, seas that were once stocked with bus-sized, guillotine-mouthed, armor-plated monster fish were now home to fish no larger than a minnow. In one of these extinction pulses, 375 million years ago, the world's largest reef system ever—a global patchwork of unfamiliar corals[fn5] and sponges, stretching ten times the extent of modern reefs—was reduced to nothing. As for the first forests themselves, they sequestered so much CO_2—in their trunks, their soils, and the algae blooms they spurred offshore—that by the end of the Devonian Period the planet had plunged yet again into another brief, cataclysmic ice age 359 million years ago.

It should be clear by now that *a lot* has happened in Earth history.

This brings us at last to the miserable dawn of the Carboniferous Period. But it would still be some time before the great swamp forests of the age of coal would rise from the Earth. In fact, where sturdy, hundred-foot-tall hardwood forests had once spanned the Devonian Earth, no plant now dared grow taller than about six feet. In vacant seas, vast gardens of sea lilies now bloomed across the planet where vanished giant fish might have once picked them clean—ancient limestone gardens that today lie just under Appalachia, and filter the water used by Kentucky bourbon masters. This wretched aftermath of extinction was the inauspicious start to the age of coal.

We've just elided 200 million years of Earth history, with each year on this planet as infinitely intricate in its texture and weave as any other. But we still need to whirl the planet around the sun *50 million more times* to jump from this impoverished, postapocalyptic world of the *early* Carboniferous Period to the phantasmagoric world of the *late* Carboniferous, landing finally in the alien coal swamps of Kentucky 310 million years ago.

BY NOW THE BIOSPHERE HAD RECOVERED AGAIN, AND AFTER A ruinous introduction to the planet, the Earth's forests had reached a détente of sorts with the world and were emphatically back. But the living world never reassembles from the same pieces after mass extinction, and it would take only a glance at this uncanny jungle to recognize that something was off. The trees rising high above these swamps were decidedly weird, quite unlike anything we would call "trees" today. Towering 160 feet in the air, they sported no branches except for crowning fireworks of green. But wait, the whole trees—not just the tiara of thin leaves atop, but the entire lengths of their giant pole-like trunks, and even their roots—were bright green too. On closer inspection, you would notice that the "bark" was tessellated in strange emerald tiles that made food for this "tree"—the singular lepidodendron. Curiously, the whole thing photosynthesized. The tremendous green poles supported their stature not with a core of hardwood like any sensible tree would, but with what were essentially rigid exoskeletons painted green with photosynthesis.

That these forests would have been so unfamiliar to us owes to the fact that this was a failed experiment: nothing from this line of life—what are colloquially known as club mosses—would ever make trees or forests again. Today you might come across club moss on a damp walk in the woods, timidly poking a few inches above the leaf litter, but you probably wouldn't notice them. For a few brief geologic moments in the Carboniferous, though, they were giants, some 400 times taller than their tallest modern relatives, shooting into the sky at the end of their decade-long lives like titanic dandelions. Reaching the top of the swamp forest, they finally cast their spores to the tropical winds and quickly died, yielding the skyline to similarly uncanny tree ferns and gigantic horsetails. As a result, the Carboniferous rain forest floor wouldn't have been perpetually shaded by canopy, like rain forests today, but rather awash in sunlight, a strangely open landscape, save the airy colonnades of these giant green telephone poles. If you could find a dry spot to bivouac in the Carboniferous, the makeshift giants would have made for a death-defying camping experience. The copses of the expired insta-trees didn't care what happened next, and with a tremendous crack the lepidodendrons crashed into the mire, briefly interrupting the searing, unending drone of insects. Taking the sunlight energy and CO_2 they captured in life down into the muck with them, it was these fallen trees, these stumps, these endless swamps that would someday become coal. It's this fossil sunlight that would someday power the Industrial Revolution, and this ancient CO_2 that would accompany its release. Meanwhile, in the hollows of the broken tree trunks, tiny reptiles plotted the coming age of land life, by now fully divorced from their ancestral seas. As you marveled at these brave creatures—your ancestors—a dragonfly would alight on your shoulder with a distressing heft, sounding something like a rusty lawn mower. It had a two-foot wingspan.

Time machines, should they ever be invented, will inevitably find the Carboniferous to be one of the least popular tourist destinations. It was a horrifying time to be alive, with not only dragonflies the size of seagulls, but scorpions the size of golden retrievers and millipedes the size of alligators. Today it's not unusual to find what look like tractor tire tracks ominously skittering across slabs of Carboniferous sandstone, the trails of these monster insects.

Strangely, it's all the coal itself that explains this singular epoch of insect gigantism: burying all those carbon-rich trees and swamps of the Carboniferous Age in the Earth supercharged the atmosphere above with oxygen. Today insects are constrained in size by their unusual respiratory systems, which forgo lungs for a network of tiny tubes that invade every alcove of their bodies, ventilating their guts with oxygen and exchanging it for CO_2. This isn't a solution to breathing that scales, but in the Carboniferous, so much organic carbon was buried in what one paper calls the "largest tropical peat mires in Earth history" that oxygen soared to extraordinary highs, topping out at 35 percent of the atmosphere, compared with today's 21 percent. Drawing more oxygen from each breath,[fn6] the insects of the Carboniferous were thus able to cast off their metabolic shackles and grow to their hateful proportions.

Oxygen rose so high from all this coal burial, in fact, that roaring, impossibly hot fires would have been an ever-present feature of this planet as well—even tearing through these drenched rain forests. This doesn't happen on our world: wet organic matter only combusts above 25 percent oxygen. As a result, this was not only an age of coal (the fossil remains of these swamp forests that were later metamorphosed by heat and pressure deep in the Earth) but of fossil charcoal as well—or simply, wood charred by fire. Even though this was a planet already well over 4 billion years in progress, fires were a new phenomenon on Earth, and the legacy of these global wildfires persists in widespread fossil charcoal deposits of the Carboniferous, the lustrousness of which belies the extreme heat of these ancient blazes, even in the ever-wet tropics. Like the rumors of giant bugs etched in the rocks, these rain forest embers point to the period's outrageous oxygen—a forensic record of the planetary capacitor vigorously charging as gigantic quantities of carbon fell into the crust.

The Carboniferous sun, meanwhile, would have been dimmed by the unending wildfire smoke and filtered through an enduring ochre haze. As it set each day it would limn the haze a dazzling fuchsia, reminiscent of Monet's colorful impressions of the acrid London fog some 300 million years later. The resemblance wouldn't have been a coincidence: both landscapes, the Palace of Westminster in the age of the Industrial Revolution and these alien coal swamps of deep time, would have been shrouded by the same smoke, billowing from the same burning swamp forests of the Carboniferous.

While the giant bugs might have loved all this oxygen, surprisingly, *too much* of it can actually be a bad thing for life, jamming the components of photosynthesis. After all, the archaic enzyme Rubisco, which turns CO_2 into life, had evolved on a far more ancient planet with virtually no oxygen at all and is ill-designed to handle the reactive gas. As a result, even today plants struggle to accommodate high oxygen. Same goes for us: oxygen reacts, burns, generates cell-destroying free radicals. Breathe air with 75 percent oxygen for a few days and you will die. And on the other side of the photosynthesis equation, burying too much CO_2 as was happening in the Carboniferous, can ultimately rob plants of their food (by draining CO_2 out of the atmosphere) and the planet of its greenhouse blanket, potentially plunging it into an icy Armageddon, like Snowball Earth. The Carboniferous planet was therefore teetering on the edge of cosmic catastrophe. While equatorial Appalachia and Western Europe might have sported balmy rain forests as CO_2 continued to fall, elsewhere on the planet, massive mile-thick ice sheets were advancing on an Antarctic Africa as a result. Because of this, the rampant wildfires sweeping through these rain forests and billowing CO_2 back into the atmosphere would have actually been a helpful check on the planetary hazard of runaway cooling (in addition to curbing the rather surprising threat of perilously high *oxygen*).

Nothing is ever simple in the carbon cycle, though, and there were still other feedbacks working at cross purposes. These same ancient wildfires, by forging all that charcoal at such exceptional temperatures, made this carbon especially resistant to further breakdown, ensuring that much of it *couldn't* be respired back to CO_2 by life. Instead, by flash-frying it and packaging it as indigestible charcoal on the surface, the raging fires prepped this carbon's export to the geologic record. Having survived the surface world, this charred wood took the CO_2 down with it as it was buried, a positive feedback to ever more cooling. Even today, the durable black carbon produced by modern wildfires and delivered to the oceans by rivers represents an important all-but-permanent carbon sink for the planet. It's the kind of subtle, counterintuitive finding in Earth science that doesn't lend itself to catchy headlines while the Amazon is burning down.

But none of this is sufficient to explain the truly preposterous amount of coal dumped into the rocks of the Carboniferous Period, a bounty that some 300 million years later would launch the Industrial Revolution and inevitably set our world on its own careening trajectory. To explain the remarkable abundance of the age, then, one theory holds that it was what was *missing* from this world that gave rise to its extraordinary strata. Given that trees were a relatively recent invention on the planet, perhaps the requisite fungi hadn't yet evolved to break down wood. The plant polymer lignin is what lends wood its woodiness, and it has a ridiculous chemical structure: endless chains and convolutions of C, H, and O that provide wood its exceptional rigidity and make it almost impossible to eat—except by the fungi specially evolved for the task. Where today virtually everything in a rain forest is eventually broken down and respired back again as CO_2 by this kind of recycling by life, perhaps this wasn't the case in the Carboniferous. If so, when the trees died, they would have just sat there on the forest floor, unmolested by life, awaiting their unhurried burial by sediment and export to the far future. Like the artificial plastic polymers new to our own planet today, synthesized from fossil fuels—and now accumulating in bottle-strewn landfills and rafts of plastic in the ocean by the megaton—life would have similarly struggled to break down these novel woody polymers. If this was the case—if the fungi that *could* break down wood didn't exist yet—then perhaps these ancient forests were able to escape the dissolving maw of life in just this one brief window of evolutionary time, and sink into the fossil record along with all the carbon that they had captured from the air. Maybe that's why the hills from Wigan Pier to Harlan County are bar-coded black in Carboniferous coal.

Unfortunately, though, this theory doesn't hold up. For one thing, if true, that is, if *none* of these Carboniferous trees were decayed or eaten, and instead *all* of the CO_2 captured by them in life ended up getting buried as organic matter and permanently stored in the ground beneath, these ancient forests would have drawn CO_2 in the air down to *zero* in less than a million years. This would have plunged the ancient Earth into an icy planetary catastrophe that isn't evident in the rock record. The thought experiment, though, helpfully illustrates that it's both sides of the equation—not only CO_2's capture by plants, but also its salutary *release* back into the environment by the rest of the living world feeding off them—that has kept our planet habitable for hundreds of millions of years. It also now appears that wood-eating fungi had, in fact, evolved by the Carboniferous. After all, life just doesn't let free energy sit around for long without evolving a way to break it down and use it, no matter how tough or unappetizing. Finally, much of the coal that feeds the fires of our modern world is made of the bizarre quasi-trees like *Lepidodendron*, which didn't have much lignin in them anyway. Taken together, and despite the pop-science appeal of the idea, it seems as though biology once more might not be the control knob on this world-changing episode, and our attention must turn again to tectonics.

This was the dawn of a storied age not only for terrestrial life, but for Earth's very geography as well. In the late Carboniferous, the supercontinent Pangaea was just starting to assemble from the planetary jigsaw. In the tropics, where Africa was colliding into the East Coast of North America and Europe, titanic mountains soared. This was a Himalayan-scaled range that spanned the equator, one that would eventually run all the way from the British Isles to West Texas. The colossal belt of snowcapped peaks would have weighed leadenly on the crust, producing vast low-lying "foreland basins" in the tropics, where this crust sagged. The down-warping crust would have continually provided a low space for sediment eroding off these mountains to accumulate, just as the Himalayas today weigh down the crust, providing room—from the Indus River Valley to the Sunderbans—for sediments to pile up at its feet, in some places over four miles deep. And as ice sheets at the poles waxed and waned over the Carboniferous, they sent seas soaring and falling over the millennia, repeatedly exposing vast expanses of new dry land to accommodate these swamp forests—only to have them buried again by sediment when the seas inevitably rose. So it was only by dragging down the crust with one of the largest mountain belts in Earth history, and painting such an expansive swath of subsiding land in tropical swamp forests, in an age of rising and falling seas, over millions of years, that the Carboniferous produced a singular epoch in the rocks striped with coal. And it was in these endless, subsiding tropical lowlands, on a world long before dinosaurs, that the Chekhov's gun of the Industrial Revolution was finally loaded.

BY BURYING SO MUCH COAL IN THE GROUND, IT NOW APPEARS THAT the teetering Earth system of the Carboniferous nearly *did* tip over into a Snowball Earth, more than 300 million years ago. Burying all this carbon in coal not only dropped atmospheric CO_2 to a level similar to today, but it did so on a planet warmed by a sun that, in the adolescence of its stellar evolution, was still somewhat dimmer than our own, by about 2.5 percent. This was extremely dangerous. As coal formed in the vast tropical lowlands, draining CO_2 from the skies, and epic polar ice sheets swelled at the poles on this ancient Earth as a result, the threshold for Armageddon drew appallingly close. As he writes in a self-explanatorily titled paper "Formation of Most of Our Coal Brought Earth Close to Global Glaciation," the Potsdam Institute scientist Georg Feulner claims that the Earth's escape from Snowball Earth in this age of coal was "surprisingly narrow."

This would have been the third paroxysm of Snowball Earth over Earth's vast 4.5-billion-year history. But where the previous two struck an ancient Earth home only to microbes, this one would have announced its arrival on a dazzling new world replete with things like trees and reptiles and sharks. Had it happened, it would have incontestably been the worst catastrophe in the history of animal life—almost certainly ending it altogether. As the ten-foot-long millipede *Arthropleura* diligently worked the twilit forest floor of an Appalachian jungle without flowers—in smoke-filled, CO_2-burying swamp forests that stretched from Des Moines to Donetsk—the Earth hung in the balance. The age of dinosaurs, much less that of mammals, would have been stillborn in this alternate past, on a planet sterilized by runaway ice sheets. Once again the age of animal life appears to have played out on the knife's edge of the carbon cycle.

Thankfully, though, CO_2 would remain just barely high enough in the Carboniferous to ward off this fatal reprise of the ice. But if burying all this coal very nearly caused the planet to tip over into an icy, planet-killing apocalypse, what, one wonders, would digging it all up in a matter of centuries and lighting it on fire do?

IN POCAHONTAS MINE NO. 1, IN THE SOUTHWEST CORNER OF VIRGINIA, the black halls are cold and damp, and the ceilings are coffered in *Lepidodendron* logs. When this mine opened in 1882, whole trunks of the 300-million-year-old trees, called kettlebottoms, lost in time, would occasionally fall out of the ceiling, killing coal miners—the victims of unlikely ancient tree falls beneath the Earth. But falling trees were the least of the miners' worries. Within two years of opening, a coal dust explosion killed 114 workers—the entire night shift. More explosions at Pocahontas followed in 1901 and 1906—the year before an explosion at the Monongah Mine in West Virginia took 361 lives. The alien Carboniferous Period, after an eon of slumber in the deep, was being roused, and by the turn of the century 1,600 miners in the US were being killed every year in the project of rousing it, mostly in workaday mine wall collapses or from "blackdamp," a suffocating miasma of CO_2.[fn7] By the time a canary farm grimly opened in downtown Pocahontas to service these tunnels, there was no turning back; the lucrative promise of the Pocahontas coalfield—a massive cache of fossil energy that spread far into West Virginia—had already been sufficiently demonstrated to its financial backers. This alien world long before Pangaea, haunted by gigantic bugs, was being summoned back to the surface world to participate in a strange and devastating afterlife.

The Pocahontas coalfield had been born in the mind of the Confederate Army cartographer Jedediah Hotchkiss, who spied the black seams eroding out of the hillsides while serving in Robert E. Lee's western Virginia campaign. Two decades later, as the country feverishly industrialized, it was time to make money off it. Hotchkiss returned to the region to aggressively promote the land to a group of Philadelphia investors, who began buying up the tracts in the name of the Norfolk & Western Railway. This wasn't just a railroad, though. As the historian Steven Stoll writes, Norfolk & Western, like the other railroad concerns of the time, was a "massive amalgamation of capital, a liquidation machine whose vortex drew in the entire landscape from treetops to minerals a mile deep …. Mills, tanneries, steam skidders and labor camps spread out behind its locomotives like the wakes that follow ships." In 1883 the Pocahontas coalfield was connected to a node of this rail system, the tendrils of which were then working their way into every mountain hollow of Appalachia, finally bringing this enormous geologic reservoir of fossil sunlight—previously considered inaccessibly remote—to the market. Once the coal found its way down these rail capillaries, winding out of the hinterlands and into metabolic centers of industrial civilization to be respired, the next few decades of exploitation and environmental devastation were all but a fait accompli.

This once isolated, energy-rich region had been put in contact with a vast global engine of energy dissipation. This was a new force on the Earth, one that gathered geologic energy like a hurricane from the far corners of the planet and drew it toward the eyewall of the metropole to power the engine of industrial capitalism, dissipating the ancient energy as heat and CO_2 undoing millions of years of photosynthesis all at once. This energy was used to transform the Earth—from its crust to its atmosphere, and the biosphere caught between—on a scale, and at a pace, with very few analogues in planetary history. And before long, Appalachia, and its extraordinary stockpile of fossil carbon from the Carboniferous, had been drawn into this widening gyre.

GROWTH, IN ANY COMPLEX SYSTEM, IS IMPOSSIBLE WITHOUT ENERGY. If you don't feed a growing child, they won't grow; empires don't swell in power during years of crop failures and famines; bacteria starved of food cease to multiply, halting their spread across the petri dish; hurricanes that run astray of their warm ocean waters weaken and disband. In our universe, even maintaining order requires continuously dissipating sources of free energy to beat back the encroaching tide of entropy.

Humanity's growth has always relied on photosynthetic energy. This is the energy embodied in plants and photosynthetic plankton, and which flows and dissipates through the entire biosphere, including the crops, livestock, and fisheries that have long powered farmers, beasts of burden, artisans, soldiers, and slaves—keeping the whole human project far from equilibrium. But for millennia the slim photosynthetic surpluses of agrarian societies had meant that 90 percent of the population was inescapably tied to the land to produce these small, diffuse amounts of plant energy. Both humans and the animals they conscripted did work by metabolizing these crops back to CO_2 to power muscles, but the activity of human societies was always limited by how much plant life could grow on the Earth's surface.[fn8] It was constrained by the fundamental fact that virtually all the energy that flows through the biosphere enters through photosynthesis as it transforms CO_2 to sunlight-charged organic matter.

Had humanity remained reliant on only the photosynthetic power available on the Earth's surface, the extraordinary exponential growth of the industrial world that achieved takeoff in the nineteenth century would have been energetically impossible. Even the pioneering economists of capitalism at the time, like Adam Smith (who published his landmark *Wealth of Nations* the same year, 1776, that his University of Glasgow colleague James Watt released his first commercial steam engine), failed to anticipate the coming explosion. The kind of ceaseless economic growth that we now take for granted was unthinkable to Smith, who, like all economists of the day, saw societies ultimately hemmed in by the rigid constraints of agriculture and population growth. He failed to anticipate the new energetic possibilities empowered by Watt's machine and its descendants, metabolizing millions of years of photosynthetic energy in the Earth's crust. As one Industrial Revolution scholar writes, Smith, along with the economist David Ricardo and their famously dismal colleague Thomas Malthus, were "adamant that growth must be limited and that the end state following a period of growth might well be as miserable as the situation prevailing at its beginning." This perspective was hard-won by centuries of experience, if not millennia.

But the avalanche of ancient photosynthesis from the depths of the Carboniferous cast off these energetic shackles, allowing the industrial world to grow vastly more powerful, in the very literal physics sense of the word *power*: that is, it enabled vastly more work per unit time. And where plant life had long released its energy, not only through food but through fire as well—providing people the heat to endure harsh winters or to make iron and steel, beer and bread—coal miners now called forth the warming bonfires of millions of years of forests. Without this, there wouldn't have been nearly enough energy to bring about the burgeoning modern world. "The creation of a national rail network," the late Cambridge University demographer Tony Wrigley wrote, "would have been physically impossible without the transition to fossil fuels as the principal energy source because of the quantity of iron and steel needed for rails, engines and rolling stock." But because reserves of coal forests were effectively limitless, having accumulated over millions of years—as opposed to trees on the surface, which grow on a decadal time span, pulsing with the seasons and limited by geography—this growth could take on an exponential quality.

This rapidly expanding system seemed to be self-reinforcing. Coal was burned to move trains,[fn9] and coked to make steel and railroads to get those trains to more and more outcrops of coal, to burn in still more trains, to get more coal, to make more railroads, to get more coal, and so on and on. As a result, between 1810 and 1910 human coal consumption multiplied 100-fold. Where miners chiseled about 10 megatons of coal out of the rocks in the early nineteenth century, dissipating the fossil energy in newfangled steam engines and iron-smelting blast furnaces, they were producing a gigaton at the turn of the next.

This geologic eruption of energy soon swamped the amount of photosynthetic power available to humans at the Earth's surface and advertised itself in empires sprawling overseas, an explosion of consumer goods churned out by clanging factories vivified by ancient sunlight, unprecedentedly deadly wars increasingly fought in coal-forged iron exoskeletons that were powered by fossil fuel, and industrial cities that sprang up overnight. By midcentury, coal provided Great Britain twice as much energy as could have possibly been provided by all the trees on the island, even if the entire country's surface area was covered in forests; it provided four times as much energy by the 1860s, and eight times the entire surface area of the country by the 1890s. By resurrecting Earth history, industrial societies soon slipped the surly bonds of the surface world's energy constraints. Where in 1800 humanity dissipated 20 exajoules of energy—98 percent of which came from burning photosynthetic stuff that grew on the planet's surface—by 1900 the industrial world had more than doubled its energy consumption to 43 exajoules, with the planet's stores of coal supplying much of this excess power.

By that point even economists couldn’t help but take note of the new metabolic regime that had arrived on the planet: as Émile Levasseur calculated, with some measure of awe, the 40 million citizens of France in the 1880s were astoundingly able to do the work of almost 100 million laborers thanks to the steam engine. Today, after another century and a half of fossil-fueled near-exponential growth, a million years of ancient photosynthesis captured in Carboniferous coal forests suffices to sate the appetite of only about three years of human consumption. And now, as a result of this still-surging dissipation of geologic energy, 8 billion people can do the energetic equivalent of 500 billion tireless preindustrial laborers. A tractor combine can do, in one sweep of the field, the work of an entire peasant village; a backhoe, the work of a throng of corvée labor; a truck, that of a monthslong caravan. But unlike an actual agricultural workforce or teams of livestock, “[w]e didn’t pay for the creation of these armies of workers,” as the ecological economist Nate Hagens writes about this virtual labor force of fossil plants, pressed into service by industrial civilization, “only their liberation.” As with life itself, once this planetary-scaled reservoir energy began to be unleashed, new, increasingly complex structures—political, economic, and social—emerged to facilitate and even accelerate its dissipation.

Because life is fundamentally sustained by energy—in the food we eat, and by expending it to turn as much of the landscape as possible into an environment conducive to human survival and habitation—the explosion of geologic power in the past few centuries has by now produced living standards in much of the developed world that surpass those of the most pampered sun kings. Life expectancy, literacy, even human height have soared in much of the world during this spasm of abundance, all powered by a one-off, massive, and ongoing discharge of the planetary battery. Unfortunately, though, burning plants—fossil or otherwise—doesn’t just unlock energy; it also unavoidably releases CO_2. But setting aside for now those dire and possibly existential environmental costs, perhaps there’s a case to be made on humanitarian grounds, as some have, that it has all been worth it so far—a case that perhaps comes more easily to those of us in the rich industrialized world, perched at the privileged terminal nodes of this global dissipative structure and shielded from the exploitation that liberated this energy from the Earth. After all, the liberation of these fossil armies, since its beginning, has come at an extraordinarily high human cost.

HERE IN THE POCAHONTAS COALFIELD, IN THE RUGGED SOUTHWEST corner of Virginia, in a little over a decade, 36 million tons of coal—picked out of these mine walls and carted to the surface by miners who never saw the sun in a working day—would stream out of Mine No. 1 to power the explosive industrial growth of the late nineteenth century. Roused from its stony slumber, the millions of years of ancient photosynthesis would be ignited in the furnaces of elegant White Star Line steamships and hauled around Cape Horn; they would be shoveled into the bowels of US Navy cruisers fueling at remote Pacific coal depots as Manifest Destiny spilled overseas.[fn10] Following Pocahontas's success, an avalanche of northern money—from J. P. Morgan, Philadelphia banking families, Boston financiers, and even English investors—poured into the region, riddling the Appalachian hillsides with holes and sending ever more men in to retrieve the old sunlight by the billions of tons. Because this coal was fed into a larger system of industrial production, in which the railroads and factories downstream were often financed by the same politically powerful capitalists as the coal mines—and who themselves were operating in a larger economic system in which they had to endlessly drive prices lower than competing firms—labor had to be procured as cheaply as possible. So Hungarians, Italians, Russians, and Germans, many of them swindled into signing ruinous contracts by mine agents at Ellis Island, were shipped off to company towns, then forced to work off their transportation costs underground. Others came to the mines seeking to escape lives of racial and economic terror sharecropping on southern plantations, while still others hesitantly descended from the surrounding hills.

For generations, the mountaineers of the Appalachian plateau had enjoyed a close-knit, self-sufficient (if often severe) community life in these hollows—one intimately tied to the ecology of the forests around them, and the free gifts of wild game, mushrooms, berries, fuelwood, and timber. But before coal transformed the region, the mountaineers first saw these very forests destroyed by industrial loggers, a devastation only accelerated by the introduction of coal-fired machine logging around the turn of the century. "The entire surroundings had been converted to a desert," the German forester Carl Schenck wrote in horror at the time, "dissected by deep gullies and ravines, looking like the landscape of the moon."

With the environment that had sustained them now stripped to a barren moonscape, and finding themselves priced out of farmland by distant speculators, and the land bought from under them, the mountaineers had nothing left to sell but their labor in the coalfields, often finding common cause in the union hall with their Black colleagues against the ruthless mine operators. Those mountaineers who tried to eke out a living in the ravaged highlands found themselves “impoverished and powerless within a new and alien social order,” as one Appalachian historian writes. Anthropologists who parachuted into the region in the early twentieth century puzzled over the backwardness of the people of Appalachia, ignoring the social and ecological trauma that had just deracinated them as surely as the trees. It was a cruel recapitulation of the original dispossession of this land, when Virginia land speculators like George Washington pushed for independence from Britain in part to snap up the “Indian Territory” of Appalachia to the west, with the rugged Scots-Irish settlers acting as the speartip that drove the Shawnee and other nations off their traditional buffalo hunting grounds.[fn11] But now the descendants of those settlers found themselves driven off the land as well, and sent into the Earth itself by the demands of distant industrial capitalists, whose wealth they would augment with each cart of coal they loaded.

When coal demand would soar to power the expanding needs of industrial civilization—in times of war, or in the exuberant swings of the business cycle—and labor became especially scarce, the jails of the South were emptied of Black prisoners, who were shuttled into coalfield-bound train cars against their will. Many who sought to escape their fate in the mines were shot on sight by armed private detective agencies, employed by the mine operators. Of the children who worked in the industry, some as young as five, a sociological study of the anthracite mines of Pennsylvania noted: “They are full of fun and frolic, but their curiosity sometimes leads them to injury and death. Many of them fall into the machinery and are mangled, or down the chutes and are smothered.” The same gruesome story could be told almost anywhere the strata of the Carboniferous were tapped, from the English underworld plied by the poor and detailed by George Orwell (“the place is like hell”) and the mines of Scotland, where it was “no uncommon thing to see” women who hauled coal from the pits “weeping most bitterly from the excessive severity of labor”; to the later arctic coal mines of the Soviet Vorkuta gulag worked by political prisoners, as the USSR tried in vain to keep pace with the soaring industrial production of the capitalist world at any human cost; to Battleship Island, where enslaved Koreans mined coal for the Imperial Japanese.

Where kings and landed nobles had long derived their power from control over agricultural lands—and the energy from the sun captured by and embodied by crops—control over this far vaster reservoir of ancient photosynthesis created a new variant of political and economic tyrant: the coal baron. Where peasants working annual fields of crops could only produce as much food energy as a tract of land allowed, each coal miner could now, in just a day's work, produce 500 times more energy than the calories in their diet that powered this labor. But if the incredible concentration of energy in coal—compressing millions of years of sunlight in seams of rock—endowed those who claimed the coalfields with near-despotic power, rival structures of democratic power soon began to emerge at the mine face as well. Those same exploited workers who manned the system's chokepoints came to recognize their unprecedented control over this energy flow from the Earth. Seeing as industrial society owed its very existence to this continued flow of energy, what would happen if the miners cut it off?

As the Columbia University historian Timothy Mitchell argues, it was the very physical attributes of coal responsible for launching the Industrial Revolution—its energy density, embodying eons of solar power—that also enabled the unprecedented organization of labor power at the turn of the century, structures that were "readily assembled" into a political "machine" from the massive flows of energy out of the mines. For millennia, the diffuse energy flows of agricultural societies had made it extraordinarily difficult for peasants, serfs, and slaves to assemble concentrated political power, as each atomized laborer produced only small amounts of photosynthetic energy for society at large. Coal miners, on the other hand, along with the railway, canal, and port workers, found themselves maintaining the narrow, dendritic channels through which this tremendous energy of society now flowed. And they could exert almost infinite political pressure on the entire system by threatening to interrupt it.

To wit, the wave of coal mine strikes, dockworker and railway strikes, and even general strikes, along with the emergence of a mass politics that enrolled the Great Unwashed in larger political machines that could meaningfully challenge the unchecked prerogative of aristocrats, and that swept the industrialized world at the turn of the century, were powerful enough to mitigate some of the horrors of galloping industrialization—from Germany's coal-stocked Ruhr Valley to Appalachia. Leaders of the industrialized world, from Theodore Roosevelt to Kaiser Wilhelm II, now had to contend with these novel, countervailing structures of power as a matter of political survival, granting concessions to the increasingly organized workforce—from the eight-hour workday to pensions and welfare programs. Energy, in a *physics* sense, was very literally being rerouted through the working class. Even today, the British labor vote maps almost exactly onto the country's outcrops of bedrock from the Carboniferous Period. But the growing power of miners and their ability to wield geologic energy toward social ends was perhaps best reflected in the commensurately violent response that was organized to keep it in check.

WHEREVER THERE ARE HUGE FLOWS OF ENERGY, QUASI-STABLE NEW structures—like convecting climate cells, plate tectonics, life, ecosystems, and even whole political economies—emerge to dissipate them. In southern and central Appalachia, the dominant political structures that organized around the geologic flows of energy were hopelessly undemocratic, serving no purpose other than to maximize the extraction of energy out of these mountains, and at the cheapest possible price, no matter the human or environmental toll. Of course, all of this made some people wildly rich, most of them in the distant industrializing capitals where that energy was dissipated to produce the high-value goods of a mechanizing society, or by manning local nodes of the corrupt political structure that enabled the flow. But for most Appalachians, draining the mountains of their ancient sunlight left in its wake a ruined and denuded moonscape, poisoned water where clear streams once ran, a disintegrated community life, and widespread economic devastation. Tens of thousands of people were left dead, disfigured, or emphysemic.

After the turn of the century, when a wave of unionization swept through the northern coalfields to push back against the inhumanity of the coal barons, the *nonunion* Appalachian coalfields, many of which were effectively run as "feudal baronies," suddenly gained a competitive edge on price. By hiding their human misery on the balance sheets, southern and central Appalachian coal companies were able to take over the market share of a rapidly industrializing Midwest, and energy increasingly flowed from these mines. State militias were sometimes sent in to "protect" the southern miners from labor organizers, and the union organizers themselves were often murdered in cold blood by private detectives. Law enforcement was often on the payroll, and the crimes went enthusiastically unsolved. Vertically integrating industrial behemoths like US Steel and the Ford Motor Company would come to run their own coal mines in the region—sometimes housing workers in bizarre, totalizing company towns, forcing them to buy marked-up goods in company-run stores, with private company currencies. This too kept costs down, and much of this coal would find its way into the belching forges of Gary, Indiana, and the multiplying factories of Detroit, fueling the country's explosive growth.

As in Europe, though, growing resistance to this overwhelming economic force of nature was registered in the swelling ranks of the United Mine Workers of America, and in the voter rolls of the Socialist Party of America, whose miners sported red bandanas around their necks ("rednecks") and thrilled to the tirades of barnstorming organizers like Mother Jones. The collision of these opposing if asymmetrical forces of the mine owners and their workers—each fighting over the flows of energy in society—was often explosive, eventually culminating in the largest armed uprising in the United States since the Civil War. In the Battle of Blair Mountain in 1921 in southern West Virginia, coal mine owners led by a local sheriff on the payroll mustered an army of three thousand strikebreakers, giving them the directive to "kill all the rednecks you can." Union miners marching fifty miles to Blair Mountain were met with an aerial bombing campaign coordinated by a National Guard colonel—one heavily inspired by the Tulsa Race Massacre three months earlier—as well as an indiscriminate blaze of Thompson submachine gun fire that sprayed the mountainside with a million rounds. Fighting lasted for five days and ended only with the intervention of the US Army. In 2010, Blair Mountain itself was delisted from the National Register of Historic Places and scheduled for mountaintop removal.[fn12]

SO WHAT COULD DRIVE SUCH A PSYCHOTIC TRANSFORMATION, NOT only of the social fabric but of the very physical landscape of the Earth as well? It's a question we'll revisit in later chapters, but suffice to say that Appalachia had been drawn into the gathering storm that was then transforming the face of the planet—and would soon send it off on a global chemistry experiment almost unprecedented in the history of life. But the presence of coal alone isn't enough to explain these developments. Coal, after all, had been burned on a relatively large scale as long as 3,600 years ago in western China, and intermittently for millennia before that. For some reason, though, its use had never escalated in such an explosive way sufficient to derange the global carbon cycle, as it has since the Industrial Revolution. But starting in the nineteenth century, this titanic reserve of energy from deep time was fed into a new way of organizing society, and nature itself—one perhaps only a few centuries old, and birthed out of what had long been the economic and cultural backwater of northern Europe.

This was the engine of industrial capitalism, and its algorithm of growth—one that now tapped into an almost limitless supply of geological energy from the Carboniferous Period. It had a certain ruthless, evolutionary logic to it, and like all engines, its function was to accelerate the dissipation of this energy.

It shared features with other economic and social systems around the planet and throughout history, but this was a distinctive mode of production, and it was transforming the world in extraordinary new ways. It had already long since transformed both the English countryside and cities alike, and was now busy notching mountainsides, spidering continents with steel and scoring it with canals, stringing the ocean with telegraph lines on the seabed and baited longlines at its surface, exhausting the North Atlantic of vast shoals of fish and the North Pacific of pods of whales, and exterminating not only herds of bison across North America, but also the people who steadfastly held to mobile, politically flexible, and reciprocal ways of life—cultures irreconcilable and illegible to a new market society carved into a quilt of private property. This system drew toward the smoking, clanging centers of industrial production the timber of Scandinavia, oceans of grain from the Baltic and the American Midwest, and bales of cotton from the Caribbean, the American South, India, Egypt, and beyond. It sapped rain forests of rubber, tapped by unseen chain gangs of indigenous Amazonians; chiseled out craters of copper from Michigan to Australia and iron from the Basque country to Birmingham, Alabama; and stripped remote Pacific islands of guano. It increasingly powered the industrial transformation of all this raw material wrested from the Earth, and flowing in from all corners of the globe, with a one-off planetary bonanza of free energy: the seemingly never-ending reservoir of coal in the planet's crust. It was a geologic force.

This system of organizing society uprooted people from the land that had sustained them and conscripted them into a global army of laborers who worked not to satisfy their hunger with the fruits of land held in common, but instead to produce commodities at the lowest possible price for a global market. With everyone in society dependent on the market for survival, even those at its heights found themselves drawn into this ruthless new competitive dynamic: capitalists could either make profits on their investments or go bankrupt. This was a radically new way to allocate power in society, and this global market acted like a kind of distributed algorithm, one that processed price information, fed in over telegraph wires and delivered by coal-fired mail ships from all over the planet to imperial centers. It rewarded the nodes in an international web of credit that funded those firms—in endless competition with each other—that could produce the commodities above at the lowest possible price, anywhere on Earth.

This compulsion to drive down the costs of production in service of profit brought with it ruthless labor exploitation, displacement, environmental and social ruin—in both distant commodity frontiers and city centers—and mass exposure to a market indifferent to human welfare. But this system was also marked by a ceaseless advance of scientific discovery, printed works, unthinkable technological and medical breakthroughs, an explosion of financial innovation, surging rivers of raw material processed by industrial machinery, unprecedentedly rapid and far-flung transportation, an eruption of consumer goods, artistic patronage, military hardware, artificial illumination, mass education, and cheap food. The rules of the system were enforced by armies both public and private, and rewarded both corporate barons and state coffers alike, whose tax revenues soared as the Earth's crust and biosphere were transformed into the stuff of a swelling industrial civilization. Above all, this system was premised on growth.

To power that growth, it had now found a planetary-scaled disequilibrium between the ocean of organic carbon underground and the highly oxidizing atmosphere above it—one that had taken hundreds of millions of years for biology and plate tectonics to generate. These two titanically reactive reservoirs were separated only by a thin layer of "overburden," as miners call the strata that overlie coal. This was a far-from-equilibrium state if there ever was one. Industrial capitalism, acting through the human societies it entrained, provided the erosive force, in a literal geomorphology sense, to remove this overburden—the trees, the topsoil, the human lives above—to force these two volatile reservoirs to react. Just like the unstable chemistry of the vents at the origin of life, industrial civilization now acted as a catalyst to catastrophically discharge this planetary disequilibrium, and in a matter of mere centuries.

And so that same coal, formed in those strange ancient swamps that nearly drove the Carboniferous planet into an icy tomb, was now being pressed back into service as CO_2 again, releasing millions of years of ancient sunlight to power the runaway engines of modernization, and reversing millions of years of climate evolution in the process. As industrial society tapped into this nearly limitless reserve of energy built up over the entire age of animal life, the energetic limits of what it was physically possible to do on the Earth's surface—in place for all of human history—were obliterated, and all hell broke loose. It is difficult to imagine any way to conduct such an extraordinary discharge of the planetary battery, or any social system that could absorb its convulsions, without being profoundly transformed in every way.

"You cannot expect to take fossilized solar energy for three hundred million years and let it off in a bang in a hundred and fifty years and get away with it," the geochemist Michael Russell told me. "It's ludicrous. Of course it's going to be awful."

As a result, the transformation of the Earth's surface in the past two hundred years is comparable not to anything in human history, but perhaps only to the other eon-making biological transformations of the Earth's surface, like the Cambrian Explosion or the Devonian dawn of trees. Appalachia, meanwhile, was just the latest frontier to be conscripted into the machinery of this world-transforming entropy generator. But the roots of this global engine, and of the human supereruption of CO_2 as we'll see, lay centuries earlier, in Europe, where most of the Carboniferous coalfields of Great Britain had already been opened by the seventeenth century, and where the mechanisms of this new metabolism were being perfected.

AFTER TAKING IN THE MOUNTAINTOP REMOVAL SITE, SARAH CARMICHAEL and I spent the afternoon picking *Lepidodendron* fossils from the side of the highway, alongside faded 2018 Senate campaign signs for Don Blankenship, the former coal CEO who served jail time related to a 2010 explosion that killed twenty-nine miners, and whose stated preference for government was "dictatorial capitalism." Carmichael pointed out a ribbon of sandstone in the roadside outcrop—a 300-million-year-old river channel that cut down into the rocks. Coal miners who unexpectedly encounter these ancient sandy river bends deep underground can set off roof collapses or spark coal dust explosions. But it's not only the miners who are threatened by the vagaries of this ancient environment. The acid mine drainage that leaks today from West Virginia hillsides and tints local streams orange, giving them a radioactive tang, is a product of the peculiar swamps of the Carboniferous as well.

"The coal forms in an anoxic environment, which also forms pyrite," says Carmichael. "So in the Carboniferous you'd have dissolved heavy metals eroding down from the mountains into the coal swamps, which then would be sucked up into the pyrite as it formed and locked away for three hundred million years. Until we mine it out again and release it back to the environment."

In 1990 the entire Appalachian coal industry would be threatened by this Carboniferous paleogeography. An amendment to the Clean Air Act regulating sulfur emissions from coal plants—emissions of the sort that were then laying waste to the ponds and forests of the East Coast—hit the region especially hard. Again, the chain of causation can be traced back more than 300 million years.

"With the rise and fall of sea level in the Carboniferous Period you would have inundated these coal swamps with salt water, and salt water is full of sulfate," says Carmichael. But other coal deposits, like those in Wyoming, formed in swamps far from shore (and much later in Earth's history), have much less sulfur. As a result, in the past few decades much of the American coal industry has decamped from Appalachia for the ancient jungles of the Great Plains instead, and now energy flows freely over the rails from the open-pit mines of the Powder River Basin in Montana and Wyoming.

The Clean Air Act, however, only accelerated the decline of what had long been a moribund industry. In 1948, when Appalachia was still buzzing from the high of a World War II coal boom, the mechanical "continuous miner" was introduced. A human miner who could produce a ton of coal in a shift was replaced with a machine that today mines upwards of thirty-six tons of coal in a minute, maintaining profits for coal mine owners but supplying wages for an ever-diminishing and ever-weaker workforce. As automation sapped union rolls and mine owners actively sought to crush them, other fossil fuels, like oil and gas, which didn't require coal's masses of workers, making them less subject to labor disruption, ate away at King Coal's profits. This decline was briefly reversed when the Organization of the Petroleum Exporting Countries (OPEC) cut back production, switched its pricing of oil from inflating dollars to gold, and imposed an oil embargo on the United States and its allies who aided Israel in the Yom Kippur War—all of which served to effectively quadruple the price of oil on the global market from 1973 to 1974. The coal industry revived again in response, but only for a moment. The death blow for coal in the United States might have been struck in the 2010s, when zero-to-negative interest rates, combined with an innovative new technique for extracting natural gas from shale called hydraulic fracturing, or "fracking," unleashed a debt-fueled avalanche of energy abundance, flooding the market with cheap, cleaner-burning[fn13] methane. Coal was doomed.

"You can't really tell the story about the downfall of the coal industry without talking about the natural gas industry," says Carmichael, who, ever the geologist, paused for a beat before adding, "which then takes you back to the Devonian Period again."

This decline, though, didn't have to mark an economic catastrophe. As oil began to eclipse coal in the 1960s in places like Germany's Ruhr Valley, coal miner unions and the state collaborated on a controlled demolition of the historic industry, investing vast sums to pay for early retirements, new manufacturing, and universities in the region. Energy was rerouted through the sociopolitical structure back toward the workers. But in Appalachia, where the unions had been successfully decapitated and the mine owners were king, the industry instead turned to increasingly destructive strip mining in the 1980s to tear the coal from the Earth by any means necessary. Mine owners even claimed that the very act of destroying these mountains would magically produce an economic windfall all on its own. To that end, some 500 mountains in central Appalachia have been flattened, leaving behind barren treeless mesas that, it was promised, would bring business to a mountainous region with little flat land. Many remain desolate today. One hosts a federal prison, another a defunct call center after the company it hosted left for El Salvador when its tax breaks expired, while others support strip malls and the kinds of chain stores purpose-built to siphon money out of the region.

"In West Virginia they didn't do any sort of slope stability, so you've got like hospitals and runways that are falling off the side of the mountain," says Carmichael. “Their economy is so in the hands of coal companies, and now gas companies, that there’s just really no accountability there.”

At the end of the day Carmichael brought me to the newest and last coal-fired power plant that will likely be built in Appalachia, one that wasn’t even running the day we visited due to recent improvements to the grid that rendered it superfluous. It was an obsolete marvel—the end of the line for a technology that had kicked off the Industrial Revolution. Where James Watt’s coal-powered steam engines topped out at 100 kilowatts, the Virginia City Hybrid Energy Center was rigged with a state-of-the-art 668 *mega*watt Toshiba steam turbine. “It’s the biggest thing I’ve ever seen,” says Carmichael. The plant burns low-quality waste coal, piled in great heaps behind it, with astounding efficiency. But in 2022, when 85 percent of the power plants being retired in the US are coal plants, it is an oddity, an anachronism. This is undoubtedly a good thing for the climate, but at the end of a century-and-a-half encounter with the Carboniferous Period, it’s astounding how much fossil energy and wealth was extracted from these hills, and how little was left behind.

THIS DEATH OF COAL, THOUGH, IS NOT ALL IT SEEMS. IN THE YEAR I write this, coal consumption—far from spiraling into oblivion—jumped 9 percent globally. In fact, more coal was burned this year than in any other year in human history. This all-time high is expected to be matched next year, and the year after.

More than a century after the coal boom that powered the industrialization of the US, setting off one of the most violent chapters of labor history in the process, and despite the popular narrative of coal’s long decline—the industry has, if anything, wildly proliferated, has found new seams, and now erupts ancient CO_2 into the atmosphere on a scale that dwarfs coal’s “golden age.” Today China alone consumes more than *4 gigatons* of coal in a year, more than quadruple the United States ever did at its peak. And in 2021 the country added an amount of electricity demand equivalent to the entire continent of Africa—56 percent of which came from coal.

As the rest of the world tries to catch up to the lifestyle of the average American—who uses 100 times more energy than can be supplied by metabolism alone—industrial civilization now dissipates energy to the tune of almost 600 exajoules per year. This is a 60-fold increase since 1800, and one that is now degrading the Earth's stockpiles of fossil sunlight at an unimaginable clip, and erupting almost 40 gigatons of CO_2 into the air every year as a result—100 times more than all the volcanoes on Earth. The energy from these eruptions presents itself not as spectacular explosions of molten rock, but in the dissipating glow of every lightbulb and flickering pixel, in the incandescent smoldering of every steel plant, and in the sundry operations of every industrial farm. (These farms are subsidized to the hilt by fossil energy and now keep billions of new people alive, most of them aspiring to high-energy lifestyles themselves.) Where the invention of aerobic respiration made life fifteen times more energetic than any comparable metabolism on the planet before—a foundational break in Earth history, billions of years ago—fossil fuels have suddenly made most humans more energetic than the rest of life on the Earth today by an even greater factor. And it was all accomplished in about a century, a geologically imperceptible time frame. It should not come as a surprise that, in response, the entire surface of our planet, the structure of society, the composition of the atmosphere, and the chemistry of the oceans have been transformed.

It's tempting to think that this ongoing disruption to the biosphere is unique in Earth history—a bizarre, historically contingent product of distinctly human processes. As it turns out, though, the planet has tried burning lots and lots of coal all at once before in its history. And it didn't turn out well.

WITH NO ONE ASKING IT TO, ANTARCTICA HAS BEEN KINDLY MAINTAINING a record of air chemistry for the most recent 2 million years of Earth history. As the ice sheet was laid down in laminations of snowfall over the millennia, these layers slowly sank into the ice sheet and became ice themselves. In doing so, they trapped samples of air in this hoary tomb. Paleoclimatologists retrieve this record of ancient air by coring into the ice sheet and studying its composition. What they've found is that starting precisely at the dawn of the Industrial Revolution, the CO_2 trapped in this ice not only began rising, but also started warping chemically. It started getting much lighter.

This is because carbon on Earth comes in two stable flavors: a lighter version, called ^{12}C, and one that is literally heavier, with an extra neutron, ^{13}C.[fn14] Plants are picky, so they tend to go for lighter stuff, because it's easier to manipulate in the nanomachinery of photosynthesis. As a result, plants, and stuff that eats plants—and stuff that *eats* the stuff that eats plants, like us—are made of lighter carbon than, say, the original CO_2 that billows out of volcanoes. This explains the appearance of strangely light carbon in the Antarctic ice. It was originally captured by discriminating plants—only these plants lived 300 million years ago in the ancient swamp forests of the Carboniferous. Then they were dug up by the gigaton in the past few centuries, lit on fire, pumped into the air, and entombed in the ice at the bottom of the world. And so, thanks to industrial civilization, the ghosts of tropical Carboniferous rain forests now haunt these Antarctic ice cores.

This same signal of spiking light carbon, though, is seen elsewhere in the fossil record—recorded not by ice but by rock. Such signals often herald profound hundred-millennia episodes of climate disruption, and these pepper the geologic record. Some of these spikes signal biosphere-cleansing climaxes of mass extinction as well, when the Earth has seen fit to burn lots and lots of fossil fuels all by itself—as it would in the age that followed the reign of the great swamp forests of Appalachia. Burying all that coal once upon a time in the Carboniferous nearly sent the planet into Snowball Earth. But burning it and releasing it in the next age—the age of Pangaea—would make hell on Earth.

5

CO_2 and the Age of Extinction

DANIEL ROTHMAN WORKS ON THE TOP FLOOR OF THE BUILDING THAT houses MIT's Department of Earth, Atmospheric, and Planetary Sciences, a big concrete domino that overlooks the Charles River in Cambridge, Massachusetts. From eighteen stories up, it offers a postcard view of Boston. The John Hancock and Prudential buildings jut like reeds above a scull-dotted Charles River that shimmers orange and pink. The river reflects a neon triangle that looms over the city—the Delphic all-seeing eye of the Venezuelan state-majority-owned petroleum company, Citgo. Downriver, the Longfellow Bridge rattles red with subway cars, tethering blue-blooded Beacon Hill to brainy Kendall Square. Rothman's office surveys a slimmer span, the Harvard Bridge, where a stream of cars flows over the wide reaches of the river to collect in the great schools that bustle along Storrow Drive.

"Boston," though, is a fiction. Four thousand years ago its infilled Back Bay would have instead been a tidal flat fringed with oyster middens. When the moon drew these waters high, indigenous Americans tended to birch fish weirs, frothing white with herring and alewife. Twenty thousand years ago, the view would have been even less familiar. CO_2 was only 180 ppm—less than half as much as today—and Rothman's office would have been buried under a half mile of ice. And 50 million years ago, when CO_2 instead soared above 1,000 ppm, it would have looked out over a monsoonal subtropical forest, canopy branches bending with tiny lemur-like primates above swamps of alligators and clearings that revealed the presence of uncanny monster birds prowling the understory.

A century from now CO_2 may well be almost 1,000 ppm again—thanks to the extraordinary and ongoing eruption of fossil fuels from the crust at human hands—and who knows what the view will be. When I visited, plans were being made to rename the department's lecture hall the Shell Auditorium after the oil company.[fn1]

Rothman isn't a paleoclimatologist. He's a mathematician interested in the behavior of complex systems, and in the Earth he has found a worthy subject. Specifically, Rothman studies the behavior of the planet's carbon cycle deep in Earth's past, especially in those rare times it was pushed over a threshold and spun out of control, regaining its equilibrium only after hundreds of thousands of years. Seeing as it's all carbon-based life here on Earth, these extreme disruptions to the carbon cycle express themselves as—and are better known as—"mass extinctions." Worryingly, in the past few decades geologists have discovered that many, if not most, of the mass extinctions of Earth history—including the very worst ever by far—were caused not by asteroids as they had expected, but by continent-spanning volcanic eruptions that injected catastrophic amounts of CO_2 into the air and oceans.

We've covered most of the kill mechanisms that these kinds of eruptions of CO_2 can set in motion already. As we've seen, CO_2 makes the planet warmer, so put enough into the air and much of the planet will become lethally hot to animals. And warmer water holds less oxygen, so if you dramatically warm the oceans, they'll hold less oxygen as well. This problem is compounded by warmer, wilder weather that wears down and erodes the continents, overloading the oceans with mineral nutrients washing in from land. Those nutrients then fuel explosive algae blooms that, when they decay, are consumed by aerobic microbes, which uses up even more oxygen—essentially turning much of the ocean into a stagnant pond. As a result, the oceans start to gasp and choke as the planet heats up too fast. And once the ocean starts losing its oxygen, even more sinister kill mechanisms come to the fore. Around the world, blooms of ancient bacteria begin to flourish. Bacteria of the type that thrives in suffocating oceans start to suffuse the seas with toxic hydrogen sulfide. Meanwhile, as we'll see, excess CO_2 also reacts with seawater to make it more acidic as well—

threatening the creatures that precipitate their shells out of the ocean, like corals and shellfish. And so global warming, ocean deoxygenation, and ocean acidification—three apocalyptic horsemen only beginning to stir today—were in full gallop during the majority of the worst events that ever happened to life on this planet. And all this planetary chaos was driven by an overdose of CO_2.

Marking each of these global catastrophes in the rocks are strange squiggles in the isotopes of carbon, which record this planetary system going haywire. Plotted on a graph over a million-year timescale, these brief hundred-millennia-long "carbon isotope excursions" look like stray hiccups in the stately march of geologic time. But given the coincidence of these strange signals in the rocks with the great mass extinctions of Earth history, they seem to be recording the planet falling apart. Beyond that, though, interpreting them remains one of the main pastimes of, and open questions for, mass extinction researchers. Rothman thinks that they show the planet's carbon cycle unraveling in worrying ways. To him, the shape and duration of these squiggles through the many different mass extinctions of Earth history seems to follow a consistent pattern, implying that it isn't so much the initial stressor—a gigantic series of volcanic eruptions, say—but the response of the carbon cycle itself that gives the great mass extinctions their recurring, apocalyptic flavor.

Put enough CO_2 into the system all at once, and push the life-sustaining carbon cycle far enough out of equilibrium, and it might escape into a sort of planetary failure mode, where processes intrinsic to the Earth itself take over, acting as positive feedbacks to release dramatically more carbon into the system. This subsequent release of carbon would amplify the initial warming, acidification, and deoxygenation, sending the planet off on a devastating hundred-millennia excursion before regaining its composure. And it wouldn't matter if CO_2 were higher or lower than it is today, or whether the Earth was warmer or cooler as a result. It's the rate of change in CO_2 that gets you to Armageddon. This is because the carbon cycle is happy to accommodate the steady stream of CO_2 that issues from volcanoes over millions of years, as it moves between the air and oceans, gets recycled by the biosphere, and ultimately turns back into geology. In fact, this *is* the carbon cycle. But short-circuit this planetary process by overloading it with a truly huge slug of CO_2 in a geologically brief time span, beyond what the

Earth can accommodate—whether from killer volcanoes or coal-fired power plants—and Rothman thinks it may be possible to set off a runaway response that proves far more devastating than whatever catastrophe set off the whole episode in the first place. There could be a threshold that separates your run-of-the-mill warming episodes in Earth history—episodes that life nevertheless absorbs with good humor—from those that spiral uncontrollably toward mass extinction.

To Rothman, in other words, the carbon cycle seems to exhibit all the characteristics of what's known as an "excitable system."

"The original interest in excitable systems goes back to the mid-twentieth century and some basic work on how electrical signals are carried by nerve impulses and neurons," he told me. "The idea is that you have a neuron and you put a current into it and usually nothing happens until you exceed a threshold, and it fires. It's called a neuron spike. So you reach a threshold, and [the neuron] fires a very large spike with a current that's much greater than what was originally going into it."

Similarly, if you raise CO_2 on Earth by 1 ppm instantaneously, nothing happens. But say you raise it by 700 ppm in a century and—zap! The carbon cycle might fire like a nerve impulse and cause a mass extinction. While it has been more than 60 million years since the planet surpassed such a threshold, by Rothman's calculation we are about to set the planet on just such an ancient and ominous trajectory, one that may take millennia to eventually arrive at the destination of mass extinction, but that may be all but inevitable once we've pushed off from shore.

This is why it's not particularly relevant how much higher CO_2 levels were, or how much warmer it was in past greenhouse ages of Earth history, when animal life was nevertheless happy. The sweltering, high-CO_2 days of the Cenozoic, when crocodiles roamed Cambridge, were probably not particularly unpleasant for the creatures alive tens of millions of years ago. After all, the Earth had had hundreds of thousands to millions of years to accommodate the changing planetary seasons. By contrast, instantaneously imposing that world on one inured to trembling glaciations like ours would be bad for any number of reasons, and anyone who tells you they know what even 4 degrees of warming or more in a century will actually mean—or what that will look like on a planet gripped by Ice Ages for the past 3 million years—is full of shit. Especially if they're an economist. But an even scarier possibility is that our unprecedented shove of the system might trigger internal Earth processes that, once started, keep going—and even accelerate—under their own momentum. It could be that the carbon cycle itself—that grand movement of carbon between the reservoirs of atmospheric and ocean CO_2 life, and even the rocks—goes irretrievably off on its own and launches the planet toward catastrophe.

“There are ample reasons to think the carbon cycle is a nonlinear system, and there’s ample reason to think events like mass extinctions are internally excited,” said Rothman.

As should be obvious by now, the carbon cycle isn’t just some esoteric bit of geochemistry. It’s the story of life on Earth.

“This is how the world works,” Rothman told me. “This isn’t just about the climate warming. This is why we’re here.”

Mass Extinctions

Mass extinctions aren’t just very bad things. They’re not civilization-halting pandemics, like COVID-19, that kill far less than 1 percent of a single species of primate. Nor even pandemics that kill 50 percent, or 100 percent. After all, species-specific pandemics on a planet with millions of species are a useless mechanism for mass extinction. Mass extinctions are not what happen when the world loses a quarter of its vegetation and a third of North America is sterilized, as happened only 20,000 years ago, when mile-thick ice sheets plowed over Canada. They’re not Yellowstone supereruptions, three of which have detonated in a little over the past 2 million years—each of which would have devastated modern agriculture and industrial civilization, but none of which had any effect on global biodiversity. They’re not even the result of far more massive eruptions than Yellowstone, like Colorado’s La Garita caldera, which 28 million years ago detonated the American Southwest in an unthinkable explosion 5,000 times more powerful than Mount St. Helens. The ancient eruption still drapes the region in ash and lava and stands as a footnote to an otherwise rather forgettable stretch of Earth history called the Oligocene.[fn2] However devastating, the Earth merely shrugged.

This is part of the bargain of living on Earth. Life wouldn’t have made it this far if it were vulnerable to the sorts of routine indigestion that are part of the workaday operation of a volcanic planet. After all, as we’ve seen, plate tectonics, by regulating the climate, recycling minerals, and storing carbon in the crust, are an important reason why Earth is habitable in the first place. And from a geological perspective, something so bad that it only happens once in a million years—about the smallest unit of time in geology[fn3] —effectively happens all the time. This is why mass extinctions are not what happen when the sea level rises or falls hundreds of feet, as it has dozens of times in the past few million years. Or even what happens when an asteroid large enough to create the massive impact crater that is Chesapeake Bay strikes the Earth, as one did 36 million years ago and had virtually no impact on life at all.[fn4] While organized society, partitioned by national borders and riven by ideology, might be liable to disintegrate under such geologically routine shocks, the biosphere is tough. And while we've been blinkered as a species to such cataclysms by the extremely short institutional memory of recorded history (at only a few thousand years long), we nevertheless wake up to find ourselves on a volatile planet, with a history far more violent and forbidding than the past few unusually equable millennia have led us to believe. Perhaps we shouldn't provoke this fitful Earth. As the historian Will Durant noted, "Civilization exists by geological consent, subject to change without notice."

But while ours is a sturdy planet, resilient to all manner of unthinkable insults to which it's regularly subjected, once every 50–100 million years, something truly very, very bad happens. *These* are the major mass extinctions—those rare anti-miracles when conditions on Earth's surface conspire to become so vile everywhere that they exceed the adaptive capacity of almost all complex life.

Five such times in the history of animal life this devastation has reached (and in one case far exceeded) the somewhat arbitrary cutoff of wiping out 75 percent of species on Earth, and so garnered the status of "major mass extinction." These are known in the paleontology community as the Big Five (though dozens of other minor mass extinctions of varying severity mar the fossil record as well). The most recent of the Big Five (and by some measures the fourth worst) struck 66 million years ago, a global catastrophe sufficient to end the age of gigantic dinosaurs, then in progress for over 135 million years. It left behind a 110-mile crater, one discovered in 1978 under Mexico's Yucatán Peninsula by geophysicists working for the Mexican state oil company Pemex. The size and shape of the crater implied that a 6-mile-wide asteroid instantaneously put a 20-mile-deep hole in the ground, followed, three minutes later, by an (extremely temporary) 10-mile-high mountain range of exploding molten granite that surged out of an unearthly Mexican kingdom of steam and chaos. The speculative global effects of the impact—which include a biosphere that might have been set on “broil” for twenty minutes on the last day of the Cretaceous, then subjected to a decade of darkness—were similarly sublime. Whatever happened, 76 percent of animal species were taken down in the maelstrom. But even raw numbers fail to capture the qualitative horror of the major mass extinctions: species that *survive* these catastrophes might nevertheless see 99 percent of their individual members die.

By comparison, the devastation wrought by humans on the rest of the living world is relatively mild, perhaps clocking in under 10 percent. Well, at least for the time being. According to an influential 2011 *Nature* study by paleobiologist Anthony Barnosky, if we keep it up at our current rate of extinctions, we could jump from our (still horrifying) ranks of a minor mass extinction into the sixth major mass extinction anywhere from three centuries to 11,330 years from now—either of which would represent a geological eyeblink of time, indistinguishable to future geologists from an asteroid strike. Even worse, there could lurk tipping points along the way, in which the world’s remaining species fall away almost all at once, less like the attrition of soldiers on a World War I battlefield and more like the nodes of a power grid failing in concert during a network collapse.

Given how catastrophic the impact of humans on the biosphere has been *already*, it’s chilling to think that the crescendo of our mass extinction might still lie in front of us. Also, it’s worth pausing to reflect on just how outrageous it really is that humans could eventually engineer a global die-off on par with these Götterdämmerungs of deep time. In an admittedly oversimplified conceptual model, the University of Chicago paleontologist David Raup once estimated that sterilizing half the planet would not even come close to causing a major mass extinction, and that instead, to reach levels of destruction comparable to the catastrophe that wiped out the dinosaurs one would have to sterilize the entire planet except for an area circumscribing New Zealand.[fn5]

But if it was the asteroid that did in life at the end of the Cretaceous, it’s the only major mass extinction convincingly linked to an impact, and so in some ways is the least instructive of the great catastrophes in Earth history.

This is because it now appears as though the cosmic doomsday at the end of the Cretaceous was a one-off nightmare in a tapestry of a deep time threaded with disaster. Asteroids (or comets, for that matter) do not seem to play a central role in the history of animal life. Instead, the rest of these many apocalypses—including the four other major mass extinctions—seem to be strange biogeochemical events, brought about by the intrinsic workings of the planet itself, as its churning mantle, its buckling, life-coated crust, its swirling oceans, and its atmosphere all conspired to push Earth's surface outside the narrow envelope suitable to complex life. And the majority of these global disasters (both minor and major mass extinctions alike) involve the eruption of huge CO_2-spewing volcanic expanses, which, quite unlike anything in human history, can vomit millions of square miles of basalt lava across entire continents and are called, somewhat unimaginatively, "large igneous provinces." The lingering existence of one such vast volcanic province at the same time as *even the dinosaurs' demise* hangs uneasily over that extinction as well.

It's not the demise of the dinosaurs, though, to which we should look for warnings for our own future, but rather a period tens of millions of years before they ever even arrived on the scene. As the Carboniferous Period drew to a close, one stretch of time stands alone as uniquely instructive—uniquely hapless, volatile, and deadly—when it comes to CO_2 overdoses. The great swampy age of coal, where we left off our tour of Earth history, gave way some 300 million years ago to the ensuing Permian Period (299–252 million years ago), which was followed by the Triassic Period (252–201 million years ago), as the Earth was entering the age of Pangaea. And as it did so, the planet repeatedly lost control of its carbon cycle and suffered 90 million years of mass extinctions, including two of the biggest global catastrophes of all time—both CO_2-driven nightmares. In one case, it nearly died. It was felled, in the words of paleontologist Paul Wignall, by "a climate of unparalleled malevolence." Almost half a billion years after the fading supercontinent Rodinia nearly destroyed the world with ice—a catastrophe that somehow gave rise to animal life—and almost 100 million years after that animal life came onto land, the Earth was once more entering the age of a supercontinent, for the first and only time in the age of animal life. And as the Appalachian State geologist Sarah Carmichael told me, "Bad things happen when you have a supercontinent."

Pangaea!

In 1912, three bodies lay dead in a tent in Antarctica, marooned on a barren, moonless plain, oceans away from home. Outside the tent was starlight, wind, and the creaking rumor of a sea far beneath the Ross Ice Shelf. Polar night had long since set in, it was −70°F, and gusts whipped this canvas mausoleum in the dark, as it had for months. One of the corpses had been a British navy captain in life—one whose mismanagement had doomed his tentmates on a foolhardy race "to secure for the British Empire the honour" of reaching the South Pole. His body now lay half out of his sleeping bag, his face translucent and anguished, hard as iron. Beside him was a diary.

"These rough notes and our dead bodies must tell the tale …. We shall stick it out to the end but we are getting weaker, of course, and the end cannot be far," the frozen pages read. "It seems a pity but I do not think I can write more. R. SCOTT Last entry. For God's sake look after our people."

Next to the body of Robert Falcon Scott in his forsaken tent at the bottom of the world was a thirty-five-pound bag of *Glossopteris* fossils—leaves from a warm Antarctic forest that had draped the continent 260 million years ago. Where Welsh coal from the Carboniferous had powered his expedition's ship through the ice, these fossil trees were from tens of millions of years later in Earth history, near the end of the Permian. Scott had collected them on Antarctica's Beardmore Glacier and hauled them till the last.

That same year, a German meteorologist, fresh from a trip to the other end of the world, announced a scandalous hypothesis. On an expedition to Greenland, a dashing young Alfred Wegener had watched as ice floes spilled out of the fjords and crashed together to form mountainous ridges. He then watched them rift and break apart as drifting islands of ice. He surmised that entire continents acted the same over geological history, alternately crashing into each other to form the world's great mountain ranges, and breaking apart into lonesome landmasses that wander the face of the Earth. And once upon a time, these continents must have been united. How else to explain the interlocking coastlines of the continents, plainly apparent to any school-aged child? Or a fossil record that once saw ancient creatures hop easily between now-disparate landmasses separated by vast oceans? Wegener predicted that Antarctica must have been connected as well, and the fruits of Captain Scott's ghastly death march proved this hunch correct—even the remote terra incognita at the South Pole had once joined Wegener's speculative ancient supercontinent in one global landmass. Yet, as with many scientific visionaries, Wegener was well ahead of his time, and the German was ridiculed for his outlandish theory of "continental drift" until his own untimely death in Greenland in 1930. There he's entombed today, somewhere deep under the ice cap. As for Scott, when springtime arrived and daylight returned to the southern continent, a shipmate who had stayed behind in the safety of the expedition's winter quarters on the coast set out and found Scott's tent buried in a snowdrift, orphaned on the wastes. "The sun shines lovely over this place of death," he wrote.

Scott and his men—as his fossils attested—had died on Gondwana, a vast southern landmass comprising one half of the far vaster supercontinent of Pangaea that once straddled the globe 250 million years ago. But the *Glossopteris* trees whose fossil leaves Scott collected—a forest biome that once stretched from South America, to Africa, to India, Australia, and Antarctica—along with almost everything else on Earth, would be destroyed by the greatest catastrophe in the history of the planet.

PANGAEA IS POPULARLY THOUGHT OF AS THE MYTHIC HOME OF THE dinosaurs, and the monumental Pangaean red rocks of the American Southwest—sculpted by wind into improbable bridges and jutting out of the sagebrush like rusting spaceships—call giants to mind. But this isn't quite right. In fact, the age of Pangaea represents the strange, afflicted prologue to that more familiar story. Instead, the supercontinent at its height was a volatile, forbidding place that only gave rise to the ensuing, 135-million-year reign of dinosaurs as it began to break apart. During the prior 90 million years of Pangaea's apex, spanning the Permian and Triassic Periods, and even bleeding into the early Jurassic, Planet Earth would endure seven mass extinctions: 273 million years ago, 260 million years ago, 252 million years ago, 249 million years ago, 235 million years ago, 201 million years ago, and 183 million years ago.[fn6] Of that litany are two *major* mass extinctions—chartered members of the Big Five—while the one that struck 252 million years ago, the aforementioned greatest catastrophe in the history of animal life (the End-Permian mass extinction), is so extreme it almost needs its own category. Once Pangaea was fully assembled, it seems as though the

Earth had all but lost its ability to regulate carbon dioxide altogether. As a result, animal life barely made it through.

We pick up our story as the once-disparate island continents had all but assembled into a supercontinental whole. Roaming coastal New Mexico and Oklahoma in the early Permian, one would encounter the arching spinal sails of 220-pound dimetrodons[fn7] sunbathing on marshy floodplains, mud-caked in swamps, lying in wait, ambling across tidal flats, and picking off fish from the banks of estuaries. Just offshore, beyond the barrier reefs, resplendent eddies of plankton perennially shaded the surf blue-green and softly fell to the seafloor, drawing millions of years of solar power into the deep: this was the birth of the vast oil and gas fields of the West Texas Permian Basin, the largest shale oil reservoir in the US. You've likely put some of this plankton in your car, or felt the warmth of the Permian sun that it captured. Forbidding mountains loomed over these Texas seas—a forgotten spur of the Appalachians—and stretched all the way to Norway, stitching Morocco to New England and Venezuela to the Gulf Coast.

Meanwhile, the color of the planet when viewed from space was getting drabber, as the vast swaths of Carboniferous coal rain forests that once stretched, as one study puts it, from "Kansas to Kazakhstan," now found themselves in the aridifying interior of a supercontinent. As the once-verdant tropics gave way to khaki and umber tones, a somewhat mysterious minor extinction 273 million years ago (Olson's Extinction) kicked off this age of troubles. *Dimetrodon* and its compatriots saw their beloved marshy hunting grounds give way to a tropical American Midwest, which grew unendurably hot, arid, and inimical to life. This hellish band of extraordinarily harsh deserts and unthinkably acidic salt lakes would continue to haunt equatorial Pangaea for perhaps the next 40 million years, spanning some 5,000 miles. Meanwhile, at the South Pole, the great ice sheet of the Paleozoic Ice Age was shrinking, shuddering with Earth's orbit. Before long—poof! The world would lose its polar ice sheets altogether, perhaps for the next quarter-billion years. Earth was inescapably entering the Greenhouse.

260 Million Years Ago: The Capitanian Extinction

After the fall of *Dimetrodon*, the terrestrial fossil record decamps the tropical Pangaean hellholes for places like Antarctic South Africa, or on the other side of the planet in Perm,[fn8] Russia, and the world at last resembled one that would appear familiar to us … if you squinted hard. At the very least, where the archaic ecosystems of early Permian operated quite unlike our own—with more carnivores than herbivores on land[fn9] —the middle Permian had somewhat more modern food webs, even if they were populated by phantasmagoric ogres. There was the bizarre antler-faced *Estemmenosuchus* and the genuinely frightening *Anteosaurus*[fn10] (seemingly patched together from the spare parts of a hippo, a tiger, and a *T. rex*). These strange creatures were the aptly named dinocephalians, or "terrible heads." While they weren't dinosaur-scaled, they made up for size in their sheer unattractiveness. In fact, there's a strong case to be made that the 14-million-year stretch of the middle Permian is the ugliest period in all of Earth history. One struggles to describe or even find analogues for these creatures—alternately snaggle-toothed, humpbacked, postured like foundering ships, with the exaggerated, unwieldy heads of sports mascots. They seem to be feinting toward a lifestyle more mammal than reptile, but they still likely lacked hair, and in artists' reconstructions they seem to be as confused to find themselves brought to life as we are to find their bones in the rocks.

The dinocephalians were early members of the clade known as therapsids. Broadly speaking, you are a therapsid like *Anteosaurus*—though in the same way that birds are dinosaurs. We descend ultimately from the tiny surviving branches of this once-lush ancestral Permian tree. But like that of *Dimetrodon*—and this will become a theme in the chapter—the dinocephalians' reign, at only a few million years, was brief in geologic terms.

In the middle of the Permian, the waters of a placid tropical lagoon in western China's Yunnan Province were limpid and pulsing with life. The reefs were healthy: tentacles probed iridescent sponges, and dense thickets of lamp shells yawned under the survey of saw-mouthed sharks. Then the whole thing exploded, blowing the sharks, sponge reefs, and everything else thousands of feet out of the water and to smithereens. This destruction had been mindlessly set in motion millions of years earlier, and thousands of miles below, when a superhot blob of mantle—kissed by the Earth's blazing core—was sent on an inexorable ascent to the surface. And now it had arrived. In the moments before what had become a 500-mile-wide blob of magma pierced the planet's eggshell crust, cracks in the ocean rock allowed seawater to pour in and encounter this glowing emissary from the middle of the Earth. The resulting interaction blew the landscape to pieces. Giant tables of limestone and marble chaperoned the doomed fish toward the heavens in a plume of steam and ash. Before long, this tropical tableau would ooze with steaming piles of lava. Across the shallow seas, from Sichuan Province, China, to northern Vietnam, this vast platform of carbonate (think modern-day Bahamas) was similarly detonated and sunk in lava. In some places this lava piled up more than three miles deep, and perhaps spanned hundreds of thousands of square miles.

But the damage wasn't limited to ancient Asia. The hapless airborne sharks were the first harbinger of a global mass extinction 260 million years ago—one that took down not only sea life all over the world but the dinocephalians as well. It was so destructive that some paleontologists insist the disaster warrants changing the Big Five mass extinctions to Six. But since this mass death shares a common cause with, and virtually all the same symptoms as, the next five mass extinctions to come in the age of Pangaea—from global warming and widespread wildfires, to ocean acidification and deoxygenation—we'll wait to tease out the grisly details of how exactly these kinds of eruptions can end the world. And unfortunately for the Permian world, there would only be an 8-million-year reprieve before the mother of all mass extinctions.

252 Million Years Ago: The End-Permian Extinction, or the Great Dying

At the very end of the Permian Period, enough lava erupted out of Siberia and intruded into the crust that it could have covered the lower forty-eight US states a kilometer deep.[fn11]

A kilometer deep.

The rocks left behind by these ancient lava flows are known as the Siberian Traps. Today the Traps produce spectacular river gorges and plateaus of black rock in the middle of Russia's boreal nowhere. The eruptions that produced them, and that once covered Siberia in 2 million square miles of steaming basalt, are in a rare class of behemoths called large igneous provinces (LIPs). We met a smaller one in that last extinction.[fn12] LIPs are by far the most dangerous thing in Earth history, with a track record far more catastrophic than asteroids. These once-an-epoch, planet-killing volcanoes are of a different species entirely than your garden-variety Tambora or Mount Rainier or Krakatau, or even Yellowstone. Imagine if Hawaii was created not over tens of millions of years and scattered across the Pacific, but in brief pulses in less than a million years, and all in one area (and sometimes emerging through the centers of continents). LIPs are the Earth's way of rudely reminding us that our thin rocky surface, and the gossamer glaze of green goo that coats it, sits atop a roiling, utterly indifferent planetary drama. It's one in which titanic currents of rock draw down entire ocean plates to the center of the world to be destroyed and reborn. When this process suffers a hiccup, LIPs gush out of the crust like tectonic indigestion, leaving gigantic swaths of the Earth buried in volcanic rock. Depending on the pace and size of these eruptions, if they're big enough and fast enough, they can destroy the world.

These kinds of colossal eruptions, geologists have discovered in the past three decades or so, are coincident with four of the Big Five mass extinctions, and perhaps even all five—as well as the majority of the twenty or so smaller, still Earth-changing minor mass extinctions. They are of a totally different scale than anything with which we're familiar throughout the entire evolutionary history of hominins, and that's a good thing.

At the end of the Permian, in the greatest mass extinction of all time, these eruptions would have featured terrifying explosions, no doubt inducing brief volcanic winters and withering acid rain. There was also widespread mercury poisoning, and toxic fluorine and chlorine gas, which would have been familiar to suffocating soldiers in World War I trenches. But these were mere arrows in a volcanic quiver that also featured a bazooka. Most importantly—and most unfortunately, for life—billowing out of the Earth in the biggest catastrophe in history was a planet-deranging amount of carbon dioxide.

BEFORE DOOMSDAY, OUR EXTENDED LINEAGE'S PLACE AT THE TOP of the planetary pyramid had seemed secure. Despite suffering blows earlier in the Permian, therapsids from our now heavily pruned branch of the family tree retained their dominance over the terrestrial world, and a new class had ascended to the throne. While the hideous dinocephalians might have already exited the world's stage, in the moments before the end of the world, dread saber-toothed therapsids like *Inostrancevia* came to prowl the fern tree understories of Pangaea, stalking ponderous, heavily armored reptiles. With dinosaurs still tens of millions of years in the future, these protomammals ruled with the confidence of a dynasty that had dominated the planet for over 50 million years, through good times and bad. Vicious, capable, athletic, modern, they were obsolesced only by the coming apocalypse, not by some evolutionary shortcoming on their own part. There was no hint that they were about to be nearly exterminated—or that the rare survivors from their ranks would be relegated to an afterlife of almost 200 million years dodging giant reptiles. But theirs would soon become a world of ghosts.

When geologists dated the biggest continental floods of lava known in Earth history to the global near-death experience at the end of the Permian that would erase this world, the story seemed complete. But curiously, as the Siberian lava has been dated ever more precisely, it turns out that wasn't until 300,000 years into the eruptions—and after *two-thirds* of this lava had already erupted, flooding the northern reaches of Pangaea in steaming rock miles thick—that this worst mass extinction of all time actually began. This is strange. These volcanoes would have been pumping out all the usual nightmare stuff this entire time, putting industrial polluters to shame—and doing so for hundreds of millennia *before* the mass extinction began. There would have been uncountable, unthinkably violent eruptions, and noxious storms of acid rain. But the biosphere is tough. And as bad as it was turning a third of Russia into a volcanic hellscape, it doesn't explain why, after all those countless centuries of misery, life suddenly winked out en masse, even at the bottom of the ocean, on the other side of the planet.

"You can rule the lavas out," says Seth Burgess, a geologist at the US Geological Survey, about the initial eruptions as a mechanism for the mass extinction. But something about these Siberian volcanoes must have dramatically changed after 300,000 years when the world quickly disintegrated. So what was it?

The planet started burning fossil fuels.

AT THE DAWN OF THE INDUSTRIAL REVOLUTION, AIR BUBBLES TRAPPED in Antarctic ice began recording our atmosphere growing increasingly overwhelmed with light carbon from the industrial-scale immolation of Carboniferous forests in British furnaces. At the end of the Permian, some 252 million years earlier, and at the very moment of the greatest extinction of all time, the very same signal appears in rocks all over the world as well, and even in the fossil tusks of our distant protomammal cousins. Carbon isotopes—in the air, in the oceans, and in rocks made of life—suddenly get dramatically lighter in the very last moment of the Permian. Like the panicked transmissions of a flight recorder recovered from the seafloor muck, these isotopic swings in ancient limestone—teased out today by grad students on mass spectrometers—record the global life-support system going haywire.

"The carbon cycle is business as usual, and it's not until right before the mass extinction starts that you get this wild swing [to lighter carbon isotopes]," says Burgess, "which is indicative of a huge pulse of carbon entering the ocean-atmosphere system."

This is one of Rothman's squiggles in Earth history, a flux of carbon into the system so massive that it overwhelms the planet's ability to regulate itself and pushes the world out of equilibrium. All on their own, volcanoes emit lots of CO_2: as much as 40 percent of the gas from a venting volcano can be carbon dioxide. But after Siberia had been smoldering at the surface for countless generations, something far more menacing began to cook below. Colossal thousand-foot-thick sheets of magma, stymied in their ascent to the surface, instead started spreading sideways into the rock far underground, like incandescent rhizomes, baking through the underworld. This is when everything went to hell. These massive magma roots were burning through an old layer cake of Russian rock eight miles thick. The quarter-billion-year pile of strata had accumulated in the vast Tunguska Basin: the remnants of bygone salt flats and sandstones, but more catastrophically, carbon-rich limestones and natural gas deposits from ancient seas, and coals from ages past. The magma cooked through all these fossil fuels and the carbon-rich rock underground on contact, and denotated spectacular gas explosions that shattered the rock far above, erupting at the surface as half-mile craters that spewed carbon dioxide and methane into the air by the gigaton.

After hundreds of thousands of years of familiar surface eruptions, the volcanoes had suddenly started burning through the subterranean world on a massive scale and began acting like enormous coal-fired power plants, natural gas plants, and cement factories. "The burning of coal," one scientist writes of the End-Permian extinction, "would have represented an uncontrolled and catastrophic release of energy from Earth's planetary fuel cell.” The Siberian Traps suddenly started to emit far too much CO_2 and far too quickly for the surface world to accommodate it.

The ancient planet, when introduced to this titanic slug of CO_2 behaved just as atmospheric science and geochemistry would predict. It was, in the clinical language of geochemists, a “major carbon cycle perturbation.” “Perturbing the carbon cycle” is a decidedly unsexy-sounding way to bring about the apocalypse, so let’s spell out the grisly details. One of the best ways to understand how something works is to watch it fall apart. And just as a functioning biosphere is mediated by a carbon system in balance, breaking the Earth system illuminates its weak points.

Here’s a plausible sequence of events at the end of the Permian. First, and most simply: the excess CO_2 trapped more energy from the sun on the surface of our planet—a simple physical process that was worked out by physicists more than 150 years ago. And so the world helplessly warmed—models and proxies both point to about 10°C of warming over thousands of years—pushing animal and plant physiology alike to their limits. It’s also a simple physical fact about our world that for every degree it warms, the atmosphere can hold about 7 percent more water, so, as the temperature climbed and the water cycle accelerated, storms began to take on a menacing, drowning intensity. As the ocean warms as well, it holds less oxygen. Unfortunately, though, living in hot water is hard work—it’s more metabolically demanding—so the luckless animals in it required *more* oxygen to live, not less. Thus, as the ocean got hotter and more stagnant, the creatures in it began to fall away, and the seas began to empty. Making matters worse, the carbon dioxide in the air diffused into these gasping seas as carbonic acid (H_2CO_3). The entire global ocean became more acidic as a result, and the water was robbed of the chalky carbonate dissolved in it, and which animals used to build their shells. In these souring seas, the creatures became brittle and sickly, or even failed to form shells in the first place. It’s the same reason that some kinds of plankton foundational to the ocean food web today, like fluttering sea snails around Antarctica, have already been found pitted with holes in our acidifying polar seas. And so it was likely in the greatest disaster of all time as well, when things that needed to make carbonate shells to live—things that looked like clams and corals—got wrecked.

As this sea life was decimated, the global marine food web began to teeter and collapse. Meanwhile, the ecosystem on land was being destroyed by wildfires (themselves spewing even more CO_2 into the air) and lashed by violent CO_2-supercharged storms, which gnawed away at entire mountain ranges. The terrestrial wreckage washed into the ocean, blasting the coastal seas with decaying vegetation and minerals weathered out of the land, like phosphorus, that acted as plant food, fueling massive algae blooms offshore. Avalanches of dead life fell to the seafloor in the vast Panthalassa Ocean and left behind sickly carbon-rich black shales, the product of what have been called the "marine snowstorms" of the End-Permian.

The oceans, already wanting for oxygen from the heat, now began to suffocate in earnest as the explosive algae blooms died and decomposed, robbing the seas around them of even *more* oxygen.[fn13] Even worse, when the oceans lost their oxygen like this, phosphate, iron, and ammonium (that is, more plant food) leached from the seafloor as well, which only further fueled these exploding blooms of algae, blooms that continued drawing down oxygen with them in death—a positive feedback to more suffocation.[fn14] As the CO_2 continued to issue from the Siberia Traps in massive and unrelenting belches, the planet became unceasingly hotter *still*, and the oceans didn't have a chance in hell. CO_2 was now pushing the planet outside the limits of complex life. And just as these lifeless, anoxic, hot seas began to spread, a specter from the Earth's ancient past was renewed on this dying planet.

Unlike most life on Earth with which we're familiar, primitive anaerobic bacteria, having evolved eons ago on an all-but-breathless world, don't need oxygen to burn their food. For some, sulfate will do the trick. And on this rotting, suffocating world, this microbial life became ominously ascendant, breathing out hydrogen sulfide (H_2S) as exhaust, just as they did in the low-oxygen glory days before the age of animal life. In catastrophes like this, when the ocean loses its oxygen, the playing field tilts in these ancient microbes' favor once more. Unfortunately, hydrogen sulfide is mercilessly toxic, instantly killing humans (and creatures like us) at concentrations as low as 700 ppm, as it sometimes does today in manure pits, or around oil pads like those in Texas's Permian Basin. And so this dark cloud of primeval life spread insidiously through the deep, and even into the shallows. The world was now very, very hot, very stormy, almost totally denuded of vegetation, with acidifying, anoxic oceans that belched unsparingly poisonous gas from an ancient microbial metabolism that killed anything that came near it.

On the other side of the planet from the eruptions, once-forested polar South Africa became so denuded of life that rivers that once happily curved and twisted—their banks anchored by living plant roots—now rushed straight over the scoured landscape in braided, sprawling arroyos. Unearthly hot and dry seasons razed the forests with fire, then alternated with apocalyptic superstorms that washed it all away. Robert Falcon Scott's vast Antarctic forests of *Glossopteris*, which had formed thick seams of coal before the extinction, vanished in the catastrophe. In the aftermath coal would all but disappear from the fossil record for the next 10 million years. As paleontologist Paul Wignall writes, "it would have been possible to walk all over the world in the earliest Triassic without ever encountering a tree." The animals that had stocked these now-vanished forests for millions of years—those vicious therapsids and their armored reptile quarry—vanished as well. In the rocks, fungal spores strangely appear in the fossil record all over the world, heralding the collapse of the biosphere. Even insects, whose sheer numbers usually cushion them against mass death, struggled to hold on.

In the oceans, the extinction was even worse. Trilobites, the scuttling sea bugs whose calcite eyes and intricately louvered armor had littered the seabed for hundreds of millions of years—by the uncountable hundreds of trillions—disappeared forever. So too for the giant sea scorpions. Squidlike spiral-shelled ammonoids, which long bobbed under the surf, saw their ranks culled by 97 percent. Reefs built by archaic corals, lamp shells, and sponges—the kinds of reefs that had limped through previous mass extinctions and recovered—were wiped off the face of the Earth. In all, up to 96 percent of life in the ocean might have vanished.

While the heat devastated life at the poles (unable, as it was, to flee to higher latitudes as the planet roasted), Earth's searing midsection had become plainly unearthly. As CO_2 sent global temperatures soaring, the ocean in the tropics became as hot as "very hot soup,"[fn15] perhaps sufficiently hot, even, to power outlandish 500-mph "hypercanes" that would have laid waste to the coasts. In the continental interiors the temperature would have leaped even farther off the charts. In papers describing the event, academic prose drops its pretensions to sobriety and strains to capture the world of the End-Permian mass extinction. Paragraphs are peppered with alarming phrases like “lethally hot” and “postapocalyptic greenhouse.” In the planet’s most miserable hour, much of its surface came to resemble less Earth as we know it than the feed from a lander probe on some hopeless and barren exoplanetary outpost. Earth, in its darkest hour, was losing its Earthiness.

In fact, the postapocalyptic ocean was so vacant that carbonate reefs all over the world came to be built again in the recovery not by animals like the archaic corals and lamp shells that were driven extinct, but by calcified mounds of bacterial slime.[fn16] Everywhere. Even a short hike from my apartment in Boulder, Colorado, brings me face-to-face with this stromatolite rock from the end of the world, left behind by foul microbial mats. In the Colorado Front Range, where Earth history has been lifted out of the ground, tilted sideways, and ornamented with ponderosa pine, one encounters this hummocky red rock laid down, layer by layer, by microbes in a deathly sea 252 million years ago. It’s wedged between more prosaic sandstones from the Carboniferous before it, and the dinosaur-trampled beach sands of the Mesozoic after it, hogbacks of which loom like a backstop behind Denver—the geology of happier times. But the implications of this brief wedge of bacterial rock, and a global ocean momentarily dominated by mounds of calcifying slime, are truly frightening. In the history of complex life, it marks the stromatolites’ only heyday since the eternal Precambrian Age—a throwback to an ocean of the Boring Billion before animal life, and a preview of the one that will someday come long after the age of animals is over.

Before long, almost every living thing on the planet was dead. The interiors of the continents were silent except for hot, howling winds that swept over the wastes—a dry desolation that alternated with punishing, unearthly storms that smelled like death. The oceans, whose open seas once flashed iridescent with shoals of bobbing spirals and tentacles, and whose nearshore reefs were once dappled fire-engine-red to ultraviolet by life, were now putrid, asphyxiating, empty, and covered in slime. Every gear of the grand, intricately interlocking biogeochemical machinery of this planet became jammed, decoupled, or spun hopelessly out of control. Complex life, as a subset of this global geochemical churn, unraveled as well. All from adding too much CO_2.

If there is a geologic precedent for what industrial civilization has been up to in the past few centuries, it is something like the volcanoes of the End-Permian mass extinction.

"The causes might be different but the results are going to be the same—the Earth doesn't care what caused [the huge injection of CO_2 into the system]. It's going to react similarly, which is pretty damn frightening," says Burgess. "The End-Permian mass extinction is unique in Earth history. Nothing else is as severe, and it's not even close."

NOW LET'S PULL BACK FROM THE BRINK. HOWEVER SIMILAR TO THIS era our modern experiment on the planet might first appear, it's worth acknowledging, even stressing, that the End-Permian climate catastrophe was truly, surpassingly bad. And on a scale unlikely ever to be matched by humans. Upper estimates for how much carbon dioxide the fossil fuel–burning Siberian Traps erupted, ranging up to *120,000 gigatons*, defy belief. Even lower estimates, of say 30,000 gigatons, constitute volumes of CO_2 so completely ridiculous that matching it would require humans to not only burn all the fossil fuel reserves in the world, but then keep putting ever more carbon into the atmosphere for thousands of years. Perhaps by burning limestone for fun on an industrial scale for generations, even as the biosphere disintegrates. As it is, industrial civilization could theoretically generate about 18,000 gigatons of CO_2 if the entire world pulled together on a nihilistic, multicentennial, international effort to burn all the accessible fossil fuels on Earth.

But while the sheer volume of CO_2 generated by the Siberian Traps dwarfs our present and future output, that total was achieved over tens of millennia. What is alarming, and why it's worth talking about the Siberian Traps in the same book as industrial civilization, is that *even* in comparison to those ancient continent-spanning eruptions, what we're doing now seems to be unique. It turns out that the focused, highly technological effort to find, extract, and burn as much of the world's fossil fuel reservoir as is economically feasible, as fast as possible, has been *extremely* prodigious at getting carbon out of the crust—even compared to the biggest LIPs in Earth history. In fact, the best estimate is that we're emitting carbon perhaps ten times faster than even the mindless, undirected Siberian volcanoes that brought about the worst mass extinction ever.

This matters, again, because it's all about the rate. There's almost no amount of carbon you can pump into the atmosphere that, given enough time, Earth couldn't buffer itself against. Volcanic CO_2 is supposed to enter the system. Without it, none of this works: the climate wouldn't be habitable, life would run out of raw material, and oxygen would run out. But everything in moderation. To maintain its homeostasis, the planet continuously scrubs CO_2 from the atmosphere and oceans so that it doesn't build up and cook the planet. But this process is very slow on a human timescale. It buries this CO_2 in coals, oil and gas deposits, and, most importantly, ocean sediments that turn to carbonate rock over millions of years. When more modest-sized eruptions inject a massive slug of CO_2 to the atmosphere, threatening to overwhelm this process, the Earth has several emergency handbrakes. The oceans absorb the excess carbon dioxide, becoming more acidic, but in their millennial overturn they bring these more acidic surface waters to the seafloor on the downdraft of the planet's great ocean currents. There they dissolve the seafloor's carbonate sediments—the massive carpeting of tiny seashells at the bottom of the ocean, laid down by life over millions of years—and buffer the seas in the exact same way that a Tums[fn17] settles an upset, acidic stomach. This is the first line of defense in the carbon cycle, and it works to restore ocean chemistry over thousands of years. At the same time, and over the longer term, as it gets hotter and wetter from high CO_2 rock weathering on land accelerates in lockstep. As we've seen, this is a planetary guardrail against runaway warming as well, and eventually, over 100,000–200,000 years these forces work to restore the carbon cycle and coax the Earth back from the edge. On a world without humans or especially catastrophic LIPs afoot, these feedbacks usually suffice to rescue the planet. The excess CO_2 is removed, transmuted to rock, and commended to the ages; the temperature eventually falls; and the pH of the ocean is restored over hundreds of millennia.[fn18]

So it's not just the amount of CO_2 that enters the system that matters, it's also the flux. Put a lot in over a very long time and the planet can manage. But put more than a lot in over a brief enough period of time and you can short-circuit the biosphere. That is, if you put so much CO_2 into the air in such a short period of time that these planetary guardrails fail, all hell breaks loose. The planet can bend, but it can also break. And unfortunately, the rate at which humans are currently injecting CO_2 into the oceans and atmosphere today far surpasses the planet's ability to keep pace. We are currently at the initial stages of a system failure. If we keep at it for much longer, we might see what actual failure means.

In the End-Permian mass extinction, the planet didn't merely surpass this carbon cycle threshold, but indeed seemed to have tied a brick to the gas pedal and kept going for thousands of years. The devastation of the planet at the end of the Permian was so complete that it didn't matter if you spent your life in the foothills of the Trans-Pangaean Mountains and had never known the smell of sea salt on the air, or if you made your living in the abyss, clinging to the seafloor of the globe-spanning Panthalassa Ocean, miles down, having never known a photon of sunlight. You were almost certainly going to go extinct, no matter where you were, or who you were. Once Earth started burning through fossil fuels on a massive scale, it was all over. When the magma met the old carbon underground, CO_2 exploded out of the Earth, the atmosphere heaved with heat, the oceans roiled and rotted, and life all but disappeared. That was the end of the Paleozoic Era, in progress since the dawn of the Cambrian.

249 Million Years Ago: The Smithian-Spathian Extinction, 232 Million Years Ago: The Carnian Pluvial Event, 201 Million Years Ago: The End-Triassic Mass Extinction, and 183 Million Years Ago: The Early Jurassic Extinction

We'll speed through the next several mass extinctions in the age of Pangaea, including one (the End Triassic mass extinction) that was among the worst in Earth history. By now we've made the kill mechanism of CO_2 clear,[fn19] and I'm sympathetic to readers for whom the various unsung biomes of the ancient Earth begin to blend together. It is still worth dwelling, though, on just how profound the scale and how alarming the frequency of suffering was on Pangaea until the supercontinent's eventual, merciful schism.

The blasted planet that followed the greatest catastrophe of all time was extremely boring. Two hundred fifty million years ago, a globetrotting ecologist could have traveled to the ends of the Earth only to find, everywhere, the same impoverished ecosystems. The same stupid clam in this tidepool was the same stupid clam you'd see in all the other tidepools, thousands of miles away, all around the planet, everywhere you went. The same torpid little piglike *Lystrosaurus* picking at weeds in Antarctica was the same ugly creature you'd find gnawing on the same miserable shrubs in China, Russia, India, or Africa. Life was drab, simple, generalist, weedy. The world was deserted, dismal, and insanely hot. And as if it weren't dismal or hot enough, the temperature suddenly soared off the charts once more, pummeling the dreary, shell-shocked planet only 2 million years after the end of the world[fn20] —perhaps a small parting gift of CO_2 from the Siberian Traps before their volcanic plumbing shut off for good. Sea surface temperatures in the tropics jumped again, perhaps to a mind-boggling 113°F—much hotter than a hot tub—and fish abandoned the middle of the planet altogether, as did whatever animals existed on land in the tropics. The sweltering oceans continued to pulse with anoxia for *5 million years* after the apocalypse, further strangling the recovery.

Then, 17 million years later, it happened *again*. About midway through the Triassic Period, 232 million years ago, an LIP started burbling from below what is now British Columbia. CO_2 soared, the global temperatures spiked perhaps 5°C again,[fn21] and, strangely, much of Pangaea was punished with a million years of heavy rain. At the same time, the dreadful, deepest interior of the supercontinent somehow seemed to grow only more arid and desolate still.[fn22] Dozens of lineages that somehow navigated the Armageddon at the end of the Permian finally surrendered to extinction in this warming spike.

Curiously, though, this particular climate spasm is more remarkable for the new world it brought into being. When the rains relented, the first true mammals appeared, along with the first crocodilians, the first turtles, modern conifers, and modern corals. Finally, a previously unimpressive lineage of reptiles, which evolved a few million years earlier, explosively radiated in the wake of the torrents and suddenly spread across the Earth. The dinosaurs. The strange effects of this pulse of CO_2 serve as a reminder that whatever world comes after our radical experiment on the climate today—which might see a similar amount of warming—may well be as different as the world midwifed by these Triassic rains was from the one that came before.

Surely that was the end of Pangaea's troubles. A new theater had been built, a new cast of actors had taken their places, and the stage debut for the dinosaurs hinted at their eventual, wildly successful stage run. Then, 31 million years later, it happened *again*, this time nearly killing the world once more. The age of Pangaea, it turns out, couldn't come to a close without a fittingly apocalyptic finale. Deep in the heart of the supercontinent, cracks had begun to appear—cracks that would someday trace the familiar coastlines of the continents but that now filled with deep lakes from the rains of the "Pangaean megamonsoon." It was from these cleaving lineaments in the crust, spreading at the rate that fingernails grow, that the breadth of the entire Atlantic Ocean would eventually be born and the supercontinent would come to a fractured end. But first, some 201 million years ago, this vast fissure would open up from below and flood the planet with 3 million square miles of lava once more. In the process it would incite one of the greatest mass extinctions in Earth history, another member of the Big Five in good standing: the End-Triassic mass extinction.

This time the scythe of extinction carved through a cartel of strange crocodilians[fn23] that had dominated the Triassic world, clearing the way for the dinosaurs to take over in the Jurassic world to come. Meanwhile, modern stony corals, which had just evolved and exploded in a burst of reef construction around the world—reefs later uplifted in continental collisions and exposed today in places like Austria, where the craggy alpine profile of Triassic corals shade Salzburg—suddenly disappeared for hundreds of thousands of years when the oceans turned foul and the living world disintegrated again. Reefs wouldn't fully recover from the global cataclysm for 20 million years.

Where sweeping vistas of volcanic rock in Siberia mark the Armageddon at the end of the Permian, the dusky cliffs of the New Jersey Palisades under the George Washington Bridge serve as a tombstone for the Triassic. These giant sills of frozen magma, which once fed the massive eruptions that once more ended this world with CO_2 now sit across the Hudson River from the hollow steel skyscrapers of Manhattan, watching over Interstate 80 traffic jams, unimpressed. You can find the same rocks in Nova Scotia, Spain, Brazil, France, Morocco, and all across what was the rending middle of the supercontinent. And just as in the End-Permian apocalypse, other rocks spanning the extinction record a huge pulse of light carbon entering the earth's oceans and atmosphere—perhaps from sheets of magma burning through fossil fuels in Brazil that pumped CO_2 out of the Earth and into the air as effectively as any modern oil multinational. Extraordinary warming, ocean acidification, and deoxygenation predictably followed once more, and mass extinction swept the globe.

In the aftermath, mammals, whose ancestors had once ruled the world, would have to be content to dart around in the shadow of monsters, waiting for an asteroid before making another play at global dominance. But that nightmare was still 135 million years in the future, and the dinosaurs were about to settle into one of the most successful runs of any clade in the history of land life.

AS THE EARTH BEGAN TO SETTLE INTO ITS HALCYON DAYS, OUR TOUR of Pangaea nears its end. Although it is by far the most popular period of Earth history among the general public, the placid age of dinosaurs—from an Earth system perspective—is a bit of a snooze. As the supercontinent began to disassemble, it would be a functional eternity until the next major mass extinction. For some reason, the prospects for life on Earth had considerably improved. The planet wouldn't escape disasters entirely during this stretch, but after Pangaea broke up, none of these calamities would progress to major mass extinctions during the long Pax Dinosauria. In the early Jurassic, for instance, and on a world in which *Dilophosaurus* had only just forgotten the nightmare crocs that terrorized its ancestors, another massive LIP disgorged, not far from Scott's final resting spot, at the junction where Antarctica, South America, and South Africa would ultimately rift apart. CO_2 shot up, and the warming seas did indeed suffocate once again, embalming dead ichthyosaurs and ammonites in black shales that today line Banff forests and crumble out of the seaside cliffs of England's Jurassic Coast. But still, this wasn't nearly the calamity of those darkest days of Pangaea. And although there would continue to be LIPs and CO_2-driven episodes of global warming like this—and even disasters of ocean anoxia that peppered the age of dinosaurs every few tens of millions of years—the planet seemed to have entered into a new, more stable phase. As the continents continued to drift over the next 200 million years, riding the incandescent currents of the mantle far below, landmasses that once comprised an epic whole of Pangaea came to stake out their own patches of a more disparate globe. And nothing even remotely like the horror of the

End-Permian, striking during the supercontinent's consolidated apex, ever happened again.

SO WHY EXACTLY WAS PANGAEA SUCH A MISERABLE, NEARLY FATAL arrangement for complex life? After all, large igneous provinces occur throughout Earth history, and they don't always drive mass death. For instance, in our age of mammals alone two have gone off,[fn24] and neither came close to ending the world. Why, then, was this supercontinental world so singularly unable to maintain the life-support systems for planet Earth? Perhaps it was just a coincidence that the planet nearly died several times over while the continents were all united as one colossal island, all gathered to one side of a vast ocean planet for the first time in a half-billion years. Or perhaps, as the Penn State geologist Lee Kump argues, it is no coincidence at all, and the unusual configuration had broken the planet's "climate regulator." It turns out Pangaea was already perched at the edge of failure, struggling to manage its CO_2 even before the catastrophic eruptions ever began to detonate.

Breaking Earth

Thirteen hundred feet under Northern Ireland, Jason Hopps, the engineer and manager of the Kilroot mine, kills the lights of his rusted Land Rover to prove a point. It's not just dark down here, it's oblivion. Somewhere above us sheep mow fields of green, bedewed with the thick air of the Irish Sea. But down here in the massive catacombs, there is not a photon to spare. When he turns the lights back on, the world is salt. Pink, brown, translucent crystalline caverns, saline snowdrifts lining the walls.

"It's all salt. Salt floors, salt walls, salt ceilings."

Headlights on, we wind through some of the forty miles of open tunnel, arranged like a city grid—one that has been slowly etched out of the bedrock over the past six decades, deep under the Carrickfergus countryside. Driving down the lightless salt boulevards of this lifeless salt neighborhood, we occasionally pass a salt cul-de-sac. At the far end of one, a lonely hydraulic scaler picks at the salt walls with outstretched drill bits, like a titanic praying mantis. Older parts of the mine are backfilled with ash from coal-fired power plants.

When I tell Hopps I'm from Boston, he asks if I know the Mahoneys and then catches himself. "That's probably a pretty common name there," he laughs. The comment is jarring, reminding me that the living world still exists out there, and that we're not on a mining outpost in deep space. We *are* in the Triassic Earth, though, deep in the cauterized, supercontinental wilderness of Pangaea—a landscape that was so hostile to life it might as well have been another planet.

This ancient palace of salt, the rubble of which one can find each winter scattered on the icy roads of London and New York City, is part of a vast stratigraphy of similar "evaporites"[fn25] all over the Earth from the age of Pangaea—from Northern Ireland to North Dakota. In southern Kansas salt and gypsum crumble out of Martian-looking rocks, pointing to a tropical environment that had grown almost unimaginably inhospitable by the middle of the Permian.

Just up the Northern Irish coast, in Larne Lough, the engineering company that built the *Titanic* eyes a similarly massive vein of Pangaean salt to excavate to store 17 billion cubic feet of natural gas—just as they do in the huge Permian salt caverns of Kansas.[fn26] Salzburg,[fn27] Austria, meanwhile, sells its Pangaean salt in upscale boutiques and as blocks upon which to cook steaks. Simply put, salt was all over the supercontinent. This was an ailing world, when Pangaea's vast, interior was drying up, and covered not with rain-fed lakes and rivers and trees and life, but with ephemeral acidic playas crusted with gypsum and salt, evaporating seas, and stark red deserts unvisited by complex life. Everything was evaporating. It was aridity on a scale unlike anything else in the age of animals.

Where there were once lush tropical rain forests and amphibian-stocked swamps, there were already, by the middle of the Permian, what geologist Neil Tabor terms "the wastelands of tropical Pangaea"—a desolate, crimson hellscape with temperatures as high as 163°F (73°C). If you found yourself marooned in this inferno and dying of thirst, the only succor was shallow, ephemeral lakes of acid, scattered over some 80,000 square miles of roasting wasteland. These unearthly playas were so caustic that some of them had *negative* pHs, an extreme phenomenon today only seen in industrial acid mine drainage.

For tens of millions of years, in other words, much of tropical Pangaea was an incredibly hot, arid, and barren hellhole: a red desert dotted with extremely acid salt lakes and devoid of life. It was this giant arid interior that had major implications for the carbon cycle and for life on Earth.

OVER THE LONG TERM, THE CLIMATE STATE IS PRIMARILY SET BY THE give-and-take of outgassing volcanic CO_2 on the one hand, and weathering rocks, which buries that CO_2 as carbonate rock like limestone, on the other. If these sources and sinks were mismatched by as little as 5 percent from each other, then the biosphere would be destroyed in less than 5 million years, either by lethal high-CO_2 heat or a zero-CO_2 catastrophe. But somehow these sources and sinks have stayed almost miraculously balanced over the long run. As we've seen, rock weathering provides Earth the thermostat that has (mostly) prevented it from veering off too far in either direction. But dropping into the middle of one of those vast, arid, and unearthly supercontinental interiors of Pangaea, you might notice something (besides the extraordinary heat). More precisely, something that's missing: there is no weather.

As long as we live on a plate-tectonic planet, volcanoes won't ever stop, and they won't stop venting CO_2 to the atmosphere. These emissions are more or less steady—and thank goodness for that—even if they're sometimes punctuated by especially prodigious volcanic belches that send the temperature soaring. But even when the planet does flush with excess CO_2 it's not necessarily a bad thing. Yes, it gets warmer, but the global water cycle ramps up as the planet warms. And the more CO_2-charged rainfall that falls on the rocks, chemically weathering them and carrying that carbon off in rivers and on into the ocean, the more of this former carbon dioxide from the air is stored in the ocean, transformed by life in carbonate shells and reefs, and buried on the seafloor, ultimately as limestone … and so CO_2 levels fall again. But what if the weather never reaches those rocks? If CO_2-laden rain never falls on the rocks, then there's nothing for them to react with. And if the rocks don't weather, and the CO_2 isn't ultimately sequestered, then there is no climate guardrail. The volcanic CO_2 isn't removed from the system, and it will continue to build and build in the atmosphere until disaster strikes. And when disaster does strike, and CO_2 keeps rising, there is no lifeline.

Once you've foreclosed huge areas of the Earth's land surface to weathering by simply starving it of rainfall—as happened in Pangaea's viciously arid interiors, hundreds of miles from the coast—then the lines of this emergency brake are effectively cut. Moreover, when you bunch all the land on Earth together in one place, as a simple matter of geometry you reduce the amount of coastline on the planet, which means that you also shrink the expanse of shallow coastal seas into which you might have dumped all this extra carbon offshore, in the skeletons of reefs and in sediments, and sea creatures made of carbon. CO_2 having nowhere else to go, climbs and climbs in the air as a result, and the world warms.

What if the rainfall is not reaching the rocks, but there's also nothing to weather down to begin with? The titanic mountains that were thrust up when Pangaea was being assembled were now—tens of millions of years into the supercontinent's lifespan—senescent, worn down, and unimpressive. Worse, virtually no new mountain ranges were being built up anywhere on the planet, and burying CO_2 through rock weathering only works if the planet is continually providing fresh new rock to weather, by pushing it into the sky to erode. If you don't push any new mountains into the sky to be weathered away, and the land that does exist has just been sitting there getting the shit kicked out of it for millions of years and becoming chemically inert, it stops weathering. The rocks of Pangaea had been weathered and weathered until they couldn't weather anymore—shielded by an unreactive patina of exhausted soils. The old rocks were spent, and on this flattened supercontinental world, the CO_2 had nowhere to go but into the atmosphere, as this geological climate guardrail failed.

As a result, Pangaea could barely handle the steady background emission of volcanic CO_2 without overheating, much less a geohistoric jolt to the system. So when the mind-bending volcanoes of Siberia started issuing their reports from the northern wilderness, the planet was more than just doomed. Earth didn't need much to push it over the edge to begin with, much less some of the most extensive volcanic eruptions in the last half-billion years. But perhaps the most deadly feedback to the carbon cycle in the age of Pangaea was due to something that hadn't evolved yet.

Today large parts of the seafloor are draped in chalky carbonate snowfall, the product of massive swirling eddies of planktonic life that pulse the seas green with the seasons, underwrite the oceanic food chain, and die in a marine snowfall that piles up on the seafloor at the rate of 200 megatons per year. These creatures build their shells from the carbonate in the ocean and bring that carbon with them to the bottom of the sea when they die (and eventually become limestone). When that seafloor is occasionally thrust above the surf and into view—as in the staggering white cliffs of Dover, England, which are made entirely from the carbonate-plated skeletons of phytoplankton—it provides a tiny peek at just how massively productive this carbon sink is. But coccolithophores didn't exist yet in the Permian. In fact, the first calcifying phytoplankton wouldn't show up until midway through the Triassic, and wouldn't become important in the ocean until well into the age of dinosaurs. As a result, when the sea acidified in the age of Pangaea, there was no thick carpet of carbonate at the bottom of the ocean waiting there to neutralize it—no antacid. Nothing to stem the misery.

These biological innovations among the smallest creatures in the ocean would reshape the entire chemistry of the oceans. Newly ballasted in minerals, this heavier plankton, which evolved only after the Permian, likely sank faster and farther into the deep before oxygen-burning microbes were able consume it. This pushed a naturally low-oxygen layer of the ocean much farther into the deep, helping to ventilate the shallow seas and making them less prone to suffocating bouts of anoxia when the waters warmed and these stifling waters shoaled. Paired with the fracturing and fragmenting of the ancient supercontinent, this might even be enough to explain why the last 200 million years have passed almost completely free from mass extinctions, even in the face of occasional LIP eruptions and climate misadventures. It also provides one of the greatest reasons for hope in our own time. With its disparate island continents, an ocean outfitted with a vigorous biological pump of carbon to the depths, and a blanket of carbonate on the seafloor, our planet today seems far more resilient to shocks to the carbon cycle of the sort that would have ended the world hundreds of millions of years ago. Our oceans and atmosphere have been remade by life and by plate tectonics. Perhaps we're safe. Or perhaps this seeming safety is illusory and we are at the threshold of summoning these very same global catastrophes back from their slumber in the eons.

“ALL SCENARIOS FOR … CENTURY’S END EITHER EXCEED OR ARE commensurate with the threshold for catastrophic change.”

I had highlighted this line in Daniel Rothman’s paper “Thresholds of Catastrophe in the Earth System,” a crumpled copy of which I had brought with me in my book bag to his MIT office. It was an alarming conclusion. Rothman was essentially arguing that parsing this or that carbon-emissions scenario for 2100 was almost irrelevant from an Earth-system perspective. That’s because the scale of carbon emissions generated by industrial society, even in the most optimistic scenario, was sufficient to push the carbon cycle far out of equilibrium and into mass-extinction territory. Rothman’s work is among the most pessimistic and grim I have ever come across, but I was drawn to it because it highlighted the perhaps neglected long tail of possible catastrophic outcomes in what is almost a no-analogue episode in Earth history. Even if such scenarios are unlikely, and will take thousands of years to play out, if the stakes are potentially cosmic, it only underlines the urgency of our situation, and the humility with which we should go about pushing this system much further.

“This won’t surprise you, but the current rate at which CO_2 is rising—or the injection of carbon into the oceans, which is my concern here—is nearly two orders of magnitude greater than these terrible events in the past,” Rothman told me, referring to the other major mass extinctions marked by similar carbon cycle chaos. “For the End-Permian mass extinction, we’re probably like one order of magnitude faster, but the End-Permian is worse [than all the other mass extinctions].”

In other words, humans might be pushing on the carbon cycle far harder than even the worst chapters in Earth history—catastrophes that were largely driven by massive injections of CO_2 into the system. And by Rothman’s lights, once you give the Earth system a sufficient push, it fires like a nerve impulse—or in this case, an unraveling planet. “System-wide instability may then follow” is the unsettling if somewhat enigmatic language of his papers.

So what about the good news? Well, we don’t live on a supercontinental world anymore, and the carbon cycle has only become more resilient to calamity since the throes of deep time. Since the age of Pangaea and the rise of a modern biological carbon pump in the ocean, it does seem as though the carbon cycle has recovered its equilibrium far faster during more recent planetary misadventures, compared to the nightmares of yore. Where, early in the age of animal life, it took the Earth hundreds of thousands, even millions of years to recover from shocks to the carbon cycle, in more recent geological history the planet has seemingly regained its composure over just tens of thousands of years. And more recent episodes that saw massive injections of CO_2 into the system—like a sweltering global warming event that came about only 56 million years ago, at the dawn of the age of mammals—were smothered in the cradle on this new, more modern, more resilient planet before they could destroy the world. These were events that might have otherwise progressed to major mass extinctions in the age of Pangaea. Besides, the ancient volcanic events that wreaked havoc on the planet all those years ago seemed to have injected far more carbon dioxide into the system in total than humans ever could. For this reason, some geologists think that we could never come close to inflicting a CO_2 catastrophe on the planet on par with the End-Permian.

But even with all these caveats, Rothman still pegs the human disturbance of the global carbon cycle on the wrong side of a major mass extinction. After all, as mentioned at the outset of this chapter, it is just not the total amount of carbon emitted that's important—crucially, it's the flux. While catastrophes like the Siberian Traps eruptions injected preposterous amounts of CO_2 into the Earth system, they also played out over thousands to tens of thousands of years. If you want to overwhelm the system in a shorter time frame, though, and shove the carbon cycle dangerously out of equilibrium, you need a much more intense infusion of CO_2 into the oceans and atmosphere—faster than biology or weathering can save you. The modern global industrial effort to find, retrieve, and burn as much ancient carbon buried in Earth's crust as possible in a matter of mere centuries might be up to the task.

"Today's event is one hundred times more intense, but one hundred times shorter [than past major mass extinctions]," says Rothman. "Therefore they fall on the same curve. So that's the answer to how awful are things today. I say, well, it's not an order of magnitude or two orders of magnitude worse, [but because of the rate] it's about the same.

"The mass extinctions are the response of the system when a threshold is significantly exceeded," Rothman adds. And he thinks we're most likely going to exceed it.

So what happens when you exceed the threshold for catastrophe?

“If you exceed the threshold, what happens is the system goes off on a trajectory,” Rothman said, ominously.

In other words, the mass extinction is inexorably set in motion. He uses the analogy of a pendulum being swung violently out of equilibrium—eventually it will come back to rest once friction does its slow work, but it may take a while. He views the carbon cycle similarly: if you push it far enough out of equilibrium, like we might be doing now, it will eventually swing back to rest, and order will be restored, but only after tens of thousands of millennia of planetary chaos, as positive feedbacks that release ever more CO_2 are tripped and the planet struggles to regain its composure.

Surveying current conditions, Rothman thinks it may be the ocean, which stores vastly more carbon than the atmosphere, that bites us. He speculates that in the millennia to come, warming and acidification might cause the huge amount of organic carbon stored in the ocean’s plankton, and the marine snowfall it produces, not to sink as far into the depths before being inevitably consumed by the rest of the life in the ocean—as now happens up to hundreds of meters below. If this happens, and all this life gets respired in the upper ocean, it could potentially belch a truly gigantic amount of CO_2 back into the atmosphere, violently amplifying the ongoing warming and carbon-cycle chaos. If he’s right, this means that the global destruction set in motion this century will unfurl not merely over this century, but over tens of thousands of years as the ocean component of the carbon cycle destabilizes. Lots of other Earth scientists have proposed other worrying positive feedbacks to CO_2 and warming that might have been triggered in the ancient mass extinctions—from thawing permafrost in the tundra, to stores of frozen methane at the bottom of the ocean. Thankfully, though, those gigantic stores of carbon seem resistant to catastrophic release.

Not all geochemists I talked to were convinced by Rothman’s interpretation of the mass extinctions. The future effects of acidification on today’s species of calcifying plankton are also far from known. The Earth could prove more resilient in the long run than we know. It’s a chance we can choose to take. But very few if any Earth-system models are run for many millennia, and with a full suite of carbon cycle feedbacks built in. In many ways we are truly jumping into the unknown, and Rothman is coy when pressed for further details on how exactly this planetary carnage unfolds. His papers are similarly vague and foreboding.

“The upshot is that an unstable trajectory would reach its maximum extent roughly 10^4 years after the threshold is breached,” one reads. “But how that process plays out remains unknown.”

Rather than providing the grisly details, Rothman says he has simply identified what seems to be a threshold in the Earth system, and a characteristic response of the planet—one recorded in the ancient rocks. We are on this same dark path now, his work implies, one that will play out in full over the next few thousand years.

As Rothman spoke, it was getting late; the sun had set over Boston, and the glassy John Hancock Tower had lost its twilight alpenglow. Fluorescent lights flickered down empty MIT hallways, the HVAC hummed as stale cups of coffee sat in empty offices getting staler, and posters peeled off cold university walls. Before I left, Rothman told me that he was originally motivated to work on the problem of mass extinctions after reading the American paleontologist Norman Newell’s landmark 1963 paper, “Crises in the History of Life,” which sought to revive the unfashionable nineteenth-century idea that catastrophes, in fact, existed in the fossil record, even coining the very term *mass extinction*. After compiling the trajectories of 2,500 animal families over the vast sweep of Earth history, Newell found that there existed a handful of surprising, sudden, and simultaneous extinctions in the fossil record, among unrelated groups that cut through all branches of the tree of life, at all latitudes and longitudes, of all shapes and sizes, which spared no part of the planet, from the deep sea to tropical rift valleys to the poles. Nothing was safe.[fn28]

“Newell posed the question ‘What causes mass extinctions?’ And he gave an answer: they are what happens when the environment changes too fast for life to keep up with it. So I thought, that’s the answer!”

Mass extinctions aren’t necessarily about volcanoes or asteroids. They are about change.

“A colleague of mine, an applied mathematician, wrote to me the other day. He said, ‘Is it really true that CO_2 today is about the lowest it’s been in three hundred million years? And that it’s only been this low ten to fifteen percent of the last five hundred million years?’” Rothman said.

“I said, ‘Well, yeah, that’s our best understanding.’ And then I thought I should say something more. I said, ‘Look, what really matters is when it changes quickly.’”

Part II

6

The Fall of CO_2 and the Rise of the Modern World

FIRST, A WORD ABOUT ICE AGES. IT MIGHT NOT SEEM LIKE IT, BUT WE LIVE in an ice age. Mile-thick ice sheets on Antarctica and Greenland might strike us as the default setting for Earth, but in fact this is strange. Most of Earth history has passed with virtually no ice on the planet at all. In the age of animal life, the poles have been far more likely to be carpeted in lush forests than smothered in ice sheets like today. Surprisingly, to find a climate analogue cold enough for our modern ice-capped planet of the past 3 million years or so, you have to go all the way back to the ice-capped world of the age of coal, some 300 million years ago. Otherwise the planet has been perfectly happy to be a much warmer, much higher-CO_2 world.

This fact is often plucked from its geological context by clueless, bad-faith political actors and used as a cudgel to downplay anthropogenic climate change, but it remains true in its broad strokes. What this elides, though, is that it took tens of millions of years to get from the sauna of the age of dinosaurs to our own modern *low*-CO_2 world, and humans, along with the rest of life on Earth today, are evolved and adapted for our icehouse planet, not the ancient high-CO_2 greenhouse climate into which industrial civilization is potentially launching us all in the next few decades. This, among many other reasons, is why modern climate change is scary. Along with everything else alive today, we are fundamentally ice-age creatures and our parochial corner at the end of history, an astoundingly volatile age of ice.

Ice ages, like our present one, are rare. As such, they demand an explanation. So why is CO_2 so strangely *low* today, relative to Earth history? The answer, as we'll find, is tectonic, geologic, even biological—an epic drawdown of carbon dioxide tens of millions of years in the making.

WE LEFT OFF JUST AS PANGAEA WAS BREAKING APART AND THE DINOSAURS were embarking on their unprecedented 135-million-year-long run of dominance. This is such a vast time span that, over the course of it, the continents, inching across the planet at the rate of fingernail growth, had more or less assumed their current positions by the time the asteroid hit.[fn1] Indeed, if you were an alien regularly visiting the Earth over the past 380 million years, since fish crawled out of the swamps, you might be tempted to summarize this as "the dinosaur planet," not the mammal world, and certainly not a human one—our reign so far amounts to little more than a rounding error at the end of time (and one with not particularly promising prospects). Dinosaurs are one of the great success stories—if not *the* great success story—in the history of land life on a planet where extinction is the rule. If you include the current age of birds (which of course you should, since there are twice as many bird species today as mammals, and birds are indisputably dinosaurs), then dinosaurs have flourished for more than 60 percent of the history of terrestrial animal life. Evolutionarily modern humans, by way of contrast, have been around for 0.05 percent of that span. Written history: 0.002 percent.[fn2]

Most adults, however, having discarded their interest in dinosaurs in adolescence, fail to grapple with the astonishing reality of their existence. This wasn't an airless museum diorama of giants, but a world every bit as vivid and immediate as our own—these creatures experienced billions of drama-filled summer days, placid nights in the desert under strange constellations, afternoons sheltering from warm midday storms, and chilly mornings with feathers puffed bracing for the frost, on this very same planet, around the same sun. A curious megalosaurid plays with a crab on a Wyoming tidal flat 164,567,218 years ago. Some 92,657,095 years later, the aurora catches the dilated eye of an Antarctic ornithopod through the eaves of a monkey puzzle tree. Every year, month, and hour in between happened as well. Everywhere on Earth. The existence of this nigh-endless and unbelievable backstory is a sublime fact about the strange world we live on, and a psychologically destabilizing one for those who would exalt our millisecond tenure on this planet above all others. That said, there are lots of books about dinosaurs, and this isn't one of them.

As for the Earth system, which is what we're interested in after all, it was time to settle into a groove. After volcanism blew the world apart at the end of the Triassic, 201 million years ago, the planet entered an interminably equable age. North America pulled away from Africa, and the rift valley lakes that once separated New Jersey from Morocco grew into a proper ocean. The massive expanses of basalt that had once erupted onto the middle of this rending supercontinent weathered apace and helped draw down the CO_2 that had nearly ended the world. But this Mesozoic planet would remain far warmer than any known to humanity. The Earth was stitched together by more extensive subduction zones, which seeded the planet with arc volcanoes that steadily fed more CO_2 to the atmosphere, keeping CO_2 perpetually high and this dinosaur world cozy—maintaining what one paper calls a "long-term trend of climatic stability for at least 100 million years." This vast era of stability comprised the mild but wonder-filled Jurassic Period, one that saw titans trample the Earth and the first flower bloom, and was capped by the truly sweltering mid-Cretaceous Period of popular imagination, 90 million years ago, when CO_2 topped 1,000 ppm, sea level was hundreds of feet higher, and dinosaurs stalked steamy rain forests at the foot of the Transantarctic Mountains.

Then, 66 million years ago, just as the biosphere was negotiating its way through yet *another* gigantic LIP eruption, this time in India, the largest asteroid to hit Earth in a billion years ran aground in Mexico at twenty times the speed of a bullet and the Mesozoic ended one fateful, unimaginable spring day. *Tyrannosaurus rex*, *Mosasaurus*, *Triceratops*, all manner of ammonites,[fn3] giraffe-sized flying pterosaurs, immense reefs of strange bucket clams, most mammals, most birds—all went extinct.

As this provisional planet gathered itself from the apocalypse, small, archaic mammals tentatively came out from the shadows, wondering where all the giants had gone. But what *had* been left behind was the dinosaurs' sultry, high-CO_2 greenhouse world. This was largely the planet that our ancestors inherited once the asteroid had done its ghastly work. A 10-million-year hangover set in as this liminal Earth, populated by unfamiliar, undersized mammals, struggled to come into being. Then CO_2 suddenly soared even higher still, temperature spiked even higher, and the world was remade yet again. Soon after, however, the planet would begin a 50-million-year-long cooling trend that would define the age of mammals as CO_2 was gently sequestered in the crust over tens of millions of years. This long chill would fundamentally shape the evolution of life on Earth, culminating in our surprising modern world, one marked by trembling ice caps. It's this journey on which we now embark, starting with the extraordinary fever that gave rise to a world of modern mammal lineages, then an epic decline of CO_2 and the long slide into our icehouse world—a trajectory that would somehow, startlingly terminate with the explosion of 500 million years of carbon out of the crust in the last nanosecond of Earth history.

"THIS IS THE LAST GASP OF WHAT LIFE WAS LIKE BEFORE EVERYTHING changed forever," said University of Florida paleontologist Jonathan Bloch, ponytailed and weatherbeaten from weeks in the sunbaked Wyoming badlands, as he picked at an outcrop of dusty fossil soils. Hacking into the rocks with pickaxe and rock hammer, I easily found palm fronds, gar scales, alligator bones, and the strange teeth of hippo doppelgängers in the layers just below the most profound warming event in the age of mammals. But above us were hundreds of feet of startling candy-striped, purple-red to yellow clays that marked this paroxysm of extreme warming that transformed the world. The episode is thought to be among the best analogues in the fossil record to study for insight into how our biosphere might similarly respond to future warming. That's because it's marked by a scale of warming comparable to that which we might eventually bring about, and it occurred on a world far more modern than the ancient disasters we've discussed in previous chapters. By the beginning of the age of mammals, there was a continental configuration similar to that of today—unlike the dysfunctional world of Pangaea—and a more recognizable roster of animals. But in other ways this was a far different Earth, and the planet heading into this ancient pulse of CO_2 was doing so from a much different starting state than our own. This world, in the long aftermath of the dinosaur Armageddon, was already extraordinarily hot.

This was also still a jungly world, and it entrusted these jungles to the Earth as vast swaths of coal. On the other side of the Bighorn Mountains, where Wyoming levels off and tan grasses stretch to the horizon, a network of artificial canyons has been chiseled out of the Great Plains, including the largest coal mine in the world, the North Antelope Rochelle Mine. Coals from these same jungles from the dawn of the age of mammals are plucked from deep in the Powder River Basin, where they're dumped from skyscraping tipples into a never-ending parade of train cars headed to the great metabolic furnaces of the United States. This early Cenozoic coal contributes about 10 percent of the United States' CO_2 emissions—displacing, in recent years, the far more ancient, seawater-tainted coals of Appalachia that powered the country's industrial rise. At the time of writing, the operator of the North Antelope mine, and the world's largest private coal producer, Peabody Energy, posted its highest quarterly profit in more than two decades, a return that also flowed into the accounts of its creditor Goldman Sachs. Across state lines in Colorado, the abundance of fossil fuels from this window of Earth history proved so tempting that it was "fracked" with a nuclear bomb in 1973, to try to wrench gas from the carbon-rich rocks of the early age of mammals, as the federal government sought to turn its atomic swords to plowshares.[fn4] Elsewhere, in Germany, coal, especially from around this age, when Europe would have resembled the Florida Everglades, still represented the country's primary electricity source as late as 2022. In Colombian coal mines from just before the warming pulse, Bloch has found the ominously gigantic vertebrae of *Titanoboa*, a 42-foot-long monster snake, dwarfing any other snake in Earth history. It would have slid past equally titanic crocodilians amid unimaginable humidity and drowning tropical rainstorms. Due to the particulars of snake physiology, size is a proxy for heat, and a snake as preposterously large as *Titanoboa* could only survive in rain forests far warmer than any in existence today. Though the mammals had inherited this greenhouse world from the dinosaurs, in the early days of their reign, when menacing reptiles still shared top billing with flightless birds nearly seven feet tall, it wasn't clear that this new age had been entrusted to the mammals at all.

But, as if the balmy world of the early age of animals wasn't hot enough, 56 million years ago—and 10 million years after the dinosaurs' celestial apocalypse—submarine volcanoes in the North Atlantic started burning through vast fossil fuel deposits and carbonate rocks at the bottom of the sea. The ignition injected the CO_2 equivalent of all currently recoverable fossil fuels into the oceans and atmosphere in less than 20,000 years, propelling the planet perhaps 5°C warmer still. This was the moment "everything changed forever," in Bloch's words. With global temperatures *before* this millennia-long eruption of CO_2 already around 12°C warmer than today, it's difficult to imagine how uncomfortable this planet would have been for ice-age creatures like ourselves. While, far to the north, the Arctic hosted crocodiles, palm trees, and sand tiger sharks splashing in 85-degree seawater at the poles, farther to the south the tropics themselves might have become uninhabitable, with seas as hot as a Jacuzzi—far too hot for animal life. Gatorlike creatures called champsosaurs went extinct here in Wyoming, and Bloch tells me that one layer of the warming event is called the "All-Hell-Breaks-Loose" section. Evidence abounds from around the world of extremely violent storms and megafloods during this ancient spasm of climate change, while some models spit out sci-fi temperatures for the continental interiors at the height of the planetary fever that push 140°F.

But global warming doesn't simply make the planet uniformly wet. When it gets warmer, more water evaporates from hot soils, meaning that while there might be more water in the atmosphere to fuel powerful storms, this can come at the expense of the parched land. In fact, in our own time, where parts of the tropics are expected to be subjected to punishing, unprecedented torrents of rain, elsewhere droughts are expected to dramatically worsen. Similarly, here in Wyoming, these croc-infested subtropical wetlands vanished in the warming pulse and, in the rocks above us, were briefly replaced for over 100,000 volatile years by tropical dry forests of the sort that wouldn't have looked out of place in Madagascar. That's not all that would have looked at home in Madagascar: in 2018, in these candy-striped badlands, Bloch found what at the time was the first primate fossil known from Earth history, a miniature lemurlike Wyoming treehopper whose ancient teeth weathered out of the clay. Also crumbling out of the badlands, weighing in at only ten pounds, were the very first horses in Earth history—dog-sized colts who were forced to get even smaller still to beat the extreme heat—as well as an adorable little creature that hopped around like a tiny kangaroo: an unlikely forerunner to all sheep, goats, camels, pigs, cows, deer, giraffes, antelopes … and whales. A new world was throttled into being by this climate paroxysm. Our world. And while countless new forms blossomed during the fever and in the new world in its wake, other animals were reshuffled all over the Earth.

"You hit the Paleocene-Eocene boundary and it's mass immigration time," says University of Colorado paleontologist Jaelyn Eberle of the hothouse blast, when warming pushed waves of animal immigrants up and over land bridges and into new continents. Their movement was aided by Beringia, which linked Russia to Alaska, while a land bridge on the other side of the continent linked Europe to North America across a narrower North Atlantic, via an unbroken stretch of humid forests across Iceland and Greenland. Future warming in our own time will similarly scatter species all about the Earth, if they can make it. Unlike in the early Eocene, though, many will be thwarted in their exodus by the human infrastructure that parcels the Earth, from networks of highways to sterile seas of corn. Drown the Florida Everglades of today and its crocodiles likely won't make it all the way to the unspoiled arctic bayous—they will run into the levees and fortifications of drowning Florida exurbs. Other creatures will run into industrial sacrifice zones, coastal wetlands drained and draped in golf courses and pavement, and, farther inland, a wire-tangled patchwork of ranchland.

While this ancient fever of 56 million years ago provided a surprisingly transformative jolt to evolution on land at the dawn of the Eocene Epoch, all the CO_2 had a somewhat more sinister effect in the oceans. As we've seen, when too much carbon dioxide reacts with seawater too fast, it acidifies. It's a problem in our own time that is projected to be a nightmare past midcentury—an existential crisis to calcifying sea life like planktonic snails, coral reefs, and oysters. Unsurprisingly, then, this blast of CO_2 caused the world's coral reefs to collapse. It took 200,000 years for ocean chemistry to recover, and it did so only on the stately timescale of global rock weathering, as the continents were lashed by punishing rains for tens of thousands of years and washed their alkaline salve into the seas.

For some reason, though, this ancient climate convulsion wasn't worse. Scientists have long puzzled over why this cresting fever didn't progress into Big Five mass extinction territory. And it could be that biology saved the day. Unlike in the Permian apocalypse 200 million years earlier, much of the world's seafloors had by now been frosted in layers of chalky calcifying plankton that were able to dissolve in the face of these corroding seas. This would have helped buffer against the acidifying flood of CO_2 by returning alkalinity to the seas. Oceanographers pulling sediment cores from the bottom of the ocean, from Greenland to Namibia, sometimes encounter records of this event in the chalky muds, which suddenly blush red with clays for tens of thousands of years as the seafloor dissolved. Perhaps it was this new geochemical emergency brake at the bottom of the ocean, one made possible by life, that kept the planet from tipping into a true mass extinction.

NEVERTHELESS, THERE WOULD BE LITTLE RESPITE FROM THE HIGH-CO_2 hothouse early in our age of mammals. Even after this feverish blip 56 million years ago had ended (after some 200,000 years), the planet stayed extraordinarily warm for millions of years to come, a persistent sultriness that was occasionally punctuated by hundred-millennia-long heat-wave spikes—sweltering carbon-cycle aftershocks that racked the Earth as it tried desperately to find some sort of equilibrium. Eventually this planet would begin to cool off and begin to resemble our own, but it's hard to convey how unfamiliar this hothouse world some 50 million years ago would have been.

"It's an extinct ecosystem," says Eberle, pulling up pictures in her Colorado office of tree stumps eroding out of a barren, windswept hillside. The fossil stumps are all that's left of a swampy cathedral of redwoods, whose canopy naves were once filled with flying lemurs, hippos that weren't hippos, giant salamanders, outrageously large bipedal birds, alligators, and tapirs. Astoundingly, Eberle's pictures are from the farthest north landmass in Arctic Canada, the tip of Ellesmere Island—a stone's throw from the North Pole. Eberle has been helicoptering to these desolate reaches for decades to study this vanished world, an ancient polar rain forest now whipped by indifferent arctic winds.

"Yes, we make comparisons to southeastern US cypress swamps with alligators and soft-shelled turtles and things, but the fact of the matter is that these are conditions that don't exist on Earth today," she says.

That is, there is no modern analogue for a swampy rain forest teeming with reptiles that nevertheless endures months of arctic twilight and polar night. One can speak about this world in relation to our own only in metaphor. Meanwhile, Antarctica was home to monkey puzzle trees and marsupials.

What's worrying about all this is that while CO_2 levels were certainly higher than today in the torridly warm world of 50 million years ago, it doesn't seem like CO_2 was all *that* high. In fact, the Eocene has long proven something of a headache for climate modelers, who have struggled to make the planet as warm as the fossil record tells us that it most assuredly was.

"The estimate we have for the early Eocene is 600 or 700 ppm CO_2" the Smithsonian paleobotanist Scott Wing told me as he hammered away at Wyoming siltstones that cracked open to reveal broad banana leaves, basking in the sun again for the first time in over 50 million years. This is an ominously low estimate given that CO_2 on our modern world just jumped from 280 to 420 ppm in a little over a century and a half, a rise that accelerates by the year. In the face of higher CO_2 plants reduce the number of pores on their leaves, and the ancient plants in Wing's Wyoming rocks have tellingly fewer pores than today's. "If that's correct, you have an Eocene-like world with what a reasonable estimate for what CO_2 will be in 2080 or something. That's palm trees growing in the Arctic."

Elsewhere, the testimony of these ancient CO_2 levels is recorded by the rocks themselves. Today your baking soda is mined from an unusual mineral called nahcolite, found in Eocene lake deposits. While you might leaven your bread with it, nahcolite doesn't form on our current world: it can only precipitate when atmospheric CO_2 is at 680 ppm or higher. But the Eocene strata are rich with the stuff, as it might become again in the coming centuries. Other proxies for CO_2 point to slightly higher numbers for this greenhouse world, around 1,000 ppm or more. But regardless, this range has long embarrassed computer modelers who have needed to ramp up CO_2 to over *4,000 ppm* to reproduce this feverish planet on their computers. Eberle recalls a conversation she once had with a climate modeler who half-jokingly asked whether she could put hair on the alligators whose fossils she plucked from the Arctic. According to their models—models that struggled to make this world as hot as the fossils demanded—alligators were simply not supposed to be at the North Pole. "If the modelers can't back-predict the early Eocene," Eberle says, "how can they possibly predict what's in store for us?"

Obviously, this ancient planet is far more extreme than anything being predicted by the end of the century by the United Nations or almost anyone else. After all, this arctic rain forest world was about 12°C warmer than our current world, while the current global ambition, enshrined in the Paris Agreement, is to limit future warming to less than 2 degrees, and possibly only 1.5 degrees, by 2100. Part of what explains this jarring disparity—between the volatility of Earth history and the modesty of future warming estimates—is that most climate projections end at the end of the century. In Wing's words, it is "as if the world ends in 2100 and we don't have to think about what happens after that." That is, feedbacks that might get you to Eocene-level warmth under high emissions would play out over much longer timescales than a century, a millennium, or even several millennia. Had the Han Chinese—who 2,000 years ago burned coal, smelted iron, and drilled for natural gas—fully embarked on an industrial revolution of their own, we might be well on our way to the Eocene already. But the other, much more worrying possibility is that we're missing something big.

"This time period makes the climate modeling people really uneasy," Wing says. "Their first reaction was 'It can't be true.'" But whether the models can be made to agree with the fossils or not, Earth history is the only experimental record we have of how the planet actually responds to carbon dioxide. There might be some missing feedbacks that make the planet much warmer than we're able to capture in our models. Planet Earth is intensely complicated, and it would be surprising if we were perfectly integrating all the chaos of a pulsing biogeosphere into lines of computer code. Maybe the missing piece involves the release of more methane from a warming biosphere than we anticipate, which might awaken on a global scale when you turn half the planet into swampland and would serve as a potent additional greenhouse gas. Maybe it's something about the response of plankton to warming that we're not capturing: around the world today, phytoplankton breathe out cloud-seeding mists of dimethylsulfide into the air that, if diminished from warming or acidification, might see clouds diminish as well, bathing the seas in sunlight and heat, and warming the planet even further. Or maybe it's something in the physics of clouds themselves that we don't understand, and that isn't captured in our models for the future.

"It's the difference between the uncontrolled, messy experiments on the planet that have actually happened, versus the controlled, orderly experiments on a computer," says Wing. "It's a reason to be very concerned about the future."

But even the climate models are starting to catch up. In 2019, one of the most sophisticated models ever run, by researchers at the California Institute of Technology, simulated global temperatures suddenly leaping by an appalling 12°C by next century at 1,200 ppm of CO_2—an extreme and extraordinarily bad, but not physically impossible, carbon emissions pathway—bringing the digital world in line with the Eocene one. And in the same year, the University of Arizona paleoclimatologist Jessica Tierney, along with colleagues from the University of Michigan, was similarly able to reproduce the warmth of this ancient greenhouse world by plugging in more sophisticated models of how water behaves at the smallest scales.

Today the San Francisco fog reliably rolls in, stranding bridge towers high above the marine layer like birthday candles. These clouds are a mainstay of west coasts around the world, reflecting sunlight back to space from coastal California to Peru and Namibia. But under higher CO_2 and higher temperatures, the water droplets in them will likely get bigger. This not only makes the clouds themselves less opaque by letting more sunlight through, but also might just wipe them out altogether. Stratus cloud decks that today send warming sunlight back into space could simply rain out, allowing a flood of sunlight to beat down against the midlatitudes. Maybe that's why the Eocene was so outrageously hot.

"The takeaway is that the clouds themselves are breaking down under high CO_2 and it's a positive feedback," Tierney says. "So this is the scary part. The sensitivity of global temperature to the doubling of CO_2—that number itself goes up, and up, and up at higher CO_2."

While it would go on to take 50 million years to descend from this Eocene world to our own ice-capped world, we could *potentially* launch our ice-age planet of today back into the CO_2-driven greenhouse of the early mammals in a matter of centuries to millennia—a geologically subatomic timescale. It is almost impossible to overstate how catastrophic this would be. In fact, such climate shocks are virtually unknown from all of Earth history, with the possible exception of the once-in-an-eon nightmares like the End-Permian mass extinction, 252 million years ago. And given that the sun has slowly been getting brighter over Earth history, at least one study claims that if humans lose their minds and truly burn *everything*,[fn5] industrial civilization could make the planet even warmer than it was in the aftermath of that apocalypse 250 million years ago. Luckily, though, the early Eocene still represents something close to (if not somewhat beyond) an absolute worst-case scenario for possible future warming. For now, wondering what the planet would look like at around 1,000 ppm CO_2 can be tentatively set aside as a thought experiment, rather than any kind of reasonable forecast for the future.[fn6] One hopes.

AS THE EOCENE DREW ON, THE FIRST BATS WOULD TAKE FLIGHT ON this balmy planet—joining insects, birds, and pterosaurs in the rarefied class of flyers in the planet's history. On the archipelago of Pakistan, a small-hoofed woodland creature that would become the ancestor to all whales pranced on all fours through the forests. It might have been encouraged to wade back into the ocean by the tepid bathwater seas of this high-CO_2 greenhouse world, where it would evolve into giants. Soon enough, however, the planetary fever would break and CO_2 would begin a long, mostly gradual 50-million-year ebb from the atmosphere—one that culminated, at last, in the icy nadir of the past 2.5 million years. But what launched us on this journey to our strangely cold, *low*-CO_2 present?

NEAR THE END OF THE AGE OF DINOSAURS, INDIA, LONG ESTRANGED from Africa and the august but fading supercontinent of Gondwana, slowly inched across the Proto-Indian Ocean and eventually smashed into Asia. And it is in this singular tectonic collision that scientists have long sought the cause of the gradual decline in CO_2 from the greenhouse highs of the dinosaurs to the recent miles-thick ice sheets of the Pleistocene ice ages. The arrival of India to Asian shores some 40 million years ago not only would have quieted CO_2-spewing volcanoes along the beckoning, churning subduction zones that were shut off during this collision, but en route India would have dragged huge swaths of basalt that blanketed the subcontinent through the stormy tropics—old provinces of cooled lava that only 20 million years earlier might have helped kill the dinosaurs. As we've seen, the exposure of this kind of rock to equatorial climes would have helped draw down CO_2 as well.

But most importantly, this collision thrust the titanic Himalayas and Tibetan Plateau toward the stars to be continually weathered and eroded away. Weathering rocks—that is, breaking them down with CO_2-infused rainwater—is the planet's most effective long-term mechanism for removing carbon dioxide from the atmosphere, one that modern geoengineers are frantically trying to reproduce in the lab, for obvious reasons. And pushing up mountains is the best way to expose fresh rocks to weathering and erosion. In fact, one of the reasons offered for why the early age of mammals was so incredibly hot *before* this collision might have been, in the words of one study, "because Earth was flat." To put it another way, there might have been a curious lack of mountains pushing into the sky during the blistering early Eocene, meaning there was little fresh rock to weather and draw down CO_2 and so the surface world simmered. But as the Himalayas rose—and still rise—the colossal mountains have lost twice their height to weathering and erosion since birth, a plodding demolition that translates to a lot of CO_2 transferred from the air to the rocks over the long age of mammals. Moreover, by encouraging the signature monsoon rains of the Indian subcontinent with their topography, the Himalayas would have accelerated this Earth-changing erosive demolition.

This remains the textbook explanation for the long CO_2 drawdown and cooling trend over the past 50 million years or so. But like any grand theory, the story has become somewhat more muddled in recent years, and the cause of this long decline of CO_2 over the age of mammals remains the sort of unsettled, PhD-minting question that fills the programs of geoscience conferences and fuels academic rivalries. For one thing, if the Himalayas really were as effective at burying CO_2 as claimed, it would plunge the planet into Snowball Earth in less than a million years. It also turns out that it is not only weathering, but the *kind* of rock that's weathered, that matters for drawing down CO_2. Granites and basalts are ideal, but some kinds of rock—like the giant stacks of ancient carbonate seafloors that stripe the Himalayas—just aren't good at driving additional CO_2 burial when eroded. Other kinds of rock, such as those rich in coals or oil, can actually *release* CO_2 back to the air when exhumed from the Earth and exposed to the oxidizing ruin of the surface world. And while the rise of the Himalayas remains a plausible driver for the long chill, other planetary processes also likely helped drain CO_2 from the skies of the cooling Cenozoic, from arcane changes to the chemistry of the seafloor over this age, to the burial of heaving storms of life—whether CO_2-capturing plankton blooms offshore or plant matter washing in from land—pulsing and shifting across the Earth's surface over tens of millions of years.

Adding to these huge CO_2 sinks, the more recent subducting, buckling, tectonic mess that lifted Indonesia and its neighbors from the sea over the past 20 million years or so also lifted vast tracts of weatherable ocean crust to the air, exposing it all to the withering assault of tropical rainstorms.

Over tens of millions of years, then, the stately march of plate tectonics, the vagaries of seafloor chemistry, and even life itself have driven this long-term climate change, in our case toward a lower-CO_2 colder world. As we'll see, humans now threaten to undo this entire epic, geologic-scale climate evolution of the Cenozoic in mere decades.

THE FIRST BIG SALVO IN THIS LONG, STEPWISE DECLINE FROM THE greenhouse of the dinosaurs to our icehouse world struck 33.6 million years ago. Where the world had been largely ice-free for tens, if not hundreds of millions of years, Antarctica, the venerable long-forested and temperate land of the far south, suddenly gained a gigantic ice sheet—half as big as the one we know today. It would keep this ice sheet, in fits and starts, for the rest of the age of mammals, to the current day. Though this planet was still dramatically warmer than our own, this shockingly sudden polar freeze dropped sea level 130 to 160 feet. For more mysterious reasons, it also dropped by a kilometer the depth in the oceans at which the chalky snowfall of plankton from the sunlit surface would dissolve, the so-called snow line in the oceans—a profound change in the dynamics of the global oceans that doubled the expanse of the seafloor where this kind of carbon-rich limestone ooze could pile up.

But the sudden shift from a jungly world to one with ice at the South Pole also marked perhaps the largest extinction event of the entire age of mammals—that is, until the human blitzkrieg tens of millions of years later. In North America, the titanic and titanically bizarre rhinoceros-like brontotheres, sporting forked snouts like dowsing rods, suddenly disappeared. In Europe, 60 percent of land mammals disappeared as well, either from the deteriorating climate or an invasion of immigrant animals from Asia spurred on by the climate change. In Antarctica, six-foot-tall penguins vanished for some reason, and in Asia, the fangy mesonychids—a vaguely wolflike clan that defies any easy comparison to anything alive today, and that had ruled the greenhouse world for millions of years—faded away. Old monstrous whales, once mistaken by nineteenth-century naturalists for dinosaurs, went extinct. Half the mollusks on the American West Coast disappeared, replaced by cooler tribes, and Patagonia lost its tropical flora.

This sudden chill (in fact, a plunge that played out over 400,000 years) was long thought to be the product of Australia and Antarctica's tectonic schism. After more than a billion years of marital bliss, the two continents began to pull apart, and the ocean found the low ground between and rushed in. For the first time, Antarctica was closed off from the wider world in a clockwise embrace of ocean currents and, in its newfound isolation, was plunged into the cold. There it has been marooned and alone at the bottom of the world in its perpetual winter ever since. But for paleoceanographers, this theory, that it was this new seaway encircling the continent that refrigerated Antarctica, increasingly appears inadequate to the task.

Instead, it seems as though the dramatic dawn of ice on Antarctica was primarily the consequence of that same, long-term background waning of CO_2. Ice sheets can dramatically, and nonlinearly, respond to seemingly small changes in temperature once a threshold is passed. And it seems as though the gradual background cooling, then millions of years in progress, finally passed just such a checkpoint 34 million years ago, when CO_2 had fallen below about 750 ppm, kicking off the sudden freeze. These dramatic thresholds in the climate system can be hard to discern in the planetary offing until they transform the world. But the climate, in its past, has tended not to respond linearly to changes in CO_2 but instead with stepwise leaps like this one, into entirely different states—a lesson we would be well advised to consider in our own time as we chart a climate path into the unknown.

But perhaps the most momentous change encouraged by this falling CO_2 happened on the smallest scale. Just over 30 million years ago, as Antarctica was getting accustomed to its new ice sheet, an entirely new form of photosynthesis—some 3 billion years into its history on this planet—called the C4 pathway, evolved independently in some two dozen lineages of plants in response to this climate change. The C4 pathway is more efficient than older kinds of photosynthesis in the drier, lower-CO_2 world, and these plants would eventually come to dominate the signature ecosystem of our particular moment in geologic time: grasslands. While it would still be tens of millions of years until this new ecosystem would take over the world, grasslands tentatively spread on this cooling planet, and grazers, with long teeth to wear down this tougher, silica-barbed grass, found Earth an increasingly amenable place. And the new seaway around Antarctica wasn't entirely without consequence for life on Earth: the creation of a colossal current around the polar continent blasted the ocean with silica (as it does to this day) and drove hurricanes of diatom plankton blooms that lit the ocean aflame with life. Baleen whales may have split off from their toothy cousins to profit from this new abundance. The introduction of giant whales to the planet, as we'll see, would someday have huge implications for the movement of carbon through oceans. After all, no part of this grandly convoluted Earth system can ever truly be disentangled from any other.

AS THE PLANET MAKES PEACE WITH ITS NEW, STILL RELATIVELY MODEST ice sheet, we pass over the ages. A pageant of strange beasts appears along the way. The oft-overlooked, 11-million-year Oligocene Epoch features dinosaur-scaled hornless rhinoceroses plucking leaves from the canopy. The largest bird in Earth history, the albatross-like *Pelagornis*, soars over the South Carolina coast, hoisting itself up thermals on a rigid twenty-one-foot wingspan. On the ground, countless mammals, all equally hard to place on a family tree, seemingly test out every possible gradation between horse, rhino, and tapir. As western North America awkwardly tries to swallow the Pacific Ocean's Farallon plate beneath it, the Southwest explodes in some of the most violent volcanic explosions in Earth history. But life goes on, and now we find ourselves blurring over almost 20 million years of Earth history, if only out of narrative necessity.

WE PAUSE FOR A MOMENT IN THE MIDDLE MIOCENE, 16 MILLION years ago. A baby LIP burbles in the Pacific Northwest, one that left behind giant stacks of lava, in some places more than two miles thick. These eruptions now lend columns of basalt to the kiteboarder-infested canyons of the Columbia River gorge in Oregon, but 16 million years ago, and after CO_2 had more or less flatlined for millions of years, these volcanoes briefly pushed CO_2 up to about 500 ppm, and so warmed the planet for some 2 million years. We alight here on our journey because this is the last time in Earth history when

CO_2 was as high as it might be again as soon as 2050. It's a level that today represents something close to the most ambitious and optimistic scenario for human carbon emissions. As a result, this mid-Miocene warming event is being frenziedly investigated by paleoclimatologists to help divine our future. But this is still a shockingly unfamiliar planet. Who knows how much we might warm the planet in the coming decades to centuries, but 16 million years ago this level of CO_2 translated to a planet about 8°C warmer than our own.

So what did the world look like the last time CO_2 was as high as it will be in the near future? Well, in the height of this mid-Miocene warmth, there are forests of dawn redwoods north of the Arctic Circle, a verdant Greenland that lived up to its name, and tropical forests on the Arabian Peninsula. There are Louisiana-like forests in the Gobi Desert and rain forests in the middle of the Australian outback. As for fauna, there are turtles and parrots in Siberia, and chameleons, alligators, and giant turtles wandering into Romania. But we're still so distantly marooned in time from our own world that most of the creatures that inhabit this leafy planet range from the flatly unfamiliar to the uncannily so. There are big cats that aren't cats, and rhino-sized "hell pigs" that aren't pigs. There are sloths that live in the ocean and walruses that aren't related to walruses. There are woodland rhinos that live in the Arctic and, though it's the waning days of their reign, an old guard still rules: Earth's largest meat-eating mammals ever lithely roam these forests—African juggernauts like *Megistotherium* and *Simbakubwa*, not related to any living mammals today. They tear giant hyraxes and early elephants apart with bladed mouths.

Look closer, though, and you'll find an unassuming gibbonlike treehopper present in these woodlands too. Our origins begin to come into focus. In a few million years, the descendants of this creature would give rise to all apes, and eventually humans as the planet cooled further. But we flatter ourselves to presume this animal was somehow straining toward humanity: natural selection can't "see" the future, and this creature was perfectly adapted to its own warm, fully realized Miocene forest world.

Though by now we're a full 50 million years after the extinction of the dinosaurs, this was still very much an unrecognizable biosphere. Just take a look at a South America traipsed by crocodilians sporting skulls, as one paleontologist writes, the size of "a moderately large tyrannosaur," giant reptiles favored by the warmer climate. Meanwhile, vicious ten-feet-tall "terror birds" roamed the forests of Patagonia—creatures that were apparently never told that the dinosaurs had gone extinct. Capping off all this uncanniness, the rivers of the Amazon flowed backward, gathering in great pools at the foot of the Andes and spilling out into the Caribbean. A vast seaway stretched from Western Europe to Kazakhstan and spilled into the Indian Ocean, while California's Central Valley was open ocean. Though it might hold clues to the climate of the twenty-first century, this was still a world staggeringly distant in time.

Perhaps the most striking thing about this world, though, is that with CO_2 at only 500 ppm, sea level was still an astounding 157 feet higher than it is today—the product of a disturbingly churlish Antarctic ice sheet. These high seas are one of the more concerning features of the middle Miocene, as it seems as though small changes in Miocene CO_2 from just below today's levels to up around 500 ppm saw Antarctica alarmingly shed what today would amount to 30–80 percent of the modern ice sheet, sending the world's oceans soaring. In other words, in the Miocene, Antarctica seemed exquisitely tuned to small changes in atmospheric CO_2 in ways we don't fully understand.

It might be surprising that we have such a precise understanding of the shifting sea level over the past few tens of millions of years, but much of this understanding comes from the petroleum industry, which has long been motivated to find an explanatory framework for the blankets of sediment that drape the continental shelves and produce fossil fuel from ancient life. It is no small irony, then, that the foundational plot of sea-level rise and fall over geologic history is called the Exxon Curve, developed by the pioneering oil industry geophysicist Peter Vail.

But these high seas wouldn't last. As we've seen, the planet has ways of cooling off after a fever, and this was a planet ultimately destined for the deep freeze. After this blush of warmth, CO_2 continued its regularly scheduled fall, and the still-adolescent Antarctic Ice Sheet suddenly gained confidence, swelling so much that sea level suddenly dropped by a mad 180–246 feet, and never looked back.

As the planet continued to cool, the very metabolism of these oceans was transformed as well. Today there might be more fish biomass in the so-called Twilight Zone layer than in the rest of the ocean combined. It's a nearly lightless realm up to 3,300 feet down, stocked with outlandish bioluminescent creatures, with one fish, the twinkling bristlemouth, representing the most abundant vertebrate on Earth, numbering in the quadrillions. Fish here are so abundant that their sonar-reflecting swim bladders led US Navy sonar operators in World War II to puzzle over a supposed “false bottom” that spanned the world’s oceans. Though this spectral world of the abyss might seem like some Lovecraftian relic from an unspeakably primordial age, much of the Twilight Zone menagerie seems to have evolved only as the world continued to cool as the Miocene drew on. Just as food left out in a sweltering room goes bad faster than it does in your refrigerator, the snowfall of organic gunk from the ocean’s sunlit layers was now able to sink farther into the deep in these cooler waters without being consumed by the ocean’s vast microbial metabolism, and could now subsidize this tenebrous, glittering world of the abyss—one that stays with us to this day.

As CO_2 continued to ebb from the surface world, the world grew ever cooler and drier, and grasslands finally became ascendant. As they gained a foothold, and forests retreated, we bid adieu to the inhabitants of that ancient and august warmer world, the last vestiges of which were now receding into Earth history. The blade-jawed friends of *Megistotherium* vanished, ceding the stage to meat-eating arrivistes—cats, dogs, bears, and hyenas. But most importantly for our story, some of those tree-hopping primates had come down from the canopy, perhaps in response to retreating African forests, and by 6 million years ago the skeletons of our ancestors testified to a more-than-passing familiarity with bipedalism.

By the end of the Miocene, the last true blush of warmth had long since passed and the world was inexorably steered toward the ice ages. But today you can still breathe some of the lost CO_2 of the Miocene. Much of it has been liberated from its geological slumber. As tremendous diatom blooms in the oceans drew down Miocene CO_2 they were interred in ancient seafloors that now form coastal rocks that host teetering California mansions. These life-rich rocks, slick with ancient plankton, would eventually turn a rural ranching backwater named Los Angeles into an oil boomtown, stock Indonesia with the oil reserves sought by Imperial Japan, and create the gigantic, empire-making oil fields of the Caspian Sea. The ongoing ignition of all this ancient Miocene life in combustion engines and power plants may eventually help send the planet back to the bygone climates of millions of years past.

AS THE EPOCHS DRAW ON, WE PASS BY STILL MORE WONDERS, UNSEEN and undreamt-of by humans. At one point the Mediterranean pinches off from the ocean and slowly dries up, salty and dead, before catastrophically refilling when the Atlantic comes crashing through the Strait of Gibraltar, pouring mile-high waterfalls off Malta. We pass countless dazzling solar storms that would roast modern electronics and fry civilization, and routine volcanic detonations that would do the same. Finally we arrive in the Pliocene Epoch, only 3 million years ago, and CO_2 has fallen at last to around the level of today's warped atmosphere: 400 ppm. This is the last time CO_2 would be this high until September 2016.

Alarmingly, this world is still 3–4°C warmer than our own, sea level remains stubbornly 70 feet higher than today, and there are still beech trees and bogs at the foot of the Transantarctic Mountains, not far from the South Pole—the so-called last forests of Antarctica. Meanwhile, Lucy the Australopithecus roams an East Africa still far more forested than today. This is not yet an ice-age planet, and we remain well outside of the evolutionary envelope of our modern world—the familiar CO_2 level notwithstanding.

Despite the fact that we're now little more than 3 million years out from the present day—in a book where we've moved at a pace of hundreds of millions, even billions of years at a time—this is still a deeply unfamiliar planet. The Rockies are 3,000 feet shorter, and North and South America are still separated by narrow straits where Panama should be. As a result, animals now iconic to South America, like big cats and tapirs, haven't yet migrated down from North America, where they evolved. Meanwhile, there are still truly odd sights to be found, reminding us just how far removed we are in time from our world—like red pandas and rhinos in Tennessee. And though, by the middle of the Pliocene, these creatures breathe the same CO_2 as today, on the shores of Nunavut in the Canadian High Arctic—where today windswept and barren tundra spreads to the horizon—evergreen forests still hang over an ice-free Arctic Ocean. This is because the Arctic always gets the brunt of the heat when the planet warms, an effect called "polar amplification." It's why maps of modern warming are crowned in a worrying fog of maroon. In the Pliocene, with the same CO_2 as today, though it's only 3–4°C warmer globally, it's a full 10–15°C warmer in the long twilight of northern Canada than today. The pine and birch woodlands at the top of the world are filled with beavers and gigantic forest-dwelling camels. Occasionally this hyperborean world erupts in wildfire, a phenomenon echoed by today's blazes that increasingly sweep the Far North as we try on Pliocene CO_2 for size, like the Quebec wildfires newly choking New Yorkers with acrid smoke.

Elsewhere, West Antarctica's ice sheet is likely absent, and Greenland's, if it exists at all, is still shriveled and pathetic—crowded in on all sides by lush taiga and expanses of tundra, patiently awaiting its takeover in the coming ice ages. Such a small ice sheet makes sense from a modeling perspective: the modern Greenland ice sheet is expected to eventually disappear somewhere between 0.8 and 3.2°C of sustained warming, and we've already sailed past 1°C in our own time. Meanwhile, Europe sports subtropical swamps and warm-temperate forests of magnolia, and dugongs patrol the Mediterranean.

Though a common projection for our own warming world is that the dry places will get drier still, the Pliocene seems to defy this rule of thumb for reasons not yet fully understood. Though by now far drier than the high-CO_2 steam bath of the Eocene or even middle Miocene, the Pliocene is still a far wetter world than our own, especially the subtropics, where—in the Sahara, the Australian outback, Chile's Atacama Desert, the southwestern US, and Namibia—lakes and woodlands jostle deserts out of the way. Meanwhile, the Congo might have been overtaken by a gargantuan but shallow lake that would have steeped the interior of Africa. As with most puzzles in paleoclimate, this ancient wetness might come down to inadequacies in how we model clouds, which are under no obligation to behave in physical reality as they do in simplified lines of computer code—and which might have made this ancient world more uniformly warm and wet. Today, vast and vigorous convecting cells of air hoist ocean water aloft in the tropics, drenching the equator as they soar, only to descend hundreds of miles away as bone-dry limbs of air that produce the world's great subtropical deserts. But on the more homogeneously warm Pliocene planet, this great engine of the climate might have been more sluggish, and the world more uniformly wet as a result.

It is *especially* wet on this world where the ancient seas drown our modern coastlines, which would be so far underwater 3 million years ago—at our current level of CO_2 400 ppm—that you'd have to take great pains to avoid getting the bends if you tried scuba diving down to them.

Today, traveling east through Virginia, North or South Carolina, or Georgia, midway through your drive you'll pass over a curious if gentle hundred-foot drop. This is the Orangeburg Scarp, a bluff hundreds of miles long that divides the broad, flat coastal plain of the American Southeast. It comprises the eroded and smoothed-out rumors of once-magnificent sea cliffs. Here waves of the Pliocene high seas chewed away at the middle of the Carolinas—an East Coast Big Sur. This ancient shoreline is strikingly visible from space by the change in soil color that divides the states, and it is visible on much closer inspection as well: to the east of this strange drop-off, ancient *Megalodon* shark teeth and whale bones litter the Carolina low country. Though warped over the ages by the secret workings of the mantle far below, these subtle banks ninety miles inland nevertheless mark the highstand shoreline of the Pliocene, when the seas were *still* up to 70 feet higher than today. But even within this warm Pliocene Period, the sea level leaped and fell by as much as 60 feet every 20,000 years to the tune of the Earth's sway in space. The planetary wobble rhythmically splashed sunlight over the high latitudes and provoked the unstable ice sheet in Antarctica, marked as it was by a volatile temperament, toying with the ancient coastline like a marionette.

So this is the Pliocene, a world that finally shares our modern-day CO_2 but little else. Where projections of future warming tend to end in 2100, as if our geochemical short-circuiting of the planet will have the same shelf life as a new steel bridge, the Pliocene illuminates just what sort of truly long-term changes we might inevitably set in motion if we maintain the deranged and anachronistic atmosphere we've already engineered—to say nothing of the more alarming Miocene or even catastrophic Eocene levels of CO_2 that we might ultimately reproduce. The jumbled climates and oceans of deep time remind us that whatever is coming in our own age, it won't be like our planet, only more so. It will likely be as unfamiliar as these lost worlds—whether the extraordinarily ferocious heat of the Eocene or the more subtle but scarcely less transformative uncanniness of the Pliocene. As the great ice sheets of our world melt, and darker forested land takes over the world's tundra, feedbacks may eventually kick in to launch our planet into a different state altogether, one that might resemble imperfectly one of these strange bygone worlds.

AS CO_2 CONTINUES TO FALL ON OUR JOURNEY, WE DRAW EVER CLOSER to the present day. Panama rises from the sea, diverting ocean currents that once flowed freely around the planet, and tenuously connects the two Americas. As this land bridge emerges between worlds, it introduces North and South American animals separated from each other for untold ages, profoundly reshaping life on both continents. Invaders from the North—horses, dogs, and bears—have the upper hand in their new southern continent, while giant ground sloths, eight-foot-tall terror birds, opossums, armadillos, and porcupines make the crossing up from the South as well and thrive in their new digs. Tapirs and camels, which originated in North America, live on today as llamas and alpaca only in the South, a relic of this ancient Pliocene union.

Then, by 2.6 million years ago, CO_2 had fallen far enough in its long decline from the greenhouse of the dinosaurs that the Earth finally tipped over the edge and the ice ages began in earnest. The Pleistocene, which would last until agriculture was born, finally begins. Swelling and retreating ice sheets, up to two miles thick, would now repeatedly overrun the northern continents for millions of years—dropping and raising the seas by hundreds of feet, and plunging the Earth in and out of stupefying planetary winters. This monumental shift in the climate state is marked in sediment cores pulled up from the ocean floor by drill ships far offshore today, seafloor muds that bear pebbles dropped into the deep by flotillas of ancient icebergs in the North Atlantic.

The Earth had passed yet another threshold, and from here on out would be subject to deep 40,000-year, then 100,000-year-long glaciations, paced by the planet's wobble in space. These planetary wobbles didn't matter much for most of Earth history, when the planet was wearing a much thicker CO_2 blanket, but in our geologically recent, knife's-edge, low-CO_2 ice-age world, they've been enough to trip runaway, transoceanic, carbon-cycle, and ice feedbacks—feedbacks that have thrust the planet in and out of the deep cold. These were violent alternations between deep glaciations, like the one that ended only a geological moment ago—when gigantic ice sheets smothered a third of North America—and brief, balmy interludes, called "interglacials," like today.

As the planet first plunged into this deep cold, extinction swept the seas. The 65-foot-monster shark *Megalodon* had already vanished by the dawn of these ice ages, while the massive seabird *Pelagornis* would no longer cast its yawning shadow over the waves. Seas that had once been populated with a universe of small baleen whales—residents for millions of years of the shallows that submerged the broad, flat edges of the world's continents—saw their habitat destroyed when swelling ice sheets and plummeting seas sent their coastal shallows over the horizon and off underwater cliffs. A third of the big life in the oceans was culled by this initial chill of the ice age.

But a new roster of life would come to adapt to this strangely dry, cold, low-CO_2 planet—a fitful, ice-capped climate unlike any since the alien world of the Carboniferous 300 million years earlier. This roster would include the largest animals in Earth history: colossal baleen whales, evolved to live off huge fat stores on their oceanwide journeys in pursuit of the more uncertain and shifting plankton blooms of the ice ages. Meanwhile on land, in monsoon-tempoed fits and starts—across Kenya, Ethiopia, and Tanzania—the grasslands advanced. While the Serengeti might seem every bit the ur-landscape of prehistory, open grasslands like this, dotted with huge herbivore grazers and sports-car-tuned predators, only became a regular, if intermittent, feature of East Africa in the chilling, drying end of the Pliocene.

It was during this descent into a still cooler, drier, more uncertain climate that Lucy the *Australopithecus* went extinct, while a highly adaptable creature with a broad-based diet named *Homo* evolved—an odd bipedal creature exceptionally suited to the volatility of the low-CO_2 ice-age Pleistocene. Ultimately this line of primates would include an outwardly unimpressive but highly social species shaped by fire, *Homo sapiens*—one that would someday wipe out the ice-age megafauna, remake the surface of the Earth, tap into the planet's half-billion-year reserves of geologic energy, threaten to undo this entire 50-million-year climate trajectory of the age of mammals in mere decades, and potentially launch the planet back into the high-CO_2 greenhouse of the dinosaurs. But that day was still far off.

AS WE FINALLY PASS INTO THIS AGE OF ICE, AND GLOSS OVER THE final 2.6 million years in our journey to the present day, we might be surprised to see that the world fails to get any more familiar, no matter how seemingly close we get to the end of geologic time. This final 2.6-million-year chunk remains an unspeakably vast chasm of time that elides the passage of dozens of ice ages and brief, warm interglacial springtimes like today, hundreds of millions of sunrises over windswept vistas of ice, moonrises over forgotten savannas, three Yellowstone supereruptions, thousands of megafloods, the last of the venerable giant terror birds, the glacial creation and destruction of innumerable islands and moraines, and the thrumming destruction of miles-thick bedrock scraped off by the fickle ice sheets. The interglacial we happen to find ourselves in right now—at some 11,700 years and counting—is much like the dozens of similar interglacials that have come and gone in the past few million years, when the punishing ice of the Pleistocene temporarily retreats to the poles for a few thousand years. But unlike previous interglacials, all of recorded human history has unfolded in this one—in this extremely temporary, bizarrely stable, mild-climate way station at the end of time.

Though we've been blinkered to the astonishing fact of our wild climate by the short institutional memory of humanity, this is the geological context for our world. We live on a frighteningly unstable icehouse planet at the end of tens of millions of years of dwindling CO_2 and cooling. And interglacials, as the name implies, don't last forever. The ice sheets inevitably return, sometimes reaching as far as Kansas, sometimes covering Boston and Chicago in almost a mile of ice for 100,000 years, before rapidly disintegrating yet again, and retreating to Greenland and Antarctica for still more thousands-year-long springtime reprieves. This titanic cycle has repeated itself dozens of times in only the past 3 million years, an astonishing metronome of continental ice that has been paced by (only mild) changes to sunlight reaching the Northern Hemisphere in summer—changes that, when the summer light is low, allow ice to build up year over year, or alternately, when slightly more summer sunlight reaches the Far North, nip that same ice in the bud each year when the warmth returns—a rhythm guided by the planet's periodic wobble in space.

As we make our way toward the present, these pulses of unthinkable continental glaciations get deeper and longer, and the ice sheets themselves get thicker and more recalcitrant. Sea level helplessly pulses up and down, sending coastlines racing out across the continental shelf, only to be drawn back in again over these plains like a shroud. As Earth's blanket of CO_2 continues to ebb, the fickle signal of northern summer sunlight now freely plays with these continental ice sheets and holds the planet in the balance.

AS WE LEAP FORWARD, MAKING OUR WAY TO ONLY 20,000 YEARS ago—the height of the last great ice age of the Pleistocene, and a moment ago, geologically—there are now modern humans on every continent. Yet the world remains completely unrecognizable. Where the early Eocene near the start of our journey might have been 12°C warmer than our world, at 1,000 ppm CO_2 this ice-age world is now 6°C *colder* than our own, at only 180 ppm. What this means in practice is that an *Antarctica's* worth of ice now rests leaden atop North America. Underneath this empire of rime, from the Hudson Bay to Indiana to Boston, it's pitch black—utterly, bedrock-scouringly dead. Similar sheets smother and sterilize northern Europe and as a result, sea level is now over 400 feet lower than today. Mammoth-trampled Florida bulges like a swollen thumb, as listing icebergs drift and founder in the mud off Miami. Stands of stunted spruce of the sort that wouldn't look out of place in northern Quebec carpet the midwestern US in alpine splendor, even sneaking all the way down to the Gulf of Mexico, pruned by mastodons. The Rockies are carved up not by wildflower-dappled mountain valleys but by majestic, overflowing rivers of ice and rock. The Pacific Northwest, edged by the American Antarctica, is a harsh and treeless place, as storm tracks that today feed temperate rain forests around Seattle and Vancouver with rain have been shoved hundreds of miles south by the advancing ice world. Instead the water-bearing jet stream slinks low into California—the land of dire wolves—and Nevada and Utah fill with cold rains.

During World War II, at Topaz, the desolate Japanese American internment camp near the Utah-Nevada border, prisoners combed the bleak, greasewood-patched flats of the Sevier Desert for unlikely seashells, fashioning miraculous little brooches from tiny mussel and snail shells to while away their exile in the wilderness. The desert seashells were 20,000 years old, from the vanished depths of the giant Pleistocene Lake Bonneville—the astonishing product of these diverted rains. This was once a Utahan Lake Superior, 1,000 feet deep in places, haunted at its shores by mammoths, saber-toothed cats, and gigantic short-faced bears, and joined by endless other pine-accented lakes scattered across what is today's bleak basin and range. Elsewhere, the retreat of the seas made most of Indonesia a peninsula of mainland Asia. Vast savannas and swamps linked Australia and New Guinea, the Persian Gulf was dry land, and of course Russia shared a meaty tundra handshake with Alaska. Beringia, as this ice-age connection is called, is sometimes described as a tenuous "land bridge" connecting the continents. But in fact this was a giant, unbroken swath of mammoth steppe, twice the size of Texas, and it is almost entirely underwater today—home to innumerable human stories lost to time, and now picked over by king crabs in the deep. Meanwhile, much of the Alps had long since surrendered to polar desert, and a trimming of forest embroidered the Mediterranean. There were reindeer in Spain and glaciers in Morocco. And everywhere loess, loess, and more loess. This was the age of dust.

Ice is a rock that flows. Send it in massive sterilizing slabs across the continents and it will quarry mountainsides, pulverize bedrock, and obliterate everything in its path. At the height of the last ice age, along the crumbling margins of the continental ice sheets, the rocky, dusty spoils of all this destruction spilled out onto the tundra. Dry winds carried this ocean of silt around the world in unthinkable dust storms, piling it up in seas of loess that buried the central US, China, and Eastern Europe in featureless drifts. In Austria, not far from the site of the voluptuous 30,000-year-old figurine *Venus of Willendorf*, are the remains of a campground of the same age: tents, hearths, burnt garbage pits, and hoards of ivory jewelry—all abandoned in the face of these violent smothering haboobs. In this dry, dusty, low-CO_2 world the Sahara swelled and dunes overtook much of southern Africa as well. Ice cores from both Antarctica and Greenland record a world ten times dustier than today.

This parched Pleistocene world would appear duller from space, hosting as it did a quarter less plant life. Not only were there hundreds of gigatons less carbon locked up in vegetation, but there was also less than half as much carbon in the air as today, with CO_2 in the atmosphere registering only a paltry 180 ppm. In fact, CO_2 was so low it might have been unable to drop any further: photosynthesis itself starts to shut down at such trifling levels, a negative feedback that might have left more CO_2—unused by plants—in the air above, and acted as a brake to the runaway deep freeze. Nevertheless, all that vanished carbon from the surface world must have gone somewhere, and one possibility is that the very dust storms then sweeping the world showered the seas with iron, seeding carbon-rich blooms of plankton that sank to the bottom of the ocean, sucking down the CO_2 that might have otherwise warmed the planet by hiding it in the deep.[fn7] In fact, CO_2 fell so low, and the planet grew so cold only a few thousand years ago, that the latest generation of (admittedly oversensitive) climate models struggle to keep this Earth—from only a geologic moment ago—from tipping over into a Snowball state when re-creating it.

So this was the strange world of the most recent ice age, one that just passed in our immediate rearview. It's so recent, in fact, that today most of Canada and Scandinavia is still bouncing back up from the now-vanished ice sheets that just weighed them down. Mammoth meat still thaws from the ice. We often refer to "the Ice Age" as if the Earth endured one monotonous, wintry slumber, but in fact when we pull back to survey the entire 2.6-million-year span of the Pleistocene, we see just how mistaken this idea is. The planet has actually plunged in and out of dozens of punishing ice sheet advances, with the most recent ice age terminating only 12,000 years ago, with humans committing creation stories to text not long after. As we'll see, this immediate prelude to recorded history was a violent thaw (though no more so than the dozens of other deglaciations of this Pleistocene) as CO_2 awoke from the oceans, sea levels leaped hundreds of feet, and the climate convulsed for millennia. It's an astonishing record of a low-CO_2 planet on edge. As a species, we have no institutional memory of this precarious, wild world from which we just emerged only moments ago, and which long shaped our ancestors as well as the entire biosphere around us.

We currently find ourselves in the most stable climate window in at least the last 650,000 years, and we take the current shorelines, the climate, the flora, and the fauna of our favorite places on this temporary interglacial planet to be eternal. But in fact, all of recorded human history has taken place within the extremely misleading stability of the past few millennia, on an otherwise wildly irascible ice-age planet. This truly weird and equable climate plateau of the past 6,000 years or so—the only world that the blinkered written word has ever known—is nestled within this forgotten, far vaster, far more volatile and frightening climate prehistory. It's this ice-age climate that shaped us as a species, along with all other life on Earth today.

Our species was raised in a harsh glacial nursery, and our modern world awoke in a brief unstable intermission from the cold, not unlike the many interglacials that had come and gone before. This temperamental ice-age planet of the past few million years is so utterly unlike the greenhouse world of the early age of mammals, much less that of the dinosaurs that began this chapter—a journey that took tens of millions of years of climate, planetary, and biological evolution to climb down from—that one's mind reels when imagining what it would mean to suddenly return to that ancient high-CO_2 future in mere *decades*.

“In the past 100 million years, CO_2 has ranged from maximum values in the mid-Cretaceous to minimum levels at the last glacial maximum,” Jessica Tierney writes in a 2020 review paper. “Going forward, we are on pace to experience an equivalent magnitude of change in atmospheric CO_2 concentrations, in reverse, over a period of time that is more than 10,000 times shorter.”

If we managed to pull off this preposterous feat, she writes, “It will be at least 500,000 years, a duration equivalent to 20,000 human generations, before atmospheric CO_2 fully returns to pre-industrial levels. Our planet will recover, but for humans, and the organisms with which we share this planet, the changes in climate will appear to be a permanent state shift.”

“We’re an ice age species,” she told me. “The Pliocene, the Eocene, these weren’t worlds that were inhabited by humans.”

7

The Rise of the New Metabolism

AT THE END OF DOZENS OF ICE AGES, IN THE VERY LAST SUBLIMINAL movie frame of Earth history, an ocean of light carbon buried in the planet's crust over hundreds of millions of years suddenly exploded out of the crust as CO_2 as it had in only a handful of cataclysms deep in geologic history. This eruption began modestly at first, long smoldering from the island of Great Britain like the initial grumblings of a caldera, before catastrophically erupting in the nineteenth century. Sills of this pseudomagmatism spread across the English Channel, then erupted from continental western Eurasia. Soon a truly massive eruption from this same swarm burst out of the Earth across the ocean, covering North America in a miasma of smog, networks of steel, and outcrops of cement. Smaller eruptions continually belched from smaller outposts across the world, from Japan to Scandinavia, sending burning embers drifting across the world's oceans, and by the early twentieth century these ignitable tendrils in the crust had spread across the entire planet, occasionally producing short, particularly explosive and violent reports (from 1938 to 1945, for instance). But these flareups seemed only to accelerate the pace of this strange volcanism, and by the end of the century CO_2 billowed uncontrollably from the Earth's crust, issuing everywhere from the huge deposits of fossil carbon left by the ages underground.

Where the motivating force for the wholesale movement of carbon from the crust into the atmosphere and oceans for billions of years had always been volcanoes—that is, the dissipation of radioactive heat inside the Earth, the churning gears of plate tectonics, the convecting, chthonic cells of mantle, the subducting slabs of ocean crust, the blobs of hot rock launched to the surface by the planet's white-hot metal core—this more recent and ongoing eruption of CO_2 has a much stranger, more novel cause. Still, we live in a physical universe, so there must be a physical explanation for the current supereruption of carbon out of the lithosphere. Instead of tectonic forces launching it out of the ground, though, this supereruption has been—strangely enough—biologically mediated, drawn out of the ground from above, by some great extractive force. But whatever its cause, the effect will be largely the same as that of the many other carbon-cycle convulsions of Earth history.

This and the next two chapters will attempt to explain—pulling together everything we've learned so far in the book—how an ice-age creature, whose evolution was shaped by fire, eventually came to be a geologic force on par with large igneous provinces, one that now emits 100 times more CO_2 every year than all the volcanoes on Earth. This strange creature—a few of them in any case—has pulled off this ignominious feat by detonating, in a matter of centuries, the geologic battery of energy in the Earth's crust that had been charging for over 500 million years, since the dawn of animal life.

"THERE'S A REASON WHY YOU START USING TOOLS, AND IT'S PROBABLY because everything sucks," University of Denver archeologist Nicole Herzog told me about the astounding ice-age trajectory of one unassuming lineage of primates that today numbers over 8 billion souls. In chapter 6 we touched on the dramatic climate context for our evolution—the last 2.6 million years of seesawing ice ages of the Pleistocene—a far colder, far drier climate state than any the Earth had seen for hundreds of millions of years, and one that had taken tens of millions of years of declining CO_2 from the hothouse of the early age of mammals, to finally arrive. In this chapter we'll see just how this volatile, punishing, low-CO_2 ice-age climate intimately shaped our evolution and pushed us toward entirely new ways of living on Earth. We are, after all, ice-age creatures, and though the earlier epochs we explored are perhaps relevant to our climate future, they were not part of our species' past.

Almost as soon as the planet plunged into the ice 2.6 million years ago, a new genus of primate, *Homo*, appeared on the scene and started to exhibit some odd behavior. While using clumsy stone tools and eating meat were already novel pastimes among a select few hominins, this one systematically, and with rather more care, began to churn out sharp-edged tools, lovingly flaked from volcanic rock. With these it butchered large carcasses, profiting off the spoils of carnivores with which it shared the savanna. This was unusual behavior for primates, which, ever since their unassuming, tree-hopping arrival in the fever of the early Eocene, tens of millions of years earlier, were far more comfortable eating fruits and insects, or gnawing on leaves like gorillas. Sometimes they caught and ate small reptiles and mammals, and every so often each other. But they didn’t carve the tongues out of antelopes, or compete with dangerous big cats for fresh-killed wildebeests or elephants, like these ones did. As hundreds of millennia passed, and the wild metronome of ice sheets plunged Earth in and out of planetary winter, the biological contours of this ice-age planet took shape around them and in turn shaped these strange new creatures. Grasslands, steppes, and tundra haltingly expanded at the expense of forests, and on this temperamental, low-CO_2 world our ancestors were pushed out from under the trees and into the margins of ever-more-fragmented mosaic landscapes.

Strangely, *Homo* would survive these climate shudders of the early Pleistocene, while big-jawed, big-brained, big-bodied Australopiths died out, despite inhabiting the same world. “They also landed on the savanna and somehow they didn’t make it—they didn’t innovate,” said Herzog. “The agitating question is just, why? Why us and not them?”

The most important of these innovations was the command of an increasingly common feature on this colder, drier, lower-CO_2 world: fire. As far as we know, in all of Earth history, ours is the only lineage that has mastered the control of it. And nothing about the rest of human history—even including the modern explosion of energy and CO_2 from the crust—is explicable without this astounding fact about our kind. Lavoisier demonstrated that fire and respiration are the same reaction, and burning organic carbon to CO_2 for energy is what keeps animals like us alive. But ever since our origins, we have entrained fire as a kind of *external* metabolism as well, and doing so gave us a massive energetic safety margin during the increasingly volatile and unpredictable environment of the Pleistocene. “Why,” Herzog wondered, “was it us that went all in on fire, and not any other primate?”

FOR SOME 430 MILLION YEARS, FIRE HAS BEEN A PERSISTENT IF FLUCTUATING feature of planet Earth. This is, nevertheless, a surprisingly tiny percentage of Earth history. For the first 90 percent of Earth history, the planet's face was entirely untouched by flame. This is because it takes three things to make a fire.

The spark. For all of Earth history, lightning has been zapping the surface, today at the rate of 100 strikes a second. And for the entire history of fire on planet Earth, the vast majority of fires have been set by these bolts from the blue. Astoundingly, though, even at the rate of 100 strikes a second, human pyrophilia today far outpaces lightning, with 84 percent of the world's blazes set by people (not counting the controlled fires in every fossil-fueled power plant, furnace, and engine on Earth).

Then you need oxygen. There are no fires without a lot of oxygen in the air, so, as a result, there weren't fires for most of Earth history. As we've seen, though oxygen is the by-product of making plant matter from CO_2 through photosynthesis, to build oxygen up to appreciable levels, vast seas of this organic carbon have to be buried in the crust as fossil fuels. Every breath you take is the gift, not of plant life today, as the intuitive cliché goes, but rather the impounding of some small fraction of this life in the crust, leaving behind an oxygen surplus that has taken hundreds of millions of years to accrue. And fire on the surface doesn't catch until oxygen reaches about 16 percent of the atmosphere. As discussed, this level of abundant oxygen is a very late feature on the surface of our planet, appearing only in the age of animals, once tectonics and life conspired to bury enough carbon in the crust below.

But even this isn't enough. The third and final ingredient in fire is the fuel—something to burn. Seeing as rocks generally don't burn, fuel was available only once life emerged onto land. As a result, the first fires appear in the fossil record only about 430 million years ago, once all three ingredients had arrived on the Earth's surface. The first fires tore across humble, pioneering patches of plant life no taller than your ankle, as well as burning through rather more astonishing, and truly bizarre, 25-foot columns of fungus that briefly held sway on this desolate world before trees. When the tiny plants suddenly gained confidence, though, and shot up toward the sky as trees some 50 million years later in the Devonian Period, the first forests tentatively spread across the world, and fire came into its own. From then on, fire would be a familiar phenomenon on Earth's surface for hundreds of millions of years, and charcoals in the fossil record testify to its presence, for the most part, ever since.

We've seen fire sweep through the high-oxygen rain forests of the Carboniferous and raze the terrestrial biosphere in the End-Permian mass extinction. Fire also played a crucial role in the rise of flowering plants in the Cretaceous, which remade the living world with more than just the fruit they bore. Flowering plants cycle through more water but are also more flammable, and, according to the paleobotanist Claire Belcher, their incendiary nature helped clear out the less fire-adapted ancient plant world that came before. These plants then established themselves firmly enough that their prodigious water recycling was sufficient to make their own weather—as testified to the clouds wisping off modern rain forests—counteracting this flammability and giving rise to the closed-canopy jungles familiar to the age of mammals. As the world lost its CO_2 blanket over this age of mammals, though, and the world chilled, grasslands, having invented a new form of photosynthesis adapted to this drier, lower-CO_2 world, haltingly but inexorably took over much of the Earth. As they did so, these prairies and savannas were now given to explosive grass fires. This was the context of our evolution.

"I think climate change is *the* fundamental piece of the puzzle," said Herzog about the continual innovation in our lineage and the rise of this strange fire creature.

> What really prompts people to innovate—or any organism to innovate—is finding yourself on the tail end of a fitness curve. You find yourself out on the edge, in other words. And at some point, there's no other direction to go in order to survive. So that sort of pushes you into a new space where you have to develop a new technology. And it's costly to do that. And if you don't have to, you probably wouldn't. So all of this ecological change is happening and up to that point, there are lots of different types of hominins, and lots of apes, and we're all sort of living in this place together. But the ecological change makes this habitat super fragmented. And for some reason, humans end up out on the edge, where all of this aridity is happening. So our ancestors are in contact with fire-altered landscapes, probably much more frequently than chimps who seem to have maintained a foothold in the forested areas.

Even today, though, chimps show a surprising comfort around fire. In her fieldwork in the Sahel of southeast Senegal, Herzog marvels at the animals' lack of concern as they patiently wait at the edge of grassland flames that yearly consume up to 75 percent of their habitat—the majority of them set by local villagers to prevent more devastating fires, promote local millet production and livestock agriculture, and reduce the threat of hiking through shoulder-high grasses filled with snakes and other unseen hazards to life and limb. Once the fires have swept through, the chimpanzees have their pick of cooked critters on a charred landscape cleared of predators, just like the baboons in the forests of Uganda who have been seen foraging for cooked grasshoppers and shoots after fire passes by. Herzog imagines our earliest ancestors might have evinced a similar curiosity about the novel fire-scarred landscape—seeing fire less as a seasonal catastrophe than as a potential caloric windfall.

But where chimps might merely poke over the ashes, our lineage at some point took the next step in promoting and even domesticating the flames—perhaps picking a smoldering stick from the edges of one grassland, and lighting the grassland in the next valley over. The energetic returns to foraging in this sparse landscape suddenly soared. Slowly and tentatively at the dawn of the Pleistocene, and then suddenly around 1.9 million years ago, with the rise of the even brainier *Homo erectus*, and as grasslands reached their peak, we had become the fire ape.

Today the entire industrialized world is held up by the energy released by oxidizing organic carbon back to CO_2. Fire, in other words. We just happen to be burning all the plant matter in Earth history that we can get our hands on, buried in the crust. But it's a trick we learned early, and it's essential to our nature. We are what's known as an "obligate pyrophile." We need fire to survive. We are "wholly dependent on fire for survival and reproduction," in the words of a study led by the archeologist Christopher Parker. In fact, as we'll see, it's written in our very anatomy.

It is impossible to exaggerate the significance of this cultural innovation. For one thing, fire allowed us to remake the world in our image. There's nothing new about this. Since the dawn of animal life, over a half-billion years ago, animals have set about remaking the landscape in ways that have benefited their own reproduction and survival. Just before the Cambrian, it was marine worms digging into the muck, aerating the seabed, possibly wiping out the blob creatures that placidly lived there, and completely remaking the ocean landscape. In our own time, a modern exemplar is beavers, who build dams, flood forests, and skillfully construct little island castles of lumber to protect themselves from predators, remaking the world to make it more habitable for more beavers. This is called ecosystem engineering, and it has been transforming the surface of the planet for as long as there has been animal life. Humans are merely the ultimate ecosystem engineers, and fire has long been the most powerful tool in our arsenal. We can't survive in a world without fire, so instead we aided its spread as far and wide as we were able. Just as there is no meaningful distinction between a forest waterlogged with beaver ponds and a supposedly more "natural" landscape, so too did the human landscapes of the Pleistocene become hopelessly entangled with the natural world.

In our case, fire allowed hunter-gatherers to reimagine the landscape by creating mosaics of fire-adapted berry bushes and nut trees, and heathlands of shoots and saplings that lured game for the hunt. It concentrated people around these hearths, increased the pressures of sociality. An engine of sorts was gathering here, drawing calories closer to human settlements, energy that was dissipated in the flickers of flame and the fires of metabolism. But the main reason fire was so important to ancient humans is the same attribute that today sends the planet careening toward a climate catastrophe: turning photosynthesis—whether living or fossil—back to CO_2 releases energy. Fire is hot. You can cook with it.

Perhaps the most telling piece of evidence testifying to the importance of fire for our kind was the attendant suite of changes to the anatomy of the now-varied group of hominins, various species of which spanned not only Africa but also Eurasia as the Pleistocene drew on, from Spain to Indonesia. With small guts but big, energy-hogging brains, these creatures were apparently performing much of their metabolism *outside* of their bodies in order to subsidize their intelligence. This is because there simply aren't enough hours in the day to forage and digest raw foods sufficient to support brains as energetically expensive as our own, contra many a modern paleo-diet guru. Cooking is the only way. This is because brains are ravenous for glucose, an energy-dense sugar synthesized by photosynthesis that we combust with a perpetual metabolic bonfire of oxygen to keep us conscious at every waking moment. But to get at glucose, we first need to cook our plants—wheat, barley, rice, millet, lentils, peas, chickpeas, bitter vetch, cassava, sago, yam, taro, plantains, breadfruit, sweet potato, etc.—in order to explode little packets of starch within them and unravel the long, dense polymers stitched together from glucose. This unlocks the sugar for a much easier metabolism, allowing us to skip the painstaking, energy-intensive step of breaking down these starches in our guts ourselves, and allows us to extract far more energy from these staple foods at lower cost.

Cooking meat, meanwhile, unfolds the involuted ribbons of protein, and avails them to the digestive onslaught of enzymes. It makes the engine of metabolism more energy-efficient. Where chimps might spend six hours a day chewing, and an anaconda might spend weeks motionless as it digests a capybara, humans have outsourced much of this digestive work to fire. This increase in efficiency also allowed humans to significantly reduce the amount of time spent gathering food. Indeed, one study estimates that in order for a gorilla to get enough energy in its diet to subsidize a brain like ours, it would have to spend more than two extra hours a day foraging for food, in a twelve-hour day most of which is already spent foraging. The math just doesn't add up without the boost from fire, and any animal that tried to make it so would starve.

For much of our evolutionary history, we might have also outsourced much of our digestion to the metabolic fires of fermentation, and even putrefaction, enlisting lactic-acid bacteria to do much of this hard work for us as well. Grinding our food was yet another shortcut to easier digestion. Because of these auxiliary, *external* metabolisms that we've developed over hundreds of thousands of years, humans haven't had to spend as much metabolic energy breaking down our food, and as a result, we now have comically undersized molars, chewing muscles, stomachs, and colons.[fn1] This left humans with an energy surplus—one of the first of many. And the energy from the photosynthesis in our diets that *wasn't* spent on the development, growth, maintenance, and function of large digestive systems was now freed up to invest in the augmentation of our brain—especially in ways that supplied an even greater return on energy down the line.

So having freed up all that spare solar power from our diet by outsourcing our digestion to fire, we used the balance of calories to grow and power gigantic electric brains, which use up to a quarter of the energy of our resting metabolism, "compared to 8–10% for other primates and just 3–5% for other mammals," the Harvard anthropologist Richard Wrangham writes. So what are big brains good for?

In good times, if the environment changes slowly enough, evolution by natural selection can suffice in enduring the winds of geologic change. The environment selects for adaptive traits over generations, a process that might ultimately give rise to entirely new body types and behaviors more suited to the new world. But when the world becomes increasingly *un*predictable, when past performance is not predictive of future results, and the conditions this month, this year, or even this decade are entirely unpredictive of the next, this kind of generational adaptation can be simply too slow to keep pace with catastrophic environmental surprises. Mass extinctions provide the most extreme demonstration of this principle, when it applies to the whole world. But there are ways around this evolutionary speed limit that animals can adopt to widen the margin for survival and to thrive in unpredictable times. One of them is called "cognitive buffering."

Today we find our own human world—one in which wetlands can be transmuted to farmland, then paved over by city blocks in a matter of years—filled with increasingly clever, adaptable creatures like crows. These animals have massive brains, shifting flexible diets, and an astounding capacity for memory and recognition. They are sociable and even use tools. Instead of programmatically following instincts sculpted by natural selection—the external world represented in their bodies only as a kind of phenotypic reflection of it—crows are problem solvers, able to improvise entire suites of behaviors to match the challenges of an uncertain age. Astoundingly, in recent years Western science has begun to take seriously aboriginal Australian reports it had long dismissed: that hawks intentionally spread fires by carrying burning sticks to new fields, something it was long thought only humans did. As the ecologist Rob Dunn writes, "The rise of smart birds around us is a measure of just how unpredictable we have made the world."

Similarly, in the convulsing low-CO_2 ice world of the Pleistocene, when armadas of icebergs repeatedly emptied into the North Atlantic and gigantic lakes of glacial meltwater collapsed, rerouting ocean circulation and upending the global climate in mere years, animals had to eke out a living on a more unpredictable planet. Familiar life-giving monsoon rains could unexpectedly decamp for distant latitudes, forests could give way to grasslands that caught fire, and fertile riverbeds could suddenly run dry and be overtaken by sand dunes in decades. But even within the depths of the ice ages themselves, temperatures sometimes swung wildly—the product of dyspeptic ice sheets whose instabilities repeatedly throttled this planet. Ice cores in Greenland testify to regional 10–15°C jumps in mere decades during the deep freeze, while under the North Atlantic seafloor today, huge dumps of iceberg-rafted debris hint at episodes of disintegrating ancient ice sheets that jammed ocean circulation, casually prompting the kinds of climate change that would ruin civilization. It’s unsurprising, then, that leaps in human evolution sometimes line up with many of the most severe climate throes of the Pleistocene. At one point, 900,000 years ago, during a pivotal moment of climate change, when the ice ages switched from about 40,000 years long to far more punishing and far deeper 100,000-year planetary freezes, our line was nearly wiped out—culled to perhaps only 1,280 individuals, and the hominid fossil record nearly disappears. In the aftermath of the climate disaster, though, it appears we speciated, with our line parting ways from our close cousins (and even future lovers) the Neanderthals and Denisovans.

During this volatile glacial-interglacial world—one in which CO_2 continued to reach new lows, and ice ages got longer and more severe—savannas, deserts, forests, wetlands, tundra, and mile-thick ice sheets oscillated wildly across the continents, and sea level relentlessly pulsed up and down by hundreds of feet. Meanwhile, our lineage continued to see an unprecedented, uninterrupted rise in brain volume unlike any other animal in Earth history over the same span. These were clever creatures, able to come up with novel solutions on the fly to life on this unpredictable planet—able to “buffer bad conditions with big brains,” says Dunn.

This is one theory for the stunning increase in brain volume in hominins over the convulsions of the ice ages: that it was the ill-tempered and unpredictable low-CO_2 ice-age climate itself that made us so smart. But there were other dynamics, intrinsic to the new hominin lifestyles, that selected for intelligence as well, subtly coaxing the development of what would eventually become 20-watt central nervous systems, housed in skulls so large that they would make maternity singularly life-threatening for humans.

In a world that abounded with predators, there was strength in numbers. The downside is that it’s tricky work sharing space with other foragers, defusing disputes, finding a mate in a community where you know everybody, and tracking rivalries, alliances, obligations. But all that extra cognitive firepower in the relevant parts of the brain allowed our ancestors to model these delicate complications of social life. In fact, some have argued that this very sociality might have been the main driver of the increase in human brain size. Sharks don’t have networking events, mixers, or diplomatic summits. But intelligence has all sorts of other benefits as well, and we could also now run complex simulations of the future and predict the behavior of prey. Hunting big game, for instance, requires anticipating the behavior of animals, intercepting migration routes, forcing herds into blind corners to be slaughtered or off cliffs to be dashed on the rocks, hiding a pit by covering it with branches, or otherwise developing novel strategies for their capture. In other words, it requires outsmarting other animals. But it also requires collaborating with allies and navigating the complex social coalitions that make such predatory gambits so lethal, as well as the delicate task of sharing the spoils with the community. And to a degree unseen in other primates, humans are even willing to collaborate with non-kin. As for the “gatherer” part of hunter-gatherer: foraging in teams in a patchy and ever-changing ice-age landscape, and sharing and pooling information with the group about the most promising spots, saw similar returns in calories. Our social bonds might have saved our necks in this deteriorating climate.

Just as simulating the entire climate-ocean system today requires titanic inputs of energy and brute supercomputing power to anticipate our uncertain future on this planet, so too did the mental models of the near future for our hominin ancestors. Both kinds of fires that made this computing power possible—the biochemical ones within us, and the more literal fires we conscripted and came to depend on—allowed our brains to swell, collaborate more effectively with fellow hominins, find enough food, and improvise shelters to withstand the blows of an unpredictable planet. Key to this resilience was the fact of fire’s physical warmth, emanating from hearths and qulliqs[fn2] around the Earth, which allowed our species to vastly increase its geographic range into cold regions far outside our biological tolerance. “As human biomass grows,” the paleontologist Anthony Barnosky writes, “the amount of solar energy and [photosynthesis] available for use by other species shrinks.”

It doesn’t require an imaginative leap to see how more social, cooperative hominins—with bigger brains subsidized by fire—might have ended up with more calories in reward for their cunning, and how they might have consequently come to outnumber the less social ones. Today we share this planet with several other highly intelligent, highly lethal, and highly social creatures that hunt game. Orca, for instance, have local dialects, teach each other games, and devise hunting techniques passed down through generations. They also earn their sobriquet “killer.” Unsurprisingly, they also have huge brains.

Over the course of the Pleistocene, as the oddball primates were becoming increasingly deadly and increasingly smart, an endless variety of foraging and food-processing techniques, an encyclopedic knowledge of the local botany and the rhythm of the seasons, the capacity to form flexible social coalitions, caching stockpiles of meat for an uncertain future, and the invention of a battery of increasingly helpful and lethal tools were all buffers against a more volatile world. And with more and more brainpower aided by fire, our mouths got smaller, our tongues more manipulable, and our larynxes dropped into our necks—a trade-off that made us more susceptible to choking but also unlocked yet another planet-transforming information technology: language. Human language consists of complex, information-rich signals encoded in sound waves that, even for all the guttural richness of the vocalization of the other great apes, do seem qualitatively distinct in structure, richness, and syntax. Languages across the Earth are both highly informationally efficient and (because it necessarily takes energy to process information) energy-efficient. They are optimized so as not to overtax our limited short-term working memory, and they take shortcuts that leverage the knowledge of the receiver to resolve ambiguous but more economical syntax, while still being robust to corruption from environmental noise, to put it in the parlance of information theorists.

Language, and culture more broadly, allow us to reliably transmit huge packages of information—novel behavioral and technological solutions to life-threatening problems, on a subgenerational timescale—from one mind to another, from one social group to another, both within generations and even across them as well, obliterating the timescales of evolution by natural selection. They are the ultimate information technology, infinitely refinable, mutable, flexible. They have allowed humans to inhabit, like no other animal on Earth, what Harvard anthropologist Terrence Deacon calls the “symbolic niche.” In their ability to transfer entire tranches of information between organisms, language and culture are rivaled in the history of life perhaps only by the unusual capacity of bacteria and archaea to undergo “horizontal gene transfers,” a trick that allows microbes to take on, wholesale, the capacities honed by other unrelated groups of organisms by swapping entire strings of genetic information. The novel chimeras produced by these microbial gene swaps in deep time are responsible for many of the most profound biological innovations in Earth history, transforming the planet in the billions of years before animal life. Similarly, the uniquely mutable information landscape inhabited by humans—which by now has mushroomed into symbolic writing and math, logic, legal systems, a universe of cultural practices, religions, music, visual arts and fashion, currencies and complex financial institutions, equally complex bureaucratic structures, and vast external networks of information processing humming away in server farms—likely goes a long way in explaining our world-transforming capacities.

Along with an almost uniquely human proficiency for imitating behavior, learning, and teaching, languages provide for the “ratchet” of human culture—one that allows novel behaviors, techniques, and beliefs to build on top of the ones that came before. As with our use of fire, there is little else to compare in the animal kingdom to our human powers of social learning and imitation.

“When you compare chimp infants to human infants, human infants engage their caretakers in a way that’s really different than chimps do, and that almost certainly has something to do with our altriciality,” Herzog says, referring to the unique helplessness of human infants and the unusual degree of parental care that needs to be invested in their younger years. “Chimp youngsters in the wild, they learn to feed on their own relatively quickly, and they don’t need to depend on others for so long. And we’re just set helpless wimps.” Humans need to learn from others first before we can make our way in the world, to an extent that is truly unusual in the animal world. This makes human culture additive, and increasingly able to manipulate the physical world around it to promote its humans’ success—and on a timescale that leaves genetic evolution in the dust. You need only look at your phone and then try to explain how it works for a visceral demonstration of the phenomenon. Just as barons of the late medieval era couldn’t have produced the COVID-19 mRNA vaccine, the earlier Sumerians would have been utterly incapable of laying siege to and sacking those same lords’ polygonal bastion forts, the premier technology of fifteenth-century Europe. So, too, spearpoints knapped in the cutting-edge

“Levallois” fashion of the middle Paleolithic would have been similarly ineffectual against those same conquering armies of ancient Sumer, or against an F-16 for that matter. Little had changed biologically about the humans in these examples; these were all *Homo sapiens*. What changed was culture. This is what makes human societies extraordinary.

“No single human being knows how to build a modern computer from scratch,” write Paul Smaldino, an information scientist, and Peter Richerson, a biologist.

> Indeed, no one knows how to build a computer mouse, or a lead pencil, or many of the complex tools we rely on for modern living from scratch. For that matter, hardly any of us know how to make simple two-strand twisted string from local raw materials, something that practically every adult once knew how to do …. It is often said that the human brain is like a computer. It processes information, takes in input, produces output, stores and retrieves memory. Groups of people, then, are like supercomputers. They can process in parallel, allocate resources, and divide tasks to produce faster and better solutions to problems than lone individuals can manage, and that are more than the sum of the individuals’ abilities were they working separately …. If groups of humans are like a supercomputer, then what of human cultures, which store and process the cumulative innovations and collaborations of generations of individuals? Cumulative culture allows human societies to act as super-duper computers.

This human cultural ratchet, this super-duper computer, exponentially augmented in power and scope by information technologies like the printing press, has been torqued quite a bit since the start of our current interglacial. Today, yoked as it is to the dissipation of hundreds of exajoules of geologic energy, and organized by symbolic systems like the global financial network, which now entrains legions of people in an abstract web of information, seemingly carrying the world along on its own autonomous logic, it is making the entire planet unrecognizable on a decadal timescale.

But this ratchet was already turning in the earliest days of our species. By 300,000 years ago—and after countless Pleistocene climate paroxysms—the strange primates had become far stranger still, and left behind whispers of an elaborate interior life, like the processed ochre pigments that might have stained their faces red. Primitive hand axes gave way to knapped obsidian points[fn3] from afar—the first fruits of long-distance trade—goods that they now deftly hafted to wooden spears. Perhaps most consequentially, though, hearths and fire pits became a reliable fixture of this human world. While our human ancestors might have been taming fires for well over a million years by this point, controlled fires in fire pits now raced across the human world, from rock shelters in South Africa to Neanderthal hearths in northern France. We started cooking in earnest, throwing tubers in the hearth that themselves had evolved to live underground to avoid the fires above. But this plant adaptation was no match for human ingenuity. Leiden University archeologist Katharine MacDonald called this spread of fire the "first clear evidence of the emergence of cultural diffusion in the evolution of humankind." In some places, like South Africa's Border Cave, these fires burned on and off continually for over 230,000 years.

As we get nearer the present, the symbolic behaviors among these humans appear to flourish, as evidenced by elaborate shell jewelry, sculpture, cave paintings, and face paint. Perhaps this efflorescence of culture was a product of our own success: as population density in Africa increased, and groups of humans began encountering each other, trading ideas and goods, we increasingly had to distinguish ourselves with unique cultural markers, rituals, and beliefs.

In the life-threatening climate swings of the Pleistocene, culture had literally sustained humans. With mastery over fire, ever-swelling brains, the willingness to collaborate with non-kin, the massive, horizontal information transfers enabled by language, the unique propensity toward social learning in our line, and the ratcheting of cultures that piled innovation on top of innovation—from foraging knowledge to increasingly lethal hunting techniques—humans set about transforming the Earth, long before anything like the written word was ever etched into clay tablets. And long before monumental architecture ever sprang from the margins of alluvial soils. Perhaps, then, it shouldn't come as a surprise that the carbon cycle and Earth's climate likely started changing at human hands long before the inaugural puff of the steam engine.

AT THE END OF THE ICE AGES, HUMANS WOULD CATASTROPHICALLY collide with the rest of the biosphere. As the Pleistocene drew on, and as the ice ages waxed and waned, that biosphere came to include huge herbivores who could turn solar-powered grass to flesh, and lithe predators who tore that flesh from them, dissipating it back to the environment as waste heat and CO_2. Plodding mammoths and woolly rhinos, fleet-footed saber-toothed cats and lions. In the sea came the biggest animals in all of Earth history: 100-foot baleen whales that still carried tiny vestigial leg bones stranded in thick jackets of blubber, to remind them of their origins on land in a sweltering high-CO_2 world long since passed. Just like human brains, the sheer size of the baleen whale and the huge stores of fat they now carried (energy-dense layers of organic carbon) were an adaptation to the wild and unpredictable climate of the ice ages, enabling transoceanic voyages and providing a bank of energy to draw down in times of want. The Pleistocene had selected for whales to take this energetic adaptation—getting big—to its most extreme conclusion, with blue whales today potentially representing the largest such animal possible, even in theory.[fn4] But while whales would one day have a catastrophic rendezvous with humans in the last second of Earth history, enduring one of the most unconscionable slaughters in the entire age of animal life in the nineteenth and twentieth centuries, the fire creatures had already proven themselves to be extraordinarily lethal in the tens of thousands of years before.

Today we conceive of Africa as the continent of big animals. But the very fact that that continent's modern menagerie strikes us as so uniquely impressive shows just how much disappeared in the waves of extinction that circled the Earth near the end of the last ice age. That megafauna didn't disappear completely in Africa, unlike in much of the rest of the world, likely owes to these African animals' long evolution beside the dangerous and cunning new social fire creature—unlike the unfortunately naïve mammoths, mastodons, giant ground sloths, giant wombats, giant lemurs, giant elk, elephant birds, moas, and countless other creatures that fell away upon their introduction to our kind. But even this seeming survival of the African megafauna to our day is deceiving. By 125,000 years ago—and before our extinctions had gone global—mammals in Africa were only 50 percent as large as those in Eurasia and the Americas at the time (the megafauna of these vast continents still untouched by our kind) while

Africa had already lost elephants, hippos, and monkeys even larger than their modern counterparts. Also, the extinction rate of large carnivores on the continent tracks the brain volume increase of hominins over the past few million years. In fact, one theory holds that our brain grew in response to these disappearing animals: as humans had to hunt smaller, more evasive prey, it required getting cleverer.

As humans spread out into new landmasses over the next few tens of thousands of years, this onslaught was unrelenting. Staggered waves of extinctions, not lining up with any particular climate event, eerily shadowed the heroic migrations of humans around the Earth: Australia and Eurasia first, where, already by the height of the last ice age, the sparsely forested landscape had been seemingly transformed by the human use of fire—and the Americas next, before hopscotching across remote Pacific islands and to Madagascar in historical times, where titanic elephant birds and gorilla-sized lemurs disappeared not long after human arrival. While archeologists might still puzzle over this asynchronous loss of megafauna around the planet near the end of the Pleistocene and bleeding into the Holocene, chasing an elusive criterion that apparently isn't satisfied by literal houses made out of mammoth bones, headdresses fashioned from animal skulls, or massive caches of megafauna meat left unused by groups of unprecedentedly lethal, spearpoint-wielding hunter-gatherers, the case is all but closed to the vast majority of paleontologists accustomed to a deeper time perspective.

"Mammals were able to track their climate with no problem, until they go extinct," says University of Nebraska–Lincoln paleontologist Kate Lyons, dismissing the idea that it instead might have been the very last, and not particularly anomalous, climate swing of the Pleistocene that had such a globally catastrophic effect on large land mammals. "If you compare anything about range shifts for mammals that went extinct versus mammals that didn't go extinct prior to the extinction, there's no difference," she says. "Some of the past deglaciation climate changes were larger in magnitude than this one, so there's nothing different about this one other than the arrival of humans. Were it not for the additional pressure humans were putting on these organisms, they wouldn't have gone extinct."

To understand this perspective, we need to survey the truly exceptional nature of the die-off of large land animals in the past few tens of thousands of years—the most devastating since the dinosaur catastrophe 66 million years earlier, and one that would be completely bewildering in the absence of humans. For one thing, though the climate swings of the most recent ice age of the Pleistocene and its concluding chaotic warming leap into our current interglacial[fn5] were certainly unprecedented in the context of *recorded* history, they were not at all exceptional for the wild vicissitudes of the Pleistocene (then over 2.5 million years in progress) and were similar to the dozens of glacials and deglacials that came before, which notably caused virtually zero extinctions. In fact, in one interglacial 400,000 years ago, it seems that much of Greenland completely melted at temperatures similar to our warming planet today (an eerie fact gleaned by scientists studying a rock core retrieved during a failed 1950s US project to shuttle nuclear weapons across the frozen landmass in a network of ice tunnels). And in the previous warm interglacial period 120,000 years ago, the last one before our own, the planet was even slightly *warmer* than today. Yet the world's entire suite of megafauna sailed through these spasms utterly unscathed, just as they had in the many interglacials before.

In Australia, meanwhile, where everything over 100 kilograms went extinct around 40,000 years ago, in the *middle* of the last glacial period, there isn't even a putative climate event to point to—though, tellingly, this extinction does follow the arrival of humans. Extinctions like that of the mammoths in North America also precede the vegetation changes sometimes invoked to explain them. For others, like the mastodon, their beloved spruce forests were spreading as they vanished from the Earth. The fact that devastating megafauna extinctions were sweeping equatorial South America at the same time that these cold-weather elephants were disappearing from the Arctic is similarly difficult to reconcile with a unifying climate culprit—oddly enough, though, they do occur at the exact same time that fluted projectile points join the South American fossil record. And as Lyons, along with the University of New Mexico paleontologist Felisa Smith and colleagues, has demonstrated, the sporadic wave of extinctions around the planet in this period has a selective signature completely unlike any other extinction in Earth history, almost exclusively hitting big land mammals. Unlike other mass extinctions, which are known mostly from the marine fossil record, this one all but avoided the ocean altogether.

"If you look at the selectivity of the extinctions—and even starting 125,000 years ago as *Homo sapiens* was migrating out of Africa—what you actually see is that every time *Homo sapiens* arrive, the average body size of mammals on that landmass drops, usually by an order of magnitude or more," says Lyons. "Going into an area and killing large-bodied mammals and driving them to extinction actually seems to be a signature of our species."

It is a trend that would at no point, until the current day, relent.

Along with the more ancient but accelerating renovation of the landscape with fire, this mass die-off might have marked one of the earliest detectable human influences on the climate, changing the very structure of how carbon moved throughout the terrestrial biosphere.

"There was a dramatic change in the way energy flowed through the global ecosystem," writes the paleontologist Anthony Barnosky about the megafauna extinction that took place over tens of thousands of years. "The energy that powers ecosystems is derived from solar radiation, which is converted to biomass …. After the extinction, increasing amounts and proportions of energy began to flow toward a single megafauna species, humans."

The sheer loss of the biomass of the missing animals alone might have amounted to some 5 million tons of carbon. But animals aren't monads that can be individually isolated from the biosphere—they inextricably flow from the environment and feed back on it as well, shuttling energy and nutrients around the Earth, and structure their own living world. On our modern world, for instance, in the smoking crater of industrial whaling, orca have recently, and quite unusually, taken to hunting sea otters, as the seas have been emptied of the juvenile whale prey they once tracked down. The subsequent loss of otters,[fn6] in turn, has contributed to population explosions of the otters' prized sea urchin prey, which then in turn has had the knock-on effect of devastating kelp forests in the Pacific (which the urchins devour), an important carbon sink in the ocean. It's known as an ecological cascade. Similarly, some of the extinctions and reorganizations of the biosphere at the end of the Pleistocene likely stemmed from such unpredictable network collapses. At the dawn of the Holocene, the mammoth steppe—a vast, unbroken strip of herbaceous grassland that spanned the breadth of Eurasia, extending across Beringia and beyond—was suddenly replaced by shrubs when its namesake grazers disappeared. This new, scrubbier landscape sequestered less carbon in the soils than the mammoth-trampled grasslands it supplanted, with the balance of CO_2

returning to the atmosphere. This scrubland would have also made the land surface darker and absorbed more sunlight—with both effects contributing to warming. The extinctions, then, reshaped these ecosystems, the cycling of carbon through the biosphere, and even the color of this Pleistocene surface world.

But to give some context for how extreme our current industrial experiment on the climate is, even with this sudden prehistoric ecological catastrophe, the planet only warmed by perhaps a tenth of a degree Celsius according to climate simulations, with local anomalies of half a degree—of *warming* in the case of northeast Asia, where the mammoths vanished, but *cooling* in the ocean around Antarctica, due to the Rube Goldberg teleconnections between the climate and ocean circulation. Still, as Stanford ecologist Christopher Doughty and colleagues write, "the human influence on climate began even earlier than previously believed," and so "the onset of the Anthropocene should be extended back many thousand years."

AS THE PLANET EMBARKED AT LAST ON ITS MOST RECENT ESCAPE from the glacial cold over 12,000 years ago, we finally approach all of recorded history—the last few seconds of December 31 on the geologic calendar of Earth history. This launch into our current interglacial, and the climate that gave birth to our modern world, was what's modestly characterized as a "stepwise" transition, rather than a gradual one. Though the waxing and waning of sunlight in the hemispheres over thousands of years (due to subtle changes in the Earth's orbit and tilt) follows the gradual slopes of a sine wave, the ice ages of the Pleistocene end with all the staccato jaggedness of an EKG—thanks in large part to feedbacks in the carbon cycle. The ice sheets of the Northern Hemisphere lost their grip, as they had dozens of times before, and darker land around the melting margins, exposed to the sun for the first time in 100,000 years, rapidly warmed and hastened the ice's retreat. Permafrost melted and methane bubbled up from thawing bogs. Colder, more CO_2-soluble oceans warmed and gave up the carbon they'd stolen during the ice age back to the air, warming the Earth even more. The planet had passed a threshold: plummeting albedo, soaring CO_2 and a reordered ocean had now set the Earth on an inevitable leap from one state to another, and the ice sheets ran for their lives. As CO_2 rose from a nadir of 180 ppm at the depth of the last ice age 20,000 years ago to about 280 ppm at the dawn of the Holocene some 12,000 years ago, the planet responded not with a linear climb out of the cold, but rather with quantum leaps of climate change and sea-level rise. Coral reefs marking the ancient sea level—but drowned today deep off Tahiti and Indonesia—reveal that midway through this warming, the seas suddenly jumped 50 feet or so in only a few centuries, as meltwater from the late, great North American ice sheet raged down the Mississippi River. Ultimately these kinds of jumps would contribute to what would become more than 400 feet of sea-level rise in the span of a few millennia.

This crumbling ice world often erupted in violence. As the North American ice sheet began to retreat, an ice dam blocked a river in Idaho and impounded a gigantic glacial lake six times the volume of Lake Erie. Then the ice dam suddenly collapsed, and so did the massive lake, releasing ten times the flow of all the rivers on Earth through eastern Washington State, carving out the Channeled Scablands, drowning mammoths, and leaving behind ripple marks so huge they can only be recognized as such from aerial photography. When it was all over, the ice dam re-formed, blocked the same river, impounded the same gigantic glacial lake, and collapsed again. This unimaginable cataclysm repeated itself, over and over perhaps sixty times, before the ice retreated over the horizon, and the last ice age finally came to a close.

BUT A FEW THOUSAND YEARS INTO THIS RAPID THAW, JUST AS THE planet had nearly escaped its ice-age clutches—and wandering humans had temporarily settled in sedentary enclaves in Japan and Ukraine—much of the planet suddenly flipped back again. Twelve thousand eight hundred years ago, the Northern Hemisphere plunged into a brief encore of the deepest Pleistocene. It was a stunning demonstration of just how fast the climate can flip to a different state. Forests that had been advancing on Europe for 2,000 years were banished by tundra as the temperature dropped 5°C. The culprit was the gigantic puddle of meltwater that had been pouring off the crumbling North American ice sheets across central Canada as the world warmed but was impounded in its flow to the sea by the ice. When the ice finally burst and emptied out into the Arctic, this flood of fresh meltwater shut down the overturning churn of the North Atlantic and arrested the seaborne flow of heat to the North. Much of the world shivered for a thousand years. But when the system finally rebooted and ocean currents carried the equator’s heat toward the Arctic once more, temperatures in Greenland leaped 10°C in perhaps a decade, fires spread, and revanchist forests reclaimed Europe for good. The Holocene had finally begun. While this all sounds dramatic—and it certainly was—it had happened many times before, throughout the dozens of previous ice ages and interglacials.

We've been strangely spared these kinds of climate spasms in recorded history. But today, with CO_2 at the unthinkable level of the Pliocene, last seen more than 3 million years ago (and threatening to climb far higher), and with Greenland now hemorrhaging meltwater into the North Atlantic again, we dare this wild planet to return. "The climate system is an angry beast," the late Columbia University climate scientist Wally Broecker was fond of saying, "and we are poking it with sticks."

By the time the planet had finally thawed, and this angry beast lay down again, for a short, millennia-long nap, there were modern humans on every continent but Antarctica, numbering some 2 to 4 million strong. The motley crew of other hominins with which they had long shared this planet were all gone, as was most of the world's megafauna—commemorated now in that most human of artifacts, art. From ochre paintings of giant ground sloths on rock shelter walls in the Colombian Amazon, to charcoal sketches of steppe bison in the caves of Altamira, this bygone menagerie lives on. Still, the changes coming to this world would soon dwarf these transformations near the end of the last glacial age. Almost as soon as the last ice age ended, for some reason agriculture began, independently, in several places across the world, on almost every continent—seemingly after hundreds of thousands of years of human history without it. The energy unlocked by this development would open up the range of possibilities of human organization to include some with the motivation, and eventually the ability, to take over the world. These increasingly energy-intensive political economies ultimately unlocked by this technology would, within only a few millennia, and in a story that has carried us billions of years so far, catastrophically reintroduce the surface world to the vast seas of sunlight stored in organic carbon that had been accumulating under the Earth for hundreds of millions of years—and reroute the course of Earth history.

"NOW I GUESS I'LL JUST ASK A CRAZY QUESTION THAT JUST OCCURRED to me," I blurted to Nicole Herzog as we wrapped up our discussion of human evolution. "Given what we've talked about—the sort of need to innovate when the environment deteriorates, or when you're in some marginal condition and threatened with extinction—and the fact that the climate is currently in the process of being completely deranged at an almost unprecedented pace, and that everything is going to look different for thousands of years, do you have any speculation about whether it's going to be sort of a primer for innovation and we'll adapt, or are we just going to collapse?"

I expected a polite demurral in response.

"I don't want to be a pessimist, but I think that there will be some level of collapse first," she said. "Because I think you really have to be pushed to the edge to really invest in groundbreaking innovation."

When our ancestors made evolutionary leaps in cognition, behavior, and social organization, and gained ever more refined control of fire, it likely wasn't by plebiscite or out of simple curiosity. It was because our lineage was threatened with extinction.

"You've got to be pushed there," Herzog continued, "and I feel that people aren't really motivated enough yet. I don't like to say it out loud, and in pleasant company. People are like, 'Oh my God, that's such a doomer approach.' And it's like, 'Well, we can find our way out of it. It's just probably going to be terrible.'"

8

The Long Fuse

SOMEWHERE NEAR COLBY, IN WESTERN KANSAS, I FELL ASLEEP. IT'S ONE of the biggest risks of driving on the Great Plains, along with apocalyptic summer hailstorms and Siberian whiteouts in the winter. There's a narcotic monotony to the seas of corn, aligned in neat rows, themselves carved into repeating squares on the gently heaving plains, the tempo kept by bobbing oil pumpjacks, and a horizon limned by forests of lazily twirling 3-megawatt wind turbines. Succumbing to the languid rhythms of the scenery, I began to nod off. Suddenly, though, the rumble strip shook me out of my stupor and into something like an epiphany. I sat up, realizing how extraordinary this landscape was—and how different it was from any landscape that had ever existed in all of Earth history. It was all energy. Everything I could see in my entire panoramic field of view, from horizon to horizon, and uncountable horizons beyond, had been entirely engineered to capture and generate as much energy as possible, and all of that energy was destined for one species alone, funneled into the ever-swelling human metabolism.

The farms themselves were hugely subsidized by geologic reserves of photosynthetic energy as well—powering arching central-pivot irrigators. The crops were fed with fertilizer synthesized from natural gas and harvested by gargantuan diesel-powered combines, the megatons of grain delivered to the far corners of the globe in diesel-powered train cars that ran alongside these diesel truck–dotted highways of asphalt (made from petroleum) and propelled over the ocean toward still-more-distant human mouths by marine bunker fuel. 1 KANSAS FARMER FEEDS MORE THAN 155

PEOPLE, read a billboard. I shook my head in disbelief—it's actually an astounding fact, I acknowledged in my moment of revelation. It's difficult to imagine how any more energy could possibly be captured from this landscape. This energetic landscape wasn't just an exception from the natural order, but an exception to most of human history as well.

As night fell on my drive, the enchantment failed to wear off. The whole horizon now intermittently flashed red like a disembodied city, warning planes burning jet fuel going hundreds of miles an hour not to hit the wind farms that captured still more energy, sent in high-voltage transmission lines that dangled above silent sheaves of corn, which themselves swayed above an unseen spidering network of natural gas pipelines buried in the Earth below. A geologic eyeblink ago, at the close of the Pleistocene, *all* of the energy that powered the scattered mobile bands of humans came from gathering the shifting wild resources on a changeable ice-age planet. But today, and even leaving the planetary discharge of fossil energy aside, an astonishing third of all the world's photosynthetic biomass on land now flows through human society. As a result, landscapes like this swathe the Earth, with domesticated plants alone channeling 800 *gigawatts* of metabolic energy—or the equivalent of 800 nuclear power plants—into one species alone: a primate whose numbers, after barely budging for the hundreds of thousands of years, have multiplied by over 1,300 times in the past few millennia. Even stranger, where one historian notes that "much of the world's population since the emergence of agriculture has been devoted to the production of food," today a vanishing percentage of the population works in agriculture, even as it erupts with this world-historical superabundance of calories. How on Earth did this happen?

ALMOST THE MOMENT THE LAST ICE AGE ENDED, OVER 11,000 YEARS ago, people in the Jordan Valley started stockpiling wild cereals in purpose-built granaries—signaling both a new rootedness to the landscape and the new abundance of the Holocene. And with the last vestiges of the ice sheets still retreating in the Far North 9,000 years ago, plant domestication was already taking hold in the Fertile Crescent (wheat and barley), China (rice), Mexico (maize and squash), the Andes (potatoes), and Amazonia (manioc). For the first time in history, people around the world were directly producing their food, rather than simply mapping onto an ever-shifting landscape where it could be foraged. Communities sprang up at the mouths of rivers, though farming was strikingly absent in some of the first sedentary villages. Instead these Holocene pioneers plucked shellfish, fish, migratory birds, and herds of herbivores from endlessly stocked wetlands in the more hospitable climate of a new age. Others gathered by the thousands in decentralized proto-cities like Çatalhöyük in Turkey, with plant cultivation and farming serving merely as a supplementary source of calories in a dietary repertoire that drew generously from the wild bounties of hunting, foraging, and fishing.

Strikingly, while only 5 percent of Çatalhöyük has been excavated, there is little evidence of palaces, elite graves, and other markers of hierarchy typically associated with the agrarian societies of the past 5,000 years, from Mexico to China. Lest we imagine this to be some sort of communal Eden of farming before the Fall, though, a quarter of the skulls at Çatalhöyük were recently found to be fractured (from blows to the back of the head, and mostly to women) as the population grew. “Prehistoric warfare was pervasive, constant, and deadly,” writes Harvard archeologist Steven LeBlanc. “There is ample evidence this applies to tribal farmers and to complex hunter-gatherers.”

Along with the apparent depletion of the local environment over the site’s history, and incipient signs of disease in these skeletons—produced by both an increasingly grain-based diet and denser, more crowded settlements—these wounds signaled the novel challenges that would accompany humanity’s own success in the coming millennia, as this kind of experimental farming gave way to vast and powerful grain-powered polities.

That these features of our modern world, sedentism and early farming, began almost immediately after the termination of the last ice age, after hundreds of thousands of years without them, was not likely a coincidence. The Holocene—the interglacial period still ongoing today, one that began some 11,700 years ago—has been almost miraculously stable, more so than any climate window in at least the last 650,000 years, with temperatures eerily flatlining for all of recorded history until the shocking, unfinished warming spike that began this past century. Even within this remarkably stable climate, though, relatively small regional fluctuations from one year to the next have been sufficient to bring ruin to societies that came to depend on domesticated crops—like the localized droughts that would help bring down countless agricultural empires in historic times, from the

Akkadians to the Classic Maya, and possibly even the entire Bronze Age world of the Mediterranean. But these blips have nothing on the vicious climate upheavals of the Pleistocene world that came before—routine snaps that would have dwarfed these minuscule Holocene misadventures, with the entire planet sometimes seesawing in mere decades.

Maybe this prior volatility explains the sudden arrival of agriculture at the dawn of the much tamer Holocene. With the climate temporarily pacified, great settlements could finally crystallize on the settled landscapes of a more stable world. Being rooted to a place is impossible if your meadow-ringed hut village could be buried under a sea of sand the next year, much less month. The former ice-age world, then, had been no place to stay put for millennia, or try one's hand at farming. Abortive experiments cultivating cereals deep in the ice age, like the Ohalo II settlement in modern Israel 23,000 years ago, are the exceptions that prove the rule. Instead, for all of human history, people tracked the movements of fickle wild resources as they shifted across the face of an irascible planet, opportunistically harvesting wild cereal grains, perhaps, but not starting generational farmsteads. Energy was limited in this world, and perennial mobility was the rule in the Pleistocene, with the wild ice-age climate acting as a centrifuge that would have torn apart incipient societies that had the gall to stay put, in the hopes next year might look like the one that came before.

More provocatively, though, it might have been the extremely low level of CO_2 *itself* in the former ice age that delayed farming until its sudden appearance at the termination of the Pleistocene,[fn1] when CO_2 leaped to its preindustrial level of 280 ppm. This is because, it turns out, the common climate-denying talking point that plants "like" CO_2 is true. Obviously it is. The whole living world is made of CO_2 fixed by photosynthesis.[fn2] That plants grow more under higher CO_2 is called "CO_2 fertilization," and one study estimates that the effect has already boosted "global, annual levels of photosynthesis by around 12% between 1981 and 2020." But like all climate half-truths, this cherry-picked phenomenon neglects to account for some salient countervailing facts: while our future climate will feature higher CO_2 which plants *do* like, it will also feature increased droughts punctuated by punishing rainfall, and increasingly high temperatures for plants that, like us, evolved on a cold, low-CO_2 Pleistocene world. As a result, crop yields will become increasingly unpredictable in the coming decades, breadbaskets increasingly unreliable, and maize production alone down by almost a quarter by the end of the century, even with the spur of CO_2 fertilization.[fn3]

Nevertheless, each year 40 percent of the CO_2 in the atmosphere enters plants through the pores on their leaves, where the chlorophyll complex miraculously pushes carbon dioxide chemistry uphill with solar power to make the high-energy organic compounds that underwrite, and dissipate through, the rest of the entire terrestrial biosphere. And under higher CO_2 plants can do this trick with *fewer* pores. This is something they desperately want to do, since while plants need to breathe, they also need to conserve water, which they lose through their pores to the air through evaporation. In the Amazon this water loss can be literally seen in the weather, as clouds wisp off the rain forest canopy. But in more arid environments this water loss can be deadly for plants. So, all things being equal, they will try to minimize the number of pores on their leaves. This mechanism is one reason why we have a decent idea of what ancient CO_2 levels were like: when CO_2 was higher, fossil leaves could get by with far fewer pores. And even in our own time, some surveys have revealed that, since the start of the Industrial Revolution, plants have already reduced the density of pores on their leaves by an astounding 34 percent, an ongoing planet-wide biological change with untold effects on the global water cycle.[fn4]

The upshot, though, is that when CO_2 is higher, plants can photosynthesize more easily. It shouldn't surprise us, then, that in the high-CO_2 dinosaur-haunted world of the Cretaceous Period 100 million years ago, there were forests all over the Earth. On the other hand, at the last glacial maximum, only 20,000 years ago, when CO_2 was just about as low as it's ever been in the entire geologic history of the Earth—on a shrubby, dust-tossed world with far less rainfall—vegetation was stressed, and agriculture might have been a more precarious, if not impossible, proposition. Tellingly, of the so-called founder crops that would eventually underwrite agricultural civilizations across the world—and that today support the existence of 8 billion people—wheat, barley, rice, potatoes, and yams all use the much older pathway of photosynthesis,[fn5] less adapted to the drier, low-CO_2 world of the Pleistocene. Agricultural civilizations as we know them, then, perched as they are atop a ziggurat of photosynthesis, might have been impossible to construct on the extremely low-CO_2 world that existed at the depths of the last ice age. As the historian Ian Morris writes, mobile hunter-gatherers and roaming pastoralists have occasionally built impressive megalithic sites like Stonehenge or Göbekli Tepe, but the astonishing monumental cities of antiquity were only possible through a commitment to sedentary agriculture: "Farmers were not the only people able to organize enough labor to move great quantities of earth and stone … but … the scale on which farmers operated was orders of magnitude greater than that of foragers." The difference was ultimately a matter of how much energy these societies could call up from the landscape.

This is because, like all animals, human populations are existentially supported by an energetic foundation of photosynthesis, which we respire and turn back to CO_2 to do the work of life—whether by eating plants directly or by eating animals that can transform the plants we can't digest into forms of fuel we can, like meat, milk, and fish. But there are hard energetic limits to life on Earth. Despite the unending bounty from the sun, less than 1 percent of the radiation that washes over the Earth is transformed by photosynthesis into the energy in plants, which power every ecosystem on the surface of the Earth. But only 5–10 percent of those plants is consumed by herbivores, and only 1–2 percent of that is subsequently turned into these animals' flesh. The animals that eat those herbivores, though, can only recover 10 percent of the energy embodied in the herbivores, and this is true for each step up the food chain, with only 10 percent of the energy making it up to the next level—the rest lost to entropy along the way. As we've established, nothing can be done without energy, so by the time one gets to the tapering peak of the food pyramid—the realm of the carnivores—one finds that they are far fewer in number than the herbivores on which they feed, just as the herbivores in turn are embarrassed by the planet's uncountable abundance of plants. The original solar energy that supports this entire ecosystem is rapidly lost as it gets passed up the food chain, and when it can't diminish any further, the food chain ends.

Human populations were fundamentally constrained by these ecosystem energetics for their entire evolutionary history on this planet; a fraction of a fraction of a fraction. These are the limits of the so-called organic economy, which in turn would strictly delimit the possibilities for human societies, the ambitions of which were unforgivingly circumscribed by the amount of energy they could extract from the biological reservoir of carbon on the Earth's surface. But over the course of the Holocene, intensive agriculture increasingly turned as much of the landscape as possible into human food, and more humans, decoupling the population from the variability of wild resources. "Instead of competing with other animals for edible plants and with local predators for the grazing animals that could provide meat, the whole product of enormous tracts of land was reserved exclusively for human use," writes the British historical demographer Tony Wrigley. And when there is a surplus of calories, the population can grow. Where foraging could support 0.02 people per hectare, the earliest grain states of Mesopotamia, Egypt, and China—which squeezed far more calories out of the ground—could support 1 person per hectare, a 500-fold increase. As a result, by the dawn of the first millennium, the global population had swelled from some 6 million to over 200 million, thanks to farming and the vastly augmented base of edible photosynthesis it provided. Today, supported by massive energy inputs of fossil fuels, drenched in synthetic fertilizer made from natural gas, and sprinkled with phosphate rock mined by diesel-powered machinery on the other side of the world, that agricultural productivity tops 10 people per hectare, and the global population grows by a billion people every decade and a half as a result. Quite simply, sedentary, intensive grain agriculture unleashed a torrent of photosynthetic power, exponentially swelling the carrying capacity of the Earth. And human population growth, after hardly budging for hundreds of thousands if not millions of years, suddenly rocketed upward by more than three orders of magnitude in our ongoing interglacial, the Holocene.

From this perspective, then, it is a simple story: if agriculture was impossible in the volatile low-CO_2 atmosphere of the ice ages, perhaps this is why we don't see rusting skyscrapers in the archeological record of the Paleolithic.

OR PERHAPS WE DON'T SEE THEM BECAUSE OUR MODERN WORLD wasn't inevitable. Even after the domestication of crops at the beginning of the Holocene, *thousands* of years would pass before they would come to dominate the diets of their cultivators, or before the rise of what we would consider agricultural "states."[fn6] During these thousands of years, farming existed, yet decentralized groups of early agriculturalists explored every possible riff on political organization—from trophy-head-collecting warlords, to communal villages of flood-recession farmers, to Burning Man–like seasonal congregations of farmer-foragers. Agriculture merely played a supplemental role in these societies' diets, with wild game, fish, and plants supplying much of their calories. In more sparsely inhabited North America,[fn7] this mosaic of farming and foraging and flexible, seasonally mobile ways of life persisted in a patchwork of complex societies that stretched across much of the continent until the apocalyptic arrival of Western Europeans. In fact, the idea that anyone on Earth would ever decide to double down on intensive grain agriculture at first seems curious: the bones of early farmers were smaller, lighter, and more susceptible to breaking than those of contemporary hunter-gatherers. Their diets were far less varied, while anemia, dental disease, and repetitive stress injuries were common. And intensive grain farming took far more menial labor and drudgery than did foraging on the verdant, game-trampled, fish-stocked river deltas of the early Holocene. In this light, it's difficult to see the draw of intensive grain farming at all, much less the later stratified, bureaucratic, agricultural state of antiquity, studded with self-aggrandizing monumental architecture and often seen as the harbinger of "civilization"—with Uruk in modern Iraq providing the prototype some 5,000 years ago. This was just one possible configuration, and that of rather restricted political imagination. Perhaps the rise of these kinds of agrarian empires, which would ultimately circumscribe the entire world, wasn't inevitable.

The more verdant and warmer Holocene was a veritable Garden of Eden for foragers. But this planetary bloom only masked the continuing trend, carried over from the ice age, of humans hunting down the food chain, driving smaller and smaller animals to extirpation. Meanwhile, in places like Southwest Asia, permanent settlements began to sprout on the landscape of the more lush Holocene, a way of life that opened up new demographic possibilities for humanity. "Mothers no longer had to carry babies with them from place to place," explains the historian Robert Allen. "Child rearing, in other words, became cheaper." Populations swelled. As they did, people were forced into marginal areas and made increasing demands of a quickly overtaxed ecosystem that couldn't meet them.

For most of our species history, the energetic payoff to hunting gigantic megafauna would have dwarfed that of agriculture. But now the energetic returns to dedicated farming began to look rather more attractive, and sedentary agriculture grew only more entrenched, intensified, and productive as the pressure on the landscape increased. In central Thailand, wild jungle fowl would come to pick at the rice seeds that spilled off the edge of this strange new ecosystem and soon became intimately swept up in this human world as well: chickens. Across Eurasia, animals like goats and cows were coaxed down from the hinterlands and welcomed into human society, turning inedible plants on the landscape into still more food: dairy and meat. Another energy buffer. But this wasn't the only energetic windfall produced by this kind of cross-species compact. Around 4000 BCE, in Eurasia, farmers yoked oxen to plows. As animal muscle came to replace human muscle in the fields of Eurasia and Africa, it dramatically increased the returns to farm work by outsourcing it to far more powerful livestock (100 watts of human labor versus 400 watts of animal power). The energy yield of fields and rice terraces under cultivation would swell as the ratchet of culture ceaselessly turned. The growing surpluses of photosynthetic energy produced by farming, made possible by the accumulation of culture, would open up vast new possibilities for society.

States would emerge to organize these flows of energy. Around the same time in Mesopotamia that fields began to be worked by beasts of burden and populations swelled, the monumental bureaucratic societies of antiquity appeared on the land—along with every ill associated with them, from chattel slavery and mass human sacrifice to bloodthirsty wars of conquest. While there exist tantalizing examples of early, complex, urban agricultural societies that seem shockingly egalitarian in comparison—like Mohenjo-Daro and Teotihuacan—these remained the exceptions to the rule. And though it might have taken thousands of years for agriculture to be organized into states like these, from a geologic perspective it took place in only an eyeblink after the last ice age ended and the climate settled down for a moment. The explosion of complex, hierarchical societies in Mesopotamia, North Africa, and China—and that would eventually blossom in the Americas as well, with the rise of powerful stratified polities like the Aztec and Inca, Mississippian culture, and farther afield in Hawaii—was all but instantaneous for a species that had gone hundreds of thousands of years without either agriculture or states. Perhaps there *was* something inevitable in this trajectory, after all. On the sparsely inhabited world of the early Holocene, well-fed foragers drawing from the most productive ecosystems and turning to farming only as a kind of hobby could casually ignore the demands of an aspiring autocrat, but that freedom diminished as humans transformed the world.

"Whenever the populace found it easier to do without a particular ruler—say by moving to a new location—then rulers felt compelled to govern more consensually," the New York University political scientist David Stasavage writes. Increasingly, though, as farming and pastoralism boosted populations and made increasing demands on the landscape, they crowded in on seasonal hunting grounds patrolled by hunter-gatherers. And as the once endlessly stocked wilderness was depleted in the cradles of civilization, and growing numbers of farmers and pastoralists increasingly vied for access to natural resources, the exit options narrowed. This was especially the case when these environments were circumscribed by unforgiving natural barriers, pushing up against harsh deserts or oceans. "The idea that exit options influence hierarchy is, in fact, so general," Stasavage notes, "it also applies to species other than humans. Among species as diverse as ants, birds, and wasps, social organization tends to be less hierarchical when the costs of what biologists call 'dispersal' are low."

The morbid consequences of a lack of exit options might even be encoded in the genomes of the Old World. In the moments before the first agrarian empires erupted from the alluvial soils of southwest Eurasia, roughly between 5000 and 3000 BCE, a strange and extraordinary bottleneck in the diversity of male Y-chromosomes appears, a dramatic winnowing not reflected in the mitochondrial DNA passed down by females—a result that one group of researchers claims implies a spectacular eruption of violence, as warring patrilineal clans ruthlessly extinguished each other[fn8] in escalating kin-based struggles for diminishing resources that ultimately left one reproducing male for every seventeen females. Perhaps, then, the emergence of states was a response to diminishing returns in an age of population pressure and violent competition among scattered bands. There might have been benefits for these groups, long engaged in increasingly destructive massacres over fewer natural resources, to lay down their arms against each other under the pacifying aegis of the state, and to redirect their energies inward toward intensified agricultural production and outward toward other emerging societies.

In 2019, a paper with 111 coauthors, synthesizing the work of 250 archeologists, led by University of Maryland–Baltimore scientist Erle Ellis and then postdoctoral fellow Lucas Stephens, concluded that already by 3,000 years ago the world had been "largely transformed by hunter-gatherers, farmers, and pastoralists … considerably earlier than the dates in the land-use reconstructions commonly used by Earth scientists." Another study, led by Kate Lyons, indicates that habitat fragmentation from agriculture was already sweeping the world by 6000 BCE, disrupting the planet's ecosystem in a pattern that was entirely novel over the past *300 million years*, since the Carboniferous Period.

"Through time," the researchers conclude, "as land became increasingly densely occupied and land use more intensive, opportunities for flexibility in subsistence strategies and the resilience that this supported were reduced."

On the other hand, the emergent state was a formidable arrangement, a kind of engine that drew in the photosynthetic energy over a large territory toward an administrative center and provided a powerful mechanism for scaling up. But not all plants are up to the task of state-building. As the late Yale anthropologist James C. Scott argued, the domesticated cereal grains of Mesopotamia, where the earliest states arose, were uniquely suited to that area. Crops like emmer wheat, einkorn, barley, and rye could be reliably taxed and appropriated (arguably the state's foundational activities, ones that allowed it to scale in size and complexity). This was, in turn, a function of these plants' biology: the cereals were "visible, divisible, assessable, storable, transportable, and 'rationable,'" whereas other staple crops, prevalent elsewhere around the world, like tubers, don't ripen on a regular harvest schedule and can be left underground, evading the would-be tax man's watchful eye. Also unlike cereals, those crops spoil easier, hindering them from being stockpiled in state granaries or transported great distances—all qualities that foil the appropriating designs of a centralizing bureaucracy. With grain, however, the yearly harvest was visible and quantifiable to the state, as was the labor of its farming populace. As such, this labor could be coordinated by the state on a vast scale to build irrigation infrastructure that could call still more crops from the Earth, as well as granaries where they could be stockpiled and redistributed when the rains failed—a buffer against starvation that unleashed the huge agricultural surpluses necessary to unlock ever higher levels of population growth and sociopolitical complexity.

Still, the nascent state was a precarious experiment. No one had ever done it before. The earliest states were extremely unstable and prone to collapse before a new kind of social technology matured: bureaucracy. Bureaucracy might not strike us as a technology today, but early versions of these states that saw initial booms in population and territory without developing the administrative capacity to carry out society-wide campaigns of information gathering and processing on the populations and lands they ruled—monitoring agricultural production, conducting cadastral surveys, collecting taxes, effectively amassing resources and labor—were subject to quick disintegration. In some places, though, these exigencies would lead to the first writing systems, like cuneiform—or their symbolic equivalents, such as the Incan knot language of *quipu*—which were mainly used for bookkeeping and organizing the information flows of the state. In ancient Sumeria, for instance, where the earliest writing system developed, “The topics of the surviving tablets in order of frequency are barley (as rations and taxes), war captives, male and female slaves,” the late anthropologist David Graeber wrote. And Sumerian administrators managing vast industrial temple complexes enabled by the agricultural surplus invented a kind of money to keep track of these state resources—an information technology that took the form of silver shekels, each of which was equivalent to one bushel of barley. The first money, in other words, was a kind of unit of energy, captured by photosynthesis and embodied in organic carbon—a strange relationship that would be echoed through the age of petrodollars. Some of these shekels found their way into the local marketplaces, where they were used as a medium of exchange. And in these new commercial markets, the uniquely symbolic world that humans had inhabited since their origins would ultimately find wildly fertile ground in the artificial world of finance and someday transform the planet.

If weak bureaucracy posed a hurdle to early agrarian state-building, then disease posed a rather more lethal one. These new denser settlements—which packed people together with livestock—would have provided crucibles for epidemics, perhaps explaining the many mysterious state collapses in the early days, as plagues swept through these polities like a thresher through the grain harvest. And the more these settlements eventually came to depend on the calories from their fields, the same devastating effect was just as easily achieved by frequent agricultural blights or livestock disease. As such, these boom-and-bust early states, many of which often also featured brutal repression, slavery, and spectacular eruptions of human sacrifice and violence—from Zhengzhou to East St. Louis—would have been plainly unattractive to egalitarian farmers or foragers as long as exit options remained. Increasingly though, there was little escape.

ULTIMATELY, THE SUCCESS OF THIS NEW FORM OF SOCIAL ORGANIZATION came down to numbers. Today there are roughly 8 billion more people alive than there were only a few thousand years ago, and the balance didn't come from forager societies. In fact, there is a curious problem in the field of demography called the "forager paradox." Simply stated, the populations of hunter-gatherers today, living in far more marginal and unproductive lands than they typically would have in prehistory, nevertheless grow at greater than 1 percent per year. This is a problem because even if this growth rate were halved it would suffice to swell a society of 100 to almost 50 billion people in just four millennia. But for two hundred millennia, human population didn't grow at all. It turns out, though, this isn't much of a paradox at all: yes, human life for its entire existence on the planet has been likely marked by this unflagging propensity toward growth, but as one 2019 study found, "periodic catastrophes over human evolutionary history are necessary" to explain this pattern. Paroxysms of climate, disease, famine, and warfare have perpetually checked human ambition since time immemorial, periodically decimating populations after blossoms of growth.

Before long, however, the benefits of increasingly complex agricultural societies began to advertise themselves, pushing back against this ageless pattern. Where for millennia since the end of the ice age, the population of the Near East had predictably waxed and waned with the climate changes of the early Holocene—as it did for all of human prehistory prior—by 2000 BCE this pattern had effectively begun to decouple. The grinding advance of increasingly adaptive social organization, widening trade networks, agricultural intensification and surpluses to buffer against downturns, expanding infrastructure and irrigation, and the unceasing accumulation of culture had finally arrived at a point by the mid-Holocene where human survival here wasn't as tightly moored to the caprice of the climate.

Foraging was soon in decline all over the world, as the world's polities came to be increasingly concentrated around food production—and in the case of much of Mesopotamia and North Africa, hemmed in by harsh deserts, where lush wetlands and grasslands had long existed. As Earth's wobble dragged monsoon rains up and down Africa over the millennia, the formerly green Sahara dried out and concentrated people around the Nile, with ecology itself pressuring people to make ever more intensive demands of the land. Like the crucible of a hydrothermal vent, or the bottleneck of Snowball Earth, perhaps it was this coercive ecology that cornered people into investing far more energy into agriculture than they otherwise ever would, in increasingly effective ways. Whatever the cause, what the Egyptians would create here in North Africa, as these planetary seasons changed, would be unprecedented.

Where the reach of previous empires was limited by the borders of their ignorance, petering out beyond the horizon—and sometimes collapsing when rulers made unrealistic demands of unseen crop yields in the provinces—Egypt was perhaps uniquely capable of gathering information on a scale vast enough to organize the huge agricultural energy flows that produced its unprecedented grandeur. While the climates of most ancient societies were driven by complex and distant processes that were impossible for people at the time to understand or predict, the success or failure of Egyptian harvest was due solely to the Nile's annual flood. Stone columns lining the Nile River and hidden in temple sanctums, called Nilometers, that were carefully annotated and even more carefully watched by bureaucrats, recorded the high-water mark of the Nile each flood season. This single piece of data illuminated exactly how much of the vast floodplain would be inundated that year and the resulting harvest across the kingdom, and was used to precisely and judiciously calculate taxes, paid in grain. While it might have been the fertile soils of the Nile Valley that produced the bounty of plant energy that powered the state, it was this capacity for large-scale information processing that explains Egypt's precocious scale, coordination, and sophistication. It's a fact that doesn't fail to astonish with repetition that the Great Pyramid of Giza was the tallest building on Earth for almost 4,000 years. And it's still by far the heaviest—embodying the more than 2.5 terajoules of gravitational potential energy that it once took to raise it.

Because these early states had to centrally store the surpluses of grain that supported their large populations, along with associated stockpiles of gold and silver, they had to be guarded. This meant defenses, walls. But it also meant that gaining someone else's photosynthetic surplus would swell your power even more, and vice versa. Just as the first animals in Earth history quickly found that they had to ornament themselves with protective shells, puncturing teeth, watchful eyes, and increasingly complex nervous systems to withstand attacks from the world's first predators, or to become predators themselves, so too for these early states—entrained as they now were in a strange new process of selection. The ecologist turned statistical historian Peter Turchin explains this new evolutionary dynamic that now operated at the polity level: "Warfare imposes a selection regime that weeds out relatively dysfunctional, poorly organized, and internally uncooperative polities, favoring those with larger populations and effective, centralized, and internally specialized institutions." Or, as the maxim goes, war made the state, and the state made war.

This demanded even more food, produced even more efficiently, to make up for the nonfarming activity required by the state: to feed miners wrenching copper from the earth, and armies marching on foot. Whether or not more egalitarian and anarchic farmer-forager societies provided more benign arrangements for their constituents was beside the point—they would have been overwhelmed, or absorbed into these sociopolitical superorganisms, unless they could escape into increasingly rare hinterlands. But as this world grew smaller, and agrarian states met each other on the battlefield, the existential stakes of war made it a crucible for technological innovation—from metallurgy to the Manhattan Project—the fruits of which only enabled further imperial expansion. The spread of chariot warfare and composite bows, for instance, might have precipitated the rise of the unprecedentedly large states of the Bronze Age, from Turkey to China, including the world's first so-called mega-empire, New Kingdom Egypt, which adopted the horse-drawn tactics of their former Hyksos conquerors—and became the first state in world history to top 1 million square kilometers.

Taking new territory meant acquiring more land to grow more food to support a growing population, and more labor to work this land, and an ever-swelling tax base and source of new military conscripts. In an age before fossil fuels, territorial expansion and agricultural intensification were the only ways for societies to wring more energy from the landscape to keep pace with a swelling population. But barring the occasional epoch-making technological improvement, the limits to the latter quickly loomed for agrarian societies. This was the expansionary material logic of empire that propelled it outward—with many empires in history exhibiting the same S-shaped growth curves seen throughout nature, from the skyward trajectories of sunflowers to the fractal growth of snowflakes. The word *superorganism* even begins to feel like less than a metaphor for these polities: for the most part, larger animals can only support themselves with larger range sizes, and for empires, control over a vast geographical area and the ability to draw plant energy from distant imperial breadbaskets across diverse climate regimes similarly served as a buffer against regional agricultural shortfalls. “We have every reason to think about imperial projects as ecological phenomena,” writes the anthropologist Alf Hornborg.

These new political structures were engines generating huge surpluses in calories—agricultural tribute that powered the project of empire. In a process that Hornborg has even described in terms of “the thermodynamics of imperialism,” these societies maintained the unequal flows of energy from the periphery to the core by some combination of coercion, dispensing spoils and privileges to regional elites, and often encouraging beliefs about the supernatural services provided by the divine elite at society’s center. It was a template that would prove durable, from the Inca emperor (divine son of the Sun), to Imperial China (the Mandate of Heaven), to the French Sun King (ruling by divine right).

But expansion, in turn, necessitated a more distributed nervous system—more administration, to put it less sexily—to hold together these larger sociobiological bodies. As the anthropologist Charles Spencer writes, drawing from examples in Mesoamerica, Peru, Egypt, Mesopotamia, the Indus Valley, and China, “The successful annexation of distant areas, those farther than a 1-day round trip from the capital, required … [delegating] partial authority to functionaries at distant outposts—in short, [they] had to bureaucratize.” Others have purported to identify a maximum limit to this imperial body size: a roughly fourteen-day journey from the capital using the fastest transportation available, beyond which the ties to the administrative center became too tenuous, and the centralizing flows of energy and information broke down. In the Andes, for instance, the Inca employed relay runners, the *chasquis*, carrying messages from the royal court encoded in *quipu* knots, traversing more than 150 miles a day on foot and reaching the outermost frontiers of the empire in two weeks, over the 25,000 miles of roads and rope bridges that held the state together. In Eurasia, however, where the nomadic art of horse riding had been imported from the Pontic-Caspian steppes into the agrarian heartland, far vaster states could form—peaking with the almost unbelievable sweep of the Mongol empires, which saw horseback warriors announce their presence from Vienna to Java. But even these continent-sweeping, grass-powered empires often found themselves overstretched, and it's no coincidence that "the only global empire[fn9] there ever has been," as the sociological historian Michael Mann calls the United States, would have to wait until the age of fossil fuels and globe-spanning electronic communication, when these limits of time and space were finally obliterated and the whole planet could be traversed fast enough that it could cohere under one integrated economic world system whose markets were protected and enforced by one global military superpower.

While this day was still far off, by 2,500 years ago farming had spread around the world, and by around 2,000 years ago—with over 90 percent of humanity living in Eurasia and North Africa—up to three-quarters of the global human population now belonged to one of four agrarian empires, with Rome and Han China circumscribing half the world's population, and the Parthian and Kushan empires enveloping another quarter. The world was getting smaller, and there was increasingly little escape for nonagrarian ways of life.

Even the so-called barbarians who came to parasitize these grain states from the periphery—from the Germanic tribes on Rome's borders, to the Xiongnu tribal confederacy that nipped at the edges of Han China—were products of this new world, and they often settled into increasingly symbiotic tribute relationships, arrangements that encouraged reciprocating advances in fortification and warmaking technology. Sometimes these seminomadic bands of cavalry even conquered and took over the apparatus of the state itself, and the erstwhile raiders suddenly found themselves having to learn the ropes of administration over vast sedentary populations. Writing[fn10] about the eventual circumscription of the entire world by large-scale agricultural societies, the Stanford historian Walter Scheidel notes, "Even a trap that was slow in closing was, in the end, a trap."

As the people who lived in agrarian societies came to be domesticated as totally as the animals and crops they utterly relied on for survival, one animal alone was transformative above all others.

“Horses tripled human power, speed, and range for four millennia,” the Oxford historian James Belich writes. “They provided half of all work energy in the United States as late as 1850, as much as humans, steam, water, and wind combined …. It is hard to think of a biotechnological development between the origin of farming 10,000 years ago and industrialisation 250 years ago that outranks the horse-human alliance.”

Enlisting draft animals to do work for us, as Eurasian and North African societies had been doing for millennia, had come to generate a growing elaboration of material culture: there were harnesses to be made, horseshoes to smith (from iron that needed mining), and animals to stable. This came to require its own specialized workforce, and these new forms of labor made demands of the photosynthetic surplus as well.

This was energy efficiency in the age before fossil fuels, but it operated by the same principle, drawing more mechanical work out of the same amount of photosynthetic carbon to burn to CO_2. And the stakes were life-and-death. Where there were once simple wooden plows awkwardly and inefficiently affixed to undersized livestock, wasting energy in the form of waste heat and friction, within millennia there would be highly energy-efficient, steel-tipped moldboard plows pulled by Herculean teams of draft horses bred solely for tractive power—amplifying the work that could be done with photosynthetic power, and the food that could be cultivated, by orders of magnitude. And just like in our own time, these energy savings didn’t result in people doing the *same* amount of work with *less* energy, but increasingly *more* with *more*. As a result, over the past 10,000 years, on all continents, and vastly predating the rise of fossil fuels, there has been a “globally convergent” trend of “cultural evolution toward more energy-consuming political economies with higher carrying capacities,” as one study finds, a ratcheting of energy consumption achieved by appropriating “ever-larger proportions of the biomass produced by ecosystems.” And when one society makes a jump in energy consumption, it tends to ripple across continents, boosting all societies in synchrony since the end of the ice age.

Seen from this perspective, human society is, unsurprisingly, continuous with a far more time-honored trend in the biosphere stretching back billions of years. Life on Earth has seen a similar, stepwise, ratchetlike increase in energy consumption over time—a trend that has held just as true for the world of microbes as it has for the plant and animal worlds, with organisms increasing their metabolic rates by eighteen orders of magnitude since the dawn of life. “The principal directional trend at this larger scale is an increase in collective power,” writes the paleoecologist Geerat Vermeij, “an expansion of the living world away from thermodynamic equilibrium, driven by the cumulative effects of selective competition for locally scarce resources.”

This pattern has held in unexpected ways: when smaller mammals came to replace the energy-hogging dinosaurs in the wake of the asteroid, the long-term trend in energy escalation would surprisingly remain unbroken. “In this case, gigantic individuals were replaced by social groups that acted as large individuals,” Vermeij explains. “The relatively small individuals in these social species have higher per capita metabolic rates than their earlier counterparts.”

However unsettling the implications, human societies have not escaped this ancient pageant of ever-escalating power on Earth. The archeological record left behind of this swelling energy use is visible by its thermodynamic twin, increasing waste over time, all over the Earth: from mountains of pottery shards, riverbanks of discarded oyster shells, and piles of bones from thousands of sacrificed turtles, to the modern islands of landfill found on the outskirts of any metropolis, and the CO_2 waste of all of industrial society, dumped into the open sewer of Earth’s atmosphere.

“The evolution of the human species, which thanks to technology and the exploitation of novel energy sources is the most powerful species (as measured by per capita and collective energy demand) yet to have evolved in the history of life, is consistent with these trends,” writes Vermeij.

HOWEVER MENACING AND FORMIDABLE THE RISE OF INCREASINGLY powerful states might have appeared to the inhabitants of the ancient world, and however far from equilibrium the escalating ratchet of agriculture and technology had pushed these societies, they were still composed of humans, hopelessly at the mercy of the vagaries of the natural world. And the more populations swelled, and the more dependent they became on an ever-growing agricultural surplus—the more energy they captured from the surrounding environment—there was only farther to fall when nature didn’t comply with their ambitions.

To take one example: In Ptolemaic Egypt, when distant volcanic eruptions stalled rains over the Ethiopian highlands, the life-giving Nile ran low. When summer floods failed to arrive, and the harvest failed, long-simmering distrust between Egyptians and the ruling Greeks spilled over. Revolts ensued. People sold their land. Priests made desperate decrees to try to restore order. But after further unseen eruptions in 46 and 44 BCE and several more Nile failures, Ptolemaic Egypt struggled with unrest until the end of the dynasty, stumbling toward the suicide of Cleopatra in 30 BCE. In the aftermath of the defeat, the agricultural surplus of Egypt would instead be absorbed by, and swell, the Roman Empire for the next seven centuries—Italian soils having been exhausted by centuries of intensive farming. This was the zero-sum world, constrained by photosynthesis on the Earth's surface, before our own age of supposedly "endless growth" unlocked by the ancient plant power of fossil fuels. And the Greeks' plight would have been familiar to the rulers of Egypt more than 2,000 years earlier—when the tomb of one governor cried that "the whole of Upper Egypt died of hunger and each individual had reached such a state of hunger that he ate his own children"—or indeed familiar to the Roman Empire several centuries later, when Nile failures hastened that ailing superpower's stumbling disintegration.

ROME WAS ONE OF THOSE PRECOCIOUS PLACES—LIKE OLD KINGDOM Egypt or Teotihuacan in Mexico—that, every once in a while, explodes into history with an unexpected and unapologetic grandeur and scope that seem centuries ahead of schedule. At its height the Roman Empire boasted a population of some 75 million subjects, or 25 percent of the Earth's population, living in over a thousand cities, with a million in Rome alone—a figure perhaps three times as large as the next-most-populous city on the planet, and which wouldn't be matched by anywhere on Earth for seven hundred years. The empire spanned from North Africa to Britain to the Persian Gulf; Roman merchants plied the Indian coast as Roman coins flowed inland and on to Ceylon (Sri Lanka), while Roman diplomats sent royal embassies to China. It guarded a 6,000-mile frontier with the capacity to call up an army almost a half a million strong—something that wouldn't be rivaled again in Europe until late-seventeenth-century France. The titanic Roman grain ships that streaked the Mediterranean and kept the empire's gargantuan granaries topped off for centuries wouldn't be surpassed in size again until the nineteenth century.

None of this has anything to do with the moral or ethical worth of the Roman project, which was deeply depraved on any number of normative axes, especially by modern sensibilities. Indeed, Rome's rise, like that of other empires around the world, was accompanied by a psychotic explosion of violence and enslavement, while its environmental record is still visible in ice cores—its mining industry alone producing pollution on a scale unmatched on Earth for centuries. And there has long been a tendency in archeology, only recently reversed, to focus on highly unequal, stratified cultures that left behind monumental architecture and other markers of so-called civilization, on the often explicitly stated assumption that they were self-evidently "superior" to those cultures without them. With those caveats in mind, Rome's scale nevertheless needs explaining. It's this escalation of sheer size and power over everything that came before that distinguishes the Roman Empire, and not, for instance its rather meager technological innovations—the Romans having largely borrowed from existing technology. And ultimately, then as now, Roman society looked the way it did thanks to an extraordinary and unprecedented capacity for energy capture.

"Human societies are a living system," the University of Oklahoma historian Kyle Harper told me. I had been drawn to Harper's work explicitly for its interdisciplinary perspective. "So I don't think thinking about the origins of life, what cells do, or the dynamics of biological evolution are sort of merely analogies for human social development. I think it's a continuum. But human societies are different. We're an offshoot of evolution, and the basic rules of evolution still apply. But we also make up new rules, because there's new ways of relating information, of transmitting information, inheriting information. But it's an offshoot of biological evolution that generates new kinds of rules within the system of rules that have governed life for four billion years."

While the Roman rise came at an opportune time of disorganization in the Mediterranean world, and was prosecuted with a fanatical militarism that allowed Rome to "concentrate unprecedented state violence," as Harper writes, its ultimate success could be just as easily, if more prosaically, attributed to photosynthesis. While historians have long sifted through the biographies of Roman leaders and military campaigns, the local climate, helplessly responding to shifting pressure systems in the distant Atlantic, was meanwhile bringing vast new tracts of farmland online. "If we *only* count the marginal land rendered susceptible to arable farming in Italy by higher temperatures, in the most conservative estimates, it could account for more than all the growth achieved between Augustus and Marcus Aurelius," Harper writes about the centuries of accommodating climes that nurtured the empire's rise. "In such perspective, human toil can seem vain. The hard fatalism of the farmer is made of such stuff. The supreme sway of the climate is nothing if not humbling."

Planetary processes sharply delimit what's possible, no matter how ambitious a program of state-building is. As Marx wrote, while humans "make their own history … they do not make it as they please." Still, Rome was skillful in capitalizing on these centuries of good climate fortune, and its ability to capture and redirect energy from the environment toward growth was unprecedented—typified by the 120-kilometer aqueduct in modern Tunisia that irrigated some 10,000 square miles of parched land and allowed Roman North Africa to bloom; or Barbegal, the astonishing 30-kilowatt water-mill complex in southern France, vastly more powerful than eighteenth-century European water mills, and called "the world's first industrial complex." But it was its massive granaries, ubiquitous across the empire, that gave it a gigantic energetic buffer of organic carbon during climate shocks and allowed it to dispense 80,000 tons of wheat annually to a growing Roman public, and even enact price controls during grain shortfalls.

"The increase in food storage facilities at all scales, from domestic to urban, over this period is absolutely astonishing," Harper told me. As population exploded, so too did these granaries in lockstep, along with the institutional systems to effectively manage them. And the Romans themselves were under no illusions about this photosynthetic and energetic base to their power. By the end of the fractured empire, and chastened by centuries of intermittent climate shocks, the reverence toward grain had become downright bizarre, with the emperor ceremonially inspecting the grain inventories of Constantinople.

"If this were to happen today, we would not believe the creepy weirdness of it," Harper says. "Like if Joe Biden was like sitting in piles of grain and you know bathing in it, and pouring it on people, and going down to see the grain rolling in from Nebraska and welcoming it to the cities of the East Coast. It would be pretty weird."

When the wheel of climate fortune finally turned, invaders and paroxysms of disease wracked the empire, its energy stores were sapped, and the expansive civilization found itself overstretched. “The empire might be construed as an organism with batteries of stored energy and layers of redundancy that permitted it to endure and recover from environmental shocks,” Harper writes. But as the Roman climate chilled and agriculture retreated, tax revenue on this harvest fell, emperors could no longer pay the garrisons on the frontier, and the empire could no longer coordinate the flows of energy it once drew from the Earth to support this bloated organism. The empire had demonstrated what was possible for the age, in an economy powered solely by photosynthesis on the Earth’s surface and subject to the whims of the natural world. But when the Vandals sacked Carthage in 439, the half-million tons of grain yearly sent back to the metropole from what is modern Tunisia ceased, it wouldn’t be long before the aqueducts crumbled, and the superorganism and its constituents were starved of energy. The point here is not to litigate what caused the fall of Rome, or even whether it fell. Its decline had no shortage of potential authors, and there are libraries of books that explore each. The point is simply to underline the unseen flows of energy that support any society, and the particular vulnerabilities inherent in extraordinarily energetic ones. Over the decades, further insults from distant frontiers and an apocalyptic eruption of plague brought down what was left of the rotting western Roman façade. In the aftermath, the Roman world became vastly simpler. By some estimates the population of Rome itself had declined over the centuries from some 1 million at its height to perhaps just 30,000 souls by the dawn of the early Middle Ages. And while this ensuing period in Europe has been reclaimed as far more equitable for its inhabitants than the imperial world that came before, and more culturally vibrant and innovative than the outdated characterization of “the Dark Ages” implies (in fact, Europe in these centuries saw slow, steady, unassuming—if revolutionary—advances in agriculture, metallurgy, clothmaking, boatbuilding, and architecture that would seed its explosion in the second millennium), by other measures the region suffered the largest reversal in social development in all of human history.

“What happens after the first plague pandemic is truly, truly a dark age,” says Harper. “It’s something you’re not allowed to say, but like, nobody’s literate, the cities are abandoned. The old world is mostly gone.”

In the centuries after, English poets marveled at the astounding ruins the Romans left behind—giants must have built them.

BECAUSE OF THE CULTURAL RATCHET, HOWEVER, NO SOCIETY EVER truly started over again. And elsewhere in the Mediterranean and Southwest Asia, civilization thrived in late antiquity, as vital new actors appeared on the scene, like the Muslim conquerors fired by a new messianic religion who exploded out of the Arabian Peninsula and feasted on the decayed husks of the Roman and Persian empires. A world away, meanwhile, China would host the world's largest cities for the next thousand years and, as we'll see, would at one point nearly launch its own industrial revolution under the Song dynasty. And eventually, when medieval Europe saw centuries of equable warm weather return and agriculture boom, half the forests were cut down, the population doubled, giant sturgeon went missing from the rivers, towns and cities crystallized on the landscape and jostled for control of trade to distant ports, Gothic cathedrals sprang from the Earth, and papal power reached its commanding height. Where the borders of the great horseback empires of the Eurasian steppe swelled and shrank as verdant years passed into droughts, Genghis Khan was propelled out of Central Asia by Mongolia's wettest period in a millennium—rains that nourished grasslands and horses alike—arriving on the doorstep of Europe, China, and the Middle East to titanic, history-changing cultural effect.

In North America, meanwhile, where there were no horses from which an aspiring empire could project power over large distances (prior to the arrival of the Spanish horses), and where there was a population density low enough to escape the violent, centralizing authoritarian states that *did* occasionally rise from the floodplains (like Cahokia in St. Louis), comparatively more egalitarian, mobile, and decentralized farming, free from the coercive controls of a powerful state, remained a viable lifestyle for far longer than anywhere else—indeed, until the European epidemic apocalypse and subsequent genocides of the sixteenth to twentieth centuries. It is intriguing that, later, when Spanish horses made their way into North American native societies through trade networks in the early sixteenth century,[fn11] they were quickly adopted by a succession of Southern Plains groups who dominated the region, echoing the steppe confederations of Eurasia: first the Apaches, who quickly destroyed and absorbed rival Jumanos, and then the Comanches, who pushed the Apaches aside to become what has been called an “empire” in the Southwest by the late eighteenth century.[fn12] The Comanches did not merely withstand colonial encroachments from the Europeans, but they dominated the Southwest in a vast trading and raiding empire, wringing tribute and slaves from settlers and rival tribes alike, bleeding Imperial Spain’s colonial American project dry. They became the preeminent power over a vast swath of the continent for more than a century, fending off the encroachments of the swelling United States’ continental empire for decades.

In the meantime, the cultural ratchet kept turning. The continued proliferation of labor-saving livestock, energy-concentrating farm implements, and yield-boosting techniques like multicropping swelled the photosynthetic base even more, meaning that ever more energy could be redirected into activities *other* than meeting the metabolic needs of human and animal labor. In biology, this amount of free energy that can be spent on activities other than meeting a creature’s base metabolic requirements is known as “scope”—the surplus that allows a peacock to grow an all but functionless if flamboyant tail, for instance. As agriculture advanced, this *scope of societies* was growing as well, and societies were increasingly able to pour their huge metabolic surpluses—unlocked with the sickle and flail, the seed drill and moldboard plow—into the construction of temples and churches, roads and bridges, natural philosophy and scribing. Hugely labor-intensive irrigation systems unlocked still *more* photosynthesis, providing a 30-fold return in energy investment and reaching their apotheosis in the elaborate “floating gardens” of the Aztec Empire’s *chinampas*, or in the astounding canal and reservoir systems of the “hydraulic empire” of the Khmer in Cambodia. In Eurasia, vast east-west trade networks made it so that where bronzemaking and chariot technology once diffused east into Asia, someday, thousands of years later—along with luxuries like silk and porcelain, and curiosities like gunpowder—these networks would carry rather more practical inventions west, like wheelbarrows (arriving in Europe only in the thirteenth century) and superior draft horse harnesses, which might have been invented in China but could find their way onto the fields of Poland, on the other side of the continent, further augmenting European agriculture.

Nevertheless, until the explosion of ancient photosynthesis from fossil fuels, the amount of work that could be done on the Earth was still stubbornly limited to the solar power available on the planet’s surface, in the carbon-hydrogen bonds of plants.[fn13] There were only so many electrons to go around, and only so much organic carbon for mammals like humans and horses to oxidize and turn into mechanical work, waste heat, and CO_2. Societies continued to bump into the punishing limits of a world of energetic scarcity.

"A society that approaches the limits of population growth can invest in clearing forests, draining swamps, irrigation, and flood control," writes Peter Turchin, drawing from the similar boom-and-bust experiences of eight ill-starred societies in human history. "All these measures will result in an increase in the carrying capacity. However, at some point there are no more forests to cut or swamps to drain." If population continues to grow, he writes, the rivets of society can start to pop: "As population density approaches the carrying capacity … there are shortages of land and food, and an oversupply of labor. As a result, food prices increase, real wages decline, and per capita consumption, especially among the poorer strata, drops." Revolution might ensue.

As wetlands were drained, as unpromising soils were tilled by ever-more-powerful teams of draft animals, as furrowed fields expanded and populations swelled—now held aloft by far more photosynthetic energy than ever before—there was also much farther to fall. As the demands of their swelling populations grew, and the ratchet of culture ceaselessly turned, people increasingly landed blows on the natural world. Already by the first century, the Mediterranean was badly overfished, as Roman nets burst with bluefin tuna and anchovies, and moray eels were caught by the thousands for feasts celebrating Julius Caesar's military victories. North Africa was emptied of its lions for gladiatorial contests, and Syria was deforested for Roman timber. This phenomenon knew no borders or hemispheres: with the population boom that followed Jamaica's settlement in 600 CE, its nearshore reef fauna was decimated. Around the same time, prized white sturgeon in San Francisco Bay grew scarcer to native fishermen, as sea otter and cormorant breeding colonies were wiped out. Across the world, fur seals were nearly extirpated at the skillful hands of Maori fishers in the centuries that followed, while the forests of Aotearoa (New Zealand) were halved and the Moa went extinct. Sometimes this man-made environmental toll would feed back on the society to devastating effect—as with the Maya people's deforestation of the Yucatán, which might have helped bring an end to their own classical period, and saw their monumental city centers empty and populations drop from 3 million to some 100,000 in just a few centuries.

BUT EACH OF THESE EXAMPLES MARKED *LOCALIZED* EPISODES OF environmental turmoil by the swelling human project. Sometimes human catastrophes grew to such a scale that they were sufficient to be registered by the entire global carbon cycle itself, and possibly even changed the planet's climate in the centuries before the eruption of fossil fuels. From the 1500s until about 1610, ice cores register a slight dip in CO_2. This might be in part because by 1600, disease had already perhaps killed 90 percent of indigenous Americans, or some 56 million people, and fallow fields of Meso-America, and the abandoned terraces of Peru, instead sprouted dense forests that sequestered carbon in them by the gigaton. In the desolate aftermath of this slew of European diseases, which hitchhiked across the Atlantic with similarly murderous explorers, the Americas, tenantless and forsaken, drew in almost as much carbon dioxide into these new forests, in a little over a century, as is today released by humanity every year.

But this is not the only historical human catastrophe that might have had an effect on the climate. The University of Munich geographer Julia Pongratz evaluated the climatic effects of four historical cataclysms, including this Great Dying of the Americas, but also the European Black Death of the fourteenth century, the fall of the Ming dynasty at the hands of the Manchu in the seventeenth century, and the Mongol invasions of the thirteenth and fourteenth centuries. However, she found that only the last, which similarly might have killed tens of millions of people and allowed vast swaths of Eurasia to regrow in the conquests' wake, was sufficient to affect the atmosphere's store of CO_2. Still, in either case, these effects, while global, are rather small, and by Pongratz's estimate even this mass death was sufficient to lower atmospheric CO_2 by less than 1 ppm.

But this focus on individual pulses of human-induced carbon-cycle disruption—apocalyptic as they were to those who experienced them, and registering incalculable human misery in only tiny changes to the composition of the atmosphere—might be missing the forest of more gradual anthropogenic climate change for the trees. This is because something funny happened before the Industrial Revolution exploded onto the world. Or rather, it didn't happen. The planet didn't drop back into an ice age. And maybe it should have.

AT THE VERY MOMENT THE PLANET WAS ABOUT TO EMBARK ON A global industrial chemistry experiment unprecedented in at least 66 million years, the Earth was also in the same orbital configuration—one that brought low sunlight to arctic summers—that in previous interglacials had been sufficient to kick off carbon cycle and ice albedo feedbacks that would plunge the Earth back into the depths of a proper Pleistocene ice age. At the end of every warm interglacial of the Pleistocene like our own, as the Earth carried on its celestial rhythms and summer sunlight grew more faint in the high latitudes, snowpack that failed to melt in the warmer months instead accumulated over the years into a proper ice sheet. As their glinting white surfaces reflected more and more solar radiation back to space, these glaciers would swell even further. Feedbacks drew CO_2 from the surface world into the depths of the oceans, and the oceans themselves would fall by hundreds of feet, gifting their fathoms instead to the ever-swelling ice sheets on land. Then the dusty, cold world would return for over a thousand generations.

For some reason, though, the next ice age never arrived in this, our Holocene interglacial. The planet, as you can tell by looking around, didn't start plummeting back into an ice age thousands of years ago. The high-latitude summer sunlight of today remains close to its "minimum at present, and there are no signs of a new ice age," as one paper coyly puts it. This is strange. In our own time this fact is somewhat easy to explain, since we've injected thousands of gigatons of planet-warming CO_2 into the air. But if this ice-age threshold should have been tripped *before* the Industrial Revolution, how did we make it this far without returning to a frozen planet in the first place?

Some have invoked the entire story we've been telling so far, one that spans millennia of land clearance and farming, the expansion of agriculture and the unprecedented reworking of the Earth by humanity in the Holocene, as responsible for averting this deep freeze. This is because, starting around 7,000 years ago, at a time when in similar interglacial periods CO_2 and methane should have begun a gentle millennia-long fall, they instead curiously began to rise in the atmosphere. Seven thousand years ago, the great old-growth forests of Europe were already being clear-cut for agriculture, a massive deforestation that, like similar land clearings taking place in China and India around the same time, released vast emissions of

CO_2 as these trees and soils decomposed and gave up the carbon they had captured in life, and were replaced instead by farmland, able to hold far less carbon. This was also around the dawn of widespread rice irrigation, which began in China 5,000 years ago and had spread across southern Asia by 1,000 years ago, degassing prodigious amounts of carbon to the atmosphere in the form of methane (which is both a more powerful greenhouse gas, and one that degrades to CO_2).

Because early agriculture was so much less efficient than more modern forms—benefiting as it has from millennia of life-or-death cultural ratcheting—it required much more land than it does today. In fact, in both China and Europe, farming required about four times as much land clearance 2,000 years ago as it did by the dawn of the Industrial Revolution. This early deforestation over many millennia, as agriculture expanded across the Earth, might have produced an astounding 343 billion tons of carbon emissions—equivalent to about nine years of our outrageous modern carbon emissions—long *before* the Industrial Revolution, potentially increasing CO_2 by 24 ppm. While the modern British countryside, with its rolling meadows and winding hedgerows, might strike us as "natural," by 1086 the natural landscape of the country had long since been destroyed, with 85 percent of the arable land in England turned over to farming and the remaining 15 percent mostly preserved as patches of forest for aristocratic hunting parties. In some vast areas of the Earth's surface, such as Peloponnesia, India, China, and England, the landscape had already been deforested "three or more times as civilizations … prospered from the newly cleared agricultural land and then collapsed," the ecologist Charles Hall notes.

The Earth scientist Yanyan Yu estimates that as farming took hold in the rich loess soils along the Wei River—a legacy of the dust storms of the Pleistocene—"between 8,000 and 4,000 years ago, the estimated population grew from 4 thousand to 1.55 *million*," while the area of land under cultivation increased 60-fold around the same time span. Even in our own age, with CO_2 erupting from the Earth's crust like it has only in a handful of mass extinctions, this kind of land-use change on the surface still accounts for almost a quarter of our current carbon emissions, and uncertainty in the trends for future deforestation still holds the door open for some truly worst-case climate scenarios.

Ironically, though, these overlooked carbon emissions starting early in civilization's infancy—a product of the accidental geoengineering of antiquity, and the growing circumscription of much of the world by agriculture—might have kept CO_2 high enough, and the planet just warm enough, to prevent a catastrophic plunge back into the Pleistocene ice ages. In fact, recent modeling work shows that if CO_2 was a mere 40 ppm lower just before the Industrial Revolution—240 ppm, instead of 280 ppm—the planet would have already passed the point of no return, and we'd already be well on our way back into the nadir of the planetary chill again. Instead of the so-called Little Ice Age in Europe, where centuries of cold that lasted until the early nineteenth century saw Swiss mountain glaciers swell and horse-drawn coaches trot across a frozen Thames to visit the lively frost fairs,[fn14] London would have been long ago sacrificed to tundra as planetary winter gripped the British Isles. Ice sheets would have long since plowed over Scandinavia, its reindeer already safely emigrated south to repopulate the more familiar Spanish ranges of the Pleistocene. Soon a third of North America would be crushed by ice sheets up to two miles thick, and the seas would plummet hundreds of feet, connecting Britain to mainland Europe again, North America to Northeast Asia, and Borneo to Vietnam by dry land—returning the planet to the dusty, dry, brutally cold low CO_2 of the ice age.

What effect this would have had on the trajectory of human society and the burgeoning world system of the early modern period, no one knows—though it seems catastrophic. Instead, before the first chunk of coal ever coaxed a single piston stroke from an engine, the planet "narrowly missed" the inception for the next great glacial age. But what happened next would ensure that the planet would not only *not* drop into an ice age, but would actually be at risk of launching instead into a far more alien climate from millions, even tens of millions of years in the past. Also, because of the contingent accidents of history, and the rather less contingent facts of geography and geology, it was in Britain too that the groundwork for this industrial explosion of ancient photosynthesis had been laid for centuries. A previously obscure society was once more straining against the limits of energy on Earth's surface, and would soon burst through them, unleashing 500 million years of sunlight stored in organic carbon.

9

The Start of the Supereruption

A QUARTER-BILLION YEARS AGO, AT THE END OF THE PERMIAN PERIOD, the Siberian Traps had already been burbling lava for some 300,000 years. Then they qualitatively changed the style of their eruptions: they started burning through fossil fuels on a massive scale. The catastrophe that followed is visible far from where the eruptions occurred, in places like Meishan,[fn1] China. The fossil record there, in layers of seabed-turned-to-rock, is thronging with the traces of ancient life—bivalves, coiled shells that once hosted tentacles, and a seafloor thoroughly burrowed and reworked by life. In the layers of rock a few inches above, however, this seafloor suddenly goes silent, giving way to a sparse, mucky wasteland, hosting a monoculture of dull, if hardy, clam survivors here and there. The sediments also record a massive spike in light carbon over this interval—the distress signal of a planet racked by CO_2 from the distant volcanoes—and a lethal shock of warming.

Humans have been around for almost 300,000 years too, and though we've registered our growing impact on the planet and its carbon cycle over this entire span, it's only in the last few centuries that there has been a qualitative phase change in our relationship to the Earth: we also started burning fossil fuels on a massive scale. Something like a stratigraphy of this incipient catastrophe is already visible, to the attentive observer.

When I visit family in midcoast Maine, I see it. Everywhere are the landscape markers of the retreat of the last ice age—giant boulders implausibly dropped in the middle of the woods, and granite outcrops scraped and polished by retreating glaciers. Above this bedrock are the strata of the region's indigenous legacy, and the abundance of the precolonial ocean, which makes up the landscape's very topography: oyster middens, giant mounds of oversize shells, crumble out of the riverbanks, a sign of millennia of native feasts. This age is interrupted, though, at the end of the peninsula by a fort, announcing the arrival of fishermen from the alien world of Bristol, England. Already by the sixteenth century, English and Basque fishermen were well acquainted with these waters off North America. And as this colonial settlement grew, it was fortified, then perennially fought over by the French, along with Wabanaki allies, who responded to English aggression with raids by birch-bark canoe and guns.

As these European imperial rivalries increasingly spilled across the world's oceans and eventually swept up the entire planet, the seas were knitted together by trade and empire. At this local fort, for instance, archeologists have found German and Venetian wares, coral from the Caribbean, and elephant ivory from West Africa. But this story could be reproduced in almost any coastal community one cared to look as the world got smaller: Arabic coins show up in Sweden, narwhal tusks arrive in Nagasaki via Dutch traders, spices from Indonesia began flavoring food in Antwerp, and Chinese-produced porcelain, colored with Iranian cobalt and paid for with Peruvian silver, found its way on to London dinner tables. The entire planet was growing increasingly connected by the oceans. And like most ocean-facing colonial settlements, plugged into these growing international trade networks, these Maine waters would soon be all but drained of fish.

By the second half of the nineteenth century, inshore cod stocks disappeared, and Maine fishermen had to venture farther offshore. Even the ancient oyster middens were mostly mined out in the nineteenth century for chicken feed. A long-standing indigenous practice of coastal whale harvesting in New England meanwhile became impossible, as whaleships leaving from Nantucket were soon forced to round *Patagonia* in the perpetual chase for new, unexploited whale pods. It was in the context of this intensifying, escalating technological race for fish and whales that on-land industrial civilization hit a vein of Earth history—coal first, then oil—and an ocean that had been fished by humans for tens of thousands of years was ravaged in a matter of decades. For humans, forever limited on the seas by the rhythms of nature, the winds, the great seasonal tuna migrations, the monsoon, there were now no limits at all. Boats could fish around the clock in all seasons in all oceans, plowing the seabed with ever-more-powerful dredges and purse seine nets that would eventually come to swallow mile-long schools of fish whole—schools that now could be tracked down year-round, ever farther from home. Railways opened the vast inland markets, and ice kept fish fresher, longer. As the human population was unlocked by agricultural revolutions and soared in number, more ocean life had to be wrenched from the sea and fed into this increasingly global metabolic network.

When cod catastrophically collapsed at the end of the twentieth century with the arrival of an international fleet of factory trawlers on the great North American fishing banks, and the ensuing race for the last fish, these seas briefly gave way to a sea urchin export boom, which was itself quickly overfished. And now there's lobster. Lots of lobster—its unusual abundance due, in part, to the disappearance of the cod, which once ate the juveniles—producing a monoculture ecosystem offshore that one marine biologist told me more resembles a "feedlot." Today the Gulf of Maine is one of the fastest-warming bodies of water on Earth, as carbon dioxide billows unceasingly from the crust, and lobster populations face steady declines in the decades to come.

Similar transformations have swept the entire Earth over the past few transformative centuries, leaving behind their own local stratigraphies. Geologically, though, this has all been instantaneous. In the outcrops of Meishan, the "brief" 30,000-year End-Permian extinction interval comprises only a few inches of rock. Similarly, to geologists of the far future, the end of the ice age will be scarcely distinguishable from the extinctions in our own time, or the ongoing industrial pulse of CO_2 into the oceans and atmosphere. The past few centuries really have been comparable in their transformational rapidity only to those other great paroxysms of Earth history.

THE REASON SO OFTEN GIVEN FOR WHY THE WORLD WENT OFF ON such a dramatic excursion over these past few centuries, and why the planet looks the way it does today—why the ocean is emptying and warming, the atmosphere is filling with CO_2 and the plastic in the ocean will soon outweigh the fish in it—is not some new continent-spanning volcanic eruption, but something called capitalism. And if large igneous provinces like the Siberian Traps are rather messy affairs, capitalism is a bizarrely ordered new geophysical force on the planet. But for an institution that is blamed by some for nearly every ill of modern society—and virtually every benefit by others—pinning down a definition of it can prove elusive.

Is it commerce? If so, then perhaps it dates back to trade networks for obsidian and iron pigments in the Pleistocene. (Virtually no one argues this.) Is it markets, or interest-bearing debt? If so, then the very earliest Mesopotamian civilizations qualify, more than 5,000 years ago. “The first capitalists on record were Sumerian merchants long ago in the third millennium BC,” claims the Oxford historian David Abulafia. “Loans at interest, business partnerships, trading contracts assigning risk, and other indications of a commercial economy with many of the attributes of mercantile capitalism abound.” Abulafia is likely in the minority in assigning such an early date to capitalism. But why? Imperialism is often bundled with it too, but as we’ve seen, that institution is equally as antique, and universal, as slavery. On that score, rebellions led by enslaved Africans made to work on sugar plantations rocked both eighteenth-century Haiti and ninth-century Abbasid Iraq. Perhaps it’s private property, which has existed in places as long as the written word, and has had adherents from ancient Babylonians to native Californians. If it’s wage labor, then the Greco-Roman world had a version of it.

Perhaps capitalism is just a sufficient accumulation of financial innovation and sophistication. If so, then the commercial Islamic world of the Middle Ages was already pretty far along—with a web of merchants seeking their fortunes everywhere from the coral trade in Algeria to slave markets from Sudan to Cyprus. This diaspora, scattered among trade networks that spanned all the way from Spain to Malaysia, pioneered many of the hallmarks of modern finance, from promissory notes to complex credit instruments, all in search of private riches. If it’s modern banking, then perhaps the birthplace of capitalism was in the fourteenth-century Italian city-states like Genoa and Venice, where the bank was a *banco*, a table in the street where the *banchieri* traded currencies and bills of exchange. If it’s the multinational corporation that does the trick, then it was an invention of the Dutch and English globe-trotting companies of the sixteenth century—or perhaps it was invented several centuries earlier by the international finance operations of the warrior-monk Knights Templar.

Typically, though, when capitalism is invoked to explain the ongoing, almost geologically unprecedented derangement of the Earth system and explosion of CO_2 out of the crust, its critics don't have these early origins in mind. Few of the masked protesters at World Trade Organization meetings would implicate the international trading houses of the Bronze Age for the current state of the world, or the astounding scale of private commercial activity, mediated by paper money, that stupefied Marco Polo upon arriving in the Yuan dynasty of the thirteenth century. So what are they talking about?

One thing everyone agrees on is that it has *something* to do with money. As discussed, humans had come to inhabit a uniquely symbolic niche, giving us the strange but highly useful capacity to understand, and believe, that an arbitrary sound can stand in for something in the real world if enough people agree that it does, or that a hieroglyph or a few tiny brushstrokes of ink can do the same. So too can a *token* represent something of value in the real world, if enough people agree that it does. Over time and across the world, this "money" has taken the form of tusk shells, lamp shells, snail shells, surf clam shells, heads of cattle, smoked eels, glass beads, axes, copper bracelets, feather coils, giant limestone discs,[fn2] lines of code, cotton cloth, and pieces of paper—from the *chao* currency of Kublai Khan to the kinds of corporate scrip used everywhere from coal company towns to Walt Disney World. And then, of course, there's gold. But, setting aside its malleability, relative scarcity, and undoubtedly cool provenance from the kilo-nova mergers of neutron stars, there is nothing intrinsically valuable about gold. It is only valuable as a medium of exchange, or store of value, because people agree that it is.

Specifically, though, the kinds of commercial transactions that we've now come to see as the basis for society—handing over a few dollars for a gallon of milk, for instance—are, in fact, very strange, alienated kinds of interactions. "It was only in the sixteenth century that markets became a primary way to satisfy daily needs and a place where prices were formed," writes the economist Kent Klintgaard. In many tribal societies, such impersonal exchanges typically would have been made only between strangers, and almost never for life's necessities. They would have seemed, and indeed *did* seem, all but alien to the inhabitants of more communal societies everywhere, where basic needs were instead met collaboratively—and paying for such staples would have seemed distasteful, even sociopathic. "For Indian nations, commerce was inseparable from relations of kinship and friendship, but also from captivity and warmarking alliances," writes the economic historian Jonathan Levy. As a result, they "noticed something different and strange about the English economic personality."

Where tokens of value *were* exchanged in many societies, they served far more social causes, like forestalling vendettas or commemorating peace treaties. Belts of wampum honoring a war compact, though clearly of tremendous value, obviously can't be sold, transferred to third parties, or chopped up and securitized. Such a concept doesn't even make sense. But in societies where trading with strangers was commonplace, like the sea-facing hubs that often serviced trade between great empires or regions, it didn't make sense to make an obligation to someone arriving in the local port from afar whom you would likely never see again. It is no surprise, then, that it was in just such a place—the relatively minor western Anatolian kingdom of Lydia, situated between the cosmopolitan worlds of the Mediterranean and Southwest Asia in the sixth century BCE—that the first standardized coins were minted. Or that the pioneers of much later financial innovations would often reside in similar geographic circumstances—along the necklace of trading posts along the Indian Ocean, or in Genoa, or Amsterdam, or London. It was in this commercial world, once obligations were transformed to *debts*—quantifiable and transferable—that the impersonal logic of the market could begin to insinuate itself into daily life.

This development facilitated positive-sum trade among large groups of strangers who might not otherwise collaborate, and stimulated the division of labor over broad geographical areas connected by trade—with middlemen coordinating the shipments of grain or timber or wine to the areas that needed (or simply wanted) them most, but which otherwise might have little, if any contact. It even provided a leap in energy efficiency: these seagoing merchants no longer had to fill their holds with heavy cargo to pay for goods at distant ports, but could instead carry comparatively light sums of money. But the potential commoditization and monetization of everything threatened to open up the entire moral universe of social interaction to the realm of bloodless calculation, the logic of the marketplace. As a result, from its earliest beginnings people have lamented the potentially socially corrosive effects of this strange technology. And inevitably, in any society with markets and money, some enterprising merchants will realize that money isn't just a way to ease transaction costs but is valuable in itself, and begin stockpiling it. For this reason, rulers in societies across the globe have perennially sought to check this independent source of power, the accumulation of private wealth.

"Though there were always urges in a market economy for some to pursue endless accumulation of monetary wealth, these urges had been repressed by political and cultural forces before modern times," writes the Johns Hopkins sociologist Ho-fung Hung. Instead of hoarding captured treasure in glimmering vaults, as petty tyrants had been doing since time immemorial, or giving it all away in extravagant potlatches, the use of money to make more money still was a peculiar activity that thinkers from Aristotle to Islamic jurists had long found unnatural. States, meanwhile, often actively sought to curtail it. Of the later Qing dynasty, for instance, Hung writes, "Because [it] was governed by a centralized paternalist state constantly fearful of social and political unrest stemming from rising inequality, merchant activities were circumscribed within the realm of market exchange, and the expansion of capital-accumulation activities was contained by a state that saw such activities as disruptive to social stability."

It was a different case entirely, though, across Eurasia in Western Europe, which, since the fall of Rome, had remained something of a backwater compared to the grandeur of the East, with living standards not reaching those once found in the Eternal City until the eighteenth century—and whose expatriates continually noted their astonishment at the advanced state of the other societies they encountered, from Shangdu to Tenochtitlán. Here a fractious and fragmented array of perpetually cash-strapped, midsized polities needed to continually raise money to engage in nearly perpetual war with each other, and so forged a unique alliance with these burgeoning private sources of wealth. As a result, the modern world was born.

WHY WAS THIS REGION SO BELLICOSE? WHERE THE VAST EMPIRES THAT spanned the rest of Eurasia might fall every now and then, many of them often reconsolidated at the same sweeping scale. In the sixteenth century, for instance, the vast majority of the world population were subjects of the Ottoman, Safavid, Ming, or Mughal empires. Indeed, a vast sweep of China had been largely unified as a bureaucratic empire for much of the past two millennia.[fn3] Europe, meanwhile, would remain an unusual, highly fractured pressure cooker of smaller political units, with only a few brief exceptions[fn4] of subcontinental union. By the dawn of the early modern period, Europeans had come to organize their societies in a kaleidoscopic and overlapping landscape of kingdoms, duchies, baronies, bishoprics, margravates, principalities, city-states, boroughs, town-leagues, and unwieldy composite monarchies, with political power ambiguously divided and shared between lords, emperor-kings, princes, popes, antipopes, dukes, knights, bishops, counts, nobles, merchant guilds, and braiding lineages of royal families. And where East Asia saw more or less peaceful periods that could be counted in centuries and sweeping bans on firearms that lasted just as long, such a thing was unheard-of in Europe, where these fractured societies were almost always trying to kill each other—by one count, fighting 443 wars between 1500 and 1800.

This difference in political formation, and more specifically, its *failure* in the case of Western Europe, might have been encouraged, at least in part, by geography. While it's important to be mindful of the bugaboo of environmental determinism, it is nevertheless uncontroversial that human societies are constrained by the geography and ecology within which they emerge. Sometimes human actors really do put their hands on the wheel of history and transcend these natural forces. And accidents of all kinds do intervene to reroute the trajectory of human affairs—whether a divine wind that sinks a fleet of invaders or a wrong turn on the streets of Sarajevo. But over the long run, could there be potential geographic features that might load the deck toward a particular outcome in a particular place? Some historians, including Stanford's Walter Scheidel, think so. Scheidel argues that regions like China's Central Plain, which saw vast fertile stretches united by similar topography and climate, connected by long navigable waterways like the Yellow River, encouraged the consolidation of large core areas. At the same time, China's proximity to the steppe and the endless incursions of pastoral horseback raiders, along a vast ill-defined border region, required perpetual scaling up and expansive state-formation to defend itself. Elsewhere in the Old World, the recurrent empires of the Middle East and broader Southwest Asia—from the Achaemenids to the Sasanians—also found themselves on the doorstep of the steppe, and the threat of horse-based incursions from the north similarly encouraged these polities to scale up and build vast consolidated empires.

Western Europe, in contrast, was uniquely carved up by imposing walls of mountains, marshy lowlands, forests, deserts, verdant river valleys, and, perhaps most importantly, fractal coastlines fringed with peninsulas and islands (about 60 percent of Western Europe is on peninsulas and islands, versus 1 percent to 3.6 percent for China, North Africa, India, and the Middle East). This geographic division was reflected in a uniquely fractious society, as power came to be durably divided between and among small competitive political units—a division that saw interstate war become a feature of the region, marking 75 percent of the years between 1494 and 1975. Japan, meanwhile, found itself so secure in its place in the world that it would abandon guns altogether, discarding them like the evolutionary loss of a vestigial appendage once the selection pressure for it was gone.

Geography might have preserved this competitive European dynamic in another way too. This is because the great Eurasian steppe ran out on the slopes of the Carpathian Mountains and the Alps. These imposing mountain ranges, the legacy of the epic continental collision of Africa and Europe beginning in the Cretaceous Period tens of millions of years earlier, might have insulated Europe's small-scale polities from a vast flank of mobile raiders that might have otherwise encouraged these societies to consolidate to repel them, or, failing that, be absorbed into a vast steppe-based empire. Indeed, when an eleventh-century proto–industrial revolution threatened to emerge in Song dynasty China—featuring everything from large-scale deforestation and coal mining to feed the forges of iron foundries, to a proliferation of printed works and even automated water-powered textile machines—the growth was rendered stillborn by continual horse-mounted trouble along its vast northern and western borders, trouble that would eventually culminate in the Mongol conquest of China. In an alternate world, though, on the other side of the continent, had the steppe extended to the Atlantic Ocean, so too might the Khans have. Instead centuries went by on this uniquely compartmentalized, wrinkled, and insulated Western European geography, where no particular power was able to gain much of an advantage, natural obstacles like the Pyrenees proved stubbornly durable as political borders, and small warring factions stayed forever at each other's throats. In this biogeographic crucible, firearm technology originally borrowed from China was ruthlessly refined, growing ever more lethal, while military organization and strategy grew both more impregnable in defense and more devastating on offense.

But while war has always been a crucible for military technology, it was increasingly becoming one for financial innovation in Europe as well. In the Italian city-states of the fourteenth and fifteenth centuries, as Tuscany erupted in wars of all against all, and Florence, Pisa, and Siena vied to outbid each other for mercenary armies—they increasingly found themselves under avalanches of debt. And it was in these desperate financial straits that government bonds were born, as the wealthy families of Florence were forced to loan money to the city government, in return for interest paid on those loans, so that it could continue to wage war against its neighbors. In independent Venice, meanwhile, there was even secondary trading of these new financial instruments, giving rise to the bond market. Here, Venetian bankers, by building on the financial innovations of the wider Islamic commercial world with which they traded (and even adopting their more efficient Hindu-Arabic numerals), pioneered the arts of double-entry bookkeeping, currency conversions, and bills of exchange. Venice put these new financial institutions to work in service of a lucrative and increasingly violent trading empire, one that erupted in armed conflict with rival Genoans for control of the Aegean and access to Byzantine markets, then the Ottomans for control of Constantinople. These maritime republics sought not only control of Mediterranean and European markets but access to the goods of the far East that were then pouring into the ports of the Levant and Egypt—the western outlet for a growing Muslim trade network that, by the thirteenth century, had spread all the way across the Indian Ocean.

The magnificent goods that flowed into European royal courts from afar loomed large in the imagination of cash-strapped monarchs. Buried in war debt, facing rebellious populations and unruly lords at home, and unable to amass wealth and power by territorial conquest, these rulers increasingly sought their wealth beyond the claustrophobic mainland. And so, as the superior Chinese silks and porcelains, the handsome calicos of Gujarat, and the spices from Malaysia were unloaded from Venetian galleys, Europe turned its sights to the source of these wonders: prosperous Asia.

Strangely, the paroxysm of the Black Death and its demographic aftershocks might have indirectly aided this European restlessness. Perhaps hopscotching three thousand miles on camel caravans before flowing down Russian rivers on grain ships to Crimea, the plague would expose the perils of widening globalization: Genoa's expanding trade empire had now come to include the Black Sea, and as a result it exported the goods of the Far North—in furs, wax, slaves, and, now, apocalypse to the rest of Europe. Starting in 1346, this plague killed half of everyone in Europe, with the population only recovering by 1600. The resulting scarcity of farm labor meant landowners were forced to pay rising wages, increase agricultural productivity through technology, or give the land over to higher-value but less labor-intensive practices like sheep grazing, to produce wool. This rendered marginal farmlands unattractive for grain farming, which in turn filled entire regions of Europe with superfluous crews of troublemaking men no longer needed on the farm. These were the perfect people to staff the mercenary armies that would soon maraud the countryside, or the motley ship crews that would soon maraud the Earth in search of Asia.

The world-remaking voyages of Christopher Columbus and his successors emerged, then, not from European strength, but from European weakness. Where the Roman Empire had once freely traded in the Indian Ocean, Western Europe was now partly stymied by the more capable and unified Ottoman Empire, which came to rule Turkey, much of the Middle East, North Africa, and the Balkans, and which dominated the overland trade routes to India and East Asia. Moreover, the Ottomans were an exemplar of competence against which Western European powers measured themselves: secure in tribute flowing from all over their empire, unified under a powerful bureaucracy and centralized state finances, wielding fearsome naval power and firepower, and ruling from the former capital of the Eastern Roman Empire. Indeed, Ottoman rulers even took calling themselves Roman emperors. “On their side is the vast wealth of their empire, unimpaired resources and practice in arms, a veteran soldiery, an uninterrupted series of victories,” wrote one dejected European ambassador as Suleiman the Magnificent laid siege to Vienna. “On ours are found an empty treasury, luxurious habits, exhausted resources, broken spirits.”

“The men who sailed forth from western Europe across the seas in the fifteenth and sixteenth centuries did not set out to create ‘merchant empires’ or ‘western colonialism,’” write New York University historian Jane Burbank and sociologist Frederick Cooper. “They sought wealth outside the confines of a continent where large-scale ambitions were constrained by tensions between lords and monarchs, religious conflicts and the Ottomans’ lock on the eastern Mediterranean.” As such, these threadbare voyages, haphazardly and speculatively financed by independent networks of wealth, left European ports with a much different disposition than, say, the spectacular fifteenth-century naval treasure fleets that set out from Ming China, voyaging all the way to Africa to advertise the glory of that hegemonic empire (and which were quickly scrapped as a wasteful extravagance for the land-based empire). No, these European ships left shore with a rather more practical aim: profit. For cash-strapped states, granting monopolies to these crews of often lethally unprepared desperadoes and taxing the windfall profits they returned (if they managed not to die en route), it was worth the roll of the dice in this war-stricken and indebted corner of the Earth. And unlike in Ming China, where overseas voyages could be forbidden wholesale from on high, in the fractured political landscape of Europe, Columbus could canvass the states of Western Europe, securing the blessing of Ferdinand and Isabella of Spain after first being turned down by Portugal.

In this desperate search for access to Asian trade—borne across the sea by the trade winds—Western Europeans stumbled upon two continents previously unknown to the Old World. But, to their great fortune, the mountainous western spine of these surprising continents had been subducting ocean crust for tens of millions of years and filling its volcanic slopes with gold and silver, including the biggest silver deposits on Earth. They only needed to take it from whoever lived there. Imperial Spain's drowning war debt and financial problems were over, or so it seemed.[fn5]

This accidental collision of the grasping and middling craggy corner of western Eurasia and an entire hemisphere of peoples and ecosystems launched Europe into global relevance, while killing tens of millions of indigenous Americans by European disease and at European hands. The mass death resulted in 7 billion tons of carbon flowing from the air to the land—as trees regrew across an American landscape once thinned by indigenous agriculture—and remade the ecology and cultural landscape of the entire world.

THIS FINAL, EXTRAORDINARILY VIOLENT, AND TRAGIC STAGE OF GLOBALIZATION would ultimately tie the Americas, and Australia too, into a Eurasian and African world linked for millennia by the exchange of goods, ideas, people, and biology. It was led by a series of increasingly modern European hegemons, first Spain, then the Netherlands and England. Each would build off the innovations of the previous empire and ultimately ensnare the entire world in a Western European web, an outcome no one on Earth could have predicted in the preceding centuries. But the die had been cast, and for aspiring powers, adopting these tools was not really optional. "Had any aspiring power—from Portugal to England to the Mughals—tried to play by the rules of 'free' markets, thinking they could avoid the cost and burdens of running an empire, they would promptly have been marginalized or eliminated from the scene," write Burbank and Cooper.

With the English, the project that would fuse capitalism and empire to all the energy in Earth history would finally come to fruition, and, after 300,000 years, humanity's impact on the planet's carbon cycle would rapidly and fundamentally rupture.

It was perhaps surprising, then, that it was the small coastal power of Portugal that would kick-start this process—a country that would remain a minor geopolitical player in mainland Europe, depending on the Spanish for protection when not dynastically united with them, as it would be for the better part of a century. When heavily armed Portuguese traders sailed around the Cape of Good Hope and blasted their way across the Indian Ocean for access to Asian riches, they shattered the long-standing Islamic commercial networks with exceptional eruptions of violence. Effectively wresting control of the spice trade, they built forts along the coasts—militarized entrepots at the nodes of long-developed trade networks, about which they knew little—intercepting these commercial flows across the Indian Ocean world by force. Their presence in the region took on the character of a "protection racket" from Mozambique to Macau.

An ocean away, the Portuguese were busy launching an even more sinister project. Starting in the fifteenth century, Portugal explored the coast of West Africa by sea, mostly seeking access to the source of the African gold trade. But soon its energy would be redirected into a far more tragic enterprise. After the Black Death wiped out sugar production across the Middle East, Genoese merchants invested in new sugar plantations on Portuguese and Spanish colonial holdings in the Atlantic, across the Madeira, Canary, and Cape Verde Islands. Needing labor for the cruel work of the plantation, and circumventing the trans-Saharan slave networks long controlled by Muslim traders, the Portuguese would begin to trade arms to indigenous rulers along the coast for local war captives.[fn6] These were the makings of an increasingly profitable and demonic arrangement: the lucrative slave-based Atlantic sugar plantation. And where the experiments with this system began in the Mediterranean and on the islands of the eastern Atlantic, with its new possessions in South America, Portugal could export this model to an entire continent and with it give rise to a new and festering ideology of racial superiority to justify its cruelty. And so the Portuguese built coastal fortifications in Angola like they would in the Indian Ocean, Brazil would become the world's biggest importer of slaves for the next three centuries, and the world economy would increasingly come to center around "African labor, American land, and European markets."

It wasn't Portugal though, but Imperial Spain that would mark the world's first truly global power, boasting trade networks that stitched together the entire planet's oceans for the first time—with Spanish galleons setting across the Pacific from Acapulco to the Phillippines, at the same time those on the Atlantic dodged pirates of the Caribbean en route back to Europe. But Spain was a relic of the old feudal order, and it was obsolete almost as soon as it had reached global preeminence. In fact, the entire millennia-old feudal system of Europe was doomed. Before long, climate change would kick out its energetic agricultural foundation.

Though the world at large experienced less than half a degree of cooling, Europe would be seized for decades by devastating winters that froze livestock in their tracks, and strange rainy summers that spoiled crops. This was the so-called Little Ice Age, which reached its depths in the seventeenth century as a series of unseen tropical volcanoes dimmed an already somewhat subdued sun—then experiencing an exceptional run almost free from sunspots. The chill drove crop failures, religious fanaticism, and apocalyptic warfare in Central Europe that took the lives of up to half its population. In Northern Europe, where grain farmers already found themselves on the edge, the cold pushed society into the wider world to instead buy grain on the international market. And it was in the estuaries of the waterlogged Low Countries along the North Sea that an emergent Dutch state would come to represent what one historian calls the "complete fusion of capitalism and imperialism."

Nominally subjects of the globe-bestriding Spanish Empire, the Dutch were quietly emerging as the shipping powerhouse of Europe. As the Little Ice Age took its share of the harvest, the Dutch looked abroad to the Baltic for grain and soon became the grain suppliers to much of a hunger-racked Europe. Importing this cheap grain freed much of its population from farmwork, and Dutch towns boomed. Meanwhile, the farmland was turned over from staple crops for subsistence, to cash crops for the market, like vegetables and flowers, and dairy and meat production. Farmers began to scatter clover seeds on their fallow fields to feed their cows through the winter, the happy side effect of which was a massive boost to the nitrogen in the soil and to crop yields in turn—spurring an agricultural revolution in Europe that could supply far more calories. And the Netherlands' weak nobility and clergy meant that much of the country's wealth could accumulate in the hands of its burgeoning entrepreneurial business class, the wealthy burghers. Relatively free from interference, they plowed this money back into commercial ventures, especially shipbuilding, through which the Dutch soon produced a new phenomenon on the waves entirely, swarming the seas with hugely capital-intensive fleets of herring buses—lumbering ships up to fifty tons, purpose-built for fish cargo, upon which entire shoals of captured herring could be salted and barreled at sea, then shuttled back to shore by carrier ships. These fish were caught not for subsistence but for sale as an internationally traded commodity. The fishing, meanwhile, could continue for months on end, as hundreds of buses chased schools of fish across the North Sea, setting drift nets at night on the Dogger Bank and diverting hundreds of thousands of tons of herring into the stomachs of Europeans. This was industrial-scale fishing. And it was hugely profitable. Having already become the fish and grain merchants to the desperate Europe of the Little Ice Age, the Dutch began to set their sights even wider.

They would soon marry this precocious maritime trading economy with still more financial innovation to bankrupt their Spanish oppressors, opening up a violent new frontier in the history of finance in the process. When the Protestant Dutch Republic declared its independence from the Catholic Spanish yoke, the Spanish responded by attacking the rich commercial hub of Antwerp, and Amsterdam welcomed the ensuing exodus of prosperous merchants who fled the city as a result. It was a momentous migration. These émigrés would provide much of the capital for the new Dutch East India Company, an organization founded to consolidate Dutch forays into the Indian Ocean and exact revenge on the Spanish-allied Portuguese with heavily armed merchant ships sent across the world to wipe out their monopoly and intercept the flow of goods and money. Along with the British East India Company, it was the planet's first modern joint-stock company, one that gave rise to the world's first stock market, in Amsterdam.

It was an increasingly modern arrangement: a multinational corporation, with limited liability, run by a board of directors, drawing from networks of private finance, the shares of which were sold on a stock market, and the returns on which were potentially astronomical. The corporation, though it had ancestors in Italy, was a new kind of social organism, one whose goals and incentives transcended those of its individual members. Soon thousands of Dutch ships swept across the seas, from Manhattan to Melaka, boasting a fleet larger than all their European rivals' fleets combined. Their extraordinarily blood-soaked command of the ocean and mastery over the trade of commodities, from nutmeg to timber to slaves, made the Dutch the wealthiest people in Europe. Their success in shattering the Portuguese stranglehold on the Indian Ocean would inspire the Netherlands' North Sea competitors as well.

Like their Dutch neighbors, English society was utterly transformed by the Little Ice Age. And this transformation shared more than a passing resemblance: England would adopt not only a Dutch ruler in 1688, but also the Dutch joint-stock model, which it would take from Hudson Bay, to New England and Virginia, to India. But every new hegemon builds on the successes of the last, and where the Netherlands' rich peat deposits powered its "Golden Age" (along with the wind that pushed its ships and mills), England would begin to pull away from the Dutch in the coming centuries as it increasingly tapped its far richer geologic deposit of photosynthesis in coal. It would apply this unprecedented explosion of energy to a new model of military-backed corporate plunder, huge outlays of public debt, protectionism, and the practical application of science to industry and war. As a result, and simply following where profit led, the English would soon trigger an almost geologically unprecedented explosion of carbon out of the crust and an attendant derangement of the planet's life-support systems unseen in tens of millions of years. And where the ships of the Dutch East India Company had to fight their own battles, the British increasingly backed up their economic interests around the world, and made claims on entire continents and peoples, with the support of what would become the biggest navy the world had ever seen.

THE ETERNAL RIVALRY OF ENGLAND AND FRANCE WOULD EXPRESS itself in a global war that came to include pitched battles in German fields, bombardments of colonial outposts on the coast of India, and sieges of bastion forts on New England lakes and along Appalachian rivers, with indigenous allies on both sides and men-of-war splintering each other's masts on the main. And as the ratchet of military technology and strategy turned, and the scale and cost of war only increased—requiring ever-larger armies equipped with ever-more-advanced technology to fight battles spilling over oceans and continents—these wars came to be extraordinarily costly, losing them even more so. War would claim upwards of 85 percent of the British government's expenses during the eighteenth century, and public debt would soar to twice the country's GDP to fund this bloodshed.

Although ruinous debt had serially brought down war-hungry European rulers before, British interest rates would remain surprisingly low, and the government was able to pay back these ever-larger loans over longer terms. This was because, whereas capricious autocrats throughout history had wrung money from their subjects coercively in times of need, or abruptly defaulted on high-interest loans to outside creditors, in ascendant England the king's authority had been increasingly constrained by the Parliament, where consent was required to fight wars and raise taxes. In such an arrangement, the Crown had to negotiate with the country's blossoming commercial class for resources, and the resulting interdependent stability and reliable prospects for eventual repayment ensured low interest rates—effectively enabling the state to recruit vast profits from the future to pay down the cost of its wars. When the Bank of England issued banknotes that were essentially chopped-up pieces of the Crown's debt in the form of IOUs, the population became more than casually interested in the outcomes of the distant wars its empire fought.

"The unusual interstate system [in Europe] which was plagued by frequent military conflict, urged state makers to compete for internationally mobile capital to finance their war efforts, thus forging a state–capital alliance unseen anywhere else," writes Ho-fung Hung. "Under such an alliance, capitalists supported state expansion by purchasing government bonds and submitting taxes, and the state offered military and political protection crucial to capitalists' accruing and securing of resource bases and trade routes."

The financialization of the economy grew apace, and by the eighteenth century, the litany of new modern institutions only grew: the huge insurance market Lloyds of London—which would insure ships, slaves, and colonial properties and remains today the largest maritime insurer in the world—was founded to make risky financial endeavors safer. Meanwhile, the similarly modern phenomenon of the “bubble” showed the system’s fatal weaknesses, bringing ruin to Dutch tulip speculators and English investors in failed joint-stock companies abroad. A similar bubble even wrecked the entire French financial system in 1720 when a frenzy of land speculation in colonial Mississippi cratered. This wave of bubbles was enabled by the reckless proliferation of discounted bills of exchange and increasingly opaque financial arrangements that, when married to the perennial allure of hype, speculation, and the quick buck, made for an explosive mix. Just as bad information about agricultural yields could bring down ancient empires, bad information about the material basis of the global economy could now bring low the increasingly financialized colonial empires. Bubbles frothed and burst and threatened to bring down the whole system, all in service of the elusive return on investment: capital seeking more capital. Karl Marx famously compared the plight of humans in his age to that of the sorcerer in “The Sorcerer’s Apprentice,” one “who is no longer able to control the powers of the nether world whom he has called up by his spells.” Financial centers soon supplanted royal courts as the locus of power in society, and long-reigning hereditary nobility lost control of this new world to the upstart capitalists.

On the home front, the very English countryside was being rapidly transformed, and it’s here in the rolling pastures of England’s South Midlands that some identify capitalism’s final consummation and total victory over society. As population rebounded after the Black Death and as the Little Ice Age cut into agricultural yields, landowners were squeezed for cash. With their interests now well represented in Parliament, they sought to dismantle a hoary feudal agricultural system that had limped on into the modern period. In the medieval fief, peasants were granted customary rights to graze and farm on common lands, and they foraged in the fens and marshes at field’s edge. The point of these farms was to support a perennial self-sufficient state of subsistence, not growth. Peasants owed lords their fealty but not profits, and for the most part, the fief was a community closed off from the vagaries of wider markets, and an all but changeless feature of the landscape. There was no incentive for peasants to produce much more than for their own needs, since any surplus was taxed to support the lavish lifestyles of their lords. But the dismantling of this old world unchained capitalism and swept this entire landscape into the broader economic competition to increase productivity and profit.

Those who aver that capitalism is the inevitable expression of human nature suffer from both a lack of imagination about the varieties of human society and a lack of historical knowledge. The transition to capitalism in the English countryside was not a gradual transition to modernity but a traumatic rift with the past. In a series of laws passed by Parliament, landowners were granted the new capacity to enclose the commons as private property. Enclosure riots swept the country as common lands were plucked from a peasantry now cast out into the new capitalist world. Agricultural yields soared as farmland was devoted to moneymaking crops and livestock, and as wetlands were drained to bring ever more farmland into production. Society, meanwhile, was upended. The Irish poet Oliver Goldsmith wrote of the enclosures that “wealth accumulates but men decay.” This was a reorganization of how energy flowed through society. Large landowners, freed from their feudal obligations to the peasantry and competing against each other to more efficiently produce high-value commodities on their land, expelled previously self-sufficient peasants into the growing cities, where they had nothing to sell but their labor. Even this was a strange new concept, one that had to be invented. “Labor is only another name for a human activity which goes with life itself,” wrote the early-twentieth-century economist Karl Polanyi, “which in its turn is not produced for sale, nor can that activity be detached from the rest of life.” Even those capitalists who purchased this human activity then had to compete with rival firms to employ it as profitably as possible, or else face extinction. Everyone was now a participant in this system, and no one could opt out.

Given that producing anything takes energy, and getting more out of less was the route to profits—and that *un*profitable businesses were ruthlessly weeded out of the system—this was an engine for generating surpluses through ever-escalating efficiency gains via technology, or by tapping into previously unexploited resources or “labor.” It delivered on its promise: agricultural productivity on capitalist farms rose in lockstep with growing food surpluses.

This resulting surplus, and profit, redounded to the benefit of the capitalist farmer—a profit denominated in money, which could be reinvested, spent on still other things, or used to buy more land, do more stuff. It was a competitive game that, in principle, was open to much of the population—not just sun kings and nobles, but sheep farmers and tradesmen too. Anyone who could accumulate enough capital could put it to work making more. In reality, though, this was still an algorithm that excluded most of humanity, and the raw materials that fed this economy remained largely produced by slaves in distant reaches of the world—mining gold and silver, and harvesting sugar, coffee, and cotton to feed the industrial machine. This way of organizing society and nature around the market required not only the creation of a new informational landscape for humans to inhabit—one of private property, and contracts, and laws—but also a physical transformation of the environment, with the construction of ports, harbors, roads, and bridges that eased the circulation of these commodities across increasingly vast trade networks and new markets.

If there were no physical or energetic constraints, this was a system designed to endlessly grow. But on the eve of the Industrial Revolution, no one could imagine a world of endless growth. Colonial powers felt themselves to be in a brutal zero-sum contest for resources, and the field of economics was still grounded in the material world. French physiocrats like Antoine Lavoisier still saw the photosynthetic base of agriculture as the source of all earthly wealth, while British economists from Adam Smith to David Ricardo to Thomas Malthus addressed themselves to problems of soil science and the caloric limits of farming and population growth. Because land was a finite resource, and the energy of society still flowed from plant photosynthesis, these economists insisted that any economic growth was self-limiting and would ultimately trend toward a level that barely rose above subsistence—a pessimistic view that was informed by history.

This preindustrial world was still run entirely on solar power at the Earth’s surface. It came packaged in the plants that powered human and animal muscle alike, in the firewood that heated homes and smelted iron, and in rivers that now powered multiplying cotton mills—rivers whose downhill release of energy was the product of (solar-powered) evaporation into the sky and rainfall. It was also found in the sun-driven winds that carried slave ships across the Atlantic to produce the cotton that fed those mills.

Today, buoyed by two and a half centuries of ever-escalating global energy consumption made possible by the one-off explosion of the planet’s fossil fuel battery, the self-appointed heirs to Adam Smith ridicule the relevance of these self-limiting energetic brakes on growth, which had applied to all societies before the Industrial Revolution. But by the eighteenth century, China and Europe both found themselves at a similar level of economic development and stagnation as they ran into the ecological limits inherent to all societies powered only by the energy embodied in photosynthesis at the Earth's surface. "As land grew scarce in the eighteenth-century cores, muscles got expensive," the historian Ian Morris puts it evocatively. With fossil fuels and steam engines, though, Europe would blow away all the energetic limits that had forever hemmed in complex societies. Up until this moment, there was a seemingly eternal ceiling on the possibilities for society, as 90 percent of citizens were required to farm, agricultural surpluses limited the size of armies, and a diminishing percentage of the population would ever learn to read.

Then, in 1776, the same year that Smith's *Wealth of Nations* was published and a new nation declared its founding before soon exploding across an entire continent, the Scottish engineer James Watt and his English business partner Matthew Boulton commercially released their steam engine, which could convert heat to mechanical motion. Though it would still take decades for the fossil fuse to ignite, when it finally did the planet would never be the same. When capitalism's imperative of ever-escalating productivity and growth became capable of yoking the mechanical work of industrial production to the reservoir of fossil solar power built up over hundreds of millions of years since Snowball Earth, it would obliterate the energetic limits to the human society.

THIS WAS THE WORLD AT THE PRECIPICE. AT THE EPICENTER OF ALL this commotion was London, a city quickly becoming a behemoth, whose growth was fast outpacing its resources. A mere second-rate conurbation in 1500, London had become Europe's largest city by the eighteenth century, requiring fuelwood from a geographical area perhaps 50 times the extent of the city limits to meet its energy needs. Soon enough England was facing a massive wood shortage. And not only for heating, but for the timber needed to build its massive warships—each of which required 2,500 oak trees—as they vied for naval supremacy on the open seas and secured the flows of global trade. This war machine required not only timber but an ever-growing supply of iron as well, which was smelted with charcoal in blast furnaces that, in the early eighteenth century, were so inefficient that to keep them sustainably fueled required six square miles of forest *each*. As a result, the island had been mostly deforested by millennia of agriculture, firewood, timber, and charcoal production.

At the same time, this, the fastest-growing economy in the world, sat atop one of the biggest deposits of Carboniferous coal in the world. This was a disequilibrium, poised for discharging if there ever was one.

Just as dynastic China sprang from the fertile loess soils left behind by the dust storms of the ice ages, which were carried and transformed by rivers into rich agricultural lands, the Industrial Revolution would spring from, and map on to, this far more ancient Carboniferous geology beneath—igniting a brushfire of geologic energy that would eventually widen into an Earth-spanning bonfire of deep time. Not only did Britain sit on top of this mother lode of coal, but it also cropped out in places like Newcastle, above tame, navigable rivers like the Tyne that were connected downstream to growing, energetically voracious cities like London, connected on the other end to the resources and markets of the wider world by the Thames. This riverine connection was crucial in an age before railroads, when "shipping [coal] the three hundred miles or so from Newcastle to London cost about the same as carrying it three or four miles overland," Barbara Freese writes in *Coal: A Human History*. If the Industrial Revolution was going to start anywhere, it would be in these unusually favorable circumstances.

This is the so-called "best first" principle: natural resources with the lowest energetic barriers to entry will be exploited first. It's a rule of thumb that explains much about the trajectory of the Holocene and beyond. It's why people hunted the biggest, most docile animals first, and fished out abundant shoals of fish closest to shore first. It's why agriculture began in the most fertile soils, and states coalesced in those places it was easiest to organize this harvest. It's why life started in those idiosyncratic crevices of the early Earth where its fundamental reactions were easiest to catalyze. It's why slower right whales were first commercially hunted by ships leaving from harbors situated nearest the great seasonal aggregations, and only later, when they were depleted, did Nantucketers voyage around Cape Horn. It's why, later, the oil industry would begin not with platforms hundreds of miles offshore, penetrating miles under the seabed, but with crude jerry-built rigs tapping gushers mere meters under the surface. It's one reason China is able to produce the latest energy source, solar panels, more cheaply and abundantly than anyone else, in factories next to, and powered by, coal mines, and worked by forced labor in western China. And it's one reason why the Industrial Revolution started here in Britain first: with its abundant geology and encouraging topography, it had a low activation energy.

However, not only were the forests mostly gone in Britain by the eighteenth century, but most of the easy outcrops of coal were gone as well. This was because coal had a history on the island that long predated the Industrial Revolution. As early as 1306, King Edward I threatened his subjects with death if they continued burning the dirty "sea coal" that they picked out of the cliffs, choking their countrymen with smog. By the seventeenth century, one Londoner would remark of his fair city that "if there be a resemblance of hell upon earth, it is in this volcano [on] a foggy day." The coal was being used for home heating, but it had also found a steady market for centuries among a proliferation of city industries, from brewers to soapmakers, who contributed to the volcanic ambience. But these were still simple applications of the fuel, using it only for the heat it gave off when combusted—something that would have been familiar even to the Roman soldiers who occupied England some 2,000 years earlier and used the coal to heat their baths. The key was converting this heat to mechanical *work*. By using the energy accumulated in these ancient plants not just to release heat as coal had always done, but to replace and automate the work done by human beings and beasts of burden who were powered by the food in their diet, or to replace the uncertain winds on the high seas, or local streams that turned waterwheels, the British would unleash an explosion of literal and figurative power.

By the end of the eighteenth century, the groundwork for this evolution to mechanical work had already been laid by water power, which animated an array of industrial machinery, efficiently grinding grain, powering sawmills, spinning wood lathes, and manufacturing textiles that enriched the empire. This work had to be done near rivers, but in theory any source of energy could move this machinery. However, the makers of the first steam engines did not have the factories in mind when they set about fashioning their epochal devices. Instead they trained their sights on a simple, practical problem: how to pump water out of coal mines.

Over the centuries, as the shallow deposits of coal had all been plucked from the surrounding strata, colliers had to go deeper and deeper into the rocks in search of more coal—not unlike the elephants who dig deeper and deeper cave systems out of the earth in search of receding veins of salt, required in their diet. But when these increasingly deep mines began intersecting the water table, they had to be continually pumped out. The first solution, as with many problems of the age, involved animal muscle, ultimately fueled by the energy captured by photosynthesis. Every energy transition is powered by the existing energy sources that came before. The buildout of a renewable economy will, by necessity, be subsidized by a massive deployment of fossil energy, for instance. Similarly, at the dawn of the fossil age, teams of horses—yoked to pulley systems—would arduously hoist water out of the mines by the bucket. But there were diminishing returns to this process: if you get less energy out of an energy system than you put into it—in both economics and biology—it cannot go on for long. Many of these "drowned" coal mines had to be abandoned, along with any profits they might have produced.

It was this problem to which the first coal-burning engines were applied, in an attempt to supplant horsepower in the draining of the coal mines. It was an English ironmonger named Thomas Newcomen who would design and build the first generation of the engines. His was a hugely inefficient[fn7] and cumbersome machine, one that ultimately proved impractical at many sites. It did, however, serve as a powerful proof of concept to the polymathic Scottish astronomical instrument engineer James Watt, who set about tinkering with this idea of a steam-powered engine in an effort to improve it.

Watt embodied the peculiarly British brand of the Enlightenment, where science was powerfully and profitably recruited to solve practical problems. With the invention of the printing press three centuries before, there was now a mechanism for faithfully reproducing and widely disseminating information—one that was unmatched in human history and allowed for the ratchet of culture and technology to furiously turn. The information technology, as Martin Luther would prove, was also lighter fuel for Europe's internal schisms and its uniquely violent and competitive state system. But on this multiplying parchment, Enlightenment ideas about natural science and rationality—hard-won after the devastation of religious wars and climate shocks that the clergy couldn't explain without recourse to ineffective witch burnings, pogroms, and exorcisms—fluttered over the fractious continent. A traumatized Europe was crisscrossed by, in the historian Philipp Blom's words, "philosophers and necromancers, Rosicrucians and mathematicians, occultists and alchemists, magicians and soothsayers, engineers and astrologers, mediums, Kabbalists, and prophets," groping for new, better explanations of the world. One of these lineages of knowledge production would prove unusually effective: the systematic investigation of the world that took nature on its own terms—a method that we would ultimately recognize as "science." It was articulated by new kinds of natural philosophers, like Antoine Lavoisier, who increasingly rejected traditional sources of knowledge derived from religious and mystical revelation, folk wisdom, divination, or the hoary invocations of sacred texts, in favor of an unyielding pursuit of dispassionate and nominally objective "reason."

"Reason must continually be subjected to experimental proof," Lavoisier declaimed in a maxim characteristic of the age. "We must preserve only those facts that are given by nature, which cannot deceive us."

This novel approach allowed for a more reliable manipulation of the physical world and lifted the continent out of centuries of arcane medieval superstition. Where ancient Rome might have mastered the practical side of engineering, its approach also made room for dubious arts of divination, from bird augury to the indigenous Etruscan practice of haruspication (divination based on the entrails of animals slain in sacrifice). In China, meanwhile, where the maintenance of imperial unity was paramount, scholars were rewarded by the state for their mastery of and reverence for old texts, not iconoclasm or disruptive new paradigms. So too for a now-creaking Ottoman Empire, which doubled down on religious conservatism at a time when Western Europe was achieving takeoff.

The dispassionate ideal of "reason" championed by Enlightenment philosophers produced information whose merit was based solely on the validation of empirical research and predictive power, rather than by appeals to authority or prophecy. At least that was the idea. But as Lavoisier would soon discover, with his head rolling around the Place de la Concorde and the French scientific academies shuttered in 1793 amid the swelling tide of revolution, the advance of science was not inevitable and self-executing.

For France, it would prove to be a particularly inopportune moment for such a setback, as Great Britain was only accelerating a relentless and practical pursuit of technology via the application of scientific principles. Its approach to science was unique in this regard. Its community of wellremunerated tinkerers was motivated less to uncover deep principles of the world like Galileo or Newton in centuries past—science for its own sake, in other words—than to make a profit. The business partnership of Watt and Boulton embodied this new attitude. Around the same time that the commons were increasingly being parceled out as private property in the seventeenth century, patent laws had been established as well, affording these same proprietary rights to the generators of information—and drawing the invention of practical devices into the competitive incentive structure of the market. This new social institution would prove highly remunerative to England's tinkerers and inventors. At the same time, heretical thinkers could find patrons across borders thanks to Europe's competitive state system, and creative émigrés were lured to the country for its innovative and business-friendly environment. So, using the intellectual toolkit of the scientific revolution, tied to networks of finance, and availed of precision boring tools originally developed for the manufacture of cannons and firearms (another legacy of Europe's unique bellicosity), capitalistic inventors could now fashion a blooming ecosystem of filigreed metal instruments and contraptions in pursuit of profit. It was not so much science itself, then, that unleashed the Industrial Revolution as it was the large pools of wealth that plowed money into its practical application.

It is unsurprising, then, that Watt's interest in steam engines was as much financial as scientific, and he instantly intuited the connection between energy and wealth—one that had been present since money's beginnings as a proxy for photosynthesis in the granaries of Sumeria. "Heat was coal; coal was money. Better to save it," writes the historian Richard Rhodes of Watt's philosophy, where financial considerations drove an unyielding search for machines that could more efficiently convert the energy in coal to work.

Watt's business partner Matthew Boulton had a somewhat more cynical understanding of the Promethean product he was selling. "I sell here, sir," he told a visitor to the Boulton-Watt workshop in 1776, "what all the world desires to have—POWER."

THE ENGINE WATT ULTIMATELY DEVELOPED WAS FAR MORE ADVANCED than Newcomen's, with a system of valves that allowed steam to alternately power both strokes of a piston—pushing and pulling with equal vigor, in a continuous motion. Instead of merely producing heat, the solar energy stored in coal could now do mechanical work. It could do just about anything. In other words, the geologic reservoir of energy in the Earth, built up over hundreds of millions of years, could now do work on the surface of the Earth and power almost any industrial process imaginable. Given the competitive economic landscape into which this new power was introduced—in which firms had access to the ill-gotten raw materials of the entire world and would be rewarded with limitless riches for transforming energy and material in service of profit—anything it could do, it would.

Somewhat blandly, the first industry to unleash this explosive dynamic was British textiles. Importing cotton calicoes from India and silks from China—as Europe had been doing for centuries—was costly. By trying to copy these industries at home, though, and subjecting them to economies of scale, Britain could finally produce such luxuries at low cost. While today countries trying to imitate the West's historic economic development are punished for raising tariffs and safeguarding local industries from exposure to the world market, one of the keys to England's success was its protectionism. Total bans on importing finished cotton cloth afforded the industry the time to experiment and learn to imitate the foreign product—an effort aided by the British East India Company's control of Bengal and an assiduous effort by British industrialists to penetrate the secrets of the region's age-old craft and import the technology back to England. This so-called import substitution—replacing Indian textiles bought on the market for those produced by local industry—is a self-amplifying economic process, where the nurturing of one new industry spawns other industries and fuels spasms of growth, and has its roots going back millennia. Some 5,000 years ago, the Uruk woolen textile industry, for instance, once powered a similar economic expansion, division of labor, and search for raw material and new markets in Mesopotamia. But those societies were ultimately limited in this economic growth by energy on the surface of the Earth. With the introduction of the steam engine, mills and factories that previously had to abut rivers—drawing energy from the solar power embodied in and released by rushing water—could now be placed anywhere, drawing instead from the seemingly endless solar power of ancient photosynthesis. Where British industry was still mostly powered by water in 1800, within seventy years steam engines would be doing the work of 40 million laborers that, if powered by a diet of wheat, would have required three times more than the total wheat production of the entire country. These engines fed energy to a multiplying variety of ingenious labor-saving, productivity-boosting devices on the factory floor, the technological fruits of the British patent system and the country's tradition of tinkering. The machines processed an unending supply of raw cotton produced all over the world—whether by slaves or sharecroppers in the United States, or former skilled Indian cotton workers now cast into the fields to grow the stuff, as the tidal wave of cheap British textiles into the subcontinent undermined the millennia-old Indian industry. The British economy could now grow as high as the coal mines were deep.

AS MENTIONED, THE ORIGINAL PURPOSE OF WATTS'S COAL-FIRED steam engines was not to spin cotton, though, but to aid the coal industry itself by pumping water out of the coal mines. This provided access to still more coal, which made coal cheaper still, which, being the very input to the steam engines themselves, made mining cheaper, which meant more coal mining, which meant more coal to feed the steam engines, which meant more coal, and so on. This system was itself an engine, a self-amplifying positive feedback to energy dissipation not unlike a hurricane. Meanwhile, where iron smelting had long required charcoal from the same disappearing forests that supplied the country's dwindling fuelwood, cheap metallurgical coal could now be transformed to coke and be fed to the fires of industrial blast furnaces. From these forges came a spidering network of railroads that now serviced a new coal-fired iron beast: the train. Trains hauled still more coal overland from the mines to industries afar, for far less energy and cost. The growing industrial monster also required a swelling torrent of raw material to feed to the machines and the people who worked them. As a result, by 1830 Britain was importing an amount of photosynthesis—in cotton, timber, and sugar—that would have required the entire island to produce otherwise.

This was a positive feedback, a gathering storm, an engine of engines, an endless profusion of more and more powerful dissipative structures on the Earth. Geologically, it arrived with the force and rapidity of a supereruption. The explosion of ancient photosynthetic energy from the Carboniferous allowed for the assembly of the strange new structures of industrial civilization on the Earth: railway networks, the steamship, industrial cities, a blizzard of new technologies fashioned from metal and transformed by ancient heat—unusual phenomena that would fall apart without this continuous dissipation of geologic energy. Within a century and a half of this eruption of fossil photosynthetic power, George Orwell could write of his society, and of this new "metabolism of the Western world," without exaggeration: "Our civilization … is founded on coal, more completely than one realizes until one stops to think about it. The machines that keep us alive, and the machines that make machines, are all directly or indirectly dependent upon coal."

This detonation of *millions* of years of solar energy powered the export of the strange economic system developed in Europe to the entire planet. It knit together the internal markets of the industrializing world, and connected them to international markets (many of which, like China, were pried open at gunpoint), remade the global countryside (away from subsistence agriculture and toward commodity production, just as had happened in England), and could potentially tap into all the natural resources on Earth. "Capitalism as a planetary system," the sociologist Jason Moore writes, "became possible through the production of a globe-encircling railroad and steamship network."

For those corners of the world isolated for a time from the turmoil enveloping the fractious western tip of Eurasia, their reintroduction was traumatic. Warlike Europeans, and their colonial spinoffs, arriving in distant ports with alien military technology, and now energized by millions of years of ancient sunlight, took the rest of world by devastating surprise. When the bejowled and implacable Commodore Matthew Perry arrived off the coast of Japan in 1853, leading four menacing, smoke-belching black American ships, he had at his command more firepower than the entire country in his sights. "We have had too long a period of peace with no intercourse with outside," the alarmed Meiji-era writer Fukuzawa Yukichi wrote in 1875. "In the meantime, other countries, stimulated by occasional wars, have invented many new things such as steam trains, steam ships, big guns and small hand guns." At the edges of empire, these encounters often resulted in the rise not of industry, but of purely extractive economies, as in Africa, where the Guyanese historian Walter Rodney noted, "All roads and railways led down to the sea. They were built to extract gold or manganese or coffee or cotton."

Soon the entire world was conscripted into this competitive, violent, and generative system, whether it wanted to be or not. Powered by the ongoing eruption of fossil fuels, it would trigger an unprecedented eruption of exponential growth—of economic value, calories, human population, material throughput, technological innovation, social complexity, environmental devastation, and destructive military power. As a result, CO_2 would roar out of the Earth with this new metabolism and the planet would convulse as it only had every 100 million years or so.

Glimmers of this planetary transformation were detectable almost from the outset. “Even by 1900,” one study led by the German climatologist Gabriele Hegerl finds, “[there is] evidence for a detectible warming driven by increases in greenhouse gases.”

The industrial engine had been turned on—the product of millennia of agricultural intensification and resource scarcity, and, in one part of the world, an endless unfurling of war, environmental depletion, ever-evolving political economies, imperial exploitation, and financial development. There were deeper roots too, scarcely perceived, in the millions of years of climate history that preceded it, and the biological evolution of one strange fire animal. As a result, the climate, and the entire planet, wouldn’t be the same for thousands, if not hundreds of thousands, of years.

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The CO_2 Supereruption

> High energy production requires an interface between reservoirs that are out of equilibrium, since it is the passage from disequilibrium to equilibrium states that generates the energy for life processes.
>
> —WALLACE BROECKER AND CHARLES LANGMUIR, *HOW TO BUILD A HABITABLE PLANET*

IN 2018, AT THE 110TH EXPLORERS CLUB GALA, AMAZON FOUNDER JEFF Bezos presented the crowd in the midtown Manhattan ballroom with his expansive vision for humanity. It was, he said, our destiny to become a multitrillion-strong colonizing force throughout the galaxy. The somewhat anachronistic club champions exploration for its own sake and admirably underwrites scientific research in that pursuit. It's based in an Upper East Side hideout whose ceilings are ribbed with wooden beams salvaged from a nineteenth-century Royal Navy warship, and whose mahogany walls are festooned with taxidermied safari animals bagged by Teddy Roosevelt. Framed above the stairways are club flags that have been to the moon and the Mariana Trench. In commensurately Victorian, sun-never-sets style, attendees of the annual gala are encouraged to wear "native garb," a dress code that produces jet-setting financier donors decked in Suomi folk costumes. At one point "Her Deepness," the legendary marine biologist Sylvia Earle, came to the stage of the annual gala to hawk Rolex watches (the evening's sponsor) to the crowd. Later at the event, Bezos was unflatteringly pictured eating an iguana—a photo that went viral and did little to reassure the public that they were engaged in any kind of common national project with the country's multibillionaires.

"I want my children to use more energy than I did in my lifetime," Bezos announced to a receptive crowd of extremely rich people and funding-starved scientists. "And I want their children to use more energy than them." Bezos explained that his plan for humanity involved putting trillions of people in space, mining asteroids for the kinds of metals that will increasingly come to dominate a renewable-energy future, and thereby slipping the surly bonds of Earth's finite resources at last. The plan would unlock unlimited growth, forever.

In 2021, Bezos showed this wasn't all idle talk, hopping aboard his own rocket ship and jetting into space. With private companies now able to accomplish what state space agencies had sixty years earlier, it was one giant step for CEOs. In a news conference the day after he was launched into low Earth orbit, he addressed his employees, whose unionization he has assiduously fought against for years, and offered what was perhaps the pithiest summary of Marx's labor theory of value ever expressed. "I want to thank every Amazon employee and every Amazon customer," said Bezos, who is worth more than $200 billion, "because you paid for all of this." At another press conference he revealed portraits of spinning ring-world habitats floating in the void, that mainstay of science fiction, which he hopes will eventually come to house most of this exponentially growing population in perpetuity. If the concept art he has commissioned is any guide, these ring worlds will also include herds of elk, meandering rivers, and Yosemite-style granite batholiths. For such a long-term vision, the design gives short shrift to the inexorable and destructive forces of erosion.

It is easy to write off such techno-utopian visions, given the fragile current state of the world, but Bezos's maximalist dream is just one logical endpoint to the economic imperatives of endless growth, and it's the only good version of it. If the economy has any relationship to the material world at all, then taking endless growth seriously on a finite planet, in an entropic universe, requires either escaping the fundamental constraints of our planet or, eventually, exhaustion and collapse. One can forgive Bezos for embracing the former.

IN THIS CHAPTER WE'VE FINALLY REACHED THE PUNCH LINE. AFTER billions of years of planetary and biological evolution, punctuated by unthinkable CO_2-driven climate paroxysms, then finally hundreds of millions of years building up a massive reservoir of flammable energy stored in the Earth's crust, and an equivalently oxidizing atmosphere above it, came at last the arrival of a highly social fire creature living in between them. This creature had been shaped by low-CO_2 ice ages, and was now governed in one part of the world by a kind of society-organizing algorithm that enabled vastly more collaboration among, and exploitation of, disparate groups of people, in pursuit of profit—a system that came to conscript the entire planet in its logic. And on one obscure island in the North Sea, a population of these fire creatures—locked in a perennial state of ruthless interstate competition with their neighbors—had exhausted their photosynthetic energy sources at the Earth's surface and turned to the fathomless supplies of fossil photosynthesis beneath it. Harnessing the mining technology and engines of a relentlessly ratcheting material culture, humankind thus came in contact with a titanic underground reservoir of hundreds of millions of years of ancient plant life bequeathed by the ages, which dwarfed the amount of photosynthetic power available on the surface world by orders of magnitude. All they had to do was light it on fire.

In other words, boom.

IT'S STRANGE THAT THERE ARE ONLY A FEW KNOWN WAYS, DEMONSTRATED in the entire geologic history of the Earth, to liberate gigatons of carbon from the planet's crust into the atmosphere. There are your once-every-50-million-years-or-so spasms of large igneous province volcanism, on the one hand, and industrial capitalism, which, as far as we know, has only happened once, on the other.[fn1] That the Siberian Traps, burning through gigatons of fossil fuels at the end of the Permian, didn't cause an industrial revolution is obviously not very surprising. It was an uncontrolled explosion of carbon energy from the Earth's crust. And the precocious use of coal in Bronze Age China—which, as in England five millennia later, featured a growing population straining against their natural resources, faced with the retreat of their forests and turning to coal to make up the difference—shows that there's still more to the story than just the use of fossil fuels themselves. After all, we do not find ourselves living in the climate produced by such an antique industrial revolution.

The Netherlands of the seventeenth century, on the other hand, might have met almost all of the social preconditions to birth an industrial revolution: it was a profit-minded society with a mastery of financial institutions and a legacy of engineering, relentlessly reinvesting profits to scale up industries like shipbuilding that allowed it to command a global market at the barrel of a gun, and subject the raw materials and labor of a vast extortionate empire to its ends. But the Netherlands didn't have the geology. Its industry was powered mostly by burning peat, not coal. It was only in England that the organization of the entire society and its natural resources around competition for profit, in the context of a globe-spanning empire that secured the flow of raw materials into this system, was supported by a near-bottomless reserve of geologic energy, powered by all the life in Earth history. The productive powers that were unleashed when this social revolution gained access to the planetary battery, built up over hundreds of millions of years, astonished and horrified in equal measure.

As coal came to assume the omnipotent role in modern life that Orwell attributed to it—its ancient solar power now transmuting iron, vivifying factories, and pushing ships across the high seas—the military-industrial research it inspired led to the discovery of the laws of thermodynamics by imperial agents like the French engineer Sadi Carnot, by British engineers fussing over sputtering steamship boilers, by bookish brewers looking to cut energy costs, and by the champions of Wissenschaft at the modernizing German university. Illumination of these principles enabled ever more efficient, effective, and powerful engines, more work for less wasted energy, higher productivity, more economic production, greater surplus, and higher profits. As the dark Carboniferous smog descended on English cities, pedestrians struggled to make out their fellow Londoners on the street from more than two yards away, and smog-choked children succumbed to rickets from the lack of sunlight. British patents, meanwhile, continued to roar off the presses. Inventors in the mold of Watt found multiplying uses for this explosion of Carboniferous solar energy, channeling it into work on the surface of the Earth through a dizzying array of new devices. The ratchet ceaselessly turned. By the time Britain, in Orwell's words, came to depend on its geology "more completely than one realizes until one stops to think about it," the chasm of time upon which the country drew to swell its economic and imperial power had already begun to dawn on its philosophers.

“We find no vestige of a beginning, no prospect of an end,” the Scotsman James Hutton famously wrote about his country’s geology while contemplating the coastal rocks of Siccar Point, where ancient strata laid down by rivers on land sat atop tilting rocks from the bottom of the deep sea. It was a heady period, and the veil of understanding was lifting not only on the abyss of deep time but also on thermodynamics, biology (evolution by natural selection, genetics), and medicine (germ theory, vaccinations). James Clerk Maxwell’s theories of electromagnetism, meanwhile, would underwrite the physics revolution of the coming century and augur the dawning age of electricity. By midcentury, the carbon cycle, along with the role that the burial of fossil fuels plays in generating Earth’s surplus of oxygen, had already been developed by Jacques-Joseph Ébelmen, a precocious French chemist “whose ideas were forgotten and were independently deduced by others only 100 to 150 years later,” according to the late Yale geochemist Robert Berner. More ominously, scientists like the Swedish physicist Svante Arrhenius were already investigating the role of carbon in the atmosphere. By 1896, Arrhenius estimated that doubling the CO_2 in the atmosphere would result in about 5 degrees of warming—a value still on the outside edge of the range of our best understanding of Earth’s climate sensitivity. (Somehow, this hoary scientific establishment of the role of CO_2 in the climate, which predates widespread acceptance of plate tectonics by a century, doesn’t embarrass modern peddlers of climate denialism.)

At the same time that scientists were beginning to understand the Earth’s atmosphere, the industrializing world was beginning its global-scale test of these calculations. By the end of the century, England was burning through underground coal that was the energetic equivalent of eight Englands’ worth of timber on its surface. For the first time in its history, human society was no longer constrained by the solar energy captured by plants at its surface. The energetic yoke of agriculture was cast off, ageless generations of farmers were suddenly freed from the land (or forced off it), and the limits to city size, to industry, to the bewildering novelties of the modern world were obliterated. Philosophy, arts, culture, and politics all struggled to absorb, or even feebly direct, the unfolding transformation of human society, and of the very planet itself.

Comparable only to the so-called Neolithic Revolution in agriculture 10,000 years earlier, the Industrial Revolution was the most transformative event in human history, and ultimately even one of the most transformative events in geologic history. Curiously, though, like the revolutionary switch to intensive, settled agriculture thousands of years before it—one that similarly called forth far more photosynthetic energy from the land—this unleashing of fossil plant energy actually led people to work *more*, not less. The new engines, and the spectacular profits they generated, needed an endless supply of coal to metabolize, and so produced an insatiable demand for backbreaking labor in the multiplying mines and steel factories of this new world. While scholars might have been dispassionately working out the eternal principles of thermodynamics in hushed study halls and under leafy stone archways, the actual application of this "motive power of heat," as Carnot called it, was taking place in the hellish boiler rooms of British steamships, by Yemeni and Somali stokers heaving coal into engines in the searing Red Sea heat.

The coal workers of the noncapitalist world hardly fared better: As the Ottoman Empire struggled to scare up workers for the mines of the "coal coast" of the Black Sea, it came to rely on uncooperative corvée labor from the local villages, often made more cooperative with whippings. By the dawn of World War I, its coal industry had assembled, according to one historian, "the largest concentrated labor force in the 600-year history of the Ottoman Empire"—one with a mortality rate 20 times higher than its European and American contemporaries. But this premodern extractive system gravely failed in competition with its capitalist imperial rivals, especially, as it would soon find out, in war.

All this energy, though, was useless unless it was made to do *work* and transform the material world around it. And so the emerging global economic system, now unchained by energy, selected for monomaniacal avatars of industrial production like Andrew Carnegie, who was strangely and single-mindedly consumed only by producing more and more steel at lower and lower cost, no matter the human toll. Carnegie's infernal coal-fired plants ran all day and night, pushed to absolute maximum capacity up to the point of failure, by an unending churn of European immigrants. When an aged Carnegie tried reconciling with his longtime business partner Henry Clay Frick, to whom he had delegated the task of violently breaking strikes to keep the plants running, Frick responded, "Tell him I'll see him in Hell, where we both are going."

The widening industrial maw also called forth a redoubling of backbreaking labor abroad to generate the raw inputs that would be fed to this machinery, and transformed by energy, like the slave- and sharecropper-picked cotton that fed the textile mills, or the sugar harvested on brutal Caribbean plantations that supplied an increasing share of the factory workers' calories. To produce these raw materials, writes the Columbia University historian Timothy Mitchell, "ever larger areas of surface territory were needed." In North America, for instance, sunlight had long powered grasses that moved 8 million bison across the plains and secured a way of life for a mobile people who depended on them. When American settlers and hunters killed 99.99 percent of these herds at the encouragement of the US Army—in an effort to push Native Americans onto reservations—the encroaching tidal wave of farmers redirected the solar energy that fell on the Great Plains into oceans of Ukrainian wheat. Avalanches of the grain poured out of the opening continent on train cars returning to the coast, where it was sent in coal-powered steamships across the Atlantic—ships that carried still more European peasants back by the millions to be swallowed by American industry, or down lightless coal mines, or to find their way too in the vast interior of a strange continent.

By the end of the nineteenth century, coal had begun to overtake fuelwood in many Western countries, as well as in a rapidly industrializing Japan—a fundamental shift in human and even biological history, with ancient photosynthesis, resurrected from the Earth, overtaking recent photosynthesis from the planet's surface as a source of energy. Yet already by the turn of the century, another fossil fuel industry had emerged—one that would ultimately eclipse coal in the great industrial dissipation of the planet's reservoir of geologic energy. If England's explosive rise was powered by coal, in the twentieth century, the centers of power, both metaphoric and literal, would emanate from those spaces on the map where the rocks entombed not ancient forests, but ancient seafloors draped in life. Power would soon flow from oil. It would be no accident that the two biggest twentieth-century superpowers, the United States and the Soviet Union, would also be its greatest oil producers.

LIKE COAL, OIL AND GAS WERE NOT UNKNOWN TO THE ANCIENT WORLD. As long as 70,000 years ago, deep in the last ice age, hunters in Syria used petroleum residue to affix spearpoints to wooden shafts. Later, the world's first major cities—like Uruk, in modern-day Iraq, or Mohenjo-Daro in Pakistan—were held together with this mortar of petroleum tar. By 5,000 years ago, native Californians were coating baskets and vessels in waterproof bitumen, Zoroastrians in the Caucasus worshipped “eternal flames” (petroleum seeping from limestone), and the Greeks wrote of oil seeps in Iraq. In the *Iliad*, Homer noted the particularly formidable use of petroleum in ancient Greek incendiary warfare—that “flame that cannot be quenched.” Genesis says Noah caulked his ark with the stuff. The Assyrians somewhat alarmingly mixed crude oil with wine and swore by it as a cough medicine. And in Arabia, two millennia before OPEC, Western powers covetous of the lucrative trade in Middle Eastern oil—which was used to grease the wheels of chariots, caulk boats, and as an aromatic resin in Egyptian mummies—ran up against an oil monopoly run by the Arab Nabatean Kingdom. The Middle Eastern wealth from this oil trade produced the magnificent city of Petra in southern Jordan, famously carved out of the region’s roseate sandstones from the Cambrian Period.

Eventually, however, the Roman general Marc Antony carved up the region, annexing the Nabateans, and gained control over these considerable oil revenues, which he gifted to his paramour Cleopatra. Cleopatra in turn leased the oil industry back to the Nabateans for an extortionate sum, hoping to use the oil revenues to fund a navy, a petromilitary arrangement that anticipated the geopolitical entanglements of the modern world by millennia. Still, it was Han dynasty–era China that was perhaps the most precocious with fossil fuels, drilling for natural gas more than two thousand years ago, distributing it in pipelines made of bamboo, and burning it under cast iron pans filled with brine to make salt. Both the salt and the fossil fuels were pulled up—and are still being pulled up today—from an underlying geologic formation from the early Triassic, a vestige of the bad old days of Pangaea.

Some 2,000 years later, in the first half of the nineteenth century, the “oil industry” initially referred to the ocean-spanning pursuit of whales, whose blubber was rendered, and spermaceti extracted, to provide illumination in oil lamps and in candles, as the oil hub Nantucket braced against the Atlantic like a Yankee Abu Dhabi. However, as right whales were nearly hunted to extinction in the Atlantic and whaling voyages were forced to venture farther and farther afield for fewer whales, the price of whale oil climbed high enough that it made “rock oil,” as it was known, fertile ground for speculation. It was a New York lawyer named George Bissell who, while visiting his alma mater Dartmouth College, came across a sample of Pennsylvanian oil in a professor's office and, inspired by those same ancient Chinese salt-drilling techniques, had an idea.

Like coal before it, this massive reservoir of ancient energy was about to be put in contact with the gathering global industrial hurricane. And once again, this next epoch-making development emerged from the nexus of finance, scientific knowledge, and practical know-how—with a consortium of Dartmouth and Bowdoin alums, a New Haven banker, and a Yale chemistry professor providing the former two, and an itinerant railroad worker named Edwin Drake, who ultimately developed the revolutionary steam-engine drilling technique, providing the latter. But the point of the venture, from the very beginning, like Watt's steam engine, was to make money. Money, even from its origins as a symbolic proxy for cereal grains in Mesopotamia, represented what the ecologist Charles Hall calls "a lien on energy." It is a concept that wouldn't be lost on later oil barons, one of whom would marvel over this strange substance, "Oil *is* almost like money." The unearthing of fossil carbon energy would make untold amounts of money for those who drew it from the Earth, while destroying the limits to how much activity could be done on the surface of the Earth.

IN THE VISITORS' CENTER OF THE DRAKE WELL MUSEUM IN TITUSVILLE, Pennsylvania, a cartoon plays every fifteen minutes about the wonders of hydrocarbons, somewhat threateningly titled "There's a Drop of Oil and Gas in Your Life Every Day." The video is hosted by anthropomorphized drops of oil and gas ("Pete Troleum" and "Ethel Leen"), one of whom strangely claims that using natural gas "helps me express myself." Here in 1859, in the wilds of western Pennsylvania, the global petroleum industry exploded into being with the force of a tornado. Just as animal muscle had powered the transition to coal, coal would midwife this coming age of oil. Where Native Americans had long dug pits along the local Oil Creek in Titusville, to gather the pooling petroleum for medicine, Edwin Drake had instead purchased a coal-fired steam engine to plow a pipe into uncooperative sands and clays to get at the stuff. The work proceeded at an excruciating pace of less than three feet a day, and soon enough Drake had run out of money from his Connecticut investors. Nevertheless, he continued to ply his hole in the Earth with a drive that locals found maniacal. Finally, at a mere 69 feet from the surface, the dark, iridescent 380-million-year-old carbon, from the algae-choked seas of the Devonian Period, began to pool in the well. Soon it was producing a thousand barrels of oil a year. A revelation. Within years of Drake's first oil strike, the entire region was denuded of trees and strewn with the smoking wreckage of wildcat drill operators and ramshackle cabins.

More than a century and a half after Drake drilled his epochal first hole in the Earth, a replica of the coal-powered steam engine still puffs and wheezes at the Drake Well site like a mechanical dragon.

"The only thing that's original is the hole in the ground," the groundskeeper told me when I visited, as the engine wheezed and clanged into the lonely Pennsylvania woods. "You guys in Boston killed all the whales. Oil was expensive, and this made it so everyone could have illumination, so it wasn't just a luxury thing." Indeed, this first oil boom at the end of the nineteenth century, however momentous, was only in service of "pushing back the night"—bringing illumination to the darkness of the industrializing world in the form of kerosene. And when this cheap oil burst onto the scene, it would come to replace whale oil, and the heaving whaleships of the nineteenth century would soon stay moored in New Bedford, Massachusetts.[fn2]

In the years after the riotous birth of the oil industry in Titusville, speculation ran amok and prices fluctuated wildly. Wild spasms of enthusiasm precipitated oil rushes, haphazardly drilled by indebted wildcatters, and the resulting oversupply led to similarly precipitous price collapses, social and economic devastation, and a landscape of derelict boomtowns and desolated, deforested, oil-soaked countryside. Elsewhere, the environmental signal of these fossil fuels was even more spectacular. An Indiana oil and gas boom in the 1880s literally burnt itself out almost as soon as it began, as mass pyromania swept the region and the countryside glowed through the night—lit everywhere by gigantic, nightmarish, and purely ornamental towers of flaming gas called "flambeaux." Townspeople enthusiastically poured oil and gas into the rivers and lit them on fire for mass entertainment. In less than three decades, this madness uselessly wasted some 500 billion cubic feet of gas and pressure collapsed in the oil fields, as did the Indiana oil and gas industry.

The rowdy business of oil needed to be brought to heel if it was literally to grease the gears of a rapidly transforming world, and the person who would do so was an austere Baptist from Cleveland, the seeming vision of capitalism incarnate, John D. Rockefeller. Rockefeller began his career in the shipping business, trading other kinds of carbon, like grain, produce, and coal. When the disorganized explosion of Pennsylvania oil erupted from the Earth, though, he spied an opportunity to bring order to the nascent industry. Ruthless in competition, untiring in his acquisition of rival firms, and almost ethereally remote, the wraithlike founder of Standard Oil cut a menacing figure. He drew as much of this ancient carbon flowing from Pennsylvania into his corporation as it could swallow and, within a decade of Drake's oil strike, Rockefeller had built a gigantic, world-beating oil refinery on the Cuyahoga River. Soon his refineries would billow from the lakeshores of northern Ohio and Indiana, the riverbanks of Philadelphia, and the bayfront of Bayonne, New Jersey, as he increasingly consolidated the American industry. He achieved this through endless corporate subterfuge—securing discounts on rail freight rates, extracting rents on rivals' shipments of oil, aggressively buying out competitors, or, failing that, cutting prices enough to destroy them and pick over the carcasses. By the end of the century, Rockefeller was the richest person in the world, his corporate octopus accounted for 90 percent of the country's refining capacity in an age when Pennsylvania oil was king, and he had amassed a vast information-processing, oil-distributing, profit-making machine in the process.

If it could be harnessed for industrial use, oil was far superior to coal along a number of axes: it burned cleaner, it was much more energy-dense, and, with the ability to put it in a barrel, it was easier to move and store. It was also favorable at a time when coal strikes repeatedly threatened to shut down entire industrializing countries and were wringing painful concessions from mine owners and governments alike. Oil did not require coal mining's armies of men to siphon it from the ground, and it could be delivered overland by pipelines—like those that Rockefeller would build streaming out of Pennsylvania to his refineries in Cleveland—circumventing a supply chain of unionized horse-drawn wagons.

Meanwhile, engineers with names like Benz, Daimler, and Maybach were experimenting across the ocean with a revolutionary new technology, the internal combustion engine, which promised to bring the explosive, propulsive power of liquid fossil sunlight within reach of a burgeoning consumer class. It was only with the arrival of cheap oil that such energy-dissipating, organic carbon-oxidizing machines could be brought to market. As the oil industry boomed, and ancient plankton gushed out of derricks from California to the Caspian, the automotive age[fn3] began. Within decades the Western world would not only be powered by oil, but also paved over by it. Asphalt, after all, is crude oil's heaviest component.[fn4]

As the industry was eventually consolidated and monopolized by centrally planned corporate giants like Standard Oil and prices stabilized, volatility was tamed and the speculative rough-and-tumble of the Titusville oil exchange—just down the road from Drake's well—shuttered. Trading in oil futures wouldn't return to the world of finance again until 1983, when the hegemony of the oil giants was broken by waves of anticolonial oil nationalization, volatility returned to the industry, oil speculation again became profitable, and a futures contract for West Texas intermediate crude was introduced on the New York Mercantile Exchange in the World Trade Center. Not long thereafter, trading in oil futures would take the spot of the formerly most popular futures commodity: the Maine potato.

Meanwhile, downtown Titusville, like Appalachia, was largely left behind by the $2 trillion global industry it spawned. On my pilgrimage to the birthplace of the oil industry, I visited the building site that once hosted the world's first oil exchange. It was now a Dollar General store, sitting across from a food bank. At the cemetery down the road is Edwin Drake himself, whose grave makes for a similarly dispiriting spectacle. "Founder of the petroleum industry / The friend of man / His last days / Oppressed by ills— / To want no stranger / He died in comparative obscurity," it reads. By the turn of the century a new world was already coming into being, one increasingly made not by the half-legendary exploits of wildcatting pioneers and half-mad backwoodsmen, but by sprawling multinational corporations.

GIVEN OIL'S GLOBAL, IF UNEVEN, DISTRIBUTION THROUGHOUT THE Earth's crust, the unchallenged dominance of the Standard Oil octopus couldn't last long. Across the ocean, this corporate imperium would be threatened by challengers who wouldn't be as easily extinguished as the wildcatters of the Pennsylvanian frontier. There were the Swedish Nobel brothers, and the French branch of the Rothschild family, who had set up in the oil boom city of Baku, Azerbaijan.[fn5] Here oil- and natural gas–fired flames had long flickered from Persian fire temples, releasing solar power that once fell on

Miocene seas, but now the petroleum erupted from a bewildering forest of oil derricks. This tar-soaked Caspian port town energized the rapidly industrializing Russian Empire up the Volga River, and increasingly supplied petroleum to the Western European market after the Rothschilds funded the construction of a railroad across the Caucasus to the Black Sea, and the Nobels dynamited through a mountain to ease the arterial flow of this energy. By the beginning of the twentieth century, this twelve-square-mile plot in West Asia supplied over *half* the world's oil.

Then there was Marcus Samuel, whose Iraqi Jewish immigrant father was a seashell merchant in East London, and whose ties to South and East Asian trading houses run by expatriate Scots, from Calcutta to Hong Kong, plugged him into the increasingly global economy. Samuel and his Shell Transport & Trading Company were recruited by the Rothschilds to provide new markets on the other end of Eurasia for this eruption of oil. There was also a Scottish outfit called Burmah Oil, which had already sprouted vast forests of oil derricks in what is now Myanmar (Burma) to replace the teak forests that were razed there to build British ships. With its contract to supply the Admiralty with petroleum, Burmah Oil would take over an oil concession in Iran as well, spawning the Crown's juggernaut in the Middle East, the Anglo-Persian Oil Company (later British Petroleum, now BP), after a gusher exploded from the desert in 1908. Once again, older energy sources aided the transition to this new energy revolution, as 6,000 mules and countless colonial laborers imported from Rangoon constructed a 138-mile pipeline through the desert, from the British concession in Iran to a refinery in the marshes of Mesopotamia at the confluence of the Tigris and Euphrates, where humans first started scaling up photosynthesis into monumental agricultural societies some 6,000 years earlier.

Finally, there was a long-struggling company, Royal Dutch, whose presence in Indonesia was the legacy of the Netherlands' centuries of commercial plunder in the Indian Ocean, and which had secured a concession from a local sultan to drill in the jungles of Sumatra. As with the Dutch East India Company before them, the British historian Max Hastings writes, the firm "surpassed all rivals in the savagery with which they persecuted local peoples in the interests of what became Royal Dutch Shell. In a 40-year war that began in 1896, some 12,500 colonial soldiers, 75,000–100,000 Acehnese and 25,000 'coolies' perished." The rise of these international firms, and their growing market share, turned the US Goliath

Standard Oil's implacable attention outward toward the globe by necessity, launching a planetary scramble for control over oil and markets abroad.

In 1878 the first oil tanker took to the seas, the Nobels' *Zoroaster*, and by the turn of the century, dozens of tankers, mostly owned by Marcus Samuel's Shell (and all bearing malacological names: *Murex*, *Conch*, *Clam*, *Cowrie*, etc.), plied the ocean. This flow of global geological energy was aided by the construction of the Suez Canal—itself the product of a joint-stock company based in Paris and backbreaking Egyptian corvée labor—which obviated the need to go around the Cape of Good Hope to reach Asia. The opening of the canal also hastened the imperial designs of the British, who had littered coal depots from Gibraltar to Port Said and Aden, and on to Bombay, a necklace of steamship-refueling stations that, along with telegraph wires that followed the same path, connected the metropole ever closer to its "Indian Crown Jewel." These fossil sinews of empire would ultimately trace the contours of what would become known to the West as "the Middle East," as it was conceptually transformed from the way station of empire to the wellspring of carbon power.

Shell's ultimate coup, though, came not in the Middle East but in the Dutch East Indies (Indonesia). The Shell Transport & Trading Company had once imported exotic antiques from the islands, but in 1907 the younger Samuel eyed what was emerging as the next major oil-producing region. Standard Oil made a failed bid to buy out Royal Dutch, which instead accepted an offer from Samuel, effectively dividing most of the world's oil between two corporate giants: Standard and this new Royal Dutch Shell Group, which soon even had the temerity to begin drilling in California and Tulsa, Oklahoma, as well.[fn6]

While it fended off formidable new international competitors, Standard faced domestic competition as well, from a diffuse legion of prospectors who all aspired to be the next Rockefeller. New oil companies—such as Gulf Oil, Sun Oil, Texaco, Union Oil, the Getty Oil Company—would blossom from the prairie to the asphalt pits that seeped from the Southern California chaparral. Meanwhile, Standard's ruthless consolidation had made it countless enemies over the years, and its pitiless tactics rankled a crusading journalist named Ida Tarbell, who had grown up in the wreckage of Titusville, Pennsylvania, and who helped turn public opinion against the corporate behemoth. In 1911 her broadsides against the corporation were validated by the US Supreme Court, and Standard Oil was ordered to dissolve. The company fissured into a family of “Baby Standards” that remain with us to this day: Standard Oil of California (Chevron), Standard Oil of Ohio (Sohio, later absorbed into BP), Standard Oil of Indiana (Amoco), the Standard-owned Continental Oil (Conoco), Ohio Oil (Marathon), and finally the titan, Standard Oil of New Jersey (Exxon), which would merge again decades later with Standard Oil of New York (Mobil) in 1998 to become the largest corporation in the United States, ExxonMobil. The company’s profits, as late as 2022, were higher than in the entire century-and-a-half history of Standard Oil that preceded it.

Rockefeller had epitomized a uniquely American, uniquely single-minded, almost ascetic pursuit of profit at the turn of the century—one resolutely, if awkwardly, fused with Christian salvation. This Protestant business theology would perhaps best be articulated by President Calvin Coolidge in 1916 when he evangelized his industrialist gospel: “The man who builds a factory builds a temple, that the man who works there worships there.” Meanwhile, a former engineer at the Edison Illuminating Company in Detroit named Henry Ford was just such a temple builder.

IN THE NINETEENTH CENTURY, WHAT THOMAS JEFFERSON CALLED the “Empire of Liberty” was finally secure in its place on the continent and embarked—as most empires do—on a period of rapid and violent territorial expansion. Waves of American settlers outpaced the military garrisons hastily built to secure the frontier, while the government gave away land and bonds to railroad companies that accelerated this flow of settlers and doubled down in its explicit ethnic cleansing campaign against Native Americans. By the late nineteenth century, this polity called the United States had come to span the entire continent—in the process replacing a shifting patchwork of native nations that had inhabited it for millennia, wiping out the ecological base to mobile ways of life, ensnaring Native Americans in debt, pressing them into foreign practices of settled livestock agriculture and Christianity, massacring those who resisted and wouldn’t move onto reservations, and pushing Spanish governors and Mexican peasants aside at gunpoint—all the way to the Pacific Ocean. The entire continent could now be integrated into one giant economic system, connected to the markets of the wider world, and its natural resources organized by the algorithm of capitalism. And it was the Great Lakes region, where a confederacy of tribes, led by the Shawnee chief Tecumseh and allied with the British Army, had failed to repel this wave of colonization in the early nineteenth century, that would turn out to be a particularly favorable nexus for industrialization. In Detroit, iron ore from Minnesota—minerals that once rusted out of a Precambrian ocean billions of years ago—could be shipped over the Great Lakes and combined with the ancient (though far younger) energy of Carboniferous coal from Appalachia to the east. The metal was transformed by this energy and molded into machines called cars, which could be fed even more geologic energy still—from other epochs of Earth history—to propel them across the vast country.

Along with Detroit's geographic advantages, it had, in its local son Henry Ford, a generational obsessive, consumed by the desire not to make profits, but to make machines. Profits were important only insofar as they could be plowed back into ever more capital investments, put ever more machinery on the ground, and provide ever-escalating amounts of energy—energy that was deployed with a ruthless and vaulting efficiency on the factory floor. Ford was concerned only with maximizing the material throughput of his factories and the output of finished, standardized machines. The workers swept up in this machine logic, and deployed as interchangeable cogs in mind-numbing assembly line routines, were kept at the line by a five-dollar-a-day salary, an innovation that turned them into the consumers of the very machines they were bringing into the world. Ford promised them not a nifty new consumer product, but in his words "a new world, a new heaven, and a new earth."

Astonishing industrial complexes, like Ford's River Rouge plant in Dearborn, Michigan, which were built at an inhuman scale and aesthetic austerity, represented mind-boggling Athenian temples to the new machine age—bleak industrial landscapes with little in the way of organic life in them, but that were, in their own way, vast organisms in and of themselves. River Rouge alone had its own "integrated steelworks, a foundry, a glass plant, a cement plant, a textile mill, a leather plant, a hospital, a motion picture laboratory … an airport, and more than 15 million square feet of factory floor space," the historian Jonathan Levy writes. It was all powered by the Carboniferous sunlight issuing from "a 100,000-horsepower coal-fired electric power plant, capable of powering a city of more than 1 million." The factories were gigantic machines that made more machines—vast productive engines that dissipated the fossil energy issuing from the Earth faster than ever.

Largely as a result of Ford's singular obsessions, the number of registered cars in the US jumped from 8,000 in 1900 to over 9 *million* by 1920. The self-reinforcing marriage of oil and cars sent company men around the world in pursuit of more customers, oil, and raw materials—unsentimentally expropriating the natural resources of the nonindustrialized world. In the rubber plantations of Brazil, for instance, Ford managers ruled over a racialized workforce in the Amazon from ersatz American suburbs. "The Ford Motor Company's supply chain was becoming a global empire unto itself," Levy writes. Taking vertical integration to new extremes, the corporation would come to own railroads, steamships, forests, and coal and ore mines across the United States, with dozens of factories churning out Model Ts around the clock and across the world, from Antwerp to Santiago.

"The machine dominates American life," an astonished American commentator wrote after touring the astounding industrial facilities then sprouting stateside. "I have felt the poetry of all this. I have felt the terror of it." That terror would soon be made manifest.

AS THE OLD POWERS OF EUROPE CARRIED ON THEIR CENTURIES OF violent interstate competition, empires accelerated their scramble for colonies abroad—both for prestige and to satisfy their mushrooming industries' near-insatiable demand for raw materials. In this quest, these countries now found themselves exponentially augmented in lethal power by that same new industrial metabolism. There were now millions of years of energy and an entire planet of raw materials and peoples to exploit.

In the race for these colonial resources, Germany, recently unified and fired by a feverish new nationalism, scaled up its navy to compete with Britain's unquestioned dominance of the seas and to challenge France's colonial possessions in North Africa, while financing the construction of a railway from Istanbul to Baghdad that would give it access to the seemingly bottomless oil of Ottoman Mesopotamia. This imperial competition abroad, the tangle of alliances across the continent, and the combustible new conceptions of nationhood and sovereignty—notions that coexisted uneasily with an older aristocratic order that limped on into the twenty-first century[fn7] —ultimately resulted in the apocalyptic clash that produced the unprecedented death toll of the second decade of the new century.

World War I has appropriately been called "the first great carbon-fuelled conflict." It shocked even the generals, many of whom had learned their strategy by way of the Napoleonic Wars, if not the Peloponnesian. The pitched battles on horseback they might have imagined were instead supplanted by the war's satanic vision of the trenches, of mangled steel forged by fossil carbon, of poison gas, and of moonscapes of cratered French countryside. As the American Civil War had already demonstrated decades earlier, railroads supplying the front lines with an unending stream of soldiers, industrially produced supplies, food, and weapons could produce staggering death tolls. The almost supernatural new powers of industrial production unleashed from the Earth that idealists thought could someday liberate human society from want were now turning the world into hell. By the end of 1918, 20 million people would be dead.

At least one thing, though, was just as true in the first decades of the twentieth century as it was in Napoleon's time. "An army marches on its stomach," the emperor supposedly said, at a time when it took 12 tons of food a day to feed the French Imperial Army and 50 tons of hay to power its horses. By World War I, this traditional energy supply still remained hugely important, even decisive to the war's outcome. Whereas Germany, subject to an unsparing Allied blockade of food imports, was continually beset by hunger and flagging morale in the Great War, Britain welcomed an endless fleet of coal-powered grain ships from the United States, Canada, and Australia to its ports.[fn8] Indeed, the dietary needs of the military only multiplied with the larger armies required by industrial warfare, but the army's stomach had also come to include the explosive metabolism of the new machines of war as well: both humans and tanks stop moving when they run out of supplies of organic carbon to oxidize to CO_2 and for the same reason. As a result, military planning now crucially took on the exigencies of supplying not only rations and fodder, but also a continuous flow of fossil energy to the front lines as its most vital task.

This challenge was complicated by the shifting metabolism of industrial warfare: before the war, the young British home secretary, Winston Churchill, spooked by the upstart German navy's advances and chastened by his own ruinous association with a strikebreaking attempt on Welsh coal miners in 1910—ordered the Royal Navy to switch from coal to oil in 1912. The US Navy would make the switch that same year. It was a momentous decision, one that sacrificed Britain's homegrown coal industry for greater speed, range, stealth,[fn9] and nimbleness on the high seas, while also rendering the military less susceptible to labor stoppages. But there was no oil in England, which meant that the empire would increasingly come to rely on the Earth's crust beneath its colonial footholds in the Middle East. Where British power in the nineteenth century depended on its globe-spanning navy, a near monopoly of coaling depots across the oceans, and the extraordinary volume of Carboniferous fossil forests it exported to the rest of the world, the rise of oil would ultimately mean the decline of this empire of coal.

The oil of Iran loomed largest for Britain, with virtually all of the country's petroleum having been conceded by a cash-strapped shah to the Crown-controlled Anglo-Persian Oil Company. The terms of the concession allowed the company to operate more or less with impunity, drilling wherever and however much it wanted, and putting pipelines wherever it pleased. Britain added to its Iranian holdings a smattering of colonially administered states and protected sheikdoms in the Persian Gulf, which it picked off a decrepit Ottoman Empire. The victors of the war would ultimately dismember the fading six-hundred-year-old dynasty entirely, nonsensically redrawing the map of the Middle East in secret and splitting its administration between the French and British.

While the French took control of Syria and Lebanon, the British gained the former Ottoman vilayets of Mosul, Baghdad, and Basra—a multiethnic and multireligious collection today known as Iraq (née Mesopotamia).[fn10] But when Iraqis bristled at British rule and violently rebelled in 1920—a revolt answered with even more violence by Royal Air Force bombers—the British cut their losses by creating an anachronistic "dynasty" in the new country and installing a friendly foreigner from modern-day Saudi Arabia, Emir Faisal, as king. The arrangement would ensure the North Sea empire wouldn't lose access to the sea of oil under faraway Mosul. It's a template that Western powers, in league with private oil companies, would try to reproduce across the region and beyond: backing friendly autocrats who would suppress democratic movements, labor unions, and revolts, while wresting lucrative concessions from the governments that delivered far more in taxes to the colonial powers than those powers did in royalties paid to the oil countries themselves. The companies would take little interest in the human lives buzzing above the rocks, except to the extent that they disrupted the flow of oil, such as when, in 1936, the crucial pipeline that delivered oil from Iraq to the eastern Mediterranean and onward to Western

Europe was sabotaged amid a wave of revolts against colonial rule, as well as against Jewish immigration into British-administered Palestine.

The United States, meanwhile—its crust generously saturated with the ancient plankton of forgotten seas—suffered no such geopolitical complications from this conversion to oil. Pockets of these ancient seas had found themselves gently cooked by the warmth of the Earth below, and rose from their graves to pool and trap in the great folds of the country rock above—rock subtly warped by the give-and-take of ancient tectonic accidents. The entire history of the continent, from the Cambrian to the Cretaceous, had generated the oil fields of Texas and Oklahoma, and in the twentieth century they would be explosively reunited with the surface world. The United States was soon the world's largest oil producer, and the surge of demand from the US Navy was fed by the perennial arrival of these massive new oil discoveries in the middle of the country. Where the British Empire's unprecedented global geopolitical dominance in the nineteenth century had been powered by coal, in World War I the British found themselves supplicants to an American upstart and its roaring spigot of petroleum. "The Allied cause had floated to victory upon a wave of oil," Lord Curzon would note, about a war in which the United States supplied 90 percent of Britain and France's oil.

THE STATESIDE OIL BONANZAS WOULD BE SERVICED BY CHAOTIC boomtowns like Cushing, Oklahoma, which went from being a sparsely inhabited cotton depot to producing 20 percent of the country's oil in just three years. To this day, the unassuming Oklahoma town prides itself as "the oil pipeline crossroads of the world." Each day, vast rivers of crude oil reach Cushing over a network of pipes, from as far afield as West Texas and Canada, and fill endless rows of massive, featureless oil tanks, their blandness belying their importance to the global economy. This crude is then sold and shipped out to refiners and chemical plants on the Gulf Coast and in the Midwest over the same network of pipelines, and at the end of each day the price of the oil in these tanks provides the standard that shapes the rest of the world's oil prices.[fn11] A Nilometer for the world's daily fossil harvest. Cushing's shabby downtown little reflects the extraordinary wealth generated by the global industry it has hosted for the past century. Meanwhile, the rivers that run through these plains are stained red by Permian mud.

Just to the south, Texas, despite having been painted black by gushers like Spindletop, still lagged Oklahoma in the first decades of the twentieth century. But the flood of American oil had only begun. In 1930, the Lone Star State's preeminence among the constellation of American gushers was finally secured with the discovery of the so-called Black Giant, a veritable sea of Cretaceous oil in East Texas that dwarfed all other discoveries that had come before—one that would upend the industry and ultimately fuel the Allies to victory in World War II. The insatiable metabolism of industry would demand (and fund) the development of new tools for finding still more oil, which would illuminate not only the ocean of oil under the state, but the history of life as well.

Three women employed by Texas oil companies—Alva Ellisor, Esther Applin, and Hedwig Kniker—would develop the field of micropaleontology, which became crucial to our understanding of the past 4 billion years of Earth history. Microscopic fossils of plankton, they discovered, could be correlated with different geologic units. "Within three years," their biographer writes, "300 micropaleontologists were employed in the oil industry, and 31 geology departments offered micropaleontology courses."

Meanwhile, the technologies produced by the crucible of World War I[fn12] gave rise to the field of geophysics, as the human sensorium came to include the ability to see through the ground and probe the deep Earth with sound waves from dynamite—a handy skill for those in search of pockets of oil. Another fossil fuel–powered technology vastly augmented by the war, the airplane, was recruited to perform aerial surveys in search of still more promising oil-rich geological structures. Like micropaleontology, or the oil-motivated study of sea level over Earth history, these technologies and the scientists who employed them pulled back the veil of human ignorance, and it's no coincidence that the very industry that could derail the planet in our own time has done so much to advance our understanding of it. There has never been another organization of human beings, however constituted, with the resources and motivation to describe, comprehend, and literally see vast swaths of the Earth at such scales of space and time as a twentieth century multinational oil company, or as the government-funded surveyors who produced continent-scaled maps of bedrock in search of "economic geology." Drilling technology, meanwhile, developed in lockstep with the scientific advances of the early twentieth century—constantly widening the rivers of geologic energy being pulled out of the Earth.

Since the end of the Pleistocene, wars and other calamities have always accelerated the development and adoption of new technologies, and World War I was no different. When cars and trucks vividly demonstrated their worth by circumventing rail bottlenecks to the western front, they were reintroduced as a practical form of transportation to a public that had largely known them as luxury items before the war. So too for the immense, unrealized promise of the newborn airplane, whose surplus production during the war kick-started a civil aviation industry. In the oceans too, oil found its civilian use, as in Japan’s diesel-powered industrial longline tuna fishing industry, which exploded in the 1920s, stringing the Pacific Ocean with thousands of miles of baited hooks and previewed the apocalypse awaiting the oceans after the next world war. The war was an advertisement for the transformative potential of these nascent technologies, and the world to come. And so investment flooded the world of oil, and the machines it powered.

As oil companies spanned the Earth, new geysers erupted in Tampico, Mexico, and along the sultry shores of Lake Maracaibo in Venezuela, where, as in Iran, miserable shantytowns housing local workers sprang up besides the opulent homes of the British and American oilmen. Repeating a pattern familiar from Pennsylvania, Burma, and everywhere else this ancient carbon spilled from the Earth, the living world around these sites was soon destroyed and the ground ran black with viscous fluid. It “covered the trees, the vines and in ever-growing streams flowed through the underbrush like black serpents,” the Venezuelan historian Manuel Tinker Salas writes. In Mexico, the racialized exploitation of workers by foreign oil companies finally became untenable, and the country defiantly nationalized its industry on March 18, 1938. Oil Expropriation Day, marking the date on which Pemex was founded, remains an annual holiday.

Mexico was exceptional, though. “By the end of the 1930s, seven large international oil corporations had acquired control of the most productive oil regions of the world,” writes the historian Giuliano Garavini. These were the baby Standard Oils of New Jersey, California, and New York, as well as Gulf Oil, Texaco, the British-owned Shell, and the Anglo-Persian Oil Company, the forerunner of BP. The world, of course, stood on the precipice of the second global cataclysm of violence of the twentieth century. The festering humiliations and traumas of the Great War and a global economic depression had bred unholy new political projects premised on the fanatical restoration of national glory through conquest and on genocidal racial superiority. Bombastic art movements like Italian futurism, which fetishized the speed, size, and power of the new fossil-fueled machine, had curdled into militaristic fascism. And naked, febrile hatred had begun to organize the entire new machinery of industry single-mindedly around the destruction of human life. But to achieve any of the nightmares dreamt up in Axis war rooms fundamentally required energy.

WITH THE ONSET OF THE SECOND PSYCHOTIC CONVULSION OF VIOLENCE of the twentieth century, the need for fossil fuel became existential, and globe-spanning military maneuvers now almost exclusively depended on (and often aimed at) securing steady flows of ancient plankton into the dissipative engines of warplanes, tanks, and battleships. It would be "a conflict," the historian Anand Toprani writes, "whose origins, conduct, and outcome revolved around access to energy (oil, coal, and food)." Organic carbon. Both Nazi Germany and Imperial Japan would stake the entire outcomes of their wars on bold offensives to capture foreign oil fields, in the Caucasus and the East Indies, and see their dreams of continental empires dashed when their fuel was cut off.

"The struggle of world politics," as a German military journal foresaw in 1934, "is today more or less a struggle over oil." But despite this insight, Germany began the war from such an energetic disadvantage that British analysts didn't even believe their own estimates of measly Nazi oil reserves. In 1938, half of Germany's oil came from the United States and Venezuela, and with the outbreak of war that share would drop to zero. Germany had enough fuel to prosecute a lightning-fast war of conquest, not a protracted multi-front slog, and certainly not one that drew in the oil-soaked juggernaut the United States.

"The German Army that went to war in 1939 was not a mechanized or even motorized one," writes Toprani. To a somewhat shocking extent, the German army, even in World War II, still relied largely on literal horsepower. "Building an army around the internal combustion engine was a non-starter because Germany simply lacked the necessary petroleum or rubber." Ironically, Germany's shocking victory over France only multiplied the country's extraordinary fuel disadvantages. By conquering much of continental Europe, Nazi Germany had effectively become a much larger entity, one that required far more energy to power it. But the accidents of geologic history had left Europe with little oil and, cut off from the rivers of fuel from beyond, Hitler's vile designs on the continent were doomed. And so the famed oil fields of the Caucasus loomed.

The "primary economic aim," the Nazi leader Hermann Göring announced about the retrospectively suicidal invasion of the Soviet Union, was "to win as much foodstuffs and petroleum for Germany as is possible." Capturing the Soviet oil fields that stretched from the Caspian to the Black Sea was the only way the Nazis could survive the energetic demands of the war and fuel their newfound empire within Europe. After that, securing the oil of Iran and then Iraq would allow it to match the Anglo-American petroleum tsunami pouring into the Allied war machine. As for that other kind of carbon energy, the Nazis explicitly planned to starve up to 30 million Soviet civilians after the invasion in order to divert the spoils of Russian agriculture to German mouths. Elsewhere, famine was stalking the war-weary world as well, not through murderous genocidal design, but callous negligence. After a typhoon destroyed crops in Bengal and a Japanese invasion of Burma cut off rice imports, the British diverted resources to the needs of the military effort in South Asia, and 3 million Bengali people starved to death.

As the USSR withstood the initial shock of the Nazi invasion and the Germans struggled to stay fueled, the Caucasus invasion was called off and the most fevered visions for world domination crumbled. Faltering oil wells in Romania were the only thing that kept the Nazi regime from collapsing completely, and by the end of the war these too were subject to relentless Allied bombing.

Thereafter, outright panic over looming oil shortages was a recurring motif among Nazi war planners, and a year later another desperate, two-fronted attempt to take the Caucasus, and then on to the Middle East's endless oil reserves, sputtered out as well, as Nazi divisions found themselves overstretched and cut off from fuel en route. There wasn't enough energy to secure still more, and the war machine starved. A lavishly funded effort to make synthetic gasoline and petrochemical products out of coal largely failed too: it was massively expensive, energetically and otherwise, requiring seven tons of coal to produce a single ton of gasoline. But its failure came only after producing untold human misery.

Prisoner No. 174517 was Primo Levi, a chemist from Torino, Italy. Levi had joined the resistance in the Italian Alps, and as mentioned at the outset of our journey, would one day see in CO_2 the source of “our millennia of history, our wars and shames and nobility and pride.” He would write, “If the transformation of carbon into organic compounds didn’t take place around us daily, on the scale of billions of tons a week, wherever the green of a leaf appears, it would fully deserve to be called a miracle.” When he was captured by fascists and ultimately sent to Auschwitz by the SS, his background in organic carbon chemistry rescued him from the gas chambers. He was put to work at the I. G. Farben plant on-site. There—and in plants across Nazi-occupied Europe—the Germans desperately tried to transmute one kind of carbon (the Carboniferous coal of Upper Silesia, mined by slave labor) to another kind (liquid fuel and synthetic rubber) to power the Reich’s faltering war effort. “This huge tangle of iron, concrete, mud, and smoke is the negation of beauty[…]” Levi wrote of the facility. “Within its precincts not a blade of grass grows, the soil is impregnated with the poisonous juices of coal and petroleum, and nothing is alive but machines and slaves—and the former are more alive than the latter.” Levi, ever the chemist, ate fatty acids made from oxidized petroleum in order to sate the ceaseless hunger of the camps. He was one of three in his group of 650 sent to the camp who would ever leave.

ON THE OTHER SIDE OF THE WORLD, SIMILARLY OIL-POOR AND RAPACIOUS Imperial Japan eyed the oil fields of the Dutch East Indies to power its continental ambitions on mainland Asia. To enable this invasion of Indonesia, and thereafter keep the sea-lanes from Sumatra and Borneo propped open for the flow of oil tankers back to Japan, they sought to decapitate the US Pacific Fleet by launching a surprise attack on Pearl Harbor, Hawaii. The gambit was a success. By mid-1942, while German war planners were increasingly gloomy that they would ever reconcile their most fevered war aims with their meager oil reserves, Japan had secured the oil fields of the East Indies. But just as “Victory Fever” swept the empire, these gains began to be excruciatingly beat back during the Allies’ island-hopping campaigns of the Pacific Theater. As a suffocating Allied naval blockade slowly cut off Japan from Indonesian oil, the imperial army would soon resort to fuel-saving kamikaze missions, including one with the largest battleship on Earth, the *Yamato,* which was given only enough oil to reach

Okinawa, where it was ordered to beach itself and fight to the death,[fn13] in this, the most desperate, oil-starved phase of the war. The Japanese countryside was even deracinated of its forests as citizens were urged to uproot the country's pine trees in a failed effort to produce gasoline from tree sap to make up for the critical fuel shortages.

In the United States, meanwhile, the largest factories in the world were set to work making as much war matériel as was physically possible. Ford churned out B-24 bombers around the clock. Chrysler, B-29s. "Between 1942 and 1944, the United States, Britain, and Soviet Union produced three times as many rifles, machine guns, artillery pieces, and combat aircraft as the Axis, and five times as many tanks and naval vessels," Toprani writes. At the same time, the two longest oil pipelines ever built were rapidly laid down across thousands of miles of North American vastness. They brought millions of years of oil from ancient Texas seas to the East Coast, flooding the Allied cause—though, scandalously, Standard Oil of New Jersey would have to be forced by the US government to end its partnership with I. G. Farben, which it maintained as late as 1942.

As the Allies relentlessly bombed Nazi synthetic fuel plants and the Soviet Union captured the Romanian oil fields on which they had relied, the planes of the Luftwaffe were grounded and German tank divisions idled motionless. The war was over, as it was energetically impossible for it to persist. Hitler's body would be doused in gasoline and burned, while Mussolini would be hanged by a mob from the awning of a Standard Oil of New Jersey gas station (named Esso after 1926). The American oil supply, on the other hand, was never threatened, with the Allies metabolizing 7 billion barrels of oil during the war, more than a quarter of the amount that had been drilled between the time of Edwin Drake's first oil strike and the start of the war. In fact, a single bomber mission in World War II consumed more energy than the entire population of Europe did during the ice ages. The United States produced two-thirds of the world's oil during the war and 133 times more than Germany the year that Nazi oil production peaked. The extraordinary new phase of energy-use acceleration in the twentieth century—today the United States uses as much oil every year as the Allies did over the entire war—began with the war won by the unprecedented torrent of geological energy out of the Earth. "America's predominance in oil proved decisive," writes historian Daniel Yergin. "And by the end of the war German and Japanese fuel tanks were empty."

IN THE AFTERMATH, THE WORLD HAD BEEN UTTERLY TRANSFORMED, both by the unspeakable violence and by the unprecedented scale of industrialization and technological innovation—powered entirely by fossil fuel. Standing astride the rubble, and above the faded old imperial powers of Western Europe, were the two biggest oil producers in the world, the United States and the USSR. While the world had discovered plenty of oil before the war, it was still a coal-powered industrial society; the postwar world, by contrast, would be powered by oil, and that oil would be purchased in US dollars—which was effectively made *the* global currency at a 1944 conference in Bretton Woods, New Hampshire, in a resort hotel built by a Pennsylvania coal baron. Even the Soviet Union would sell its oil on the global market to get access to this supreme unit of exchange, using these dollars to purchase the infrastructure, equipment, and even grain it needed to (unsuccessfully) keep up with the competitive industrialization of the West.

On Valentine's Day in 1945, President Franklin Roosevelt met with King Ibn Saud of Saudi Arabia aboard the USS *Quincy* as the ship floated just offshore from the Suez Canal. The meeting has been freighted in retrospect with symbolism and meaning—seemingly inaugurating America's uneasy partnership with the kingdom and a presence in the region that still disproportionately drives American foreign policy to this day. But the American relationship had already been forged by 1933, when the king, needing a source of largesse to dole out to political allies in the fragile young country, had given the exclusive rights to drill in his kingdom to a joint venture of the Standard Oils of California, New York, and New Jersey and to Texaco—called Aramco—in return for a large advance of gold coins. In 1939, to much fanfare, King Saud celebrated the first portentous shipment of Saudi oil out of the country, borne on a Standard Oil of California tanker.

In its extraordinary oil superabundance, Saudi Arabia is geologically unique on Earth. And, as we have seen, such extraordinary geology can indicate a correspondingly extraordinary event in Earth history. At least one group has proposed that Middle Eastern oil is the echo of an astounding if controversial planetary reorganization in the Jurassic, when the entire crust and mantle of the Earth became unmoored from the planet's axis, and all the world's continents rotated about 30 degrees in only 15 million years. In the case of the Middle East, this planetary jostle quickly dragged what had long been a tropical offshore cascade of plankton into the Earth's arid subtropics, quickly capping and sealing this bounty of carbon under the parched layers of salts that replaced these vanished seas. It was from this strange Mesozoic bounty—the greatest stockpile of ancient photosynthesis on Earth—that Europe would be rebuilt.

The religiously conservative, autocratic family government of Saudi Arabia would serve the oil companies that funded them well for decades, quashing democratic revolts and arresting labor organizers on behalf of Aramco. But trouble in the shattered political landscape left behind by the disintegration of the Ottoman Empire would continually threaten this new order. When independent Syria rejected the proposed route for Aramco's 750-mile Trans-Arabian Pipeline—which would pass through the country on the journey from the Arabian Desert to the Mediterranean—a military coup installed a new government that quickly approved it. The coincidence has been speculatively linked to the CIA ever since. When the democratically elected Iranian prime minister Mohammed Mossadegh, representing a broad coalition of middle-class Iranians, moved to nationalize the Anglo-Iranian Oil Company, the US intelligence apparatus, led by the Dulles brothers and the British MI6 intelligence agency, supported a coup by the Iranian army that installed the authoritarian, aristocratic shah, Reza Pahlavi, in 1953. "The decision to support the shah in Iran mirrored similar trends in Iraq, Saudi Arabia, and Venezuela, where the United States and Britain supported pro-Western authoritarian governments to protect the positions of private corporations," write the oil historians David Painter and Greg Brew. "Fearing that nationalist governments might fall to communist influence, the United States tended to back right-wing, authoritarian governments that were both anti-communist and generally favorable to private capital and foreign investment." When the Iraqi prime minister Abd al-Karim Qasim tried to do as Mossadegh had, a young Saddam Hussein, with contacts in the CIA, tried to assassinate him. A successful assassination and coup followed a few years later.

As smaller, rogue producers like Getty and the Italian ENI, eager to get in on the Middle Eastern bonanza at any price, offered to drill for these governments on far more favorable terms than the incumbent Standards, Shell, or BP ever had, the long-exploited countries grew more confident in wringing concessions from the multinational corporations operating within their borders. But the growing anger toward the West was still geopolitically impotent as long as the United States remained the oil-producing behemoth on the world stage, with a stateside surplus of oil vast enough to insulate itself from any Middle Eastern price shocks.

THIS WAS THE POSTWAR STORY ON THE PRODUCER SIDE. ON THE consumer side, all this added up to the biggest explosion of energy and economic growth in human history. Trends that began at the dawn of the Industrial Revolution hit the exponential part of the curve. The massive wartime machine of industrial production, technology, and logistics that had been constructed to defeat the Axis powers, and that literally turned the lights back on and lifted the United States out of a depression, had to find a peacetime raison d'être. Availing itself of a practically unlimited amount of geologic energy, this industrial capacity would turn inward, where it would transform the landscape of the continent and the lives of an aspiring middle class. Suburbs and shopping malls crystallized out of the eastern woodlands, tied together by networks of new asphalt highways, conveying refrigerated trucks to stock refrigerated supermarkets. Oil tanker trucks—stocked by bobbing pumpjacks that drained the Cretaceous from Kansas, and the Permian from Texas—unloaded ancient seas of carbon into endlessly refilling neon-illumined gas stations, which would then feed schools of "giant finned cars, that nose forward like fish," as the poet Robert Lowell shuddered in horror.

From 1850 to 1940, the cost of energy, even including firewood, had largely stayed flat, but in the decades after the war it suddenly kept getting cheaper and cheaper—even as energy consumption dramatically increased. This combination of falling price amid skyrocketing demand is some indication of the truly gigantic nature of the oil reservoirs that were then being tapped—pristine seas of crude, accumulating and untouched for hundreds of millions of years, close to the surface and easy to draw from the Earth. This was the "best first" principle in action.

This eruption of ancient energy dissipated in the flickering glow of televisions, the steady hum of air conditioners, and the roaring jet engines of commercial flights as millions of Americans moved into the middle class. By 1970, life expectancy in the United States was almost a quarter-century longer that it had been in 1900, and the country provided the template for the modern high-energy society, one envied the world over, and marked by a suite of modern technologies enumerated by the University of Manitoba environmental scientist Vaclav Smil: there were "coal mines, oil and gas fields, thermal power stations, hydroelectric dams, pipeline networks, ports, refineries, iron and steel mills, aluminum smelters, fertilizer plants, countless processing, chemical and manufacturing enterprises, railroads, multilane highways, airports, skyscraper-dominated downtowns, an extensive suburbia."

Western development agencies, seeking to export the American way to the entire planet, sent engineers and contractors to countries trying to replicate the United States' astounding twentieth-century rise. This was a world further from equilibrium than ever before, existentially held up by the ongoing explosion of geologic energy. The desert bloomed as farms were soaked in fertilizers and pesticides synthesized from fossil fuels, plowed by diesel-powered tractors, and watered with fossil fuel–powered irrigation pumps. The superabundance unleashed by this ancient energy, along with the United States' towering position in the global economy, was such that American labor and capital lay down like lions and lambs—epitomized by the United Auto Workers' so-called Treaty of Detroit with the big three automakers, an agreement that all but guaranteed a consumerist American Dream in return for the unions' pacificity. Generous pensions were promised, picket lines scattered. In this Trente Glorieuses, the growing abundance only brought into relief those left out of the postwar windfall, and a civil rights movement that had been suppressed with terror since Reconstruction began to beat back the United States' racial apartheid system.

There was a darker side to this explosion of energy, though. Vastly augmented in power, the interstate competition of the latter twentieth century between the world's two biggest oil producers and most powerful countries in Earth history—the far more powerful US and the Soviet Union—now threatened nuclear Armageddon. More inspiringly, it would send ships not across the oceans, as imperial rivalries had in centuries past, but out of the Earth's gravity well altogether, and into outer space.

If previous periods of growth had registered blows on the global environment, this one arrived with the force of an asteroid. As industrial might and military sonar were trained on the seas, 90 percent of the ocean's top fish predators were plucked from the oceans. These dwindling stocks encouraged only more powerful, more destructive fossil fuel–powered fishing fleets, in search of ever-more-distant fish stocks. Today, decades on, the global fishing fleet is twice as big as it was in 1950 but catches 80 percent less fish per the same amount of effort. Meanwhile, whaling reached its satanic peak, not in the 1860s, but in the late 1960s, as the hoary industry was mechanized and fueled by this eruption of petroleum. As late as 1972, General Motors was still using sperm whale oil in its automatic transmission fluid and lobbied Congress against a ban. And in a stunning demonstration of how eras can surprisingly overlap, and just how fast the world was changing, sperm whale oil was even used to lubricate components in the Apollo lunar modules and intercontinental ballistic missiles (ICBMs). The demand was met by a global fleet of industrial whaling ships from Norway, Japan, and Great Britain. But it was the Soviet Union, which had little practical use for the whales but had a calcified bureaucracy plagued by poor information and desperate for international prestige, and which generated senseless quotas that its captains were incentivized to exceed, that executed the most rapid and destructive phase in whaling's entire, ignominious history.

The ensuing Soviet slaughter of whales in the Southern Ocean and the North Pacific fundamentally restructured the way carbon moved through the ocean. The waters around Antarctica had long been famously red from near-infinite swarms of krill and the spouts of whales that filled the horizon, as they critically recycled nutrients through the water column. By drawing nutrients up from the deep and pooping them at the surface, whales had ignited these blizzards of sea life. But since a million whales were killed here, many of them uselessly discarded by Soviet factory ships, the vast krill swarms haven't been seen in over four decades. The huge standing mass of carbon they sequestered was lost, and the surface waters are now dominated by microbes. The world's oceans now host a fifth of the whale biomass they once did, and countless cetacean cultures were lost forever. The extinction of countless populations of these ice-age giants, even in this age of optimism, became increasingly intolerable, and an international whaling ban was finally passed in 1986, after decades of activism.

Meanwhile, the runoff of fertilizer and sewage into the shallows was robbing the oceans of their oxygen, and dead zones came to pulse from the shore—just as they had in previous planetary crises in Earth history. As the West feverishly industrialized, the world's wetlands were falling silent too, as chemicals synthesized from hydrocarbons like DDT wiped out ospreys and eagles. This was happening alongside oil spills from the nascent offshore oil industry, flammable midwestern rivers, suffocating smog in big cities, and the proliferation of hydroelectric dams that replumbed entire continents and obliterated anadromous ecosystems to provide more power still. Thus a nascent environmental movement was born. But while the movement's ire was trained on these direct industrial interventions in the environment, the very chemistry of the oceans and atmosphere was beginning to more subtly warp as well, from the billowing eruption of geologic carbon that powered this Great Acceleration.

As this new world came into being, the sun was officially setting on the fading British Empire—like the erstwhile Dutch, Spanish, and Portuguese before it. If the British had enjoyed the benefits of Iranian oil more than Iran itself had, much of this oil was shipped through a Suez Canal that similarly funneled more money into Britain's coffers than Egypt's. Both were representative of the untenable colonial arrangements that had enriched European powers for centuries, but whose days were numbered. When Egypt nationalized the Suez Canal in 1956, the British, French, and Israelis launched a failed invasion of Egypt that infuriated Washington. (The closure had the unintended consequence of giving birth to the largest oil tankers in history, very large crude carriers [VLCCs] and ultra-large crude carriers [ULCCs], invented to profitably reroute this oil all the way around Africa's Cape of Good Hope.) When Syria subsequently turned off the Trans-Arabian Pipeline for a day to demonstrate the new nature of power in the region, a British diplomat complained to President Dwight Eisenhower in a revealing admission that connected energy to money, and money to power, demystifying 6,000 years of financial and military history from Uruk to the age of petrodollars: "If Middle East oil is denied to us for a year or two our gold reserves will disappear. If our gold reserves disappear the sterling area disintegrates. If the sterling area disintegrates and we have no reserves we shall not be able to maintain a force in Germany or, indeed, anywhere else. I doubt whether we shall be able to pay for the bare minimum necessary for our defense. And a country that cannot provide for its defense is finished."

While the US might have dismissed these protests as the mewlings of an outmoded empire in decline, one that increasingly traded its homegrown heavy industry for financial services, the reckoning would soon come for it as well. In 1956, a Shell geologist named M. King Hubbert predicted that oil production in the lower United States would peak around 1970. He was correct. In 1971, American oil production peaked and then continually declined until the technological deus ex machina of fracking rescued the industry in the 2010s. And with this end of endless oil in 1970 came the end of the American Empire's insulation from Middle Eastern ire as well. After two and a half decades of unthinkable postwar growth, as cheap oil drained from the continent and was dissipated in the engines of modernity, the United States' spare capacity disappeared, along with its margin of safety. A Middle East, long trampled by Western prerogatives, could now wield the "oil weapon."

AS FAR BACK AS THE 1100S, THERE HAVE BEEN FOSSIL FUEL CARTELS. Starting with the medieval prince bishops of Durham, England, and then the Newcastle Hostmen's Guild, whose control over the River Tyne enabled them to establish a monopoly over the supply of coal to fuel-starved London, these organizations restricted supply to produce profits. With the dawning of the age of oil, Standard emerged as a cartel of sorts as well, consolidating the industry under a single banner, controlling production, naming its price on the market, and swelling its own power with the resulting windfall. But in the decades after Standard's breakup and the gigantic new discoveries in East Texas, the industry was unpleasantly reintroduced to the ruinous effect of a free market. So much oil was pumped out of the ground by a disorganized army of independent drillers that the price repeatedly crashed, and the army had to be called in to bind the invisible hand of the market and stop the rampant drilling at gunpoint. An unlikely solution was found in the Texas Railroad Commission, which oversaw the transport of the oil out of the state, and which in 1930 assumed the powers to set quotas and throttle back production in order to maintain a profitable price for producers—effectively setting the global price of oil.

Thirty years later, oil-exporting countries, stung by unilateral price cuts dictated by the oil giants, would band together to form the Organization of the Petroleum Exporting Countries (OPEC), to a similar aim. In fact, Venezuela, the odd Western Hemisphere partner in OPEC, had explicitly modeled this transnational organization on the Texas Railroad Commission. Prior to 1970, however, OPEC could not effectively wield its political power, as the world's foremost oil superpower, America, was still drenched in surplus oil. When this buffer ran out, though, as the demands of rapid postwar growth finally exhausted the energy physically available in North America, the power in the world palpably shifted to the fathomless reserves of the Gulf states, and those who maintained the spigots of this ancient energy.

In 1973, in response to US support for Israel against the Arab states in the Yom Kippur War, OPEC imposed a total oil embargo on the superpower. Within the course of a few months, the price of oil in the US would quadruple, and two decades of unthinkable growth came to a screeching halt. Yet this first “oil shock” was not simply, or even primarily, the result of the embargo, as is often depicted; instead it was rooted in the years before, and in the physical world, as oil consumption in the developed world continued to skyrocket even as it became clear that the geology of the Earth was not readily giving up vast new seas of petroleum. The postwar party was over. The world of leapfrogging growth, industrialization, and high modernity collided with a growing recognition of overconsumption and the potential exhaustion of Earth’s resources. Indeed, two years before the shocking deployment of the “oil weapon,” Richard Nixon tackled the country’s massive and growing deficits by repealing the convertibility of dollars to gold, effectively devaluing the world’s reserve currency. Given that oil was sold in dollars, OPEC acted—by dramatically increasing its prices—both to conserve its most valuable resource in a world of worrying overconsumption, and to ensure that it didn’t depreciate its economic lifeblood in a newly weakened dollar system. The embargo was merely another factor that contributed to the suddenly soaring price of what had been forever seen as an inexhaustible resource.

Faced with economic pain, the Western reaction was less than charitable. When countries like Libya, whose biggest exports before the discovery of oil included the sale of scrap metal left over from the North African Theater of World War II, were thrust into international prominence and power by virtue of the ancient carbon under their feet, Jimmy Carter averred that he “deeply resented that the greatest nation on earth was being jerked around by a few desert states.” When the US-backed but unstable regime of the shah of Iran, having failed to remake the country in his image using its oil revenues, was overthrown in the 1979 Iranian Revolution, a century of antipathy toward the West was unleashed all at once. The price of oil soared once more, as did inflation. When a be-sweatered Jimmy Carter encouraged frugality and fuel conservation to a weary American public, while pulling price controls on oil to incentivize American oil companies to invest in new production, the price only rose further. Belt-tightening makes for bad politics, though, and the electorate was restless from a decade of austerity and decline. As a result, so too rose the political prospects of Ronald Reagan.

Chastened by the previous decade of oil shocks—and the painful demonstration of the economy's existential dependence on energy—in 1980 American diplomatic and military might would become almost single-mindedly concentrated on keeping the flow of oil pouring out of the Persian Gulf. The Gulf is a narrow waterway that becomes narrower still, nearly pinching off in the Strait of Hormuz, a ninety-mile-wide chokepoint buzzing with US Navy destroyers and Coast Guard cutters, and through which squeezes a fifth of the world's oil and a third of the world's liquified natural gas. It plays in the modern world the same role that maritime chokepoints for grain, like the Turkish straits, long played as previous sites of imperial contestation. In his last State of the Union address, Jimmy Carter would announce that, in effect, meddling with US control over the distant region would be "regarded as an assault on the vital interests of the United States of America."

Carter's doctrine had a long pedigree. In 1903, the British government's official policy had been to "regard the establishment of a naval base or of a fortified port in the Persian Gulf by any other power as a very grave menace to British interests, and we should certainly resist it with all the means at our disposal." But this colonial legacy over Hormuz stretches back even further still. Today the US maintains an air and naval base at its narrowest point in Oman, across the strait from a crumbling five-hundred-year-old fort from which the Portuguese once maintained their stranglehold over the Indian Ocean.

In the case of the US, this policy has produced some strange bedfellows. During the Iran-Iraq War in May 1987, after Saddam Hussein accidentally killed thirty-seven Americans in the Gulf aboard the USS *Stark* with French-made Exocet missiles, "the Reagan administration quickly forgave Baghdad," write historians Hal Brands and David Palkki, "and Iraq received the single largest credit in the history of [the US Department of Agriculture's financing institution], totaling more than $1 billion. It also received greater intelligence sharing, increased diplomatic support in the UN Security Council, and a further loosening of restrictions on high-tech exports from the United States …. By the time a truce brought an end to the war in August 1988, Saddam had reaped considerable benefits from his continued engagement with Washington."

While honoring the American servicemen killed in the attack, Reagan explained that naval presence in the region was nevertheless necessary because the Persian Gulf represented "the starting place for the oil that is the lifeblood of much of the world economy," even claiming that the United States' very freedom was at stake if the oil was cut off. His remarks echoed FDR's secretary of the interior, Harold Ickes, who noted a half century earlier that "[w]ithout oil, American civilization as we know it could not exist." Given the energy requirements of the developed world, Reagan's lifeblood metaphor was scarcely metaphorical, and US geopolitical strategy has been organized around keeping this global chokepoint open like an arterial stent ever since.

But fears of oil exhaustion soon subsided as high prices spurred the development of vast new deposits in Alaska and Mexico, and cities of gargantuan offshore rigs dotted the North Sea in the 1980s. When the latter began to flood Britain with cheap oil, Prime Minister Margaret Thatcher used it to destroy what remained of the country's unionized coal industry.

By the early 2000s, though, when this buzz of production had begun to falter too, and newfound concerns about "peak oil" reared again, even a Texas oilman like George W. Bush was able to admit in a State of the Union address, "We have a serious problem: America is addicted to oil." The skyline of New York City had been brought low by fifteen Saudi Arabian hijackers and four other Middle Eastern nationals, organized and funded by the son of the biggest contractor in Saudi Arabia, with the financial support of officials in the Saudi government. In response, the United States spent upwards of $6.5 trillion prosecuting wars in the Middle East and Central Asia—not to punish Saudi Arabia, with which it was too economically and geopolitically enmeshed, but to ineffectually reassert its influence in the region by attacking Afghanistan and, in a total non sequitur, Iraq.

Meanwhile, at the dawn of the new millennium, an old superpower was rising to the east. China was sacrificing more than a thousand coal miners a day to energize an episode of economic growth that dwarfed even the unimaginable postwar rise of the United States—this one powered almost entirely by coal. Today China burns more coal than the rest of the world combined, and the world is burning more coal, oil, gas, and even firewood than it ever has in human history. Global GDP is still going up. And so is the temperature.

11

Energy, CO_2, and Civilization

THE LAST FEW CHAPTERS, FULL OF NAMES AND DATES AND PLACES, ARE one way to understand the story of the past couple of centuries. It is human history, in all its complexity and drama. But if we acknowledge that the boundary between the human and natural worlds is less than absolute, and in fact is profoundly continuous, we can understand this singular eruption of geologic energy in a somewhat different context—as a piece with Earth history.

As it has been doing since its beginnings, life exploits energy gradients and dissipates them to create its complexity and do work on the environment. Whatever exactly it was that happened in our recent past, what emerged was a new biological phenomenon on the planet, one that, driven by intensifying agricultural, then industrial production, catalyzed a massive discharge of the planetary capacitor on a scale never before seen in the history of complex life. In its impact and novelty, it's not an exaggeration to compare it to the Cambrian explosion that occurred some 530 million years earlier, in the wake of Snowball Earth.

In that case, the sudden radiation of animal life emerged only after the development of new, finer systems of information control and processing (gene-regulatory networks capable of coordinating and directing the development of more complex animal body plans), a division of labor within multicellular life (between different types of cells, tissues, and organs), and finally the rise of a vastly more powerful source of energy (oxygen-powered metabolism) to power these vastly more complex, active, and energetic body plans and lifestyles. The subsequent arms race that followed, between and among predator and prey, expressed itself in a staggering proliferation of biological armor, weaponry, strategies for capturing prey and avoiding predators, eyes and other complex tools for gathering information about the environment, nervous systems to process this bewildering new landscape, and even new systems for communication. But the Cambrian Explosion is most easily identifiable in the geology by a somewhat sudden transition from a rock record that, for billions of years, was made of undisturbed layers of placid bacterial muck, to one now burrowed, churned up, and utterly reworked by the energetic frenzy of animal life, and by a reorganization of how carbon and other nutrients flowed through the biogeosphere. All this hubbub might have also caused a mass extinction of a provisional placid world of bloblike creatures that came before.

Over the past few centuries—a half-billion years later—the planet was struck by a similar, if exponentially more intense, transformation of the rock record and corresponding wave of extinctions. Like the Cambrian, it came only after the development of new systems of collective information processing (the printing press, the scientific revolution), an intensified division of labor (one enabled by trade and exploitation, and enforced by capitalism), and the sudden availability of all the carbon energy ever built up in Earth’s crust. This explosion of power allowed for the reworking of the very surface and deep subsurface of the Earth, the organization of human societies at a scale and complexity never seen in history, and, among hypercompetitive industrialized countries, a subsequent explosion of violence. The ensuing arms race produced the staggering new weaponry, technologies, and systems of communication and information processing of the twentieth and twenty-first centuries.

If it was a surge in oxygen that set off the Cambrian Explosion, though, by the time the Industrial Revolution started oxygen was no longer the limiting factor on the Earth’s surface: after hundreds of millions of years of carbon burial, oxygen had long since built up to its staggering concentration in the atmosphere today. Instead, ours was something like an inverse Cambrian Explosion. A tidal wave of organic carbon was brought to Earth’s surface and *reunited* with this oxygen, and a new industrial metabolism exploited this titanic new source of free energy, obliterating the amount of activity that was possible on our planet. What ultimately emerged was a novel, planetary-scaled dissipative structure—industrial society—one that is now extraordinarily, and perilously, far from equilibrium. If each of our cells requires a million electrons a second to hold back entropy and keep us from quickly disintegrating, this global engine of industrial society now requires 18 trillion constant watts of energy to maintain itself. And like the Cambrian, whose planetary transformation is most visible today in the seafloor burrows of animal life, our own age may be most durably recorded in the Earth's crust (and observable to geologists of the geological far future) by the vast excavated subsurface beneath our own society—in our mines and bore holes, sometimes extending miles deep.

However much we privilege human history as distinct from the rest of the natural world—however unrecoverably messy and fundamentally *human* our story on this planet seems, charged as our lives are with moral responsibility and agency, and diffracted as they are through the kaleidoscope of human institutions and cultures—we are a natural phenomenon. And at the end of all this human turbulence, all this ugliness and beauty, what had been assembled on the planet was a massive novel biological system, accelerating the dissipation of the greatest energy gradient on the planet: between the sea of organic carbon under its surface and the extremely oxidizing world above it. All that carbon had begun as CO_2 eons ago, and to CO_2 it now returns, explosively released at a rate hundreds of millions of years faster than it took to capture and bury. It has been, as the climate scientists Roger Revelle and Hans Suess have called it, "a large-scale geophysical experiment of a kind that could not have happened in the past nor be reproduced in the future." The turbulence of human history swirls off in eddies from this one-off explosive release of CO_2 and waste heat.

DESPITE THIS TRULY UNIQUE AND RUNAWAY RIFT WITH THE GEOLOGICALLY paced processes of the Earth system, our artificial, technological human world is still nevertheless contiguous with, and inseparable from, the natural world. Farming, for instance, was first invented not by humans, but, perhaps as long ago as the Cretaceous Period, by ants that purposefully grow fungus on leaf cuttings. Ants engage in lots of other uncannily familiar activity as well, like continent-spanning wars. Spiders meanwhile invest energy into the construction of their webs, then see the returns on their investments roll in when they capture fluttering bits of energy packaged in insect biomass—as surely as a well-capitalized Dutch herring fleet. Conversely, the human global economy also shares some of the attributes of biological systems, for the simple reason that it is one. Plotting the data from more than forty countries, the University of New Mexico biologist James H. Brown purported to find that the energy use of a country and its gross domestic product (GDP) had almost the exact same mathematical relation as the well-known biological scaling relationship between the metabolic rate of an animal and its body mass.

"This may not be coincidental," Brown explains. "In a very real sense both animals and economies have 'metabolisms.' Both consume, transform, and allocate energy to maintain complex adaptive systems far from thermodynamic equilibrium. The energy and other resources that sustain these systems are supplied by hierarchically branching networks, such as the blood vessels and lungs of mammals and the oil pipelines, power grids, and transportation networks of nations."

If it seems crude, or unhelpfully naturalizing, to put our messy political and economic world in these kinds of biological terms, the model works both ways: there are political implications to viewing our human world as a thermodynamic phenomenon. It forces us to reckon with the fundamentally energetic basis of society. Because it takes energy to do anything, and we've come to live on a world where almost all the energy we dissipate is fossil carbon, this means that when the price of oil goes up, the price of making almost everything else goes up as well. This is inflation, the specter of which, at the time of writing, is haunting the global economy as it attempts to equilibrate with a huge surge in energy demand, after years of reduced oil production during a global pandemic. Oil production can't be quickly brought back online—or can only be done so at a considerable markup, as oil producers try to recoup their pandemic losses by keeping prices, and profits, high. Adding to these inflationary worries came an old-fashioned European war of imperial conquest—one with roots dating back to the early twenty-first or thirteenth centuries, depending on who you ask—the aggressor of which is one of the world's biggest petrostates. As a result of Russia's invasion of Ukraine, the inescapable energetic demands of Western Europe began to collide with their ideals. While decrying Russia's invasion of Ukraine, Germany was simultaneously subsidizing it to the tune of $220 million *a day* in energy payments to Russia. And when Germany's gas was cut off entirely and energy prices soared, the economic powerhouse of Europe began to deindustrialize, its citizens' real wages declined more than at any point in the past seventy years, and the German far right became ascendant once again.

Being too important to the functioning of the modern industrialized world, of course, sanctioned Russian oil made it into the global market through alternative streams, like a stopped-up river finding new courses and rivulets around a logjam—in illicit and laundered shipments, and more openly to those untroubled by Russia's aggression. The bayous and barrier islands of the US Gulf Coast, meanwhile, are sprouting with new liquified natural gas terminals and cast a fleet of tankers across the Atlantic to make up the energetic difference for Western Europe. Wheat and corn harvests aren't quite as fluid and responsive as oil and gas production, though, and shortfalls of Russian and Ukrainian grain have sent global food prices soaring.

FOR HEADS OF STATE, THEN, EFFECTIVELY MANAGING FLOWS OF photosynthetic energy in such uncertain times can be a matter of political survival. It's why imperial Chinese bureaucrats bought stockpiles of wheat for their granaries when it was cheap, and sold it to restive citizens in bad harvest years when it was expensive again; it's why the Romans littered the Mediterranean with massive warehouses of grain; and it's why countries like the United States, in the wake of the first oil shock in 1973, developed fossil fuel stockpiles like the Strategic Petroleum Reserve—which serves the same function for a fossil-fueled society that glycogen stored in the liver, or fat stored around the waistband, does for humans in times of famine, or that caches of energy-dense mammoth meat did for Paleolithic bands during the lean months. It's a buffer of energy.

"The processes that dissipate energy in our economy are physical machines such as cars, manufacturing equipment, air conditioners, and the computer server farms," writes the University of Texas economist Carey King. "These machines require energy to operate. The more mass of physical machines, or physical capital, that dissipate energy, the larger the GDP of a country."

When economic expansions are fueled by a mirage of debt untethered to the material world beneath it, bubbles can eventually burst. The University of Victoria biophysical economist Chris Kennedy recently found, for instance, that underinvestment in coal-carrying railroads during the 1920s might have cut the energy supply to the US economy by a staggering third in the Great Depression. And perhaps it's not a coincidence that oil shocks precede every recession. Oil prices spiked in the months surrounding the embargo in 1973, the Iranian Revolution, the Iran-Iraq War, and the Persian Gulf War, and recessions followed each time soon thereafter. In fact, the biggest oil price spike of all time immediately preceded the 2007–8 financial crisis as well. As global oil production stagnated—many thought for *good*—while demand soared from the new industrial powerhouse of China, the looming prospect of peak oil once more stalked the geopolitical landscape. As oil prices soared, the managers of the swelling sovereign wealth funds of oil-exporting nations sought lucrative returns on these ballooning revenues and poured money into financial markets that in turn exploded an all but fictional world of derivatives and shady mortgage-backed securities. When inflated home prices started to decline and high oil prices made the cost of living marginally more expensive, mortgage delinquency rates began to rise and the entire global economy nearly collapsed.

When there's less energy available to grow the economy, but profits must nevertheless continually rise (as they must), the wage share falls and people increasingly take out debt to maintain their current lifestyle. Eventually, however, the thermodynamic check must be cashed, and economic crisis hits. Who ultimately foots the bill during these corrective periods of crisis is called politics.

"When resource consumption stops growing, the profits have to stay up, so the wage share goes down, right?" King told me. "We're a capitalist economy. The capitalist says, 'I can't maintain profits in this kind of environment. And I can't affect the oil price, or the energy price anymore, so we need to cut wages. We can't let wages go up with inflation.' In 1980 we elected a president who said, 'I'll go along with that.' And we got Reagan. We got Thatcher."

When the long climbing trend in postwar American growth, powered by cheap oil, ceased with a financially (and spiritually) ruinous episode of military adventurism in Vietnam, the exhaustion of American oil, and then the oil shocks of the 1970s, inflation soared. In response, the chair of the Federal Reserve, Paul Volcker, jacked up interest rates to intentionally steer the economy into a recession, increase unemployment, reduce consumer demand, and stabilize ever-inflating prices at the end of that decade—in other words, to cool the economy, literally. Volcker stayed on for the

Reagan administration, which, like its kindred Thatcher administration across the sea, inaugurated a new era of hostility toward unions. And by decimating unions' ability to collectively bargain for real wages that would keep up with inflation, they staved off the dreaded wage-price spiral that would push prices ever higher. Workers, it was decided, would take the hit.

Like the sudden appearance in the archeological record of elite graves one sometimes finds when natural resources abruptly become scarce—during historical spasms of drought, cold, and privation—inequality soared in response to this new modern age of energy scarcity. And just as those earlier societies could legitimate their new unequal distribution by appealing to the supernatural powers of a chief, this modern redistribution was legitimated by appeals to the almost mystical impersonal forces of the market. But where Western leaders could paint a recession as operating according to the unchangeable mechanics of the economic universe, in the Soviet Union, which saw the return of cheap oil in the 1980s devastate its export revenues, economic pain was laid squarely at the foot of state planners, chipping away at the regime's legitimacy and culminating in the dissolution of the communist world in 1991.

While these events were known as oil shocks to the beleaguered oil-importing nations of the industrialized world, to the nations of OPEC they were economic windfalls. When oil prices spiked, the ascendant oil-producing countries suddenly found themselves under an avalanche of oil revenue. Needing somewhere to invest all this newfound wealth, OPEC sent it back to its main importers, with countries like Saudi Arabia both stocking up on American military hardware and flooding the financial centers of New York and London with petrodollars. From there these dollars were loaned out to developing countries in desperate need of capital. But when interest rates began to soar in 1979, these loans were suddenly called back in by Western investors, only at the new punishing levels of interest—devastating the economies of the Global South. As financial crises swept over the developing world, the US-led World Bank then imposed structural adjustment programs on the struggling countries, forcing them to cut public spending and open their industries to foreign investors who could recoup their losses—a scheme that even the conservative historian Niall Ferguson calls "a wishlist of … economic policies that would have gladdened the heart of a British imperial administrator a hundred years before."

“Money is power,” declared signatories of the Arusha Declaration, led by government officials of Tanzania and Jamaica, who protested the reforms the World Bank coerced its member countries into adopting. “Those who wield power control money. Those who manage and control money wield power. An international monetary system is both a function and an instrument of prevailing power structures.” Or as Thomas Hobbes put it some three hundred years earlier, “Wealth is power and power is wealth.”

Once again, today, after a decade of low interest rates, during which countries like Ghana borrowed cheaply to build infrastructure for their swelling populations, the developing world finds itself racked with punishing, unpayable, dollar-denominated debt as the United States tries to shock its domestic inflation with higher rates.

AS THE PANICKED TREASURERS OF HYPERINFLATING CURRENCIES DISCOVER, more and more money alone cannot improve material conditions. Fetishizing the tokens themselves, we lose sight of the thermodynamic basis for money until gas pipelines are cut off and inflation soars. Human welfare, then, is now mediated by the flows of energy and material around the planet, denominated in and coordinated by money.

This is why the diplomatic crises, state failures, and financial collapses of the twentieth century are all visible in plots of global CO_2 emissions—with downticks in the early 1930s (the Great Depression), 1973 (OPEC oil embargo and revaluation), and 1979 (Iranian Revolution); a flatlining in the early 1990s (collapse of the Soviet Union); and two more drops corresponding to the 2008 financial crisis and the COVID-19 pandemic. The human drama has always been registered in the composition of the atmosphere, but since we’ve enlisted fossil fuels to help us do work on the surface of the planet, our stories have become impossible to miss.

What makes climate change such a wicked problem—as implied by the human misery encoded in these downticks of CO_2—is that there is a fundamental relationship between energy and living standards. *Homo sapiens* has always existentially relied on its material culture to survive, and transforming raw material to make it useful takes energy. In the “developed world,” we live in houses that stay warm in the cold, and cold in the heat, and that are filled with right angles, wiring made of mined and refined copper from distant countries, labor-saving devices, airtight vaults filled with out-of-season produce from the other side of the world, and other highly improbable structures found nowhere else in nature. Just like the extraordinarily complex and ordered biological systems that keep our bodies alive, our human environments have become extraordinarily low-entropy as well. This trend explains the incredible explosion of human population in the past two centuries, as humans have dissipated unfathomable amounts of energy to create the improbable environments, supply chains, and products that now sustain 8 billion people. Where for essentially all of human history the few hundred thousand humans alive depended on the food they could gather in their immediate vicinity—and on the life-threatening vicissitudes of the environment in which it grew—today's world population can only meet its demand for staple crops by drawing from supply chains that draw from an average minimum distance of over 1,300 miles.

Creating and sustaining this order of the modern world takes *lots* of energy. Maintaining a house, for instance, requires constant vigilance against the onslaught of water damage and foundational rot; it requires replacing aging furnaces and water heaters; it requires swapping out lightbulbs, filled with strange corners of the periodic table, that are electrified by power lines and transformers maintained by utility companies (staffed by workers who, importantly, don't have to spend all day gathering and growing their own food) using hydraulic equipment, and connected to distant industrial power plants, which require their own maintenance regimes and supply chains, and so on. Any failure in any link of this system means your lightbulb won't turn on. It's all very improbable! Maintaining the improbability of our artificial, manufactured world—maintaining any improbable structure in our universe, for that matter—necessarily requires dissipating lots of energy. The fight against entropy is unending. But this extremely improbable infrastructure allows the privileged of our world to live in extreme, unprecedented comfort, and enjoy lifespans unseen in human history. And it's all made possible by energy.

For all the horrors of the global resource extraction economy and the military conflagrations of the twentieth century, this explosion of energy has doubled global life expectancy since the invention of the steam engine. Where no peoples in any part of the world, in any part of human history, could depend on living much beyond an average of forty years, today it's over seventy. Much of this extension is thanks to public health measures like sanitation. But sanitation entails massive public works projects and water treatment facilities—industrial-scale infrastructure projects unthinkable in a preindustrial world. They would have required untold populations of corvée labor working around the clock to construct them, and a corresponding amount of food energy to power these workers, and farmers and land to grow that food. But in our own time, even as the planet has added billions of people, malnourishment has dropped by about 40 percent. Literacy is up more than 70 percent. Globally. If one is to impugn the entire system—or aspire to tear it down—these are uncomfortable facts with which to reckon.

The biosphere has taken an astonishing blow in the past few centuries, accelerating the ongoing trends in place since the dawn of our species. This is in large part because of our success: most of the world's landscapes, and the photosynthetic energy embodied in them, have been reoriented to serve humans. Half the world's land surface has been turned to farmland. Wild animals now account for only 2 percent of the Earth's land mammals—the remaining 98 percent of the mammal biomass is humans and our livestock. We kill more dinosaurs every year in industrial chicken slaughterhouses than the End-Cretaceous asteroid ever did, as we turn as much of the biosphere as we can into food. Indeed, while humans—starting with the wave of ice age extinctions—have slashed the biomass of *wild* land mammals on Earth by over 80 percent, we've astonishingly *grown* the overall biomass of *all* land mammals by a factor of four over the same period, in livestock penned in feedlots, on the pivot-irrigated greenery of our strange fossil-fueled farms, and in ourselves (we are land mammals). However uncanny this picture may appear, though, famine has been all but eliminated across much of the world, and not only from this freakish fossil-fueled agricultural productivity, but because of our globe-spanning communication systems that can alert the world to impending catastrophes.

Anyone who argues that the average human being is materially worse off on average than their preindustrial counterpart is simply misled or in denial. This new abundance, if horribly inequitably distributed, is especially impressive given the extraordinary feat of having to provide for billions more mouths than at any point in our hunger-racked history as a species. In the developed countries, which have enjoyed the lion's share of the gains from industrialization, we are physically bigger, taller, in better health, and longer-lived than every generation before us. While rich Western society might be shot through with spiritual listlessness, tawdriness, a toxic political culture, and a vapid pop culture that celebrates mediocrity, it is not a surprise that much of the world might aspire to our lives of relative comfort.

Cities, with all the attendant benefits of city living, are massive standing waves that process material and energy. They are organisms of a sort. Huge inputs of food, energy, and material go in one end—ports, distribution centers, oil storage tanks and refineries in heavy industrial zones, high-voltage transmission wires—and CO_2 waste heat, trash,[fn1] and sewage comes out the other. The enrichment and vibrancy of living in a modern city are the more qualitative products of this flux. Like life itself, cities are organized flows of material and energy that exist extremely far from equilibrium. Cut a city off from power, or inputs of food, or pumps and valves and switching stations, and watch it wither and die.

There is even a metric used in economic and social science, the "carbon intensity of human well-being," that explores the undeniable positive relationship between human development and the amount of energy people use. (It goes without saying here that carbon isn't the relevant resource at play, but the energy it supplies, which could in principle come from any source.) The correlation eventually levels off, after one's personal security and health are secured, and quality of life even declines in hyperenergetic but somewhat dysfunctional societies like the United States. To visit a decaying Erie Canal town in western New York State or a vacant block of row houses in East Baltimore, for instance, is to see where the energy in a society has stopped flowing.

If the connection between energy and living standards still seems tenuous, consider the COVID-19 vaccine. It could not have been developed in a world without access to tremendous surpluses of energy. It required a well-fed, extremely educated group of dedicated researchers, taught by similarly well-educated specialists, who had been freed their entire lives from the obligations of producing their own food or keeping themselves warm, working in electrified labs kept sterile by megawatt-chugging HVAC systems, storing samples in ultracold refrigerators and collaborating with global colleagues over the internet—itself a global fog of electric impulses firing over hundreds of thousands of miles of undersea cables, ricocheting off networks of satellites and radiating out of radio towers—with huge energy inputs at every step of the way, from the 450-kilowatt combine harvesters that hoovered up the oceans of wheat and corn that fed them their entire lives, to the terajoule-scale raising of university buildings to work in. Fundamentally, however, inventing an mRNA vaccine is so energy-intensive because a vial of synthetically produced mRNA vaccine is an extremely improbable and low-entropy configuration. Since things only ever get more disordered in our universe, it takes a tremendous input of energy to produce such a highly ordered substance. Gathering and systematizing information such that it can become causally effective on our world in a predictable and desirable way, especially on the impossibly complex and nanoscopic scale of an mRNA vaccine, unavoidably takes lots of energy. This is the second law of thermodynamics, which shall not be violated.

This is true for all information technology. It's why your computer fan starts chugging when you have too many tabs open. It's why Microsoft is looking to sink its data centers to the bottom of the ocean, where the vast waste heat produced by industrial-scale information processing can be wicked away by the frigid deep. "Information is physical," as the IBM researcher Rolf Landauer famously said, and computation is inescapably energy-intensive.[fn2] What this all means is that, in a universe that is falling apart, there is an energy cost to processing information about your environment. This is why brains didn't show up on the planet until the arrival of the energy unlocked by the growing oxygenation of the oceans at the dawn of animal life. Life produces little islands of order and complexity in bodies and brains by pumping out even more disorder into the environment as waste heat, dutifully obeying the second law. So do computers. And as with animals, for computers, the energy to pull off such a thermodynamic trick comes from converting plant stuff back to CO_2 in the presence of oxygen. When the industrial-scaled computation of our civilization is hooked up to a grid powered mostly by fossil fuels, it transforms huge amounts of organic carbon to CO_2 and waste heat. That is why Microsoft's CO_2 emissions spiked in 2021 after declining for years—thanks, in large part, to an army of zombie-eyed gamers, now housebound by the pandemic, playing ungodly hours of *Call of Duty: Warzone*. By 2040, the information technology sector could account for as much as 14 percent of global carbon emissions.

If we are to safely navigate the coming decades, we will need to fight against this entropy, like life itself, and get much better at processing information about our environment. Consider, for instance, that one of the most computationally demanding things humans do is simulate the entire Earth system on supercomputers. As the unpredictability of the world accelerates thanks to the ongoing and unprecedented chemistry experiment that is currently underway, humans will increasingly need to understand the complexities of this climate-ocean-life system and the potential carbon-cycle responses of the planet, if we are to adapt. Even our noble pursuit of the secrets of the universe is subject to energy constraints: when Russia invaded Ukraine and Europe was struck with an energy crisis, the power-chugging Large Hadron Collider had to shut down.

This is the central crux of the problem, and why CO_2-driven climate change is unlike any environmental challenge ever faced by humanity. Doing anything requires energy. Energy, for the entire history of life, is what gives organisms greater control over their environment, provides buffers against misfortune during uncertainty, and liberates us from lives of precarity and drudgery. The chemical reaction that powers our fossil-fueled world is the same one that has powered animal life since its origins: turning the energy-dense organic carbon of life back into CO_2. The now-existential dependence of our society on this uniquely abundant, calorically dense, and uniquely dangerous form of energy—given that the carbon cycle is what makes Earth *Earth*—is why climate change is such a wicked, even cosmic problem. It's because the story of CO_2 is the story of everything. And, as we've seen, this stuff can end the world.

While the middle-of-the-road estimates for future warming are usually quoted in the press and especially by economists, the unpredictable response of the extraordinarily complex carbon cycle—one hinted at by the poorly understood rapid-warming events of geologic history—leaves open the possibility for true climate catastrophe. There are already indications that the planet may not be as accommodating to our future carbon emissions budget as we thought. "The best laid schemes o' Mice an' Men / Gang aft agley."

TODAY, HUMANITY PRODUCES MORE CO_2 THAN ALL THE OTHER SUBSTANCES we produce on Earth *combined*. It is our signature product. From a planetary perspective, human society is now, above all else, a conduit for moving carbon in the crust into the atmosphere. CO_2 is what we make. It's the exhausted end product of an industrial respiration that is busy metabolizing all of Earth history in order to transform as much of the planet's surface as possible into more human civilization. "It's almost like a biblical resurrection story," as the atmospheric physicist Tim Garrett told me. "We're merely catalysts for the resurrection of the Carboniferous."

Although such predictions rarely stand the test of time, in 2023 the US Department of Energy outlined a frightening if plausible projection of future CO_2 emissions. It predicted that, yes, the world would see renewable energy gain a growing share of the market over the next thirty years—but these gains would be more than swamped by population and economic growth, and the introduction of ever more fossil fuel–based energy to support it. This would lead our already preposterous, geologically unprecedented emissions to keep *rising* through 2050. Midcentury, it's worth noting, is around the time when not only should global CO_2 emissions not still be rising from their already preposterous nearly 40 gigatons a year, but instead they should be hitting *zero* to prevent catastrophic climate change. Another 2023 projection by the research consulting company the Rhodium Group forecasts that emissions very well could keep rising through the end of the *century*. And these are only the CO_2 emissions we can control.

The wildfires that have swept through the forests of the Rocky Mountains in recent years are unprecedented in the past 2,000 years. In Australia's 2019–20 summer, far more CO_2 was released by wildfires than the entire country emits in a year. Canada's 2023 wildfire season released more than twice as much as its own industrial emissions. Farther north, in the Arctic, soils frozen for hundreds of millennia are decomposing, releasing their methane and CO_2 to the sky. As peatlands warm up, they could add an entire decade's worth of CO_2 emissions by the end of the century, yet rarely are they included in climate models. This would almost be the carbon equivalent, though, of burning down the Amazon. Wetlands already appear to be releasing methane faster than the absolute worst-case scenarios. These carbon behemoths could be waking up in ways that would make a mockery of our best-laid emissions plans. If we get unlucky with these feedbacks from the carbon cycle, around which there is huge uncertainty, even ambitious human emissions scenarios could provide catastrophic warming.

The ocean, which has absorbed 30 percent of the CO_2 emitted since the industrial revolution, and grown 30 percent more acidic as a result, is beginning to show signs that it is growing tired of cooperating as well. In 2023 scientists noted with alarm that this vast carbon sink appears to be weakening, absorbing some 15 percent less CO_2 than expected in recent years—a trend that will only leave more of it in the atmosphere and accelerate warming. Even more worrying, in 2023 one research group claims that the global land carbon sink "collapsed," with plants and soils, on balance, drawing down virtually no carbon dioxide at all—as plant growth was offset by those same massive fires across the boreal North, and in the Amazon. One of the study's authors wrote, "In the coming years if this decline continues, we may see a rapid acceleration of CO_2 and global warming which was unforeseen in future climate model projections."

If you've read any book about climate change, or even a news story about the latest report from the Intergovernmental Panel on Climate Change, you're probably familiar with the legion of interlacing threats posed by this rapid, uncontrolled release of CO_2—and touched on throughout this book in the context of Earth history. Under even relatively mild amounts of warming, the combination of increased heat and humidity will render large swaths of the planet, now populated by billions of people, inimical to human physiology for significant parts of the year. With just over 3°C of warming, for instance, over 3 billion people might be exposed to mean annual temperatures that prevail over less than 1 percent of the Earth's surface today, mostly confined to the heart of the Sahara Desert, but that will come to cover almost 20 percent of the land.

This is a problem because humans, like all animals, have a "climate niche." Although fire and technology have allowed us to stretch this range of tolerable environments, we, our crops, and our livestock are largely restricted to a "surprisingly narrow" band of the climate, as a recent study puts it, concentrated between mean annual temperatures of 11°C and 15°C (52°F and 59°F). This has been true of human societies for thousands of years. By the end of the century, though, several billion people will find themselves outside these "climate conditions that have served humanity well over the past 6,000 years."

As the venerable ice sheets of our ice age inexorably crumble, sea-level rise will displace hundreds of millions in low-lying cities. Unprecedented droughts will desiccate agricultural heartlands, and unprecedented floods will wash others away. The belt of rainfall that girdles the tropics may shift in response to warming, leaving those on its edges—like smallholder coffee and maize farmers in Central America—subject to punishing aridification. Combined, these perils may drive billions to migrate. A hotter world makes hurricanes more powerful, malaria more abundant. Heat makes people stupider and kill each other more.

Our entire global economic system is premised on the tacit prediction that tomorrow will look something like today. Insurance, mortgages, credit, and government bonds all rely on expectations about the future. A house in the woods in the midst of the worst drought in two millennia, or a condominium on a crumbling coast, is not an insurable one. A farm in a region that may dry up next year is not loanworthy. A country drowned by sea-level rise on its coasts, charred by wildfires inland, and racked by crop failures is not bondable. Nor is one with a falling foreign exchange, a workforce sidelined by heat stress, infrastructure repeatedly wrecked by hurricanes, a failing grid overtaxed by heat, public services strained by mass immigration from abroad, and cities overwhelmed by penurious smallholder farmers escaping the drought-ridden countryside. In short, an unpredictable future is not a legible one to capital. This makes the radical uncertainty in store from climate change underrated as a source of compounding, catastrophic risk to the global economic system. Human society may respond to high CO_2 as unpredictably and nonlinearly as the ice sheets or the carbon cycle itself. Some economists may be catching up, though: a recent study estimated that global incomes would be cut 19 percent by 2050, and that such dramatic damages are already unavoidable. Things are perhaps only more dire for the rest of the living, nonhuman world.

With only 2.5°C of warming, some 30 percent of the planet’s species will be exposed to dangerous temperatures, beyond their physiological thresholds. Already the acidifying ocean has left some foundational plankton in the ocean around Antarctica pitted with holes. There are even the incipient signs of the seafloor dissolving in some places on Earth, just as it did during extreme acidification events of the ancient past. Marine heat waves unprecedented in recent geological history now sweep in pulses across the ocean, leaving vacant algae-covered hulks of white rock where coral reefs once dazzled. Though providing for 25 percent of biodiversity in the ocean, these stony gardens of life are thought unlikely to survive the century, with some 99.7 percent of them exposed to potentially fatal ocean heat in their current ranges by 2100—even if we managed to limit warming to only 2°C.

The oceans have lost 2 percent of their oxygen since 1960[fn3] —an astonishing amount—and, in the very long term, are on pace to eventually reproduce the kinds of pulses of global anoxia that were a feature of almost every mass extinction in Earth history. As we've seen, looking back at this geologic history—at the only experimental record we have of how the planet actually responds to changes in CO_2—there is little to reassure us. Ice sheets tend to disintegrate in the ancient past in response to relatively small amounts of warming. As we saw from the paleoclimate record, the climate tends to change in large jumps to new, very different, stable states—it does not tend to indefinitely, and incrementally, change at a linear pace, as it has so far in recent human history. Indeed, some climate scientists are beginning to publicly worry that they are detecting the first signs of an acceleration in warming in recent years.

The study of ancient mass extinctions perhaps counsels the most caution. We are beginning to see every geochemical symptom of those nightmares, in prefatory, ominous signals: in the graveyard of sediments spanning these ancient apocalypses, we see not only pulses of extreme ocean deoxygenation, but acidification, warming, wildfires, and even the very same signature of carbon isotopes going wild in the very same way that we find in ice cores today. Also, the speed at which we are emitting CO_2 is virtually unprecedented in the history of animal life. We are still a *very* long way away from reenacting the End-Permian, but the road down which we are headed is dark and well trod in Earth history.

Where does that leave us? We obviously need to immediately stop emitting carbon dioxide to the atmosphere. But to do that, we will need to understand just where the levers of control are on the nonlinear, complex, dynamic system that is the global economy—if there are any. Human institutions are now dominant components of the global carbon cycle, and the answers to climate change are not to be found in the realm of atmospheric physics or geochemistry. They are political and economic. Just as a string of DNA is useless when it's unmoored from the energetic powerhouse of a cell, a dire climate report detached from the levers of geopolitical and economic power is similarly inert. And perhaps nowhere is the nexus of social power and geologic energy more nakedly expressed than in a strange museum in West Texas.

AT THE PERMIAN BASIN PETROLEUM MUSEUM IN MIDLAND, TEXAS, two exhibits sit in adjacent rooms and, at first pass, seem to have little to do with each other. One is a stunning diorama of an alien coral reef from 260 million years ago. In this tranquil tableau, tentacled nautiloids probe towering sponges for trilobite prey, while armored sea lilies hang over the primordial reef wall. The next room over, a wall is decorated with plaques to George W. Bush and his father, as well as a veritable gallery of aggressively dull company men—all inductees to the museum's "Petroleum Hall of Fame." Surprisingly, the two displays are intimately linked. The Bushes commanded the most powerful position in the world, as de facto head of the most powerful species in Earth history, precisely because there was a shallow ocean in Texas a quarter-billion years ago filled with unusual creatures. This giant tropical reef system, near a nutrient-rich upwelling zone of the ocean, locked up a lot of sunlight in this Texan life. While most of this life was eaten and respired back to CO_2 by other ocean life before it ever made it to the seafloor, a vanishingly small amount made it into the rocks to be preserved. An even *smaller* amount, perhaps 1 percent, was cooked deep in the Earth millions of years later, migrated upward, and was trapped as oil by still other rocks, producing accessible reservoirs of fuel just under the surface.

"The power we employ in the greatest engine is but an infinitesimal portion, withdrawn from the immeasurable expanse of natural forces," John Maynard Keynes rightly noted. Nevertheless, even this comparatively tiny amount of energy from deep time is enough to be transformative when wielded by one species. In fact, despite comprising only 0.02 percent of the Earth's biomass, humanity has effectively doubled the power deployed by the living world through this radical new external metabolism. People like the Bushes, with access to blue-blooded networks of money and credit, took control of the release and distribution of this energy from the Earth. Manning the spigots of this primeval residue that fuels humanity's strange new metabolism can make you very rich and powerful.

This is not new. The mansions of Newport, Rhode Island, reflect the geologic energy once diverted to coal barons; the stately manors of

Nantucket, the energy from whales; the castles of feudal lords and daimyos, the energy of the land. In the twentieth century, it wasn't just the world of politics and business, but the world of letters as well that came to be populated by well-heeled men, living patrician lives scarcely conceivable to the roughnecks on oil rigs to whom they owed their social standing, such as William F. Buckley Jr. (whose father was an independent oilman who sold his concessions in Venezuela to Standard Oil) or longtime *Harper's* editor Lewis Lapham (his grandfather founded Texaco[fn4]). The same is true in the world of philanthropy: the Pew Foundation (funded by the Sun Oil fortune), the Mellon Foundation (Gulf Oil), the Rockefeller Foundation (Standard Oil), the Getty Foundation (Getty Oil).

And despite W's supposed misgivings about our addiction to oil, in Texas the oil industry remains triumphant: as humanity needs to cut its carbon emissions by 60 percent in the next decade or so to ward off a return to a high-CO_2 planet, oil production is set to expand here fivefold in the same time period, turbocharged by huge injections of capital—most recently $10 billion from Warren Buffett. In total, the world is currently planning to spend at least a trillion dollars on new oil and gas projects at a time when we're warned that even *existing* reserves can't be fully tapped if we are to keep the planet from warming to dangerous levels. But as an oil executive once said, oil, after all, "*is* almost like money."

As for another inductee into the Permian Basin Petroleum Hall of Fame, Exxon (née Standard Oil of New Jersey), it shares some features with the kinds of swashbuckling joint-stock corporations of old, like the Dutch East India Company, operating around the world from well-defended enclaves to extract valuable resources in distant territories, the inhabitants of which it has little interest in. Unlike the Dutch East India Company, though, whose goals were indistinguishable from those of the Dutch state, multinationals like Exxon, which pay almost nothing in taxes from the profits they generate in countries all over the world, can often appear more like their own independent, sovereign powers.

"I'm not a U.S. company and I don't make decisions based on what's good for the U.S.," Exxon CEO Lee Raymond once said. "Presidents come and go; Exxon doesn't come and go." As Steve Coll documented in his aptly named study of the company, *Private Empire*, Exxon has its own foreign policy, its own global intelligence-gathering agency, and its own mercenary security services—and its interests are often at odds with the US government’s. When Exxon staff in Chad, for instance, pointedly ignored the presence of the US State Department in the country, embassy cables captured a resigned diplomat admitting that they held the weaker hand, with ExxonMobil investing almost $4 billion more in the country than the US government had provided in aid. After defying the State Department by signing oil agreements in Iraqi Kurdistan, Exxon CEO Rex Tillerson would explain, “I had to do what was best for my shareholders.” But where old empires snapped up spaces on a map, sweeping productive agricultural lands and conquered populations into their ambit, oil multinationals like Exxon, whose territorial footprint in these distant countries is minimal, instead represent empires in *time*.

For empires in human history, often the only route to increased power was expansion in space, absorbing agricultural lands and labor. A similar principle exists in biology, where, generally speaking, the bigger the animal, the bigger its range size. This is because bigger animals need to draw photosynthesis from over a greater territory to support their size and metabolism. But today, corporate oil empires, drawing from concentrated ancient photosynthesis, can be geographically limited but swell their power by spreading out over vast sweeps of Earth *history*. This is why reading oil business news is like strolling through a natural history museum. In West Texas, it’s the millions of years of life in the Permian that generated this photosynthetic surplus. When it was discovered that Texas’s Wolfcamp Shale and the overlying geologic formation held seven times as much oil as North Dakota’s Devonian bounty of energy, Secretary of the Interior Ryan Zinke triumphantly announced that “American strength flows from American energy.” But the Wolfcamp is more familiar to paleontologists as the chunk of Earth history to which it lends its name: the 20-million-year-long Wolfcampian, which was the Pangaean golden age of sail-backed dimetrodons.

As a corporation that deals almost exclusively in geologic time, Exxon has long known better than nearly any other organization the climate consequences of its global empire. Its staff scientists pioneered the study of ancient sea level, with one of them even warning executives in 1977 that the continued burning of fossil fuels would lead to a “super-interglacial” in the coming decades. But Exxon is an oil company. Its raison d’être is to acquire and sell oil to customers, and its shareholders want to see profits. If it ceases to succeed in this goal, another oil company will pick up its slack.

By the mid-1990s (a century after Arrhenius's calculated temperature would rise 5 degrees if CO_2 doubled), it was clear that climate change posed the biggest threat to Exxon's business model—with the issue, according to one executive at the time, "bigger than anything else" at the company. So Exxon went on a public relations spree. In 1996, as BP ran big-budget, high-concept ads directed by Steven Spielberg, Walt Disney World opened its Exxon-sponsored ride, "Ellen's Energy Adventure." It was hosted by an ingenuous duo of Ellen DeGeneres and Bill Nye, who introduced ridegoers to the country's energy system, laundering every familiar industry talking point—hyping "clean coal," casting doubt on the reliability of renewables, and lauding offshore drilling as the culmination of human technical achievement.[fn5]

Since the ride's premiere, more than half the oil ever produced in human history was drilled, as Exxon bought crucial time to sell its product. In the meantime, the United States has regained its title as the world capital of oil, once again becoming the largest oil producer on Earth and building on its legacy of emitting more CO_2 over its history than any other country in the world (by far).

Yet Exxon tends to play dumb in court filings. "Plaintiffs' claims necessarily implicate innumerable third parties, including anyone who used automobiles, jets, ships, trains, power plants, heating systems, and factories since the dawn of the Industrial Revolution …. Everyone has contributed to the problem of global warming," Exxon attorneys meretriciously wrote in 2019, in response to a lawsuit that tried to hold the company liable for environmental damages.

But to focus solely on private actors like Exxon is to lose sight of the fact that, by the twenty-first century, the vast majority of oil has come to be owned and produced not by private companies staffed by mustache-twirling capitalists—as often portrayed in environmentalist tracts—but by sovereign governments. A private oil company might be unlikely to voluntarily leave trillions of dollars in the ground—they are after all, as Tillerson says, responsible only to their shareholders. This is why a commonly proposed, if politically far-fetched, solution is to nationalize these companies and shut down their oil production altogether. (In theory, the government is responsible for protecting the public interest when it conflicts with the private one, and it isn't bound to make a profit.) But the utter inability of even state-owned oil companies to turn off the spigots (they are, in fact, still investing in exploration for new deposits) underlines the extraordinarily difficult nature of the problem.

And it might be far more difficult still. If the existential dependence on energy of our modern world explains much of the inertia in stanching the flow of ancient carbon from the Earth, then this dynamic is ill-captured by the dominant schools of economics. As a result, and given that the global economy is now a crucial, if wildly destabilizing, component of the carbon cycle, the economic models we use to predict future CO_2 emissions may have profound weaknesses.

ECONOMISTS HAVE LONG SOUGHT TO TEASE OUT WHAT SEEMS TO BE the uncanny link between energy and the economy. But where economists have sought to do this from the perspective of *money*, energy is a physics concept. Neoclassical economics, meanwhile, was developed during the steepest part of the most extreme and atypical exponential curve in energy use in human history. Taking this ongoing explosion of energy for granted, the field made theoretically perfect markets that tended toward equilibrium by responding to consumer preferences the object of its study, while the eruption of energy that supported this burst of economic activity remained all but invisible in its calculations. This is a bit like trying to deduce the laws of gravity while actively in free fall. According to this prevailing view, though, energy is like any other commodity, and if it takes up 5 percent of the spending in the economy, then that represents its importance to the economy. Take it away and the economy takes a 5 percent hit—and because of the endless creativity of the market, alternatives will inevitably arise to take its place. But of course, that’s nonsense. Energy is the whole story. Take it away and there’s not a 5 percent hit to the economy. There is no economy, or life for that matter.

“Economists really don’t understand energy,” the University of Victoria industrial ecologist Chris Kennedy told me. “They really don’t. They see energy as a commodity, not as a lifeblood.”

Again, doing *anything* requires energy. And just as an animal dies when it’s cut off from the photosynthetic energy in its food, countless commentators in the twentieth century noted the uncanny lifelessness of their industrial machinery when cut off from its fossil lifeblood. The comparison to biological death for them seemed more than metaphorical.

For instance, when the Suez Canal—the “jugular” for oil—was closed, the British prime minister Anthony Eden howled that his country “could not live without oil and … we had no intention of being strangled to death.” Meanwhile, Eden’s supposed tormenter, Egyptian president Gamal Abdel Nasser, noted that “[w]ithout petroleum … all the machines and tools of the industrial world are ‘mere pieces of iron, rusty, motionless, and lifeless.’” In the wake of World War II, one Japanese admiral similarly marveled, “If there were no supply of oil, battleships and any other warships would be nothing more than scarecrows.” And as the iconoclastic economist Steve Keen has more recently put it, “Labor without energy is a corpse, while capital without energy is a sculpture.” The barb is in pointed reference to the work of Robert Solow, the neoclassical economist who won the Nobel Prize in economics in 1987 for his model that purported to explain economic growth. Economic growth, by his lights, was produced by the inputs of capital, labor, and a mysterious fudge factor called total factor productivity (TFP), sometimes used interchangeably with “technology,” a force that somehow increases productivity over time. While Solow’s work helpfully isolates the factors of production, it, of course, explains nothing, leaving the most interesting part of what actually drove the incredible economic growth we’ve experienced since the dawn of the Industrial Revolution unexplained, by relegating it to an enigmatic placeholder.

Most notably, energy is found nowhere in this model of economic growth.

But if, as some people claim, we can absolutely decouple economic growth from energy growth in the future, has this always been true? If the preindustrial agricultural societies had never increased their energy use whatsoever, could they have reproduced the skyrocketing growth of the twentieth century? This obviously seems ridiculous, but then what makes future different than past? Why was decoupling energy from economic growth impossible in the past but will not be impossible in the future? The alternative view is that clearly energy has something to do with economic growth, and inescapably so.

The highest growth in TFP in American history by far, according to the Solow model, was during the so-called Golden Age of Capitalism—from World War II through the postwar boom. “What is less understood is that labor productivity increased in direct proportion to the amount of energy used per worker hour,” writes the biophysical economist Charles Hall. “At that time labor productivity in the United States was two or three times that of a European worker, not because the worker worked harder or was more clever, as commonly assumed, but because he or she had big machines using two or three times more energy helping him do the job!"

Similarly, the astounding growth in agricultural productivity in the past century or so has been mostly driven by the ever-increasing application of fertilizer to farmland. Fertilizers are synthesized from natural gas, or mined as phosphate rock by diesel-powered dragline excavators (mostly in Morocco), and shipped across the world in gigantic quantities in bunker fuel-powered ships. Another powerful driver of agricultural productivity has been the application of ever-more-powerful fossil-fueled farming equipment and irrigation pumps on the fields themselves. This is all very energy-intensive. As a result, given that it's been estimated that 10 calories of fossil fuel goes into the average American calorie of food, one can uncover some rather disquieting facts about the relative miles per gallon of even, say, walking, in this oil-soaked agricultural regime. In fact, given that 30 percent of the world's greenhouse gas emissions are from the global food system, even if all other industrial emissions halted tomorrow, the planet would still soar past 2 degrees of warming just from farming alone.[fn6]

EVEN THE FIRST MODERN ECONOMIC SCHOOL, THE FRENCH PHYSIOCRATS of the eighteenth century, understood, if only dimly, that growth flowed from the activity powered by photosynthetic energy. Agriculture—the carbon energy of the preindustrial society, and which powered virtually all activity in society—was the source of a country's wealth, they argued. But two centuries on, in an age when energy became so cheap it could be virtually ignored in economic models, economists were left like alchemists in search of the philosopher's stone, puzzling over the meaning of their own abstractions, like TFP.

In 2019, a group led by the structural engineer turned economic modeler Paul Brockway at the University of Leeds proposed that this mysterious "residual" that drives economic growth was entirely explicable by economy-wide gains in thermodynamic energy efficiency—that is, getting more work out of the same amount of energy. Here's how it works: If you're a consumer, buying a more efficient boiler will ultimately save you money that can be spent on other goods and services—which in turn increases demand for those products (which themselves take energy to make) and rewards still more industrial production. On the production side, if you're a factory, then making more products for less energy allows you to expand your operations and make ever more stuff to meet this demand. If you're in commercial shipping and you use your fuel more economically, you can go farther, visit more ports, make more stops, pick up and drop off more goods—especially if you containerize your ships and drop them off at gigantic mechanized, standardized ports with few workers. Everything along this chain has become more efficient, and the material throughput has soared as a result. All along, while the entire economy has been getting ever more energy-efficient—a supposedly crucial tool in the effort to reduce our energy consumption—it has also been getting much larger. And larger economies use more energy. That extra energy can even be used to find and produce more energy to grow the economy further. And given that firms want to grow, the demand for energy will only ever rise.

If it isn't energy consumption per se that drives global economic growth, then perhaps it's how much effective work you can get out of that energy. If Newcomen's original, highly inefficient steam engine, which burned far more coal than any engine to follow, had never been improved upon, then the economic growth of the nineteenth and twentieth centuries never would have been physically possible. Yet improving upon it—by making engines more thermodynamically efficient, to the point where today a coal-fired power plant amazingly operates at a higher efficiency than *human metabolism*—made it possible to do vastly more work on the planet, exponentially increase economic growth, and ultimately burn far more coal in total. Paradoxically, using energy efficiently allows for the exploitation of far more energy. This is as true in biology as it is in economics.

"My suspicion is that power will be to macroevolution what gravity is to astronomy," the University of California's Charles Marshall recently announced to a group of fellow paleontologists, at the 2023 Geological Society of America meeting in Pittsburgh. I was in the audience for the talk and couldn't help but notice the consonance with the economic theorists I was interviewing at the same time. "The usable power for life's activities is simply the difference between the power generated or gained—photosynthesis, or wolves eating rabbits—minus the cost of acquiring those rabbits." Organisms that are able to maximize this difference—between the energy surplus they gather from their environment relative to the energetic cost of acquiring it—differentially survive, and drive the entire ecosystem toward greater power. Or, as Marshall put the point, "Organisms that capture and use more energy will have a selective advantage." This trend is unmistakable over Earth history, as both ecosystems and organisms have progressively come to consume far more energy over the course of geologic time. That firms competing in an idealized capitalistic economy, over finite resources, are subject to a similarly selective, almost evolutionary process was famously noted by the early twentieth century economist Joseph Schumpeter, who described a "process of industrial mutation," one that "incessantly revolutionizes the economic structure *from within*, incessantly destroying the old one, incessantly creating a new one." In this system, "the profit of every individual plant is incessantly being threatened by actual or potential competition." As a result, "[e]conomic evolution in capitalist economies during the last two centuries has displayed a clear direction towards higher productivity," writes the Cambridge economist Matthias Kelm.

Where many economists expect us to somehow absolutely decouple energy use from economic growth and grow ever richer in the future while using ever less energy in total, Brockway's work has rather more dour implications. Energy efficiency might only unlock further energy use. But not only does economic growth depend on energy use: economic growth will eventually slow—as indeed it already has[fn7] in richer economies—as skillfully engineered machines and power plants begin to push up against the limits of energy efficiency and ever closer to Sadi Carnot's impossibly perfect engines.

"The purpose of the economy seems to be to maximize the average rate of useful work output," Carey King writes. "The global economic superorganism selects for a distribution of organizations with traits that enable it to extract resources at an increasingly higher rate, just like natural selection selects for a distribution of organizations with traits that extract the maximum flow of energy from the environment."

SINCE THE DAWN OF THE STEAM ENGINE—AND IN FACT, FOR ALL OF human and biological history—when we've been able to do more work with the same amount of energy, it hasn't reduced consumption. Just the opposite. This phenomenon, which we've encountered repeatedly throughout the book, is known to economists as the Jevons paradox, after the nineteenth-century

English economist William Stanley Jevons. Jevons was a remarkably perceptive observer of the Industrial Revolution, then transforming his country. He noticed that the introduction of more and more efficient steam engines was attended only by *more and more* coal consumption, not less. “It is wholly a confusion of ideas to suppose that the economical use of fuels is equivalent to diminished consumption,” Jevons wrote, anticipating the next two centuries. “The very contrary is the truth. As a rule, new modes of economy will lead to an increase of consumption.”

This model has some strange implications: be virtuous and reduce your own energy consumption, and it will only make more energy available for the rest of the economy at a cheaper price, allowing it to grow further—and further from equilibrium. But this pattern has continued to hold, for the most part, ever since Jevons articulated it. Thomas Edison’s lightbulbs were 10 times more efficient than the candles they replaced at converting chemical energy to light. Today’s lightbulbs—when lit by power plants burning natural gas—are 90 times more energy-efficient still. In 2000, consumers spent 1 percent as much on light as they did in 1900, and the money left over was available to spend on still other goods, calling new demand into being that took still more energy to produce. In the meantime, lighting went from being a luxury good at the turn of the century to being so widespread today that the night side of the planet positively glows. This isn’t because the entire economy was redirected toward lightbulb production—instead the rest of the economy grew exponentially. But where lighting accounted for negligible CO_2 emissions when people were still putting Standard Oil in their kerosene lamps, today it accounts for something like 2 gigatons of CO_2 a year. Smelting iron with metallurgic coal, meanwhile, is about 20 times more energy-efficient today than ironmaking was in 1750. Yet CO_2 emissions from the industry have grown from a similarly negligible amount in the age of King George to over 4 billion tons per year today—not despite these unthinkable improvements in efficiency, but *because* of them. Energy efficiency unlocked far more energy available to then invest in further production and innovation: where in 1750 only 800,000 tons of iron were produced in extraordinarily wasteful charcoal furnaces, today 1.2 billion tons are produced in maximally efficient industrial electric arc furnaces.

Since 1975, thanks to investments made after the oil shocks, the United States has doubled its energy efficiency. But its energy consumption has increased by over 20 percent in the same span. Once again, despite the name of the phenomenon, this isn't a paradox. In China, where wood was still the main energy source at midcentury, energy efficiency jumped 50-fold in just 50 years, as the country switched to fossil fuels and modernized what had been a catastrophically inefficient planned communist economy. It's perhaps not a coincidence then that China's economic growth *and* its growth in carbon emissions since that point are unrivaled in human history.

But these gains in efficiency weren't solely products of the industrial age. They were a continuation of millennia of innovations since the Pleistocene. At the dawn of the Industrial Revolution, agriculture had already become a far more efficient proposition than it was at the dawn of intensive agriculture 6,000 years earlier: to wit, by the late nineteenth century, a team of draft horses could do almost 50 times the work of livestock in antiquity. These trends, then, didn't necessarily begin with the steam engine.

"From a fundamental biophysical perspective," writes the scientist Vaclav Smil, "both prehistoric human evolution and the course of history can be seen as the quest for controlling greater stores and flows of more concentrated and more versatile forms of energy and converting them, in more affordable ways at lower costs and with higher efficiencies, into heat, light, and motion." In fact, even restricting our attention to hot-blooded birds and mammals, the most energy-efficient species use the most energy.

Many economists, though, insist that the bogeyman of the Jevons paradox has been slain and that we can start, and indeed have started, to effectively decouple energy use from economic growth in wealthy countries. It's worth flagging that the continued relevance of the paradox is hugely contested, and that everything you've read above is hugely controversial. Studies have estimated that the so-called rebound effect, where energy savings from efficiency are eroded by the increased energy use they spur elsewhere in the economy, ranges from a frightening savings offset of more than 100 percent to a reassuringly negligible amount. I hope the latter estimates are correct. Besides, some recent advances, like the switch to LED bulbs, seem to have been such massive jumps in efficiency that they swamped any possible rebound, while efficiency standards enacted for appliances seem to have stalled load growth on the grid. But one recent review of studies estimating *economy-wide* rebound effects still found that these rebound effects will erode more than 50 percent of the energy savings from energy efficiency.

Because these are still net savings, they would lower energy demand, even if less than hoped for. But that these effects will instead prove to be negligible, along with the global adoption of renewable energy, is one of *the* hopes for the future. And yet every energy source ever introduced over the course of human history has been merely additive so far. In fact, in 2024, even as renewable capacity skyrocketed around the world, humans not only burned more coal than ever before, but also burned more *wood*—the original fuel of humanity—as well.

Nevertheless, the models used by the Intergovernmental Panel on Climate Change to plot emissions to 2100 assume that the future will be utterly unlike the past—and in fact *depend* on this being the case. “If I’m being very charitable, they have a very weak connection between energy and economic growth,” says the economic modeler Paul Brockway. “When you look at their growth predictions they say, ‘Okay, we can have GDP growth at two percent per year, and energy’s going down.’ So that’s absolute decoupling, but we haven’t had a single year of absolute decoupling in our history—so that’s a bit concerning.”

If, on the other hand, an ever-growing industrial civilization will ultimately require far more energy than projected, then we have to either take the leap of faith that 100 percent renewable energy technologies can still completely replace the even higher energy needs of a growing global economy, or we must *degrow*—since indefinitely burning carbon to subsidize our lifestyles will, without overstatement, cause the apocalypse. One corollary is that the people who think we should all stop using fossil fuels tomorrow—without bridging the gap with an equivalently scaled supply of carbon-free energy to prop up global industrialized civilization, or navigating a world historic reorganization of human society—are calling for a precipitous return to the energy ground state, aka civilizational collapse. Or, at the very least, they are calling for a profound simplification of the global economy. This is because people’s lives have literally come to depend on this new, high-energy, globalized supply chain. “Approximately one quarter of all food produced for human consumption is traded internationally,” a recent study notes. “And almost one billion people are consuming internationally traded products to cover their daily nutrition.” And the more we grow, the more energy we’ll need to keep us up at this new, incredibly high energetic perch, much less to grow further still.

Vladimir Lenin, whose communist economy, by the end of its lifespan, was pound-for-pound 50 percent dirtier than that of the United States, famously outlined his vision for a utopian society *defined* by its energy use, declaring, “Communism is Soviet power plus the electrification of the whole country.” The socialist commentator Leigh Phillips has even speculated that, had socialism been applied more broadly in the twentieth century, and material development been more widely spread, climate change would be even worse right now, though perhaps easier to address because of the system’s supposedly more democratic controls of power. This material promise of energy is why the leaders of many developing countries scoff when browbeaten by richer countries about climate change. The entire continent of Africa, for example, has contributed 2 percent of historical CO_2 emissions, and nearly half the continent has no access to electricity. Meanwhile, in 2021 alone, China’s electricity demand jumped by an amount equivalent to the power consumption of the entire continent of Africa. Seeking the same material gains as more industrialized regions by developing the geological energy beneath them seems only fair for these countries, especially after centuries of colonial plunder. “Under no circumstances are we going to be apologizing,” the energy minister of Equatorial Guinea told Reuters during a 2019 African oil conference that drew hundreds of delegates. “Anybody out of the continent saying we should not develop those fields, that is criminal. It is very unfair.”

This need for energy in much of the world is not to feed SUVs and decorative gas fire pits, but to build hospitals, install refrigeration to preserve food, and provide clean water where diarrhea kills, heating where people freeze, and air-conditioning where people are struck down by heat. These are not luxuries.

Ironically, one of the main things that climate change threatens is this very proliferating energy use—which is required to sustain a global industrial society, and to extend the material gains of the Western world to the rest of humanity. But, of course, when yoked to the energy reservoirs of deep time, this swelling power of humanity could regenerate the atmosphere of the Cretaceous. It is a tremendous pickle.

All of this presupposes, though, that there will be endless reservoirs of fossil fuel to exploit. What if there aren’t?

Industrial society has already exploited most of the easily accessible fossil fuel deposits as it has dramatically expanded. Just as fishing out the teeming cod shoals of Cape Cod Bay with baskets has given way to industrial fishing fleets burning incredible amounts of fuel to track down diminishing fish stocks at the ends of the world, the haphazard wildcat oil strikes of the last century have instead given way to unthinkably complex, unimaginably expensive, and extraordinarily capital-intensive oil and gas megaprojects, operating hundreds of miles offshore and in exceptionally inhospitable environments such as the Arctic Ocean. These projects themselves require increasingly giant investments of energy, and these declining prospects can make the tragedy of climate change seem increasingly farcical: ConocoPhillips, for instance, recently revealed its plan to implant heat-siphoning devices in the permafrost of the Alaskan high Arctic, to stabilize the melting ground just enough to drill for more oil. At the time of writing, rain now falls in the arctic winter night, and the North Pole is 50°F above normal, a forecast out of time—but one that would have been familiar to the mammals of the early Cenozoic.

One thing's for sure: the Earth isn't making any new East Texas oil fields. The golden age for discovering new oil deposits was in the 1960s, and exploration geologists have been finding fewer and fewer ever since. As the most readily accessible oil deposits have been drained, it has increasingly taken more energy to get additional energy out of the ground. Even fracking, which has been heralded as opening a new endless frontier of fossil abundance, is far more energy-intensive than conventional oil drilling, with oil production at these wells dropping by more than 50 percent after the first year (compared with the decades of gushing production from the supergiants of yesteryear). Fracking bankrupted almost every company that invested in it in the 2010s and was revived only by high oil prices in recent years (along with genuine advances in drill technology). Where the original gushers of East Texas might have provided 80 times more energy in oil than the energy required to drill them, today's oil megaprojects—along with lower-quality fuels like tar sands that require incredible amounts of energy to process—might only provide as little as three times more energy than they take to build and operate.

As easy-to-extract fossil fuels are further depleted to support industrial civilization, this ratio could continue to decline and, in an extreme scenario, ultimately require investing more energy into such projects than they are able to pull from the ground, an impossible situation to sustain. Starvation, as it's known in the natural world. If not replaced with other sources of energy at the same scale, then, the energy surplus that has made the rise of the industrial world possible would disappear. Predictions of global peak oil have perennially embarrassed would-be Cassandras (a craze in the early 2000s was quickly embarrassed by the flood of oil from fracking), and perhaps continual technological advances will keep oil companies afloat longer than industry pessimists think. Indeed, drill technology has recently made leaps and bounds in productivity. But *eventually* the eruption of liquid carbon from the ground may stop making much energetic sense (to say nothing of its climate sense) and become uneconomical. What this would look like on the ground for industrial civilization, if this energy wasn't replaced by other sources, is difficult to even imagine—a prospect that explains the recurrent, apocalyptic flavor of peak-oil mania.

Strangely, then, it's energy's very cheapness over the past two centuries that has made it so existentially valuable to the economy. With seemingly infinite supplies that were easy to access, spending on energy took up only about 5 percent of the global economy, while the rest of the economy could be dedicated to everything else that we think of as our modern industrial civilization. But when energy spending has briefly topped double digits as a share of the economy, it has meant recession, if not depression. In the future, if the majority of the economy comes to be dedicated to acquiring increasingly expensive energy—to say nothing of the CO_2 impacts of continuing to burn fossil fuels indefinitely—civilization could find itself in a bizarre version of the preindustrial agricultural era, where it is dedicated only to energy production and the bare necessities of life, as much of the rest of the modern economy becomes energetically impossible and falls away. Depending on whom you ask, we are either close to this energetic break point again, as the fracking sugar buzz eventually fizzles out and oil and gas get ever more energy- and capital-intensive to produce, or we are still decades or even centuries away from any such calamity. As someone whose brain has been poisoned by geological timescales, though, I admit to finding such debates uninteresting: even if it is centuries away, this is an eyeblink even in a human history where some archeological sites span thousands of years. Either way, without a comparably gigantic and growing renewable energy system to replace this one-off pulse of energy from all the life in Earth history, the past few centuries will appear in retrospect as the most bizarre and transient convulsion of the Earth system in its history. And if it does end up just being a momentary, if titanic, planetary pulse of energy and attendant hurricane of human activity and population growth, it is difficult to imagine a scenario on the downslope that doesn't entail a quite unimaginable level of human suffering and despair, even without taking climate change into account.

Two things, then, seem undeniable: we have to get off fossil fuels as fast as possible, but we're still going to need *lots* of energy. So what do we do?

12

The Future of CO_2

AT THE END OF OUR JOURNEY THROUGH THE PLANETARY HISTORY OF CO_2, we find ourselves in Carbon County, Wyoming, though we're really back where we started. As we saw at the outset of our journey, at some point billions of years ago, life learned to bundle up the energy gradients of the seafloor vents within itself and set out on its own. Life learned to do this—and life still does this—by capturing energy in its environment to push protons uphill across a membrane, where they're held back like water behind a dam, then let back down to power a nanoturbine that *then* generates energy in a form (ATP) that can be sent away to power the manifold activities of life. Here in Wyoming, after some 4 billion years of intervening Earthly drama, engineers are, unwittingly, reconstructing this system at a monumental scale. Not with lipids and enzymes and gooey biological stuff, but with cement and steel pipes and backhoes and copper wiring.

I happened to visit this part of the state years ago with a geologist who had come to study the End-Permian mass extinction, which pokes out of the sagebrush in dreadfully lifeless maroon rocks outside Rawlins. We had capped off our fieldwork with a road trip to the largest open-pit coal mine in the country, where the End-Permian mass extinction was being reenacted, in great rectangular craters carved out of the golden shortgrass prairie. Peering over the edge of the artificial canyons, we saw what looked like toy trucks shuffling along at the bottom, scraping bits off a gigantic black stripe that ran all along the wall above the crater floor. Down the road, a twenty-four-hour greasy-spoon diner kept the continual shift of mine workers powered, while an endless link of train cars lined up over the crater rim, destined for the metabolic nodes of civilization. Eggs, bacon, and diesel came into the Powder River Basin, and coal left in heaping train cars. At root, though, it was all energy from photosynthesis. Some ancient, some present-day. This was, of course, only a tiny component of a much larger global industrial metabolism that shuffles bits of carbon around the superorganism to be respired again to CO_2 to do work. It can't go on much longer. But today in Carbon County, the makings of the next metabolism are being planned.

Here, from a reservoir banked by that same shock of End-Permian red rocks, humans are going to build a gigantic machine, one that mimics the same system that powers almost all life. Using energy from the sun, sent over transmission lines from the gigawatt's worth of wind farms sprouting nearby, pumps will push water uphill to a new reservoir in the mountains above, only to let it back down to power a turbine, to generate electricity, which it will then send off to the nodes of civilization. The beauty of this project is that when operating—generating, storing, and releasing energy to society when it's needed—it will emit virtually no CO_2.

The energy required to build out these machines, however, is another question. This is because it takes the existing energy system to build the next one, and today we build large infrastructure projects with fossil fuel–powered heavy machinery. Just as coal was first mined with the motive power of human and animal muscle, the transition to clean energy before us will inevitably involve a huge subsidy of fossil fuels to get it off the ground—to construct the turbines and solar panels, and mine the rare earth minerals to put in them. To wit, a fifth of China's CO_2 emissions growth in 2023 came from its world-beating solar and battery industries, which are powered by the sunlight of the Jurassic, in the coal of western China. One research group estimates that it will take about 195 gigatons of subsidizing CO_2 emissions to build out the kind of global clean-energy regime that will eventually release almost no CO_2 at all. Alarmingly, this is about half our rapidly vanishing carbon budget to stay within 1.5°C of warming. This is one reason why it's so critical to jump headlong into the energy transition as fast as is feasible. The hope, though, is that the energy input required for this renewable energy transition will be like overcoming the kinetic barrier of chemical reaction, and it will unlock an energy source in perpetuity. Just like life did at its origins.

Someday, if we are to succeed in halting our global exhalation of CO_2 from the crust, strange new photosynthetic machines like this Wyoming project may dot the Earth, and eventually bootstrap more such machines into existence with their own energy—long severed from the fossil carbon that gave rise to this new energy system. To avoid the dire climate trajectories that lie ahead, we have to do the equivalent of switching out the metabolism that has been powering all aerobic life for over a billion years for an equally powerful—no, *more* powerful—one, in a matter of decades, and at a scale unprecedented in Earth history, much less human history. We might actually pull it off. If we did pull it off, it would be one of the most incredible things in the history of the planet—ever since a smaller world smashed into us some 4.4 billion years ago and peeled off the moon. If we fail, it could result in one of the greatest mass extinctions in all of Earth history, and potentially the collapse of human civilization. So those are the stakes.

After eons freeloading off the energy captured by photosynthesis and burning it back to CO_2 we need to instead capture our own. We need to become photosynthetic.[fn1]

SOME 3 BILLION YEARS AGO OR SO, LIFE EVOLVED AN EXTREMELY unusual mineral structure anchored by four manganese atoms—an arrangement found in no rock on Earth, or anywhere else in the solar system for that matter. This structure, which exists only in the photosynthetic centers of cyanobacteria and plants, is used to pull off the almost impossible task of splitting water for its electrons to make plant matter out of CO_2. It was invented, brought into being, by biology. We still don't quite understand how photosynthesis works or how it evolved. As one ecologist says, it is "the only effective and large-scale technology that has ever been 'invented' for capturing and storing that solar energy." Though life is organic carbon, made from air and water, it critically depends on a trace amount of metals, like this manganese, for the many miraculous electrical systems that manipulate electrons and shuttle them around the cell to power life—from the iron-sulfur proteins that wire our cells, to the molybdenum and vanadium complexes that bacteria use to charge up and tear apart stubborn nitrogen gas, producing much of the world's natural fertilizer. In fact, life's dependence on metals goes back to our origins at the hydrothermal vents of the primeval ocean floor.

Though we're still teasing apart how, exactly, many of these miraculous bioelectrical systems work, human society has fortunately inherited this inventive, generative capacity of biology, and, through the super-duper computer of culture, we are perhaps unmatched in our ability to generate novel arrangements of materials and deploy them to our own advantage. To that end, we've already synthesized tens of thousands of chemicals (and even dozens of elements) not found in nature, ostensibly to aid in our own flourishing. While one could extend this legacy of material innovation back to Bronze Age metallurgy and beyond, most of these new substances have been derived from fossil fuels. And many are now maligned for some of the very qualities for which they were once celebrated, like the durability of plastics that surpass the plant polymers they resemble. As Primo Levi himself once uneasily marveled about polyethylene, the unusual substance is "flexible, light, and splendidly impermeable: but it is also a bit too incorruptible, and not by chance God Almighty himself, although he is a master of polymerization, abstained from patenting it: He does not like incorruptible things."

If we're going to ditch our 2-million-year old penchant for lighting plant stuff on fire, humanity will need to train its synthetic creativity on the inorganic world. We will need to capture our own energy directly from the environment, rather than relying on plants, modern or ancient, to do the trick for us. To actually transmit and use this energy, we will need to electrify everything that it's possible to electrify. Most challenging of all, we will have to all but ditch carbon, an admittedly tall order for carbon-based life, and begin to take our cues from the strange distorted manganese nanocube at the heart of photosynthesis or the metalloproteins that electrify life. We will have to go metal. This is why humanity is getting increasingly familiar with the more exotic corners of the periodic table, from the five-ton neodymium magnets in each wind turbine to the dysprosium in electric car batteries.[fn2]

Unfortunately, these so-called critical minerals, required for the clean energy transition, are not at all like fossil fuels, the presence of which is the legacy of all the life that's ever lived. There is no Saudi Arabian Ghawar oil field of neodymium, dysprosium, or indium. All the world's solar panels, for instance, depend on super-pure quartz known only from one mine on

Earth, in the mountains of western North Carolina. New mines for minerals can take decades to permit, and financial analysts and commodity traders warn that, already by the end of this decade, there will be severe shortages in the face of unprecedented demand for both renewable energy and electric vehicles, with 99 percent of the cars and trucks on Earth still needing to be replaced. Many of these strange minerals have not been mined at scale before. Others have been mined at scale, and that's precisely the problem.

FOR SOCIETY, AS IT IS IN THE NANOMACHINES OF BIOLOGY, COPPER is extremely good at conducting electricity. This means that an increasingly electric world will almost certainly be a world of copper wire, and to effectively decarbonize, we will need to mine more copper in the next thirty years than humans have mined over our entire history. This should sound some alarms. We have, after all, already mined quite a bit of copper to date. In the Americas, people have been mining it for at least 8,000 years, since the indigenous metalworkers of Michigan's Keweenaw Peninsula began fashioning bracelets and fishhooks from it. In Eurasia and North Africa, there is an entire two-millennia stretch of human history defined by it (the Bronze Age). The Romans smelted so much copper, mined from all corners of its empire—from the Middle East to Britannia—that it polluted ice cores in Greenland, a feat that wouldn't be reproduced again for centuries until the Song dynasty did the same.[fn3] In fact, demand for the metal ran so high during the boom days of the Song that shortages prompted the invention of the world's first paper currency.

But where mining camps could still descend on Butte, Montana, at the turn of the twentieth century to collect massive nuggets of pure copper in the local streams and hack ore out of rocks that was 50 percent copper, a century of exploitation later the average grade of copper ore mined in the US has dropped to only 0.4 percent today. This means that 99.6 percent of the rock mined in the US for copper doesn't have copper in it.[fn4] This is the "best first" principle in action. The low-hanging fruit has already been picked, and the bottoms of the trees stripped bare. And as demand for copper soars as the energy transition takes off, and even these unappealing rocks are exhausted, ore grade will only decrease. Vastly more rock will need to be mined, ground down, and processed, for ever less copper … which will take vastly more energy and fresh water still, and at a time when both will be in extremely high demand, perhaps direly so. A global copper shortfall is expected to bite in 2025, then blow wide open to 2035, when the world's mining companies expect demand to double from 25 million tons a year today to an astonishing 50 million tons in just a decade, but for supply to potentially fall short by some 10 million tons in the meantime. From here demand will only grow, with a study from S&P Global finding that this "record-high level of demand would be sustained and continue to grow to 53 million metric tons in 2050." This is why, citing "declining ore grades," the head of mining and metals at one global investment bank recently predicted that "it's likely there won't be enough copper to meet decarbonisation goals in the next few decades." Indeed, by the end of the century, the human world could require as much copper in a decade as all the currently exploitable reserves on Earth.

It's also worth noting, in our brief buzzkill survey of how difficult ditching CO_2 will be, that no one wants copper mines in their backyard. Copper mining might not produce gigatons of CO_2 like the fossil fuel industry, but it does produce gigatons of tailings, slag, and waste rock, loaded with arsenic, cadmium, mercury, and lead. This waste is mostly dumped in landfills, where it leaches into the local land and water, contaminating them with heavy metals. Even more troublesome is acid-mine drainage, an all-but-unavoidable consequence of open-pit mining, when pyrite minerals are exposed to the surface world. When these rocks are exhumed and inevitably exposed to rain and hardy metal-eating microbes, a runaway chain reaction produces runoff from the mines loaded with heavy metals and water so acidic that these streams sometimes reach *negative* pHs. Nor are these harms necessarily short-lived. There are, today, Roman-era copper mines that still pollute: in the wadis of southern Jordan, desert shrubs and goats are loaded with heavy metals from this mining of antiquity—a toxic ghost of empire that haunts the desert.

As a result, even though the copper industry accounts for only about 5 percent of the metal mining world, it is responsible for the majority of its toxic contamination of the environment. This is why the US Interior Department recently blocked a huge new mine in the forests of northern Minnesota, citing the potential for "irreparable harm" to the vast Boundary Waters and even the Great Lakes downstream; why the marine biologist Carl Safina warns that the long-planned, gigantic Pebble Mine in the watershed of Bristol Bay, Alaska, "would destroy the planet's most productive salmon fishery"; and why what was once one of the largest, most profitable copper mines in the world, the Paguna mine in the tropical rain forests of Papua New Guinea, has been shut down since 1988, when the mine's extraordinary environmental destruction—and similarly extraordinary profits that left the country along with the copper—led to a local insurrection and a ten-year civil war that claimed 20,000 lives. Panama just canceled one of the world's largest copper mines in 2023, a gash in the rain forest the size of San Francisco.

Despite this troubled history, copper mining will almost immediately need to be ramped up to levels unprecedented in world history to decarbonize. Because the lifespan of solar panels and wind farms is only twenty to forty years, these machines will need to be continually rebuilt, which means a continuous supply of metals like copper. Some of the most optimistic scenarios envision recycled scrap metal accounting for just over a quarter of the copper supply around midcentury (today it stands at about 17 percent), but this would still mean new copper mines indefinitely into the future, as ore grade continuously declines.

Copper is, moreover, just one mineral in the alphabet soup of elements required by the energy transition. For nickel, for which demand is expected to double in coming years, bountiful Indonesia has proposed a new "OPEC" as China pours money into the deforested, open-pit nickel wastelands of Sulawesi. In the Democratic Republic of the Congo, Chinese-owned cobalt mines that supply that country's gigantic EV industry are ringed by heavily armed guards from the Frontier Services Group, a private security company founded by Blackwater CEO Erik Prince. The mines themselves are huge open pits dug by hand by desperate Congolese workers, producing a landscape that looks imported from a Hieronymus Bosch painting. Prince, for his part, wants in on the scramble for these kinds of EV battery minerals in Africa. This past year, he announced plans to raise $500 million to mine copper and cobalt on the continent, exhorting the West to "put the imperial hat back on."

What this all means is that clean energy will not deliver a utopia. The work of building a more just society is not entailed by building out a zero-carbon one. For better or worse, human history will go on, as will our damage to the natural world, even if the supereruption of CO_2 does not. (It's just that in the alternate scenario, where the supereruption of CO_2 continues unabated, human history may not.) The sheer scale and complexity of modern civilization at this point—the tremendous energetic and material throughput required to simply maintain it, much less grow—are such that it is simply impossible for humanity to leave little impact on the planet. Sustaining extremely low-entropy, far-from-equilibrium systems, in biology or in a global industrial civilization, requires dissipating lots of energy and generating lots of waste. Fossil fuels or renewables, then: the phenomenon of industrial civilization will continue to land blows on the natural world, no matter what is used to power it.

Let's look at these unpleasant facts a moment longer. A 2023 report, put together by a coalition of energy industry and environmental NGOs, one nominally "committed to achieving net-zero emissions by mid-century" trumpeted the feasibility of doing so, and was received warmly by renewable energy advocates as a result. Still, the report found (while studiously not phrasing it this way) that a net-zero energy system will require mining on the same order of magnitude as the current fossil fuel system.[fn5] This is not the impression typically given by advocates for clean energy—who usually cite figures that leave out waste rock (producing the ridiculous but oft-quoted stat that it will require 535 times less mining than the fossil fuel system) or account for only the materials needed for power generation, omitting those needed for transmission and storage, or some combination of the two. But I think it will ultimately prove counterproductive if we fail to prepare people for the scale of mining and reworking of the surface world necessary to decarbonize. Indeed, the same report sheepishly avers that during the transition, and without adequate protections, "local environmental impacts could be on the same order of magnitude as maintaining the current fossil fuel based system." That said, the catastrophic *global* impacts of maintaining the current fossil fuel system, even in this worst-case scenario, more than dwarf the environmental costs of such a switch.

Still, the narrative that our zero-carbon future will be a less materially intensive one grows even shakier when one adds the sheer physical mass required to patch up what increasingly appears an unbridgeable CO_2 emissions gap for the near future, with (still aspirational) carbon removal technology. The need for this technology stems from the fact that there are not yet scalable, cost-effective, zero-carbon solutions to the CO_2-emitting behemoths of cement and steelmaking, heavy industry, or heavy long-distance freight trucking and shipping. The sinews of industrial civilization, in other words. For instance, the aviation industry, the most carbon-intensive component of humanity's transportation emissions, has seen relentless growth, with no slowdown in sight, as more of the world's population gets to see their own planet. Unfortunately, simple physics seems to rule out the possibility that today's passenger planes could ever be powered by electric batteries, since, by necessity, they would have to be so huge and heavy as to render the planes unflightworthy.[fn6]

With so much of our world so hard to decarbonize, perhaps the best solution is to simply suck all of our CO_2 out of the air. To that end, one proposal to sequester our unyielding exhalation of CO_2 from the crust is to mine and finely grind up basalt rock, ship it out to sea, and dump it—effectively accelerating the natural weathering processes that we've repeatedly encountered throughout this book, and that normally operate on 100-millennia timescales. By one estimate, though, it would take 1.4 gigatons of this mined and crushed rock (twice as much as the US produces every year) to sequester even one gigaton of CO_2 or about one-fortieth of the CO_2 we put into the atmosphere every single year. Leaving aside the energy required to mine, crush, and grind up such an astounding amount of rock, every gigaton of the mineral would require 1,000 new dedicated large ships on the ocean, on top of the 10,000 similar vessels currently afloat in the world, exacerbating the effects of shipping noise on ocean life, as well as pollution from large ships—which today are fueled by the dirtiest component of crude oil.[fn7] Also, we're unsure if this method will even work. A similar approach instead using lime would require ramping up its production to the size of today's global cement industry, which is responsible for almost 10 percent of the world's CO_2 emissions. Again, this would be to draw down less than one year's worth of carbon emissions.

Then there are the still rather notional "direct air capture" machines that would simply suck CO_2 out of the air—the favorite climate solution for a tech crowd who spent the past decade plowing money into unprofitable ridesharing and food delivery apps. Unfortunately, direct air capture will never be the climate deus ex machina that many of the venture capitalists investing in the speculative technology expect it to be. Undoing even one year's worth of emissions by this method would take two years' worth of global electricity consumption. This is because the efficiency of the technology runs into hard physical limits, strictly imposed by thermodynamics. (Capturing a gas as diffuse and as scarce as CO_2, at 420 parts per *million* in the air and concentrating it all in one place lowers its entropy, and so unavoidably requires lots of energy to pull off.) “Up to 20 times as much energy is required to remove a tonne of CO_2 from the atmosphere than to prevent that tonne entering in the first place,” writes the European Academies’ Science Advisory Council.

There are diminishing returns—if indeed there are any returns at all—to such energy-intensive strategies for fighting climate change. For instance, there are well-grounded reasons—not only thermodynamic, but geologic and logistical reasons as well—to think that the price of methods like direct air capture will never drop below $600 per ton of CO_2 that’s permanently removed from the atmosphere. But according to the climate scientist Zeke Hausfather, even at a heroic $100 a ton, it would still cost $22 trillion to reverse warming by a tenth of a degree. To be able to site these carbon capture machines anywhere in the world would also require building out a global infrastructure of CO_2 pipelines comparable to today’s oil industry to carry the gas to those rare, suitable geologic storage sites where it can actually be buried. Only in this case, this vast network would carry not the most profitable product in the world (like oil), but instead one that has negative value. This doesn’t make a lot of sense to me, but I’m not an economist.

Direct air carbon capture, then, is “unfortunately only an energetically and financially costly distraction,” the researchers Sudipta Chatterjee and Kuo-Wei Huang recently concluded in a letter to *Nature Communications*. Or perhaps it’s worse than a distraction. “The risk of assuming that [direct air carbon capture and storage] can be deployed at scale, and finding it to be subsequently unavailable, leads to a global temperature overshoot of up to 0.8°C [1.4°F],” another recent study found.

In light of these ridiculous and costly climate kludges, one can forgive the sarcastic refrain one sometimes hears that we already have CO_2-capturing machines, and they’re called plants. Unfortunately, though, schemes that rely on photosynthesis at the Earth’s surface to draw down our CO_2 and get us out of our jam are, if anything, even more infeasible. You cannot fix a problem with biology that you got into with geology. They are different scales. There’s just too much carbon down there in the crust, and we’ve already burned through quite a bit of it. For a sense of the scale of the challenge: to grow enough kelp forests in the ocean to remove just 0.1 gigaton of CO_2 in a year (on a world where humans, again, emit nearly 40 gigatons a year) would require growing it in a great kelp belt 100 meters wide that circumscribes 63 percent of the world’s coastlines. Another proposal to plant 1 trillion more trees (there are currently 3 trillion on the planet) would require an area the size of the contiguous United States to chip a mere 6 percent off our needed CO_2 reductions (though depending on where they were grown, these forests could even darken the surface, absorb more sunlight, and actually make the planet warmer, all while expropriating the land from farmers and indigenous people). Losing the Amazon, the largest tract of rain forest in the world, one that spans eight countries, would be an incalculable, even cosmic loss. But it would result in about 15 years of our current emissions from fossil fuels.[fn8]

This is why we can’t capture and bury our way out of this mess. Carbon dioxide removal schemes are not even remotely up to the task of bailing us out, nor is there good reason to think they ever will be. Their sharpest proponents and most knowledgeable researchers know this, and don’t pretend otherwise. As the oceanographer and carbon removal expert David Ho recently wrote in the journal *Nature*, “We must be prepared for CDR to be a failure.” In a best-case scenario, it will only help around the edges, and only after we’ve already dropped our emissions to near zero, by embarking on an almost inconceivable effort to decarbonize the world. And without all but ceasing CO_2 emissions, carbon dioxide removal will be less than useless.

IN SUMMARY, WE’RE IN DEEP SHIT, AND THE FUTURE WILL BE JUST AS geological and material as our past, if not more so. The big difference is that this new world (hopefully, eventually) will emit virtually no CO_2 compared to the geologically unprecedented supereruption of it today. Still, given what we know about the carbon cycle, on balance it still seems like a no-brainer to try to decarbonize, despite the difficulty—as it’s the only alternative to a massive, planned contraction of the global economy.

This last idea is, in fact, the prescription from advocates for “degrowth,” who propose massively scaling back the global economy, given the continuing ecological and climate damage from business as usual. I think that most critics of degrowth fail to honestly engage with the seriousness of its proponents' ideas, the breadth of their literature, or—if there's any relationship whatsoever with economic growth and the physical and energetic world—what truly *endless* growth entails. But I also don't think a major energetic retreat today is at all desirable. The idea that the current global system, with all the current actors on board, could coordinate such a managed climb-down—purposefully reducing global GDP—in a way that wouldn't entail incalculable human suffering strikes me as fantastical. Such a retrenchment seems like it would instead unfold catastrophically. There needs to be a positive vision for the future for people to sign on to, not a prescription for austerity, no matter how righteously advanced. I can imagine the political backlash to such a program, and the reactionary forces it might empower could even make the problem worse. I think, given our current maturity as a species, a project of economic and energetic degrowth would much more resemble a chaotic, ongoing, and worsening Great Depression—the greatest reversal of growth in the twentieth century—than a reversion to some more harmonious way of living.

For better or worse, I think there's no way out but through. We have to somehow thread the needle between supplying enough energy to the vastly expanded twenty-first-century human project such that it can flourish, and do so without destroying the world. I think we have to hope that such a thing is possible, because the alternatives are too horrible. One thing is sure, though: putting industrial civilization on autopilot in its current fossil-fueled form *would* destroy the world. "A temperature increase of 5.2°C above the pre-industrial level at present rates of increase," a recent *Nature Communications* paper predicts about a level of warming unlikely this century, but easily achievable in the next, "would likely result in mass extinction comparable to [the Big Five mass extinctions in the fossil record], even without other, non-climatic anthropogenic impacts." It should be somewhat unnerving that the most advanced climate models, those with the most sophisticated cloud physics built in, and widely assumed to be "running too hot," keep spitting out warming of 5 degrees under rather middle-of-the-road emissions scenarios. The mature thing to do is to face up to the extraordinary challenge ahead; recognize, even lament the inevitable ongoing wounds to the biosphere; and—if you're convinced that it is worth doing so—resolutely press ahead.

An attitude prevails in some climate circles, especially among neoclassical economists, that everything will work out in the end, almost by some cosmic covenant—that the logic of the free market will subtly shift us into a more sustainable lifestyle on this planet. This is a fantasy. If burning fossil fuels remains profitable, and useful, then we will keep doing it, barring concerted intervention. And burning fossil fuels will most definitely remain profitable. This means that the market, left unchecked, will not deliver a zero-carbon future, or likely even a livable one.

As even the starchy *Financial Times* recently wrote, “The sheer scale of the physical infrastructure that must be revamped, demolished or replaced is almost beyond comprehension. Governments, not BlackRock, will have to lead this new Marshall Plan. And keep doing it.” The world is currently falling short of needed spending on the transition by trillions of dollars a year, and it will need to scare up some $275 trillion in the next three decades, according to a McKinsey & Company report that calls for a “fundamental transformation of the global economy.” The largest climate legislation in US history, 2022’s Inflation Reduction Act (IRA), promises a still-astonishing $369 billion of investments in clean energy. It should be lauded, given the barren desert of meaningful climate legislation over the past few decades. Yet for reasons of perhaps unavoidable political expedience, it is also an act that mandates millions of square miles of offshore oil and gas lease sales in the Gulf of Mexico, will speed up the construction of a 300-mile natural gas pipeline through Appalachia, and about which then-senator Joe Manchin of West Virginia could trumpet in a press release, “BECAUSE OF THE IRA, WE ARE PRODUCING FOSSIL FUELS AT RECORD LEVELS.”

Fossil fuel giveaways aside, the IRA’s drafters also made a huge bet on the private sector: rather than develop the state capacity to build out the scale of infrastructure necessary to decarbonize, the build almost exclusively provides inducements to private companies to carry out this extraordinary task. This is a departure from the twentieth-century model of state-led infrastructural transformation, like the Interstate Highway System, or the New Deal–era Tennessee Valley Authority, a federally owned utility that built and still runs a system of hydroelectric dams, controls floods, and generates electricity for seven southern states at rates lower than in most of the country. Indeed, to reach net-zero by 2050, the spending needed just on new transmission infrastructure alone in the United States is expected to far surpass the $600 billion price tag (in inflation-adjusted dollars) for the entire 43,000-mile Interstate Highway System. Whether this kind of spigot of tax breaks, loans, and grants to private companies will deliver us to a climate promised land remains to be seen. But with US state administrative capacity relentlessly eroded away since the dawn of the neoliberal era of deregulation and privatization of public services—a trend that came of age in the wake of the oil shocks of the 1970s—it's the only lever American lawmakers still know how to pull. And indeed, as this book goes to press, a new US administration threatens to gut climate regulations and roll back even these compromised attempts at decarbonization.

On the international stage, effective market intervention will require extraordinary global coordination. Countries that force their industries to invest the hundreds of billions of dollars needed for decarbonization in the coming years—as members of the European Union soon will—will render them uncompetitive in a global market where other actors can burn fossil fuels penalty-free. Moreover, where wind and especially solar power have been rapidly increasing in recent years, they've been more than offset by the ceaseless growth in total global energy consumption. Indeed, in human history, no energy source has ever been replaced at a global scale,[fn9] and each new energy source humanity has ever used has only been additive. Even burning wood, an activity that has defined our species for its entire 2-million-year history, is currently in its heyday. This persistence of incumbent energy sources is as much a feature of the rest of the biosphere as it is for us: the weird low-energy metabolisms of the microbes that ruled the world before oxygen billions of years ago weren't displaced by more active aerobic life, but instead still hum away in the background today, critically driving much of the world's global biogeochemical cycling.

The problem with unguided free-market competition is that, like evolution itself, it cannot "see" far into the future, nor does it "care" at all whether the practices it rewards can be sustained indefinitely. It cares about returns. This is why it's more profitable to deforest the Ivory Coast and Ghana in a few decades for chocolate plantations—to take one internationally traded commodity—that last only a generation and exhaust the land, than to conserve the forest and ensure a crop that sustains itself indefinitely. This is why the French agronomist François Ruf predicts that business-as-usual in the chocolate industry will carry on "until there is no more primary forest available in the world." The same story holds for palm oil deforesting Borneo, or beef and soy obliterating the Amazon. Imagining that such a shortsighted system can anticipate, or will be compatible with, life-sustaining global geochemical cycles that operate on multi-millennial, even million-year timescales, like the global carbon cycle, seems folly. To wit, even after a quarter century of escalating alarm about the emerging climate crisis, and endless corporate cheerleading about the clean energy transition, fossil fuel production on Earth has dropped from 86 percent of global energy in 2000 to 82 percent in 2023. We're still emitting CO_2 10 times faster than in the End-Permian mass extinction, no matter how many glitzy climate summits get thrown. This is because digging up oil and gas is still by far the most profitable industry in the world. Indeed, recent concerns that generating electricity with solar panels and capital-intensive wind farms in deregulated energy markets isn't as profitable as with fossil fuels —especially in an age of high interest rates—have only served to highlight the continued rosy future for oil and gas without concerted market intervention. Surveying the coastal dead zones, bleached coral reefs, plummeting wildlife stocks, and lifeless rivers across the world, it's clear that the market doesn't care about the planet unless it is made to.

History is perhaps most instructive on the folly of looking to the invisible hand for guidance to the future. The "market" didn't care about the human misery of slavery in North America, as financial institutions in New York and London offered generous lines of credit to slave owners for the operation of their plantations, while the slave and cotton trades enriched a web of shipping, insurance, and banking firms in the North. Slaves were seen by their owners as liquid capital assets and by 1860 were valued at $3 billion—three times more than all of American industry at the time. The cotton they harvested was fed to the coal-fired factories across the Atlantic that enriched the British Empire, launching the Industrial Revolution. With low interest rates in the South just before the Civil War, the market foresaw an indefinite future for the practice. Instead, the inhuman institution was ended only through massive intervention by human agency, a mass political movement, and the deaths of 600,000 Americans.

If climate catastrophe is avoided, it too will be the product of human agency. By political intervention. By remembering that it is humans, not algorithms or price signals, who are in charge of the system. Or at least that is the hope.

Surveying the state of global leadership, though, can be more than discouraging. Rhetoric around access to "critical minerals" has slotted easily into the twentieth-century foreign policy establishment conceptions of winner-take-all great-power competition, if not the Great Game geopolitics of the nineteenth. "China's dominance in critical mineral supply chains," the Atlantic Council recently warned, with as much unease as war planners once fretted over oil, "could leave the US vulnerable during a major conflict." For its part, China, which refines 90 percent of the graphite used in almost all EV batteries worldwide, has clamped down on exports of the mineral, citing "national security." This kind of mistrustful zero-sum geostrategic thinking is hardly indicative of a world on the path to some sort of ecumenical, problem-solving *Star Trek*–style community of nations. And *none* of the mounting, unprecedented global threats, of which climate change is merely the most far-reaching—if the most vertiginous in its level of difficulty—are at all solvable without an unthinkable level of US-China cooperation and coordination. This is just as true for the ongoing cataclysm of global overfishing or the urgent and mounting threat of global pandemics as it is for the climate. Yet military planners on both sides of the Pacific are seriously gaming out a conflict over Taiwan that would see the US lose "dozens of ships," "hundreds of aircraft," even several aircraft carriers[fn10] within the first four weeks of a Chinese invasion, with tens of thousands of US service members dying at a rate not seen since World War II. China, in these scenarios, is expected to suffer even worse losses. That anyone is even considering this prospect at the current precipitous moment in Earth history is a sign of civilizational insanity.

Meanwhile, the democratic pressure points available to the unionized coal miners and railway and port workers at the turn of the twentieth century—whose control over the flow of carbon allowed them to extract gains from capital—look increasingly diffuse and elusive a century on. Much of the world's natural resources today—its metals, grain, and coal and oil—and indeed much of the carbon cycle itself are mediated and routed through secretive commodity-trading companies. These corporations have inscrutable names like Glencore and Trafigura; are registered to international tax havens; contract ships sailing under "flags of convenience" like Panama and Liberia to avoid labor laws and even hull inspections; employ the cheapest, most exploitable international workforce; and pick up and unload much of the world's energy, food, and raw commodities in deregulated free port zones, outside of local jurisdiction. Much of the capital from this international flow of planetary resources is not taxed and reinvested in society, but pools in offshore bank accounts. The system is now so complex, unregulated, subcontracted, and opaque that sectors of the economy as mundane as teachers' pension funds—invested in private equity firms—can find themselves unexpectedly funding revolution in the Middle East with oil sales from sanctioned breakaway republics[fn11] or investing in beef cattle ranches burning down the Amazon. These are the strange and byzantine components that now must be added to schematics of the Earth's carbon cycle, alongside planetary mainstays like volcanoes, forests, and the ocean's biological carbon pump. Perhaps this system for organizing the world, capitalism, nurtured in the polders of the Dutch Republic four hundred years ago, has truly escaped containment. Otherwise, we have to hope that there is still a steering wheel left for society to grab.

IF THERE IS SOMETHING UNSEEMLY ABOUT JEFF BEZOS'S SPACE PROJECT, it might not be its ambition, though it does seem slightly absurd given the current state of the planet. Rather, the problem is that it's an aristocratic endeavor, rather than the collaborative, democratic dream of humanity. Meanwhile, just 0.8 percent of Bezos's wealth could pay to fix all the failing bridges and roads in the United States. Like Antoine Lavoisier's cutting-edge Left Bank science laboratory, funded by the sweat and toil of his countrymen, Bezos's space venture may push forward the frontiers of human knowledge while representing an increasingly unwieldy social pyramid. The brilliant Lavoisier was seen as an avatar of a ruling class whose opulence was an insult to a population struggling through climate shocks and crop failures, and who were left to pick up the tab for punishing war debts and the splendor of Versailles. And he soon found himself swept up in revolution. The French Revolution, to be sure, was a humanitarian catastrophe, one that not even Robespierre escaped with his head, but the idea it enshrined—and that could never be put back—was that sovereignty ultimately flowed from the people, not from an aristocracy or monarchy. There were no subjects, only citizens.

Today, though, things are beginning to feel awfully top-heavy, and wobbly again, as a vast share of the world's wealth has come to be controlled by a tiny minority, passively generating income and skirting taxes. The richest Americans, who produce vastly more CO_2 every year than their fellow low-income citizens,[fn12] are even getting bored with all their money, with the billionaire tech founder Sean Parker recently complaining about “all this money sitting on the sidelines” as the wealthiest Americans are just “sitting on large capital gains with low basis and huge appreciation.” Capitalism seems to be increasingly failing in its central generative activity, productively reinvesting profits, and its elites are instead reverting to those more timeworn practices of hoarding treasure and extracting monopoly rents. And where people *are* investing in new technologies, like gig economy and sports gambling apps, they seem purpose-built to extract even more money from those with the least. Recent corporate success stories like Dollar General, which blew away all other retailers in the US in recent years by opening more than 2,000 new locations, do little to inspire. This is where everyone else gets to shop, since dollar stores accounted for “more than one-third of all stores that opened in the United States in 2021 and 2022,” a recent *New York Times* story noted. Their patrons are increasingly disengaging from an electoral politics that has delivered them little, or are attracted to political arsonists promising to tear down the system. Meanwhile, up to $32 trillion of private wealth is held in offshore tax havens, along with the superyachts moored offshore to prove it. And if vested interests in the status quo are an obstacle to transforming our relationship to the natural world, it surely doesn’t help that billionaires now fund our presidential elections—and that one has been given unprecedented and unelected authority to shape the government. The stability of the social system already feels fairly precarious, and this before much of the potential climate shock has hit.

People throughout human history have valued and venerated a universe of things as varied as the landscapes they’ve inhabited. Cultures have been organized around ancestor worship, sun worship, the eternal leader, the mastery of ancient texts, the hunt, the harvest, mutual aid, the blood-soaked glory of a violent longboat raid, conquest by horseback on the steppe, nonviolence and asceticism, a life of contemplation, and mysticism and enlightenment. These values are reflected in the monuments and the material culture these societies generate—the kinship of the longhouse, the spectacle of the colosseum. Surveying the most expensive neighborhoods in the country illuminates what our society values too: McMansion and defense-contractor-filled Northern Virginia, the glass-sheathed headquarters of global finance in Manhattan, the land of VC-drenched tech startups and billion-dollar-losing monopoly plays in San Francisco, the oil and gas industry in Houston. Farther abroad, the coastal sand flats of the Persian Gulf bloom with Ferrari dealerships and starchitect-designed luxury hotels built by immigrant slave labor. These artifacts do not seem to reflect cultures with geologically durable values. If humanity is going to collectively confront climate change and endure into deep time itself, clearly something has to change.

IN THE PREVIOUS CHAPTER, I POSED THE QUESTION "WHAT DO WE do?" as if an answer were forthcoming. But this is a question I continue to ask myself. The point of this book has been to provide a mechanistic account, starting at the beginning of Earth history, of how things got to be the way they are today, at this unprecedentedly strange and out-of-control, if infinitesimal, moment in Earth history. I've hoped that in providing that account, and illuminating the path behind us, I might cast some light on the way forward. But I admit to being overwhelmed too by the sheer constellation of merging dangers, and a planetary task that, beyond requiring an unprecedented level of international coordination and sheer physical transformation, pushes back against much of human if not biological history.

Surveying the bewildering landscape of climate policy and the likely climate futures that could await us, I don't know if it all adds up: the costs of carbon, the carbon budget, carbon offsets, the levelized costs of electricity, clean-energy tax credits, metal resources and reserves, declining ore grades, absolute decoupling, duck curves,[fn13] the IPCC RCPs, the GWP of LNG, PPAs, ETSs, ESG ETFs, AMOC shutdown, climate sensitivity, cloud feedbacks, carbon-cycle feedbacks, spiraling insurance premiums, meters of sea-level rise, trillions of dollars for adaptation, energy rebound, energy-return-on-energy-invested, exajoules, exawatts, and emissions overshoot. I don't know if anyone else knows either. While one can plot out a technocratic road map to a zero-CO_2 future, these best-laid plans will unavoidably collide with a fractious and messy human world and perhaps less compliant carbon cycle. While some estimates, as in a 2022 Oxford University study, found that rapidly switching to renewable energy would return trillions in savings, even without accounting for the staggering losses avoided by mitigating climate change, I admit to being made nervous by other estimates, like that from McKinsey, that the same heroic effort to spend hundreds of trillions of dollars to transition to a net-zero CO_2 world by 2050 would instead initially deliver electricity that's 25 percent more expensive than today's by around 2040—as the world pays the up-front costs of building out this vast new global energy system—before dropping in price. Or that "the initial push for a transition is likely to cause a 10–34% decline in net energy available to society," as one *Nature Communications* study recently found. I do know that the community of people currently working on climate change comprises some of the most dedicated, sincere, and smartest people ever applied to a global problem (like all communities, it has its share of charlatans as well). I know that maintaining our current path will lead toward certain climate catastrophe, so whatever the odds of our success to alter this trajectory, there is only one way to find out, and we might as well give it our best shot. That's about where I've landed.

HUMANS CAN BE WARLIKE, SHORTSIGHTED, PROVINCIAL, AND STUPID, but our species is also uniquely capable of collaboration, learning, ingenuity, and adaptation. I spoke with some scientists for this book who instead viewed humanity deterministically. They see our relationship with fossil fuels a bit like a colony of bacteria hitting the nutrient patch on a petri dish, predictably exploding in size until it exhausts its energy source and inevitably collapses. Or like the hurricane that dissipates the energy source beneath it to enlarge itself—warm ocean water in the case of a hurricane, fossil fuel in the case of the economy—growing into a giant engine of dissipation, and then faltering when there isn't enough energy to power that engine. Maybe they're right, but I disagree. I don't think we are the bacteria on the agar plate, even if looking at human history from 40,000 feet can sometimes give that impression. I think what distinguishes us as a species—from bacteria, other animals, and especially hurricanes—is our ability to learn from each other and the world around us, transcend the limits of our biology and our environment by improvising, and organize ourselves and our societies in novel and adaptive ways. After all, it wasn't thick hides or fangs that kept us alive across an ice age that saw the global temperature swing 6°C or more over tens of millennia, but our culture.

It's true, we can also be incredibly stupid to the point that it registers at a planetary scale—think of Mao Zedong killing a billion sparrows in an effort to boost agricultural yields, but instead unleashing grain-destroying locusts that helped kill 30 million people; or his next-door-neighbor Stalin's disastrous embrace of crackpot agricultural theories that flattered Soviet ideology yet led to similar disaster. But while we can be complete idiots, and cruel beyond words, we've also sent Bach and Blind Willie Johnson recordings to the heliopause, invented extreme ultraviolet lithography, and created ballet. Maybe we can figure this out. To do so, however, will require not merely heroic technological breakthroughs and buildouts of renewable energy. It will also require that we learn to see our global economic system as part of the Earth's grand geochemical systems, not existing in a platonic realm that hovers somewhere above it.

It's not very encouraging that the only mechanism successfully demonstrated to date for globally coordinated emissions cuts is economic catastrophe: in the 2008 financial crisis, when tens of trillions of dollars of opaque semifraudulent financial instruments that had been fed into the global economy—and therefore the global carbon cycle—were revealed as worthless, economic activity seized up, and CO_2 emissions dropped by 1.4 percent as a result. The first year of COVID registered an almost 5 percent plunge. Meanwhile, efforts to *actively* manage our CO_2 emissions through financial legerdemain have been somewhat less successful: emissions trading schemes that supposedly let companies make up for their CO_2 emissions by having trees planted—or simply not cut down somewhere else—have been increasingly revealed as worthless, or even driving additional CO_2 emissions. US farm subsidies for "renewable" ethanol fuel made from corn sent global food prices soaring, indirectly launching the slash-and-burn palm oil plantations of Indonesia and their catastrophic 2015 wildfires that released a billion tons of CO_2 into the air.

In the planet's history, feedbacks in the Earth system have been strange and unpredictable, and the outsize human leg of the carbon cycle will likely make this response only more so. Ninety-four million years ago, when a large igneous province erupted at the bottom of the ocean and warmed the world with CO_2 it buried seas of carbon in the suffocating oceans. Oxygen surprisingly built up in the air above as a result—so much so that forest fires swept this more flammable Earth, as planetary knock-on effect produced knock-on effect. As human society layers ever more unpredictability on top of this systemic planetary complexity, there may be similarly strange, paradoxical, and highly unpredictable *geopolitical* feedbacks that aren't captured in either economic or climate models of future warming—but given our scale now as a species, these will be just as important to the carbon cycle as the mechanisms of the climate-ocean system itself. Some of these feedbacks could be catastrophic: a deteriorating climate, disrupted supply chains, failing agricultural yields, and high inflation could push unscrupulous leaders on a wartime footing to turn to local reserves of fossil fuels like coal, instead of relying on the complex global supply chains of rare earth minerals needed for an energy transition.

Other human feedbacks could be liberating: tipping points don't exist only in ecosystems or power grids, and they're not always catastrophic. They also exist in societies, in the kinds of informational cascades like the one that suddenly swept the dissatisfied inhabitants of East Germany and brought down the Berlin Wall in a matter of weeks. And where the "best first" principle might not be our ally with copper, we haven't been in the rare earth metals game for very long as a species, or with much urgency. Where many assume we're leaving an age of geology as we move off fossil fuels, if anything we're only doubling down on the rocks, and the crust could even prove to be our friend.

For instance, for industries like cement, steelmaking, and heavy-duty transportation that can't be electrified—and are extremely difficult to decarbonize—hydrogen could potentially step in for metallurgic coal and diesel as a zero-carbon fuel. Unfortunately, splitting water to get at it at an industrially relevant scale requires an astronomical amount of energy. For steelmaking alone, it would require doubling the planet's fleet of nuclear and renewable energy. But while making hydrogen from water, or even natural gas, in order to power society might prove to be a climate, energy, and financial boondoggle, it could be that the Earth just so happens to be making far more hydrogen all on its own than we ever thought. A billion years ago, North America tried to tear in half and stumbled, forming a failed rift across the continent that brought rocks from deep in the Earth to the surface—a trauma that still bisects the country with a stripe of deep mantle rocks stretching from Kansas to Ontario, where they emerge and loom over Lake Superior in spectacular dusky headlands. Deep belowground, though, where water meets these rocks, they produce free hydrogen in the exact same reactions[fn14] that might have given rise to life itself at the seafloor vents, some 4 billion years ago. These processes, which have been active essentially as long as the planet has been around, might have been all the while patiently making hydrogen in vast enough quantities to embarrass our industrial needs. It's just that no one's been looking for it.

Dig down a few tens of feet too, anywhere on Earth—there's geothermal heat everywhere. However, it increasingly looks as though hydraulic fracturing (fracking) technology will be the key to unlocking this near-limitless energy source. If hydrogen ever scales, it too would have to piggyback off the hydrogen pipeline technology already developed by the fossil fuel industry. It shouldn't be surprising that the tools developed by the biggest geological industry on Earth might be useful for our geological future, and climate advocates shouldn't be squeamish about taking advantage of the geological expertise and genuinely extraordinary technology developed by the oil industry. I view it as akin to birds' "exaptation" of feathers, which dinosaurs once evolved for warmth but today's avian dinosaurs co-opted for flight.

Durably living into a geologic future will require understanding the geologic biography of our living planet. And one industry has been highly motivated to understand and exploit this story. For decades, college students saddled with student debt, and majoring in geology because they like dinosaurs, have been handed six-figure checks by oil majors upon graduation, offers that often prove irresistible. But today, geoscience PhDs studying the strange life of the early Earth are being similarly sought by battery companies hoping to exploit the secrets of archaic microbes that have strange metabolisms based on iron and that once ruled the oceans in the age before oxygen. Large-scale energy storage is one of the primary challenges of the dawning electric world, and this Archean Eon technology could help us out of the very Phanerozoic Eon[fn15] problem of industrial-scale respiration of organic carbon with oxygen.

It's something of an irony that metals and reactions used by life that predate photosynthesis, and of the sort required by renewable technology, now hold the key to our future civilizational metabolism. In a way this represents a return to our most ancient origins and the primeval respiration of the deep-sea vent world. In the near-miraculous event that nuclear fusion ever scales, it would represent a leap even further back in time, to when the sun winked on as we return even closer to the origins of our story and become a kind of star ourselves.[fn16]

EVEN THE STATE OF COPPER MIGHT NOT BE AS DIRE AS IT FIRST APPEARS. In the 1920s, the US Geological Survey panicked that the country, indeed the world, would soon run out of oil. But the galloping technical advances of exploration geology and drilling unleashed the biblical flood of petroleum that transformed the Earth in the second half of the twentieth century. Almost a century later, in the early 2000s, when it appeared peak oil had truly arrived at last, high prices spurred the development of fracking, which catapulted the industry to still-swelling record production. Making estimates of peak anything has embarrassed forecasters since the dawn of the Industrial Revolution, and perhaps spiking prices will similarly wring ever more copper from the Earth by showering exploration geologists with the money to find more of the stuff, and the capital to develop new technologies to extract it. This kind of leap of faith is the tacit assumption of the energy transition. But it's worth keeping in mind that the premise that it will all work out is not vouchsafed by some cosmic law. We are not operating with a net.

The dawn of the Industrial Revolution was only a handful of decades ago. I think that most people assume that the breakneck acceleration of everything since then represents a new, enduring steady state for human society rather than a transient explosion, or one-off exhaustion of much of the world's natural resources. But exponential graphs don't skew upward forever. Drawing the lesson from the fact that it has worked out so far (all Anthropocene-sized caveats aside here) and so it necessarily will indefinitely into the future puts me in the mind of Bertrand Russell's chicken. The philosopher's unfortunate bird assumed, since its farmer had fed it and taken care of it every day, that the farmer would do so indefinitely, only to come to the barn one day to have its neck wrung. There's no guarantee that the future will look anything like the past, nor that the planet will comply with our demands of it. Booms and busts are the oldest feature of human civilization; they've just never swept up the entire world before.

The advocates for degrowth are right about one thing: truly endless growth on a finite planet is physically impossible. Simple back-of-the-envelope calculations that assume even a conservative relationship between economic growth and energy eventually spit out absurd results, like that within 400 years either the entire land surface of planet Earth will need to be completely paved in solar panels, or we will otherwise be producing so much waste heat that the oceans will boil away. Obviously neither of these will happen. Perhaps the global economy can "grow" forever without the material or energy throughput growing at all, although it's not clear what the economy even is in that case, if its growth has no relation whatsoever to the physical, material world. Perhaps people can hand things back and forth to each other forever, and the economy will grow in perpetuity. At the very least, this seems strange.

But if we continue to grow, and that growth maintains its relationship with energy—which in recent history has grown at just over 2 percent per year—then in only 1,300 years, the physicist Tom Murphy of the University of California, San Diego, calculates, society will need more energy than the sun puts out in all directions. This is, of course, impossible. Given that energy is used to transform raw material, one shudders to think of the physical footprint and the waste produced by such a society. Plus energy limits represent only the ultimate constraint—one could imagine any number of other faltering resources, like the phosphate rock we existentially depend on for our crops, that could arrive much sooner. A mainstay of science fiction, and a narrative signifier of an impossibly futuristic society, is one that builds a sphere of solar panels that entirely envelops its host star, capturing all of its energy.[fn17] But if those who envision endless growth are correct, then this impossibly far-fetched scenario is a thousand years closer to us in time than the Roman Republic. Not very far-seeing. Again, the longest components of the carbon cycle operate on hundred-millennia to million-year timescales. Endless growth isn't happening.

This means that at some point the physical growth of the economy will stop. Today our entire global society is swept up and guided by this imperative of growth, suffering economic catastrophe when that growth briefly falters. Eventually, though, our society will have to look and operate much differently, simply out of physical constraints. Again, a few centuries, much less millennia, represents a tiny slice of human history. Geologically it's a ridiculously irrelevant time span. Quantum scale. Nevertheless, the transformation of human society in a matter of centuries can be profound. Four hundred years ago, as Europe was racked by cold snaps, the feudal system was being dismantled and new systems, capitalism and the nation-state, were emerging from the devastation of thirty years of war. It's worth remembering that these were adaptations at the time, innovations on how society and nature would be organized, after the old feudal world had ceased working. What the world looks like 400 years hence, who knows, but one thing is certain: we will be in some new kind of system, one that has reached some new sort of equilibrium with the planet. Perhaps it will take the coming climate chaos to shake it out of us.

Within a few centuries, then, we will be operating under a new regime. Whether we should actively anticipate this new world and actively steer toward it right now is another question. Complex modern state societies that have managed true sustainability with their environment, like Tokugawa Japan, were largely closed off to the rest of the world and required policies of social control we would find draconian, even grotesque—up to and including the death penalty for cutting down trees as well as infanticide for population control. Instead I am tempted by the Red Queen, who famously had to keep running faster to stay in the same place. We have defied gravity for a few short centuries and need to do so for a little while longer in the hope that we will be wiser tomorrow. In the meantime, I think we should focus on both rapidly decarbonizing and democratizing, and when we have those tools in place, we can get down to the work of plotting our deep geological future.

Life itself offers the most encouragement: not all dissipative structures on Earth grow exponentially and then disintegrate as soon as they've reached maturity. Some, like the global ecosystem, have managed to persist for hundreds of millions of years. If humans truly want an Anthropocene worthy of the name, we will ultimately need to figure out how to imitate these enduring systems on the Earth, rather than mimic the one-off supereruption of a large igneous province. In the canyons of the far future, the Anthropocene may appear as a transient event, like those many other brief catastrophes in Earth history, perhaps leaving only a strange line of shale among a towering limestone edifice, a wave of megafauna extinctions, and the strange suite of chemical excursions found in this thin boundary layer. In the rocks a few meters above, and millions of years later, there would be little indication to the discerning paleontologist that we were ever here. Or perhaps our mark will be found in those rocks as well, signifying the extraordinary journey of a species that found equilibrium at last.

While today the flow of carbon on Earth is modulated by strange legal fictions, with names like Archer Daniels Midland and Aramco, and political ones like America and Argentina, one can only guess what kinds of human organizations carbon will flow through hundreds or thousands of years from now (if there are any), or what the climate of that world will look like.

Hundreds of thousands of years from now, though, as the Earth and its carbon cycle carry on their stately rhythms, our gigantic transient pulse of CO_2 will have been drawn back down into the Earth, and the ice ages may resume. Perhaps we'll even slip into Snowball Earth at some point, and that will be that. Perhaps not. Millions of years from now a large igneous province may erupt from Kenya, where it appears a plume is rising from the depths, as East Africa is pulling away from the rest. Whatever strange forms populate the planet, they will struggle through this spasm, like many before them. Perhaps oxygen will disappear, as the crust fails to be replenished with carbon. In the very, *very* long term, as the sun brightens and weathering accelerates, CO_2 will draw down below the threshold of photosynthesis, and the energetic foundation of the biosphere will disappear. Or maybe biology has a few more tricks to discover.

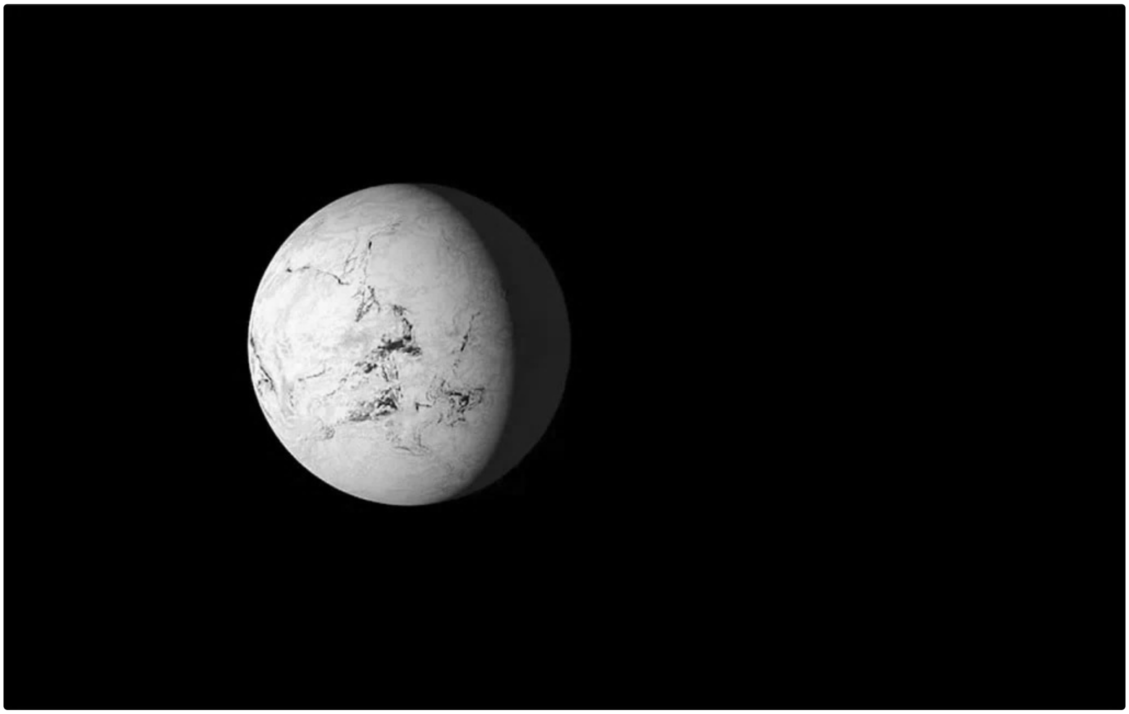
FOUR BILLION YEARS AGO, CO_2CAME ALIVE. IT REMAINS THE ULTIMATE source of carbon for all life, the primary knob of earth's temperature, and our species' signature product. Over the history of life, it has come up out of the rocks, gone into the sky and oceans; some of it has become life, and some of that life has become rock again. This CO_2 from volcanoes is mostly balanced by burial in the rocks over millions of years. When this cycle goes haywire, lots of stuff on Earth dies.

Taking the whole planet into view, the climate, oceans, and biosphere comprise an almost imperceptibly thin, blue-green film. The film itself coats a sugar-frosting of granite continents and seafloor carpets of waterlogged and rending basalt—all of it high atop a gigantic gravity-bound mess of convecting rock, glimmering green with olivine. Far below, this hard planet grades, in the abyss, to a psychedelic, white-hot heart of iron and nickel, swaddled in a liquid metal whose secret streams wrap the planet in an aurora-limned chrysalis of magnetism.

But, as far as we're concerned, everything interesting that has ever happened has taken place in that liminal realm at the tippy-top, smeared onto the surface between exposed rock and outer space. It's a vulnerable place. This curving skin is stippled with tiny mountain ranges that poke up from vast curling slicks of seawater. The mountains create weather, deflect weather, inject CO_2 into the air in little volcanic puffs. The CO_2 traps energy from the perpetual bomb in the sky and keeps the surface from surely freezing to death. The mountains are also destroyed by weather, and in their steady destruction pull CO_2 back down out of the air. Every big river you've ever seen has kept flowing since the day you saw it, day and night, whether or not you were thinking about it. Those waters have never stopped cutting down Earth's restless mountains, for billions of years. Neither has the rain. The mud-laden rivers carry those eroded mountains, and the carbon they stole from the air, to the sea, where they are turned into limestone reefs and seashells. Everywhere—draped onto the heaving continents, diffuse and shimmering through the oceans—carbon moves through life in a standing wave of endless forms: stegosaurs, smilodons, cypresses, swordfish, sparrows, snails. Breathing, dying, killing, evolving. Some of this carbon gets entrained in the crust, and oxygen rises in the air above. We use it up, relieving this disequilibrium. In the industrial explosion of the past two centuries, we've hastened this descent. Someday, who knows when, this angry eddy of dissipation will end. Who knows whether the little hurricane we've kicked up in the past few centuries will rage against the entropy for a few centuries more, or whether it will run out of fuel in the coming decades. It can't go on forever. But perhaps, just like that original flame of life—one kept kindling at the edge of the seafloor at the dawn of time, and through the throbbing jungles of the Carboniferous, to today—we can keep ours flickering awhile longer still.

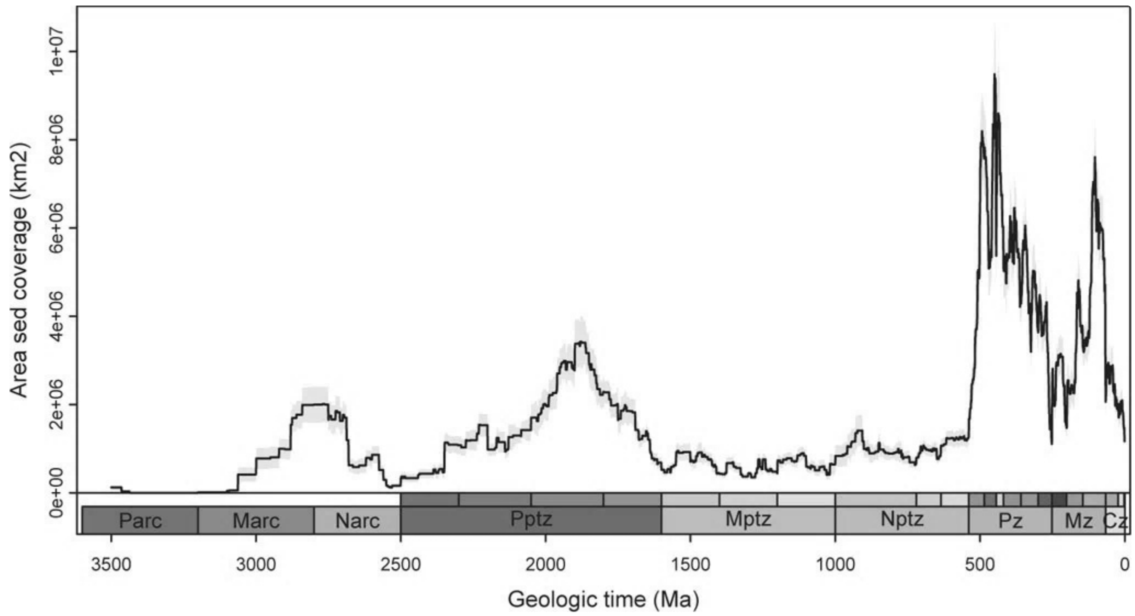
One of the spires at the Lost City Hydrothermal Field in the middle of the Atlantic Ocean. Delicate alkaline hydrothermal vents like this might have given rise to life at its origins. *Susan Lang © Woods Hole Oceanographic Institution*

Geologists Carl Simpson and Lizzy Trower in the hottest place on Earth, Death Valley, studying rocks from the coldest episode in Earth history, Snowball Earth. *Courtesy of author*



Artist’s illustration of Snowball Earth. *Courtesy NASA*

Notch Peak, Utah. Cambrian-aged limestone from the bottom of the ocean and part of the continent-spanning Great American Carbonate Bank. *Courtesy Utah Geological Survey*

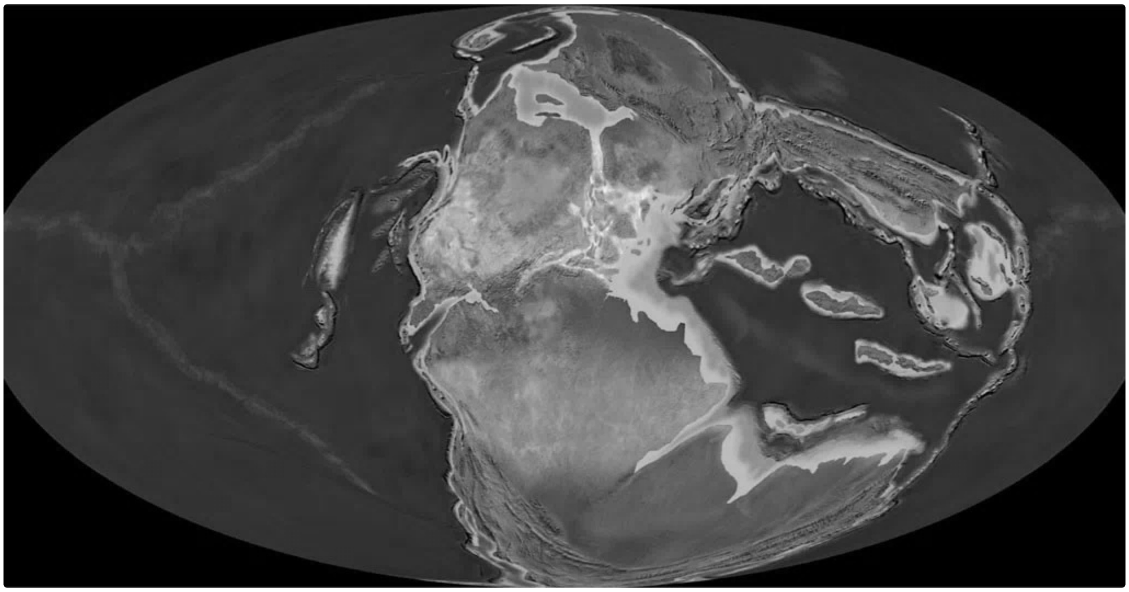


Sedimentary bedrock covering North America, by age in millions of years (Ma). Note the massive jump in strata around 500 million years ago, after Snowball Earth and just at the dawn of the age of animal life. This massive growth in the rock record, which would have sequestered organic carbon, might have helped drive the oxygenation of the surface world and the flourishing of animal ecosystems. The previous episode of growth, around 2 billion years ago, occurs suggestively near the Great Oxidation Event. Key: Paleroarchean (Parc), Mesoarchean (Marc), Neoarchean (Narc) Paleoproterozoic (Pptz), Mesoproterozoic (Mptz), Neoproterozoic (Nptz), Paleozoic (Pz), Mesozoic (Mz), Cenozoic (Cz). *Courtesy Shanan Peters*

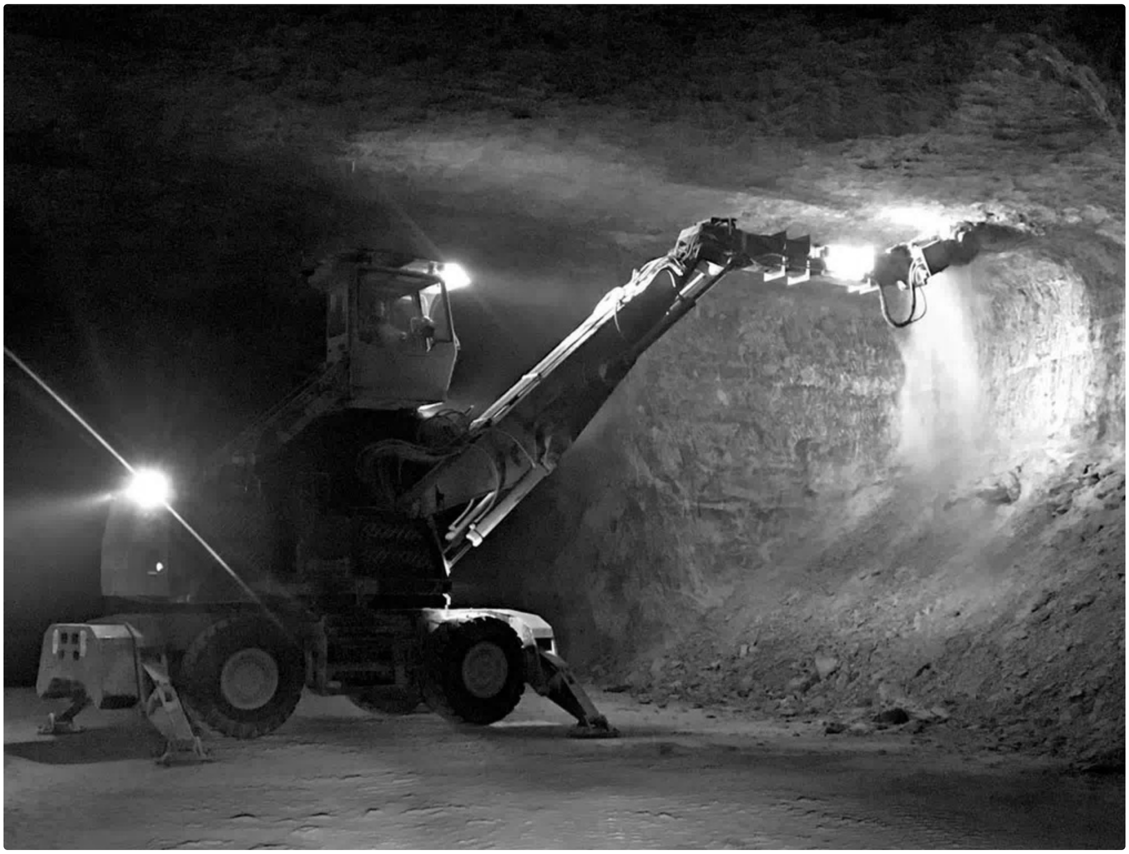
Coal mine workers outside of a mine in Kaymoor, West Virginia, 1914. *Courtesy National Park Service*

Mountaintop removal mining site, Harlan County, Kentucky. *Courtesy of author*

The Houses of Parliament, Sunset, Oscar-Claude Monet, 1903. Note the pink hues from coal smoke over London. It's a similar sky as would have haunted the Carboniferous, when the same plant matter was being burned in the period's raging forest fires, driven by its higher oxygen atmosphere—itself a product of the coal being buried at the time. *Courtesy National Gallery of Art, Washington, DC*



Earth, 240 million years ago. Pangaea during the Triassic Period, between two of the biggest mass extinctions in Earth history. © *2016 Colorado Plateau Geosystems Inc*.



Triassic salt mine underneath Northern Ireland. So-called evaporites like this, left behind from the brutally arid interior of Pangaea over 200 million years ago, span the Earth. *Courtesy of author*

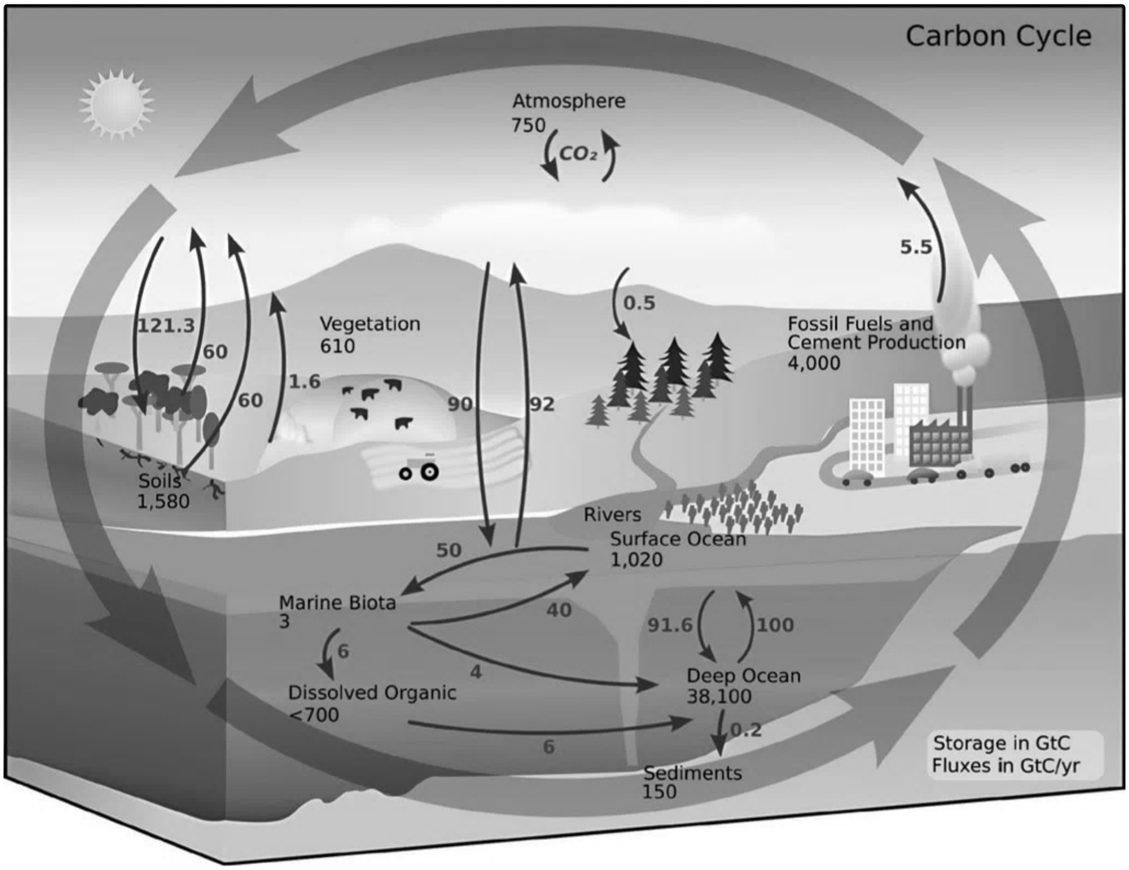
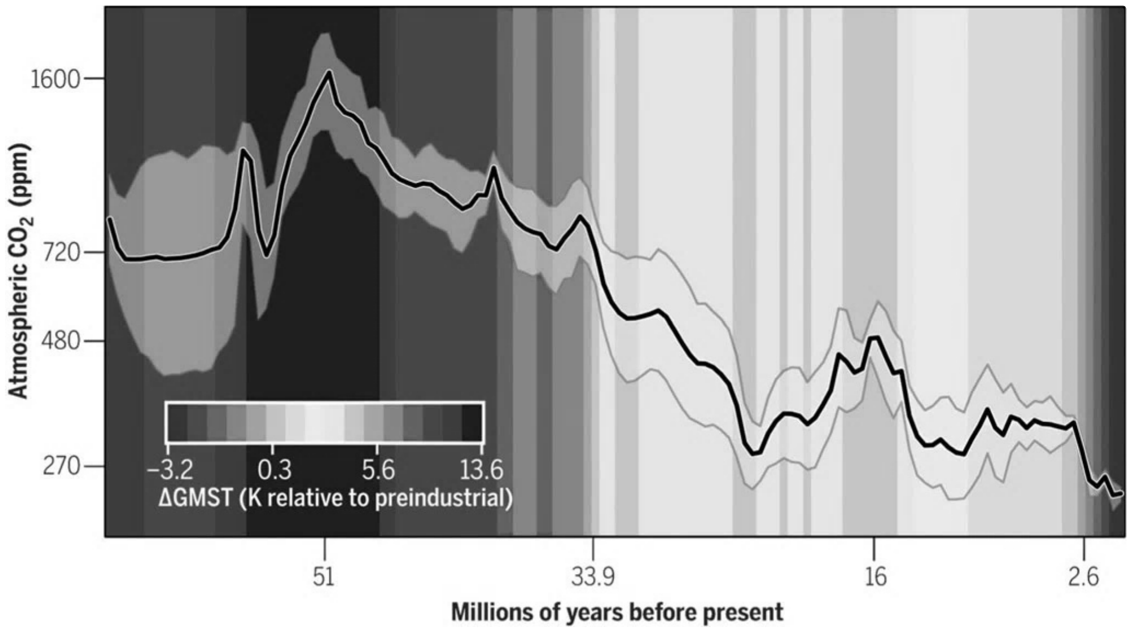
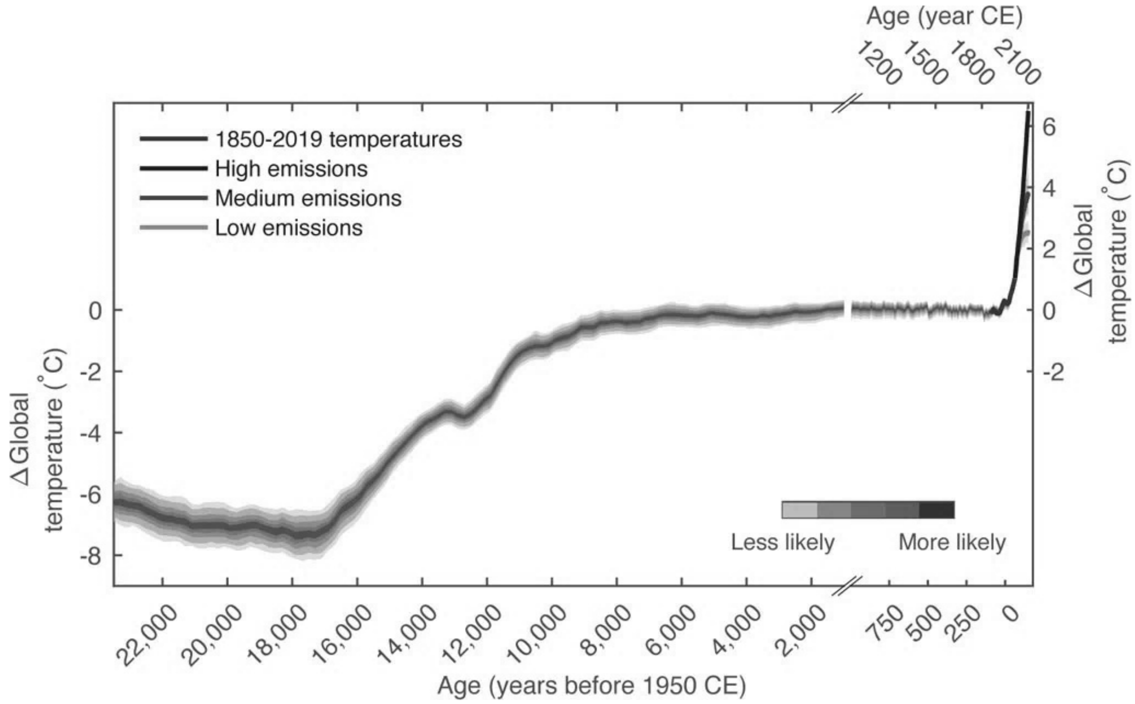


Diagram of the carbon cycle.*Courtesy NASA*



CO_2 and temperature over the Cenozoic (the so-called age of mammals) since the End-Cretaceous mass extinction wiped out non-avian dinosaurs. Note the Eocene Epoch around 50 million years ago, when CO_2 topped 1000 ppm, the planet was about 12°C(K) warmer, and there were crocodiles and palm trees in the Arctic Circle. Compare with recent Pleistocene ice ages of the past 2.6 million years, when CO_2 has recently fluctuated between about 180 and 280 ppm and temperatures fluctuated between more than 6°C(K) colder than today (as shown in next graph) and slightly warmer temperatures. Also note the sharp cooling around 33 million years ago, when Antarctica gained its (then much smaller) ice sheet. *Courtesy Gabriel Bowen*



Global temperature since the depth of the last ice age around 20,000 years ago, as well as potential temperature rise in the future. At the nadir of the last ice age (left part of graph) the planet was perhaps more than 6°C colder than today, sea level was more than 400 feet lower, and there was an Antarctica's-worth of ice on North America. CO_2 was around 180 ppm. The slight dip in temperature around 12,000 years ago, coming out of the last ice age, is the so-called Younger Dryas event, likely caused by an influx of fresh meltwater into the North Atlantic that temporarily shut down an important heat-conveying ocean current system. Note that during the 6,000 years of recorded history the climate has been astoundingly stable. *Courtesy Dr. Matthew Osman*

Forest of oil derricks on Huntington Beach during the early twentieth-century boom times for oil in the Los Angeles area, 1926. *Courtesy Orange County Archives*

Drake Oil Well, Titusville, PA. *Courtesy of author*

Rock layers from the late Devonian mass extinction on the shores of Lake Erie, with a decommissioned coal-fired power plant in the background. These are carbon-rich black shales from the seafloor of a dying ocean 375 million years ago, during the Devonian Period, that were also tapped for the country's first commercial natural gas well nearby in 1825. Rocks from this age of stagnant seas, and the trapped residue of life in them, provide much of the gas fracked in the United States. *Courtesy of author*

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Finally, every book has errors, and this book covers an inadvisably wide range of subjects—none of which I could claim to be an expert on. So if you find any errors, please do let me know—I take full responsibility for them. (Just don’t be a jerk about it.)

And thank you, especially, to ATP synthase, photosystem II, plate tectonics, and every other miraculous component of this absurdly miraculous planet that I woke up on a few decades ago. Every second that any of us is alive is an unthinkable miracle, and there’s no reason ever to be bored. And thanks again to you, Mom.

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Footnotes

Introduction

fn1 Eight minutes to an outside observer, that is; photons don't experience time.

fn2 Along with water and waste heat.

fn3 (first as carbonic acid, then as bicarbonate, then as carbonate)

fn4 And, to a lesser extent, geochemical energy available at hydrothermal vents and in rocks.

fn5 "Eminently breathable air" that Lavoisier had named *oxygène* and "fixed air," or carbon dioxide.

fn6 To paraphrase the physicist Sean Carroll.

fn7 Along with the ceaseless imprecations of the envious Jacobin leader Paul Marat, whose prescientific ideas about phlogiston and mesmerism Lavoisier had embarrassingly dismantled as the president of France's National Academy.

fn8 Today it hangs in the Metropolitan Museum of Art in New York City.

fn9 The figure is now .04 percent.

Chapter 1: CO2, the Stuff of Life

fn1 The qualification "normal matter" is necessary here because only 4 percent of the universe is even made of the kind of stuff that we even think of as stuff—the rest of the pie chart is set aside for mysterious placeholders called "dark matter" and "dark energy."

fn2 Or it will fall into a black hole, and *then* dissipate to nothing more than 10^{67} years later.

fn3 Analogy courtesy of geologist Robert Hazen of the Carnegie Institution.

fn4 There is also energy provided by the gravitation of the moon, planets, and sun, that is, the tides.

fn5 It's even why we experience an arrow of time that goes forward. There's no underlying arrow of time in the physics equations that describe our world; it's just that our brains are constructed such that we can only remember and make sense of our ordered past, but not our increasingly disordered future. It's easy to remember where you left a can of gasoline, for instance; it's impossible to know what all the random motions of every atom will be as your car hood heats the air around it. If it weren't so, we might "remember" the future.

fn6 Without replacing them with equivalently sized energy sources.

fn7 A concept later named by his successors.

fn8 For example, MIT's Kerry Emanuel: "The mature hurricane is an almost perfect example of a Carnot heat engine."

fn9 It oxidizes hydrogen sulfide from the vents with oxygen, for energy. This oxygen it uses, and which ventilates the deep sea, is ultimately produced by photosynthesis at Earth's surface.

fn10 There are some exceptions, like salt.

fn11 With some nitrogen, phosphorus, sulfur, and trace metals thrown in for good measure.

fn12 Unless, in the case of "silicone," it is joined with organic carbon.

fn13 The mineral scaffolding for all this organic carbon that is your skeleton will last awhile longer.

fn14 In perhaps a telling digression, geoscience students are often taught to remember the processes of oxidation and reduction with the mnemonic "OIL RIG," meaning: "oxidation is leaving, reduction is gaining," in reference to what happens to the electrons involved in these reactions.

fn15 Limestone that precipitates on its own in unique chemical circumstances, without the aid of life.

fn16 The inanimate *Zelig* of oceanography.

fn17 The same goes for the shark emerging like a phantom out of the darkness, which Lang captured on video on another investigation of the vents.

fn18 To Treibs's credit, he lost his professorship at the Technical University of Munich after supporting protests against the Nazi Party and was expelled after being flagged as politically undesirable, though he continued on in a leadership position at the chemical company Wacker-Chemie, which employed slave labor.

fn19 They are, perhaps disappointingly: acetate, pyruvate, oxaloacetate, succinate, and ☐-ketoglutarate.

fn20 Outside a sliver of extremely metamorphosed rock in Canada's Northwest Territories, and other outcrops scattered around the world containing a few microscopic grains of zircon, eroded out of their now-vanished Hadean parent rock and entrained in sedimentary rocks hundreds of millions of years later in the ensuing Archean Eon.

fn21 In our case, oxygen. But when oxygen isn't around, bacteria and archaea can use other "electron acceptors" like sulfate, nitrate, iron, etc.

fn22 Per the biochemists Rouslan Efremov and Leonid Sazanov.

fn23 And in the case of some microbes, by maintaining other kinds of positively charged ion gradients, like sodium. Some fermenters are also strange in that, while they do maintain gradients like this, they don't need them to generate ATP.

fn24 Called the acetyl-CoA or Wood-Ljungdahl pathway.

fn25 "[This metabolic] pathway is not only ancient, as those who have worked on it always suspected, but it is older than the enzymes that catalyze it," writes the biologist William Martin. "[It is] older than the genes that encode its enzymes."

fn26 The early atmosphere was instead likely dominated by carbon dioxide, nitrogen, and water.

fn27 RNA is so irresistible as a candidate for the origin of life because not only is it able to store genetic code, but it can also catalyze some reactions today performed by proteins.

fn28 Or, as the Austrian physicist Ludwig Boltzmann anticipated, astoundingly as long ago as 1886: "The general struggle for existence of animate beings is therefore not a struggle for raw materials—these, for organisms, are air, water and soil, all abundantly available—nor for energy which exists in plenty in any body in the form of heat … but a struggle for entropy."

Chapter 2: The Great CO2 Freakout and the End of Eternity

fn1 As we'll later see, not all forms of photosynthesis tear apart water, or produce oxygen.

fn2 Independently discovered by the American scientist Eunice Foote in 1856 and the Irish physicist John Tyndall in 1859.

fn3 In Budyko's model, by reducing the amount of sunlight hitting the Earth by only 1.5 to 2 percent.

fn4 134 feet.

fn5 There are still remnants of this plate, such as the Juan de Fuca Plate in the Pacific Northwest, that have not yet been fully subducted by North America.

fn6 Frances Westall and Shuhai Xiao, “Precambrian Earth: Co-Evolution of Life and Geodynamics,” 2024.

fn7 To say nothing of the fact that photosynthesis would stop entirely as well.

fn8 Even if the CO_2 is emitted into the ocean, the oceans and atmosphere eventually have to equilibrate with each other.

fn9 Earth emits CO_2 because of the complex interplay of its mantle chemistry and its size. If it were much smaller it would emit methane, potentially robbing life of its feedstock.

fn10 Sulfur dioxide is the next-most-abundant gas.

fn11 Calcium carbonate. While today this process is mostly mediated by life, limestone can precipitate out of supersaturated water on its own without life.

fn12 This doesn’t do much when the rocks that were originally weathered were limestones themselves, but when the rocks getting weathered are so-called silicate rocks—things like granites and basalts—then the products of these weathering reactions, when eventually transmuted in the ocean to limestone, on balance, draw down CO_2. Ocean carbonate chemistry is complicated.

fn13 Even Lavoisier had seen that exhaling (CO_2) into limewater would precipitate calcium carbonate, a beaker-scale demonstration of a similar experiment.

fn14 And other carbonate rocks, like dolomite and magnesite.

fn15 These predators were also microscopic eukaryotes, described by the paleontologist Susannah Porter as “tiny vampires” for the ominous tiny holes they left behind in their victims’ shells.

fn16 “Kitlineq” to the 2,000-odd Inuit living on a landmass larger than Great Britain.

fn17 The colloquial name for CO_2’s ice state.

fn18 24-isopropylcholestane.

fn19 A far more muted ice age 579 million years ago called the “Gaskiers glaciation,” after the glacial deposits in Gaskiers, Newfoundland.

Chapter 3: Fossil Fuel Goes Down, Oxygen Goes Up

fn1 For streamlining sake, all mentions of “photosynthesis” hereafter refer to oxygen-producing, or oxygenic, photosynthesis. There are other forms of photosynthesis, but we don’t have to trouble ourselves with them here.

fn2 Or, at least, we call them that in those places on Earth where this buried ancient biomass is economically extractable.

fn3 An extremely small amount of oxygen is produced by ultraviolet light breaking apart water molecules in the atmosphere into its constituent parts, hydrogen and oxygen.

fn4 A process that, as we’ve seen, life at its origins might have pulled off only in the extraordinarily specific conditions of the alkaline hydrothermal vent, or through even more exotic chemistry at Earth’s surface.

fn5 There are also much stranger microbial metabolisms that get their electrons from things like iron, and get their hydrogen elsewhere, but we don’t have to concern ourselves with them here.

fn6 Or at least that portion of it in the sunlit layers.

fn7 Called the Calvin cycle.

fn8 Along with our gas-powered cars, coal-fired power plants, and almost everything else in the global industrial economy.

fn9 The more we learn about photosynthesis, the more astounding the details. New papers investigating it come peppered with descriptions of impossibly fine networks of “surge protectors,” “transistors,” and other metaphors that struggle to describe this nanoscopic world by appealing to the far less advanced world of our own crude technology.

fn10 When there are no other oxidants around at all, some extreme life, methanogens, can even use CO_2 as an oxidant, passing electrons onto carbon to make methane. Things work very differently in the zero-oxygen world.

fn11 Meaning power rather literally here. In physics, power = energy/time.

fn12 And then some: because there are other large sinks for oxygen, like iron, the burial of organic carbon in the rock record has pumped something like 35 to 40 times more oxygen into the atmosphere than the atmosphere currently contains.

fn13 These are what they sound like: rocks made of layers of ancient mud and muck.

fn14 The overwhelming majority of fossil fuel is not recoverable. There is about 3,000 times more organic carbon spread out through the crust than is available in recoverable deposits of fossil fuels: 5,000 petagrams (petagram=10^{15}g) of carbon in recoverable fossil fuels (or 5,000,000,000,000,000,000 grams) to 17 zettagrams (zettagram=10^{21}g) total in the crust (or 17,000,000,000,000,000,000,000 grams).

fn15 For the purposes of simplification I'm using the term *age of animal life* as a stand-in for the wonkier *Phanerozoic Eon*. Molecular clock estimates, organic biomarkers, trace fossils, and even body fossils indicate that animals evolved just before the start of the Phanerozoic, but the explosion of animal fossils in the Cambrian Period (the first period of the Phanerozoic) and their dominance in the fossil record ever since justify characterizing this as the age of animal life.

fn16 The Bahamas have been accumulating on the trailing Atlantic edge of North America ever since Pangaea started breaking apart almost 200 million years ago.

fn17 A piece of which was then still improbably attached to roughly what is today the US Gulf Coast.

fn18 The word *structure* may not be sufficient to capture such a colossal feature of the planet, both through time and space. A more apt description might be *hyperobject*, which the philosopher Timothy Morton coined to capture phenomena too sublime and overwhelming in scope to be apprehended by our minds as simple objects.

fn19 Google "anomalocaris" or "opabinia" for the most iconic, and bizarre, of these creatures.

fn20 The earliest animals evolved under much lower and more variable oxygen conditions than today. In fact, there's even a debate among geologists and paleontologists whether it was oxygen that spurred the evolution of simple animal life, or whether the biological innovations led. The causes of the Cambrian Explosion, which likely had many fathers, remain hotly debated, but the general trend ever since has been toward higher, more stable oxygen, without which the kinds of energy-demanding animals and ecosystems we see today would not have been possible.

fn21 Some metamorphic rocks like graphite and diamond are exceptions, having been cooked and pressurized into pure carbon.

fn22 When these shales and coals are cooked deep in the Earth, they can generate oil and gas, which migrate upward through the rock and can become trapped in vast reservoirs like the Great American Carbonate Bank, if the appropriate kind of seal covers it. Otherwise it escapes into the surface world and is eventually oxidized back into CO_2 on its own.

fn23 One that took the lives of three of the ten explorers who set out.

fn24 A question that arose in an earlier draft by one reader was why didn't this destruction just cancel out the subsequent growth of the crust. The answer I received from Peters is that the continental crust had been weathering and passively eroding for an extremely long time without replenishing with carbon, and so would not have had much in it for the Great Unconformity to exhume.

fn25 Looking back, we find our planetary past peppered with these kinds of uncanny accidents, like Snowball Earth. It could be that, wherever conscious retrospection like our own exists in the universe, its practitioners will find their histories similarly strange and unlikely, and any cosmic neighbors unfathomably distant.

fn26 Except for the odd city that happens to find itself in a subsiding basin continually smothered in sediment, like New Orleans.

fn27 Piling up sediments on the deep seafloor doesn’t help much either, because dense ocean crust gets continually destroyed in subduction zones, behaving much more like the steady-state rock cycle of theory than the all-but-eternal continents that float above.

fn28 Though the Gulf of Mexico is the ocean today, it is the continental shelf and part of the continental crust, not denser ocean crust.

fn29 Like the burial and release of phosphorus from rocks and the seafloor.

fn30 This was the product of the largest continental collision in Earth history, created when East Antarctica crashed into Africa some 600 million years ago, destroying the “Mozambique Ocean” and producing a range that actually lives up to its epic name, at over 5,000 by 1,600 miles. In its erosion, it dumped enough sediment to cover all fifty states of the US over six miles deep.

fn31 These are the sorts of geochemical signals that, elsewhere in the fossil record, indicate the largest environmental changes in Earth history—like those that occur in the run-up to Snowball Earth or during mass extinctions, when the Earth system is turned upside down. What’s so strange about the Shuram Anomaly is that (1) it is so long and sustained, (2) it is so far off the map geochemically that there are few mechanisms in the Earth system capable of generating it, and (3) unlike other large “negative carbon isotope excursions” that are associated with planetary catastrophes, the

Shuram Anomaly is instead associated with the beginning of the age of animals.

Chapter 4: CO2 and the Great Age of Coal

fn1 This age of coal is focused in the middle and late Carboniferous Period but bleeds into the succeeding Permian Period as well, stretching from roughly 330 to 260 million years ago.

fn2 After humbly debuting in the fossil record of the age before it, the Ediacaran Period.

fn3 The so-called Great Ordovician Biodiversification Event.

fn4 These can be found in upstate New York about an hour west of Albany, the so-called Gilboa Fossil Forest—the first trees and forests ever. New York would have resembled Bangladesh at the time, in the shadow of what would have been Himalayan-scaled Appalachian Mountains. The fossil trees were discovered by construction workers building a reservoir for New York City in the 1920s; their steam shovels continually jammed on the fossil tree stumps, some 380 million years old.

fn5 Fragments of these 380-million-year-old corals are available on the shores of northern Michigan—the famous, and unusual, Petoskey stones.

fn6 The oxygen might have also encouraged these insects' huge size by supercharging their larval stage. The gigantism might also be partially explained by the fact that this terrestrial world was still all but free from predators.

fn7 The result of fresh coal being exposed to oxygen in poorly ventilated mine shafts and oxidizing all on its own, turning back to CO_2 and water. There was also "whitedamp," when the coal would partially oxidize to carbon monoxide. And chokedamp, firedamp, stinkdamp, and afterdamp—still other chemical varieties produced by this oxidizing coal.

fn8 And to a much lesser extent, wind and water power similarly energized by the sun.

fn9 In the UK, especially, trains were powered by coal, but in the US many of them surprisingly remained wood-fired until later in the nineteenth century.

fn10 The coal forests that once upon a time found themselves closer to the collisions that created the Appalachian Himalayas, like those in eastern Pennsylvania, were partially cooked and transformed over geologic time to hotter-burning anthracite, and had meanwhile found their way down canals and over rails into domestic stoves, and into the Union's industrial war machine—churning out munitions, and the railroads to deliver troops and supplies to the front lines—that made the American Civil War so deadly and foreshadowed the unimaginable destruction of twentieth-century industrial warfare.

fn11 Native claims on Appalachia were recognized by the British Royal Proclamation of 1763 but not by the American Revolutionary elite.

fn12 Ultimately unsuccessfully after locally led protests.

fn13 Though some studies have found that it is similarly potent for global warming, due to "fugitive emissions" of methane from natural gas wells, pipelines, and other infrastructure.

fn14 There's also the oddball ^{14}C, a sprinkling of which is constantly created in the atmosphere when cosmic rays bombard nitrogen atoms, knocking out a proton, and briefly turn them into radioactive carbon. But, unlike other carbon, ^{14}C is unstable and decays back into nitrogen. This is why it's useful for dating archeological artifacts—it disappears at a steady rate within a few thousand years—but is irrelevant on geological timescales. We can ignore it for our purposes.

Chapter 5: CO2 and the Age of Extinction

fn1 When the announcement was met with anger, Shell eventually agreed to name it the Dixie Lee Bryant Lecture Hall, after the MIT scientist.

fn2 Sorry, Cenozoic paleontologists, I know it's not that boring.

fn3 Denoted as "Ma" in academic geology papers, for *mega annum*.

fn4 The Chesapeake Bay crater was originally tied to a minor mass extinction at the boundary of the Eocene and Oligocene Epochs, but it was found to have struck 2 million years too early to be implicated.

fn5 Raup's speculation was relayed to me by the late Purdue University impact modeler Jay Melosh.

fn6 There was also a minor mass extinction near the close of the Carboniferous, called the Carboniferous Rain Forest Collapse. As Pangaea started to assemble, the vast coal forests began to fragment and shrink, and the strange, iconic flora of the Carboniferous disappeared—perhaps profoundly affecting the trajectory of the evolution of animal life on land as well.

fn7 For some reason often popularly represented in the same scenes as *Tyrannosaurus rex*, even though the two were separated by over 200 million years of Earth history and dimetrodons weren't closely related to dinosaurs.

fn8 Yes, the eponym for the Permian Period.

fn9 This inverted trophic pyramid, unique in Earth's history, was made possible only thanks to creatures like dimetrodons' reliance on eating aquatic species—with the abundance of life on land still umbilically tied to its ancestral waters.

fn10 One of its cousins' Latin names, *Titanophoneus*, literally translates to "titanic murderer." Which is a little silly.

fn11 Or more than half a mile.

fn12 We've also come across them much earlier in Earth history as well: the vast eruptions that struck just before the Earth system spun out of control in the lead-up to Snowball Earth were LIPs too.

fn13 When aerobic microbes consume dead organic matter, they use up oxygen (by definition) and produce CO_2. The oldest waters in the deep ocean today are known for their low oxygen and high CO_2, because the organic matter in them has been sitting around for so long that it's had the longest time to be consumed by other organisms.

fn14 Though in some ways a helpful mechanism for carbon burial on an overheating planet.

fn15 Another colorful formulation of paleontologist Paul Wignall.

fn16 In some places the carbonate rock after the mass extinction was not laid down by life at all, but simply precipitated out of the seawater in upward-growing crystalline fans of carbonate, as it did only in the high-CO_2 nightmare interludes of Snowball Earth a half-billion years earlier.

fn17 Tums are literally made of limestone (calcium carbonate), just like many seafloor sediments. In fact, food-grade calcium carbonate is usually mined from ancient marine limestones and chalks.

fn18 Even in the wake of the End-Permian mass extinction, extreme rock weathering on land eventually not only ended the acidification, but wildly overcorrected—swinging violently in the other direction, supercharging the ocean with alkalinity, and decorating the seafloor in bizarre crystal fans of calcium carbonate, reminiscent of the aftermath of Snowball Earth.

fn19 For a deeper dive on all of the Big Five mass extinctions in the age of animals, might I recommend my first book, *The Ends of the World*. For the extinctions of Pangaea in particular, I would direct readers to paleontologist Paul Wignall's thorough guide, *The Worst of Times*.

fn20 Known as the Smithian-Spathian extinction.

fn21 The Carnian Pluvial Event.

fn22 Deep lake, river, and humid swamp deposits briefly take over much of the fossil record during the event, indicating a huge increase in rainfall. But chunks of Triassic deep seafloor today grafted onto Japan, on the other hand, record a brief influx of windblown desert dust from deep in the heart of the supercontinent thousands of miles away, indicating the interior of Pangaea was aridifying.

fn23 For a representative sampling of the diverse croc world on offer in the late Triassic, I would google Carolina Butcher (found outside Raleigh) or *Saurosuchus*, for the carnivores. As for vegetarians, there was everything from the bipedal plant-eater *Effigia* to the pike-shouldered aetosaurs that ambled through the understory from Algeria to Albuquerque.

fn24 Since the End-Cretaceous mass extinction death of the nonbird dinosaurs 66 million years ago, the North Atlantic Igneous Province (56 million years ago) and the Columbia River Basalts (16 million years ago) erupted, both of which drove global warming, but neither of which caused a major mass extinction. We'll encounter these episodes in the next chapter.

fn25 What they sound like, minerals that precipitate out of a water body, like salt, when evaporation dominates.

fn26 Underground salt caverns make for ideal natural gas storage units because their walls are nearly airtight.

fn27 Whose name literally means "Salt City."

fn28 Newell's work picked up on an all-but-abandoned thread of thinking pursued a century earlier by the English geologist John Phillips, who noted tremendous drops in life that seemed to separate the great reigns of the trilobites, dinosaurs, and mammals. Phillips's work in turn drew on the nineteenth-century work of the French naturalist Georges Cuvier, who proposed that the ancien régime of elephants in North America—apparent from mammoth and mastodon fossils strewn across the Midwest—had been swept away in a "revolution," not unlike his country's Bourbon monarchy.

Chapter 6: The Fall of CO2 and the Rise of the Modern World

fn1 At least compared to the alien past worlds we've surveyed so far, which hosted such vastly unfamiliar geography.

fn2 0.004 percent and 0.0001 percent of Earth history, respectively.

fn3 A few species of ammonite actually seemed to have survived up to a few hundred thousand years after the asteroid impact, most famously found in the outcrops of Stevns Klint in Denmark.

fn4 The test was conducted under the aptly named Project Plowshare.

fn5 Don't worry, we won't.

fn6 For one thing, it is very difficult to conceive how a functioning global civilization, consuming terawatts of energy every year, could be maintained for long in the face of such mounting, apocalyptic warming and ecological breakdown. At some point, civilizational collapse would provide a negative feedback to further warming.

fn7 A process that is being researched to sequester CO_2 today, by seeding appropriate areas of the ocean with iron.

Chapter 7: The Rise of the New Metabolism

fn1 "Less than 60 percent of the mass that would be expected for a primate of our body weight," per Wrangham.

fn2 The seal oil and blubber-fired lamps of the arctic.

fn3 Arguably one of the first advances in energy efficiency, as these spearpoints and ax blades concentrated the energy of muscles far more effectively on their target.

fn4 According to Smithsonian paleontologist Nick Pyesnon, mathematical modeling of drag on blue whales over 110 feet long shows they would not have been able to feed effectively and would expend far too much energy doing so.

fn5 Which, despite being one of dozens of interglacials, is somewhat anthropocentrically given its own epoch, the Holocene—or possibly even *two* epochs if the Anthropocene is ultimately recognized by the International Commission on Stratigraphy as well, potentially having begun in 1950. For comparison, other epochs in the geologic record, like, say, the early Cretaceous, are almost 50 million years long.

fn6 Sea otters were also previously decimated in the nineteenth century by the fur trade, but had been recovering over the twentieth century.

Chapter 8: The Long Fuse

fn1 Just why there is no evidence of intensive agriculture in the previous interglacial, which ended 115,000 years ago and hosted modern humans, is the stuff of speculation, but among possible reasons are that the interglacial was somewhat less stable climatically, or that perhaps we had not yet hunted down the food chain enough and so had to invent it.

fn2 Normal caveats apply here about the chemosynthetic bacteria at hydrothermal vents and elsewhere, and rock-eating bacteria, which don't rely on photosynthesis.

fn3 Though wheat, surprisingly, could boom as higher latitudes warm.

fn4 Since this process works to cool trees just like sweating in humans does, at least one biologist claims that "[t]he shade of an old oak tree may not be as cool of a respite as it used to be."

fn5 The so-called C3 pathway (versus the C4 pathway mentioned in chapter 7), which evolved midway through the age of mammals.

fn6 As the Yale anthropologist James C. Scott explains, the definition of statehood can be rather nebulous—more of a continuum than a clearly demarcated category—but “a polity with a king, specialized administrative staff, social hierarchy, a monumental center, city walls, and tax collection and distribution is certainly a ‘state’ in the strong sense of the term.” These are the kinds of societies that would finally arrive beginning some 5,000 years ago in Mesopotamia and thousands of years later in China, Mexico, India, South America, and beyond.

fn7 Excluding densely settled Meso-America.

fn8 Curiously, in matrilineal social species, like orca, the opposite has been seen, with dramatically reduced diversity in the mitochondrial DNA on the female side perhaps explained by similar bottlenecks.

fn9 Mann defines *empire* as a “centralized, hierarchical system of rule acquired and maintained by coercion through which a core territory dominates peripheral territories, serves as the intermediary for their main interactions, and channels resources from and between the peripheries.”

fn10 In a critique of the bestselling book *The Dawn of Everything*, by anthropologist David Graeber and the archeologist David Wengrow, which argues that there was nothing inevitable about the rise of bureaucratic states and that the possibilities of human political organization are still as expansive as they were to the hunter-gatherer-foragers at the dawn of the Holocene.

fn11 Or, rather, were reintroduced—horses evolved in North America and were wiped out in the extinctions at the end of the Pleistocene.

fn12 By doubling down on bison hunting at the expense of gathering, the Comanches saw their diet become deficient in grains—one solution for which was to take New Mexican markets from the Apaches to trade for maize. In the words of the historian Pekka

Hämäläinen, “the Comanche-Apache wars were fought over carbohydrates.”

fn13 As well as in wind power (windmills) and water power (water mills). But even these are ultimately forms of converted solar power, as the sun drives Earth’s winds and weather, which ultimately powers the planet’s rivers.

fn14 Along with the punishing cold snaps of the Little Ice Age, the Thames was also shallower and slower than the modern river and more prone to freezing.

Chapter 9: The Start of the Supereruption

fn1 *Meishan* means “coal mountain” for the coal deposits from millions of years earlier found deep underground.

fn2 As used on the Micronesian Island of Yap. Even if one of the giant discs falls off a boat in transport, it will still retain its value and ownership, and can still be used in transactions even if it remains at the bottom of the ocean.

fn3 Walter Scheidel, citing the political scientist Victoria Tin-bor Hui: “By one count, ‘core China’—defined as the territory held by the Qin state at its peak in 214 BCE—was unified under one ruler for 947 of the past 2,231 years, or 42 percent of the time.”

fn4 Notably the reigns of Charlemagne, Charles V, and later Napoleon, and very briefly Hitler.

fn5 Not quite—they would still go bankrupt in 1596 trying, unsuccessfully, to keep the Netherlands within the empire.

fn6 The so-called Islamic slave trade would not be entirely usurped with the advent of European transatlantic slavery: “11.75 million enslaved sub-Saharan Africans were exported across the Sahara, Red Sea, and Indian Ocean from 650–1900 CE,” writes the historian Rudolph Ware.

fn7 It was 0.5 percent efficient at converting the energy in coal to work, the rest lost to waste heat. Today the average coal plant in the United States operates at just over 30 percent efficiency.

Chapter 10: The CO2 Supereruption

fn1 There are likely others as well, such as massive asteroid impacts vaporizing carbonate platforms and, perhaps less effectively, communism, which was a reaction to—and envisioned as an attempt to democratically seize the productive powers developed by—capitalism and capitalists.

fn2 But the whales' reprieve would be brief: soon oil-powered ships allowed whalers to track down more elusive whales like fin and blue whales, who were too fast for earlier wind-powered whaleships, and surprisingly the most destructive decades for whaling would not arrive for another *century*.

fn3 Perhaps surprisingly, bicycles were invented around the same time.

fn4 Asphalt had been used in Persia for paving roads for over a millennium.

fn5 Then part of the Russian Empire.

fn6 Though the company would (finally) drop Royal Dutch from its name in 2022, there remains today an unbroken line from the pallid illumination of a Shell gas station sign over a strip mall parking lot to the sixteenth-century depredations of the Dutch East India Company.

fn7 As in the Habsburg-led, multiethnic Austria-Hungary dual monarchy, or the Ottoman composite monarchy.

fn8 Barring those sunk by U-boats. Unrestricted submarine warfare was Germany's answer to the blockade.

fn9 Oil-burning ships were relatively stealthy compared to their smoke-belching coal equivalents. They also didn't require coal-steamship armies of stokers and trimmers to keep the engines fed with fuel.

fn10 Though the French also gained a small share of the country's oil as spoils from the defeated Germans' prewar dealings on the region.

fn11 Cushing is why the price of oil briefly went negative at the start of the COVID-19 pandemic, when the global economy froze and the oil in these tanks could not find buyers.

fn12 The list of technologies forged by World War I is extraordinarily extensive and includes submarines, tanks, fighter planes, skin grafts, the widespread use of wristwatches, and, perhaps surprisingly, Pilates.

fn13 The *Yamato* was sunk en route to Okinawa.

Chapter 11: Energy, CO2, and Civilization

fn1 New York City generates 14 million tons of solid waste per year.

fn2 For information theorists who know what the following words mean—a group I can not count myself among—computing an irreversible logical operation on one bit of information releases $3x10^{-21}$ joules of waste heat to the environment.

fn3 Both from warming and from fertilizer runoff and sewage.

fn4 And was also the grandfather of the actor Christopher Lloyd.

fn5 Saudi Arabia plans to take this fusion of oil and mass entertainment one step further by building an actual theme park on a string of connected offshore oil rigs, which the Saudi Public Investment Fund describes as "the world's first tourism destination inspired by offshore oil platforms."

fn6 These emissions are also due to land clearance for agriculture, as well as other greenhouse gases like methane and nitrous oxide produced by livestock and released by agrichemicals.

fn7 This is the mysterious phenomenon afflicting Western economies known as “secular stagnation.”

Chapter 12: The Future of CO2

fn1 And more, capturing energy not only from the sun—whether directly through solar panels, or indirectly through wind and hydropower—but from the Earth, through geothermal and geologic hydrogen, and even from the moon as well, in tidal energy. At the opposite scale, there is nuclear power as well, from the energy of the atom.

fn2 One partial list of the metals needed for current state-of-the-art manufacturing includes Be, B, Sc, V, Ga, Ge, Se, Sr, Y, Zr, In, Te, Cs, Ba, La, Hf, Ta, Os, Tl, Li, Ru, W, Cd, Hg, Sb, Ir, Mo, and Mg.

fn3 The Inca, meanwhile, left their own legacy of copper mining pollution, preserved in the peat bogs of Tierra del Fuego and the ice caps of the Peruvian Andes.

fn4 Globally it’s 99.4 percent.

fn5 0.3 gigatons of refined minerals and 13 gigatons of associated waste rock mined for renewable energy materials, versus 15 gigatons of fossil fuels and 2 gigatons of associated waste rock mined for fossil fuels today, according to the Energy Transition Commission’s *Materials and Resource Requirements for the Energy Transition* (2023).

fn6 Alternatively, proposals to replace fossil jet fuel with low-to zero-carbon biofuel made from plants and algae face a host of challenges, from these fuels’ comparatively very high cost to the vast area of agricultural land that would be required to produce it at industrial scale.

fn7 As with large planes, cargo and container ships cannot be electrified, though they could be powered by fuels synthesized from CO_2 and hydrogen, or landfill gas, with renewable sources of electricity.

fn8 And not dent the planet's oxygen budget at all, contra one popular motif in conservation circles.

fn9 With some possible exceptions, like the use of whale oil for illumination.

fn10 According to the Center for Strategic and International Studies' 2023 report "The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan."

fn11 As recently happened with the Pennsylvania Teachers' Union Pension Fund, as documented by the Bloomberg reporter Javier Blas.

fn12 Homes in Beverly Hills, California, produce an average of 9,480 pounds of CO_2 compared to those in South Central Los Angeles, which produce 2,425.

fn13 The mismatch between when solar is generated and when it is needed, a phenomenon that adds costs and complexity to grid management, and that, when plotted, looks like a duck.

fn14 Serpentinization, described in the first chapter.

fn15 The 500-million-year current eon of animal life and high oxygen.

fn16 There are very good practical and energetic reasons to think fusion as an energy source will not be scalable in anything like the near future, but, as many have noticed, there's a nuclear fusion reactor in the sky that operates and generates energy for free.

fn17 Setting aside calculations, like that by Ohio State University astronomer Paul Sutter, that the energy payoff on such an energy-intensive project would not arrive for tens of thousands of years.